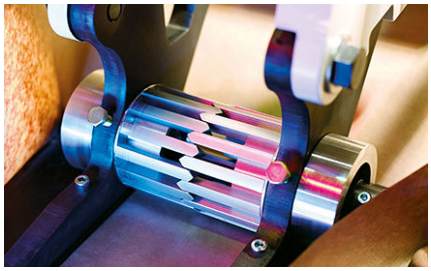
WINDMÖLLER & HÖLSCHER



Original Instruction Manual

Tube former Polytex, Filmatic S Mono winder

62386, 61559

EXTRUSION PRINTING CONVERTING

Windmöller & Hölscher KG | Postfach 1660 | 49516 Lengerich | Deutschland | Tel.: +49 5481 14-0 | Fax: +49 5481 14-2649
info@wuh-lengerich.de | www.wuh-lengerich.de
© Copyright Windmöller & Hölscher KG

Overall table of contents

Version overview

1 General information
1.1 Explanation of the folder structure
1.2 Information on the instruction manual
1.3 Qualification of the personnel
1.4 Information for the operating company
1.5 Phases of machine life
1.6 Contact details

2 Safety

2.1 Machine data
2.1.1 Technical specifications
2.1.2 Name plate

2.2 Important safety instructions
2.2.1 Essential safety instructions
2.2.2 Safety instructions for web handling
2.2.3 Safety instructions for use of the seam extruder
2.2.4 Safety instructions on the winder
2.2.5 Safety instructions on hot surfaces and materials
2.2.6 Hazardous substances
2.2.7 Longer shut-down periods
2.2.8 Residual risks
2.2.9 Equipotential bonding
2.2.10 Internet Remote Service

2.3 Workstations
2.3.1 Workstations

2.4 Safety devices: Overview
2.4.1 Polytex Emergency-Stop
2.4.2 Safety devices Polytex
2.4.3 Safety devices at the winder

2.5 Safety devices: Description
2.5.1 EMERGENCY-STOP
2.5.2 Emergency stop trip wire
2.5.3 Effective range of Emergency-Stop
2.5.4 Main switch
2.5.5 Signaling device
2.5.6 Guard door
2.5.7 Opening and closing the guard door
2.5.8 Winder Stop
2.5.9 Winding station Stop
2.5.10 Light barrier

2.5.11 Warning lamp with horn
2.5.12 Strobe light
2.5.13 Key switch
2.5.14 Safety cut-out mat

2.6 Danger signs
2.6.1 Information on the warning signs
2.6.2 Unwind
2.6.3 Draw and dancer roller
2.6.4 Web processing station with micro-perforator
2.6.5 Web processing and embossing unit
2.6.6 Gusseted tube forming unit
2.6.7 Longitudinal seam extruder
2.6.8 Main draw
2.6.9 Cut edge blocking
2.6.10 Switch boxes
2.6.11 Enclosure

2.7 Residual risks
2.7.1 Winder

2.8 Intended use
2.8.1 Intended use
2.8.2 Intended use
2.8.3 Intended use – Winding shaft
2.8.4 Reasonably foreseeable misuse

3 Machine description

3.1 Polytex
3.1.1 Machine overview
3.1.2 Unwind
3.1.3 Web processing station, outer web – Draw with dancer roller
3.1.4 Micro-perforator, outer web
3.1.5 Continuous embossing unit
3.1.6 Side gusseted tube forming station
3.1.7 Seam extruder
3.1.8 Main draw
3.1.9 Cooling zone
3.1.10 Cut edge blocking
3.1.11 Oscillation

3.2 Winder
3.2.1 Winding principles
3.2.2 Overall view
3.2.3 Functional groups
3.2.4 Vacuum cutover drum
3.2.5 Cut edge blocking
3.2.6 Winding shaft loading system

3.2.7 Center drive
3.2.8 Web guide control

4 Control elements and screen pages

4.1 Operator panels: Overview
4.1.1 Control elements
4.1.2 Polytex Operator panels

4.2 Operator panels: Description
4.2.1 Unwind
4.2.2 Web processing station with micro-perforator and embossing unit
4.2.3 Main draw

4.3 Screen pages
4.3.1 Information on screen pages
4.3.2 Home screen page
4.3.3 Production screen page
4.3.4 Display screen page
4.3.5 Parameters/settings screen page
4.3.6 I/O peripherals screen page
4.3.7 ACOPOS screen page
4.3.8 Touch screen screen page
4.3.9 Module overview screen page
4.3.10 Unwind screen page
4.3.11 Web processing screen page
4.3.12 Longitudinal seam screen page
4.3.13 Heating zones screen page
4.3.14 Temperature profile screen page
4.3.15 Longitudinal seam monitoring screen page
4.3.16 Draw screen page
4.3.17 Oscillation screen page
4.3.18 Cut edge blocking screen page
4.3.19 Alarm list screen page - Log
4.3.20 Enable screen page
4.3.21 Contact screen page

4.4 Screen pages: Operation
4.4.1 Operating concept
4.4.2 Header
4.4.3 Information line
4.4.4 Footer
4.4.5 Entering your password
4.4.6 Touch screen screen page
4.4.7 Setting up the web processing station
4.4.8 Setting up longitudinal seam monitoring
4.4.9 Enabling the software

4.5 Winder

4.5.1 Main control console +R82A for the winder
4.5.2 Screen in control console +R82A/B
4.5.3 Control elements on operator panel +R82
4.5.4 Operator panel +R85A
4.5.5 Control consoles +R84A/B
4.5.6 Pneumatic control elements in the winding station

5 Operating the machine

5.1 Unwind

5.1.1 Unwinder

5.2 Web processing station

5.2.1 Dancer roller
5.2.2 Micro-perforator
5.2.3 Embossing

5.3 Tube forming station

5.3.1 Web edge - Safe behavior
5.3.2 Tensioning the emergency stop trip wire
5.3.3 Guide elements at the infeed section
5.3.4 Side gusset tube forming unit - Infeed rollers
5.3.5 Adjusting the side gusseted tube forming
5.3.6 Adjusting the side gusset depth
5.3.7 Side gusset tube forming unit - Former plates

5.4 Seam extruder

5.4.1 Longitudinal seam extruder
5.4.2 Transporting the frame
5.4.3 Switching on Automatic mode
5.4.4 Refilling resin – Safety instructions
5.4.5 Resin monitoring
5.4.6 Thermocouples – Overview
5.4.7 Thermocouples
5.4.8 Heating elements
5.4.9 Melt pressure sensor
5.4.10 Longitudinal seam monitoring – Drag rope
5.4.11 Longitudinal seam extruder - Collecting tank

5.5 Main draw

5.5.1 Control voltage
5.5.2 Key switch
5.5.3 Machine ON
5.5.4 Jog
5.5.5 Switching on Automatic mode
5.5.6 Setting up the contact pressure
5.5.7 Main draw
5.5.8 Main draw – Compressed air supply

5.5.9 Longitudinal seam monitoring
5.5.10 Seam cooling roller
5.5.11 Replacing the seam cooling roller
5.5.12 Supplying coolant
5.5.13 Nip roll
5.5.14 Cut edge blocking

5.6 Web guide control
5.6.1 Web guide control, ekr
5.6.2 Web guide control - Material setup

5.7 Winder
5.7.1 Control voltage
5.7.2 Change winding shaft
5.7.3 Fixing the winding core
5.7.4 Starting the winder
5.7.5 Web sag on startup
5.7.6 Automatic step sequence during a roll change
5.7.7 Moving the film reel
5.7.8 Film roll delivery
5.7.9 Shut-down sequence when producing tubular film

6 Production

6.1 Messages
6.1.1 Alarms

6.2 Web travel
6.2.1 Safe behavior
6.2.2 Unwinding unit
6.2.3 Micro-perforator, outer web
6.2.4 Continuous embossing unit, outer web
6.2.5 Main draw
6.2.6 Cooling zone

6.3 Seam extruder
6.3.1 Extrusion lines

7 Maintenance instructions
7.1 Scope of maintenance works
7.2 Maintenance staff
7.3 Safety instructions
7.4 Corrosion protection measures
7.5 Corrosion protection

8 Maintenance

8.1 Cleaning the machine
8.1.1 Unwind – Web break detector
8.1.2 Web guide control

8.1.3 Sliding doors
8.1.4 Micro-perforator
8.1.5 Embossing unit
8.1.6 Gusseted tube forming unit
8.1.7 Longitudinal seam extruder
8.1.8 Main draw
8.1.9 Longitudinal seam monitoring
8.1.10 Main draw - Slide protection
8.1.11 Oscillation
8.1.12 Contact roller
8.1.13 Center drive
8.1.14 Cross cutting system
8.1.15 Hydraulic unit
8.1.16 Delivery arms
8.1.17 Electrostatic charging and discharging

8.2 Maintaining the machine

8.2.1 Winding shaft
8.2.2 Unwind
8.2.3 Unwind – Web break detector
8.2.4 Web guide control
8.2.5 Draw and dancer roller
8.2.6 Sliding doors
8.2.7 Micro-perforator
8.2.8 Embossing
8.2.9 Gusseted tube forming unit
8.2.10 Longitudinal seam extruder
8.2.11 Main draw
8.2.12 Main draw - Slide protection
8.2.13 Compressed-air conditioner
8.2.14 Equipment
8.2.15 Parallel guide
8.2.16 Contact roller
8.2.17 Hydraulic unit
8.2.18 Pneumatic system
8.2.19 Air conditioner for the switch cabinet

8.3 Lubricating the machine

8.3.1 Lubricants
8.3.2 Sliding doors
8.3.3 Draw and dancer roller
8.3.4 Embossing
8.3.5 Gusseted tube forming unit
8.3.6 Longitudinal seam extruder
8.3.7 Longitudinal seam monitoring
8.3.8 Main draw
8.3.9 Oscillation

8.3.10 Dancer roller
8.3.11 Center drive
8.3.12 Hydraulic unit

9 Doing maintenance work

9.1 General maintenance
9.1.1 Safe behavior
9.1.2 Safety-relevant components
9.1.3 Emergency-Stop switch
9.1.4 Emergency pull wire switch
9.1.5 Safety light curtain
9.1.6 Safety switch with guard locking
9.1.7 Safety switch without guard locking
9.1.8 Warning devices
9.1.9 Screws
9.1.10 Roller covers
9.1.11 Timing belts
9.1.12 Universal joint shafts
9.1.13 Proximity sensors
9.1.14 Filter
9.1.15 Pneumatic cylinder
9.1.16 Establishing a remote diagnostics connection

9.2 Winder
9.2.1 Replacing the cut-off knife

10 Basic setting
10.1 Longitudinal seam monitoring
10.2 Setting up the seam cooling roller – Setting up the idle stroke
10.3 Setting up the seam cooling roller
10.4 Oscillation

11 Putting the machine out of operation
11.1 Assembly - Safety
11.2 Assembly instructions
11.3 Dismantling
11.4 Recommissioning
11.5 Transportation
11.6 Corrosion protection
11.7 Disposal

12 Spare parts and documents
12.1 PDE - Parts and Documentation Explorer

VERSION OVERVIEW		
Version	Month/year	Remarks
1.0	10/2018	–

1 General information

1 General information

1.1 Explanation of the folder structure....13
Structure of the spine labels....13
Meaning of the pictograms....13

1.2 Information on the instruction manual....14
Contents and availability....14
Target group....14
Signs and symbols used....14
Structure and contents of warnings....14

1.3 Qualification of the personnel....16

1.4 Information for the operating company....16
Responsibility of the operating company....16
Liability and warranty....16
Utility conditions....16

1.5 Phases of machine life....18

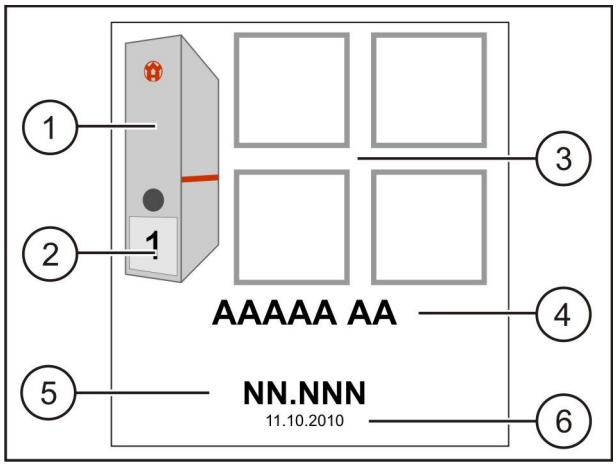
1.6 Contact details....19
Manufacturer's address....19
Service contact....19

1.1 Explanation of the folder structure

Structure of the spine labels

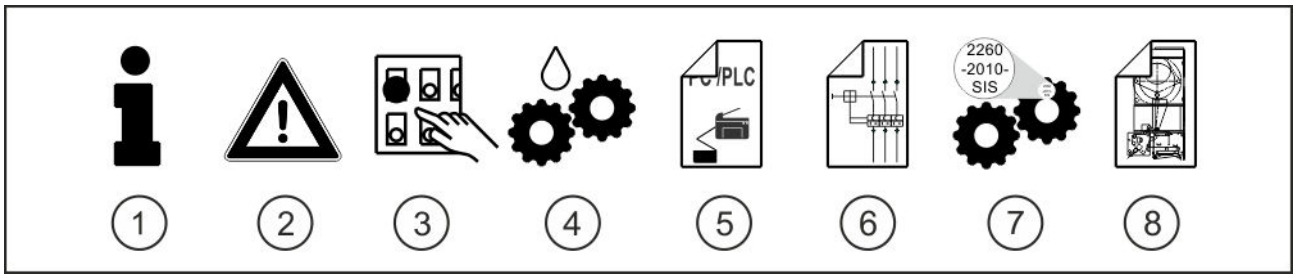
The operating instructions and additional documents are filed in one or several folders. Every folder has a spine label. The spine labels inform you of the folder contents.

(1)	Folder
(2)	Folder number
(3)	Icons; inform you of the folder content
(4)	Machine name
(5)	Equipment number
(6)	Date, as of which the content of the folder is effective



Meaning of the pictograms

The pictograms indicate the contents in the folder. Not all of the pictograms shown below may appear on the spine labels. The pictograms which appear on the spine label depend on the equipment layout.



(1)	General information	(2)	Safety instructions
(3)	Operation	(4)	Maintenance and repair (cleaning, maintenance, lubrication, inspection), repairs, basic settings
(5)	Program printout (PLC printouts, program printouts from the machine computer)	(6)	Circuit diagrams (electrical, pneumatic, hydraulic energies)
(7)	Identification of spare parts	(8)	General machine drawings and sub-assembly drawings

1.2 Information on the instruction manual

Contents and availability

Every person authorized to access the machine must have read the instruction manual. The instruction manual contains the following information:

- description of the machine and the intended use
- instructions for the safe and intended use
- instructions for preventing dangers

The instruction manual may contain descriptions of optional sub-assemblies or functions that are not available on the equipment that you are operating.

The purpose of the instruction manual is:

- to recognize and avert dangers to persons and the machine
- to reduce cost or repairs and downtimes
- to increase the reliability and service life of the machine.

In addition to the instructions given in the instruction manual national guidelines must also be adhered to. The operating company is obliged to complete the instruction manual by adding the national guidelines.

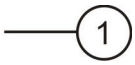
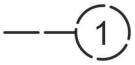
Target group

This instruction manual addresses all the persons allowed to gain access to the machine. The machine must only be operated by personnel trained for this purpose and according to the instruction manual. The instruction manual distinguishes between the following groups of personnel:

- Operating personnel
- Maintenance personnel

Signs and symbols used

Various signs and symbols are used in this instruction manual to structure information.

Symbol	Explanation
	General safety alert symbol; warns you of dangerous situations which may result in personal injury or even dath
	Note; warns you of property damage
	Information
(1)	Position number in the text; refers to a position number in the figure
	Position number in the figure; points at a visible component
	Position number in the figure; points at a hidden component
▶	Action step; signifies an action to be taken
●	Bullet point; signifies a bulleted list without defined sequence
BUTTON	Control elementsignifies, for example, a button or a key switch
'Function'	Designation of a function
'Screen page'	Designation of a screen page
italic	Action result
	Text on a gray background preceding an instruction: requirements Text on a gray background after an instruction: subsequent action

Structure and contents of warnings

The general hazard sign indicates warnings not associated to activities. Warnings during activities are in bold. Warnings are to be observed and followed to prevent dangerous situations from occurring.

Warnings are structured as follows:

- SIGNAL WORD (DANGER/WARNING/CAUTION)
- type and source of hazard
- consequences resulting from non-compliance.
- measures taken to eliminte risks.

The signal word introduces different types of dangers:

- DANGER: There is a high risk. The danger will lead to direct death or serious injuries.
- WARNING: There is a medium risk. The danger may possibly lead to death or serious injuries.
- CAUTION: There is a low risk. The danger may lead to minor or medium injuries.

1.3 Qualification of the personnel

The operating company must clearly define responsible persons and their tasks to be carried on the machine. Personnel to be trained or taught must be supervised. The following skilled personnel are allowed to carry out works on the machine:

- Skilled electrician: Only a skilled electrician or specially trained persons are allowed to work on the electrical equipment of the machine under the direction and supervision of a skilled electrician.
- Hydraulic specialist: Only a hydraulic specialist is allowed to work on the hydraulic equipment of the machine.
- Pneumatic specialist: Only a pneumatic specialist is allowed to work on the pneumatic equipment of the machine.
- Maintenance personnel: Only trained and reliable maintenance personnel are allowed to maintain the machine in compliance with the relevant instruction manual.
- Operating personnel: Only trained and reliable operating personnel are allowed to operate and run the machine in compliance with the relevant instruction manual.

The skilled personnel received appropriate training for the work to be done and know the pertaining local rules and regulations. Thanks to their special training and experience, the skilled personnel are able to carry out works independently and realize and prevent possible dangers.

1.4 Information for the operating company

Responsibility of the operating company

In addition to the general safety instructions, the following instructions for the operating company shall apply:

- Ensure that the machine's working area is clear of unauthorized persons (e.g. visitors) and that they cannot reach into machine areas through barriers while the machine is running.
- Only qualified personnel who have been trained on the machine are to be employed.
- The operating company must ensure that safety instructions such as fire fighting procedures and first aid advice are made available to his employees.
- Check the employees' educational level and provide additional training, if necessary.
- Ensure that skilled personnel supervise the works of unskilled employees (e.g. apprentices, trainees).
- Make personal protective equipment available to the employees. Make sure that the employees wear their personal protective equipment, if necessary.
- Ensure that the instruction manual is always kept near to hand at the application site of the machine.
- Check all safety devices on the machine for proper functioning at regular intervals and according to the manufacturer's instructions. If the safety devices do not function properly, the machine should be put out of operation immediately until the safety devices have been properly repaired.

Liability and warranty

The manufacturer's liability and warranty should be deemed to exclude the following:

- constructive modifications and improper repair and maintenance are not allowed
- the operating company is not allowed to install spare parts unless specified by the manufacturer
- improper use of the machine

Utility conditions

When placing the order, the operating company will receive data regarding utility and operating conditions for proper use of the machine. These conditions include requirements for proper use such as, for example, the ambient temperature. The utility and operating conditions must be heeded to ensure that the machine is kept in a technically perfect condition.

1.5 Phases of machine life

Phase of life	Description
Construction	The construction of the machine components takes place at the manufacturer's premises. This phase of life is not covered by the instruction manual.
Assembly	The assembly of the machine components takes place at the manufacturer's premises. This phase of life is not covered by the instruction manual.
Transportation	Transportation of the machine is carried out by the manufacturer or a company put in charge. This phase of life is not covered by the instruction manual.The installation and the connection to the power supply networks is carried out in an area confined by the machine operating company by personnel specially trained and authorized by the manufacturer for this purpose. The subsequent in-plant transportation of the machine and the connection to the power supply networks may only be conducted with strict adherence to the safety instructions by personnel specially trained and authorized by the machine operating company.
Initial commissioning	This phase of life is not covered by the instruction manual.The initial commissioning of the machine is carried out in an area confined by the machine operating company by personnel specially trained and authorized by the manufacturer for this purpose. During this phase, it often occurs that protective devices, covers or safety devices are not yet available or active.
Commissioning	The commissioning of the machine is carried out in an area confined by the machine operating company by personnel specially trained and authorized by the manufacturer for this purpose. During this phase, it often occurs that protective devices, covers or safety devices are not yet available or active.
Setup	The machine setup/make-ready may only be carried out by personnel specially trained and authorized by the machine operating company.
Production	During this phase of life, the safety devices are operative. Works may only be conducted according to the instruction manual with strict adherence to the safety instructions by personnel specially trained and authorized by the machine operating company.
Cleaning	Unless explicitely otherwise described in the instruction manual, de-energize and de-pressurize the equipment before cleaning. Cleaning works may only be conducted according to the specifications in the instruction manual with strict adherence to the safety instructions by personnel specially trained and authorized by the machine operating company.
Maintenance	Maintenance works go beyond the scope of routine maintenance. Repair work should only be carried out in an area confined by the machine operating company by personnel specially trained and authorized by the machine operating company.
Routine maintenance	Unless explicitely otherwise described in the instruction manual, de-energize and de-pressurize the equipment before doing routine maintenance work. Routine maintenance work may only be conducted according to the specifications in the instruction manual with strict adherence to the safety instructions by personnel specially trained and authorized by the machine operating company.

Phase of life	Description
Troubleshooting	Unless explicitely otherwise described in the instruction manual, de-energize and de-pressurize the equipment before troubleshooting. Troubleshooting may only be conducted according to the specifications in the instruction manual with strict adherence to the safety instructions by personnel specially trained and authorized by the machine operating company.
Putting the machine out of operation (applies for printing machines only)	The machine is put out of operation by the machine operating company. We recommend to contact the manufacturer before putting the machine out of operation.
Dismantling	Machine dismantling is carried out by the machine operating company. We recommend to contact the manufacturer before dismantling the machine.
Recommissioning	The recommissioning of the machine is carried out in an area confined by the machine operating company by personnel specially trained and authorized by the machine operating company for this purpose.

1.6 Contact details

Manufacturer's address

Windmöller & Hölscher KG

Münsterstraße 50

49525 Lengerich

Germany

Switchboard: +49 (0) 5481 14 – 0

E–Mail: info@wuh-group.com

Home page: www.wuh-group.com

Service contact

	GBD (Division Printing and Finishing)	GBV (Division Converting)	GBE (Division Extrusion)
E-Mail	s1-idz@wuh-group.com	s2-idz@wuh-group.com	s3-idz@wuh-group.com
24-hour service hotline	+49 (0) 5481 – 14 3333		

2 Safety

2.1 Machine data
- 2.1.1 Technical specifications
- 2.1.2 Name plate

2.2 Important safety instructions
- 2.2.1 Essential safety instructions
- 2.2.2 Safety instructions for web handling
- 2.2.3 Safety instructions for use of the seam extruder
- 2.2.4 Safety instructions on the winder
- 2.2.5 Safety instructions on hot surfaces and materials
- 2.2.6 Hazardous substances
- 2.2.7 Longer shut-down periods
- 2.2.8 Residual risks
- 2.2.9 Equipotential bonding
- 2.2.10 Internet Remote Service

2.3 Workstations
- 2.3.1 Workstations

2.4 Safety devices: Overview
- 2.4.1 Polytex Emergency-Stop
- 2.4.2 Safety devices Polytex
- 2.4.3 Safety devices at the winder

2.5 Safety devices: Description
- 2.5.1 EMERGENCY-STOP
- 2.5.2 Emergency stop trip wire
- 2.5.3 Effective range of Emergency-Stop
- 2.5.4 Main switch
- 2.5.5 Signaling device
- 2.5.6 Guard door
- 2.5.7 Opening and closing the guard door
- 2.5.8 Winder Stop
- 2.5.9 Winding station Stop
- 2.5.10 Light barrier
- 2.5.11 Warning lamp with horn
- 2.5.12 Strobe light
- 2.5.13 Key switch
- 2.5.14 Safety cut-out mat

2.6 Danger signs
- 2.6.1 Information on the warning signs
- 2.6.2 Unwind
- 2.6.3 Draw and dancer roller
- 2.6.4 Web processing station with micro-perforator
- 2.6.5 Web processing and embossing unit
- 2.6.6 Gusseted tube forming unit

2.6.7 Longitudinal seam extruder
2.6.8 Main draw
2.6.9 Cut edge blocking
2.6.10 Switch boxes
2.6.11 Enclosure

2.7 Residual risks
2.7.1 Winder

2.8 Intended use
2.8.1 Intended use
2.8.2 Intended use
2.8.3 Intended use – Winding shaft
2.8.4 Reasonably foreseeable misuse

2.1 Machine data

2.1.1 Technical specifications....23
Tube former Polytex....23
FILMATIC S Film Winder....23

2.1.2 Name plate....25

2.1.1 Technical specifications

Tube former Polytex

In the temperature range from >40 °C to 55 °C a power reduction is possible, as well as a reduction of the service life of components outside of the switch cabinet.

		min	max
Ambient conditions			
Temperature	°C	16	55
Relative humidity, at 26 °C, no condensation	%		90
Roll temperature during the processing	°C	+15	–
Web speed	m/min	–	150
The actual machine performace that can be achieved depends on the quality of the materials used, the product size and design as well as the connected units.			
Web speed during roll change	m/min	–	150
Roll diameter Unwind	mm	–	1500
Roll diameter Rewind	mm	–	1500
Flat web width	mm	–	1430
Size range Gusseted tube forming table (depending on the size choice)			
Gusseted tube width	mm	200	550
Side gusset depth	mm	30	130
Materials to be processed for the inner / outer layer (deviating material and material thickness need technical clarification)			
Material thickness	µm	80	200
Lay-flat film made of LDPE or compounds, e.g. 80% LDPE with 20% LLDPE as well as contents of HDPE or MDPE and PP compounds.			
Static friction	µS	> 0.35	
Sliding friction	µD	> 0.3	
Energy requirements			
Electrical energy (Port)	KVA	138	
Compressed air (6 bar)	approx. m3/h	5 - 12	
Emission sound pressure level L pfA1) = 83 dB(A) 1) Complies with DIN EN ISO 3746 (2011-03, page 20, chapter 8.3.4) The emission noise level is calculated according to the enveloping surface method by means of which the A-weighted emission noise level is averaged over the measuring surface. The position of the test points can be taken from the drawing of the measurement report. Measuring is carried out by taking into account the deduction of extraneous noises and the effect of the measurement environment with the machine in an operating state complying with DIN EN 13023 (2010-08, page 36, chapter J5). The measurement uncertainty is 4 dB. The specification is binding only if no sub-vendor components are integrated in the machine.			

		min	max
For further technical specifications, please refer to the installation drawing, the foundation load drawing and the cable routing diagram.			

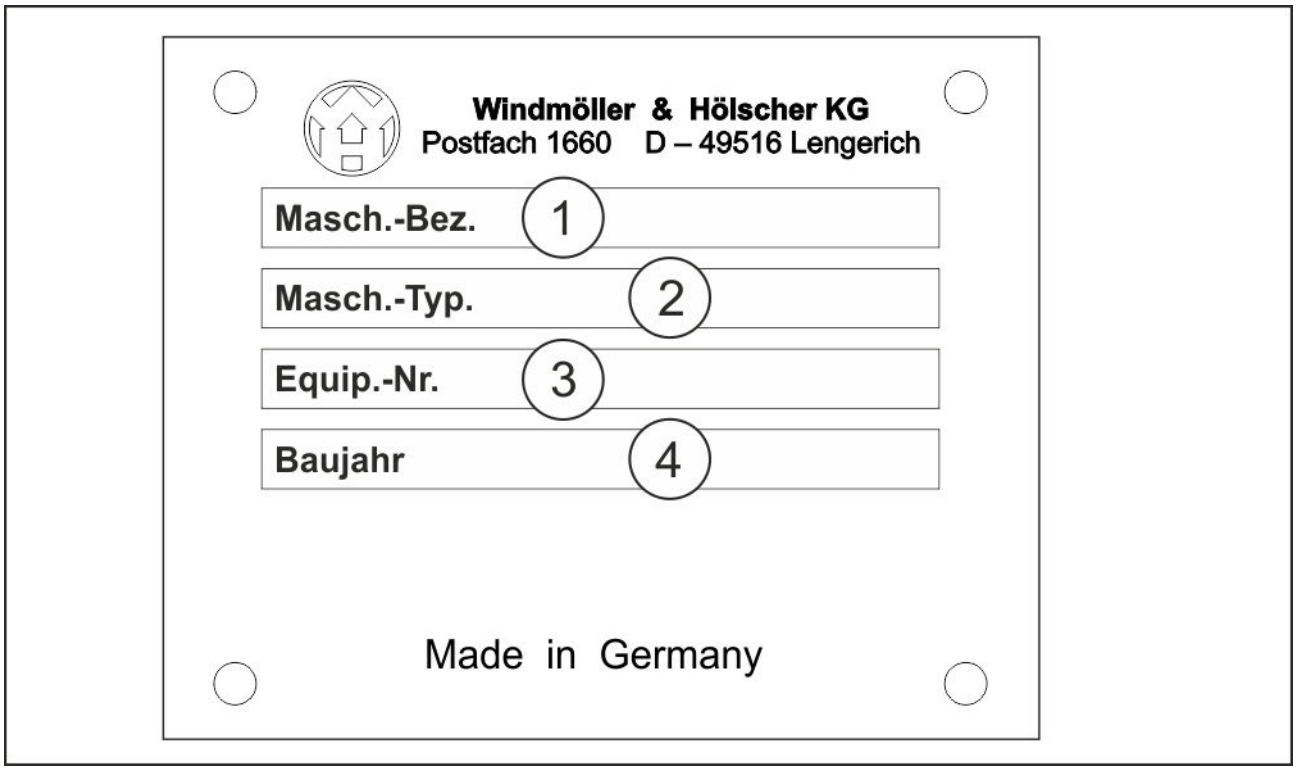
FILMATIC S Film Winder

Line speed, max.	m/min	150
Film width	mm	260 – 720
Material thickness	µm	80 - 200
Roll diameter, max.	mm	1,500
Roll weight, max.	kg	2,500
Web tension prior to the contact roller	N	30 - 500
Taper tension ratio in the contact roller drive section at max. roller diameter	%	0 - 50
Contact force between the contact roller and the film roll	N	100 - 1,000
Taper pressure ratio of the film roll at max. roll diameter	%	0 - 50
For further technical specifications, please refer to the installation drawing, the foundation load drawing and the cable routing diagram. All specifications solely apply for Windmöller & Hölscher's scope of supply and are product-dependent.		

2.1.2 Name plate

NOTE

The details given on the name plate are important for ordering spare parts. Windmöller & Hölscher can only supply the correct spare part if the machine type and equipment number are known.



(1)	Machine model	(2)	Machine type
(3)	Equipment number	(4)	Year of construction

2.2 Important safety instructions

2.2.1 Essential safety instructions....27
2.2.2 Safety instructions for web handling....29
2.2.3 Safety instructions for use of the seam extruder....29
2.2.4 Safety instructions on the winder....30
2.2.5 Safety instructions on hot surfaces and materials....31
2.2.6 Hazardous substances....31
2.2.7 Longer shut-down periods....32
2.2.8 Residual risks....32
2.2.9 Equipotential bonding....33
2.2.10 Internet Remote Service....34
Safety instructions....34
Operating company's safety and supervisory obligations....34

2.2.1 Essential safety instructions

The machine represents the state of the art and meets the latest safety standards. However, in its use, function-related dangers may result to life and limb of the user or third-parties. The machine and other property values may be damaged. Read the following safety instructions before using the machine and follow them when operating the machine. The safety instructions apply for all persons that have access to the machine, are working on the machine or are responsible for works on the machine.

Mortal danger due to unauthorized switching on and starting up the machine. Unauthorized switching on the machine may lead to severe injuries or even death. Disconnect the machine from the energy sources via the main switch and shut-off valve (electric, pneumatics, hydraulic). The main switch and shut-off valves must be padlocked in place after it has been switched off in order to prevent unauthorized switching on. Empty any memory before beginning the maintenance work.

Mortal danger from electric energy. Working on the live machine parts may lead to severe injuries or even death. Carry out work on the machine only when the respective machine parts have been switched to zero potential. Check that machine parts have zero potential, then earth and short-circuit them. Pay attention to any potentially stored energy (capacitors, batteries, uninterrupted power supply). Pax attention to external voltages from external current circuits. If neighboring machine parts are under voltage, isolate the neighboring parts.

Mortal danger due to defective electric lines. Defective electric lines may lead to severe injuries or even death. Replace defective electric lines immediately.

Mortal danger due to pneumatic and hydraulic energy. Working on the pneumatic or hydraulic machine parts may lead to severe injuries or even death. Carry out work on the machine only when the respective machine parts have been switched pressureless. Pay attention to any potential energy stored in pressure accumulators or in the lines.

Mortal danger due to processes running automatically behind the safety devices. Remaining behind closed safety devices may lead to severe injuries or even death. Do not remain behind closed safety devices. Never bypass any of the safety devices. Never bypass locked safety devices. Never close safety devices when persons are behind these safety devices.

Mortal danger due to escape routes not being present or are blocked. Paths, escape routes and emergency exits must always be kept free so that they can be used at all times.

Risk of injury due to moving machine parts. Moving machine parts may pull in and crush persons in the machine. Never reach into the machine when it is running. Wear tight-fitting clothes in the working area of the machine. Persons with long hair must wear a hair net in the working area of the machine. Do not wear jewelry in the working area of the machine.

Risk of injury due to magnetic fields for persons wearing a pacemaker, defibrillators or metallic implants. Magnetic fields are generated in the area of electric components that are operated using high energy or contain strong permanent magnets. These dangerous areas with magnetic fields are, amongst others, on torque drives, magnetic transport systems or electric drives. The magnetic fields may impair the function of pacemakers, defibrillators or metallic implants and endanger persons wearing such devices. Persons wearing pacemakers, defibrillators or metallic implants may not access the safety area during operation and must maintain a safety distance of at least one meter to these components. Persons wearing pacemakers, defibrillators or metallic implants may not access the safety area of components with permanent magnets at any time, even if the machine is not being operated.

Cutting hazard due to sharp blades. Sharp blades may injure persons. Before working on and in the proximity of cutting assemblies, cover the blade with a blade cover and wear cut-proof protective gloves.

Risk of injury due to unsuitable work equipment. Unsuitable clothing, shoes or work equipment may lead to injuries. Use the personal protective equipment were necessary or when required by regulations. Personal protective equipment include, amongst others: work protection clothing, safety goggles, safety helmet, safety shoes, face protection, breathing mask, safety groves.

Risk of injury due to residual risks in the working area of the machine. Residual risks may be present in certain areas of the machine that may lead to hazards and injuries if not operated correctly. These residual risks are marked by warnings on the machine. Observe all warnings on the machine. Keep all warnings legible, clean and intact.

Risk of injury caused by incorrect use of the machine. Failure to use the machine as intended may result in dangers for the persons and in damages to the machine. Use the machine only in a technically flawless state and according to its intended use. Read all operating instructions carefully and follow the instructions.

Risk of injury due to non-functional safety devices. If the safety devices do not function correctly, there is a hazard for persons. If the safety function malfunctions, stop the machine immediately and report to the person responsible. Operate the machine only if the safety device functions correctly. Check the function of the safety devices at regular intervals. Perform a function test after replacement of safety devices. Do not manipulate safety devices.

Risk of injury due to unsuitable climbing aids. Persons may fall from unsuitable climbing aids and injure themselves severely. Use only climbing aids and work platforms intended for this purpose that meet the safety requirements when working above head-height. Ladders, gangways and platforms should only be loaded according to their stipulated specifications. Do not use machine parts as climbing aids. Do not climb on machine parts. Wear safety equipment when working at great heights. When climbing, always face the ladder when ascending and descending, holding onto the ladder with both hands. For ladders with platforms, place objects brought along on the platform and push to the inside so that they cannot fall off the platform. Do not move mobile climbing aids or platforms when persons are located on this. Keep all handles, steps, handrails, pedestals, platforms and ladders free from soiling. Check climbing aids for integrity at regular intervals.

Tripping hazards and falling hazards. Due to confined spatial conditions, soiling or wet floors or incorrectly routed cables and hoses, there is a risk of tripping and falling that may lead to contact with dangerous machine parts and to injury. Always keep floors clean and dry. Route cables and hoses in the channels and platforms intended for this purpose. Avoid restricted entrances. Use personal protective equipment.

Risk of injury due to leaking hoses and lines. Fluids can escape from leaks and lead to short-circuits, slipping points or scalding to skin. Check hoses for leaks at regular intervals. Replace leaking hoses.

Risk of injury and material damage from machines not supervised. If only the control voltage of the machine is switched off, the machine does not have zero potential. Parts of the machine are still live and the machine must be supervised. If the machine is not supervised, functions running automatically may cause material damage that could lead to personal injury. At the end of operation, disconnect the machine from the power supply at the main switch. If the machine is not switched off at the main switch at the end of operation, continue to have the machine supervised by specialist personnel.

2.2.2 Safety instructions for web handling

Risk of injury from electrostatic charging of the web. Strong electrostatic charging of the web may occur. There is a risk of personal injury from electric shocks caused by electrostatic charging which may result in uncontrolled reactions. Keep a safe distance.Deioniziing devices and suitable earthing must be provided, if necessary. Persons fitted with pacemakers or defibrillators are not allowed to operate or work on the machine unless the have consulted their doctor and/or the manufacturer of defibrillators/pacemakers.

2.2.3 Safety instructions for use of the seam extruder

Risk of injuries and burns from escaping and dropping melt. Escaping melt may have temperatures of more than 300 °C and will cause burns to your skin when getting in contact with the hot melt. Persons may be injured from dropping melt. Wear personal protective equipment. Pay attention to the warning signs on the machine.

Health hazard from escaping melt vapors. Escaping vapors may endanger your health. Make sure that the area for the operator to stay is sufficiently ventilated. Make sure that the time spent near to the melt is minimized. Pay attention to the safety data sheets of the products when starting production.

Risk of slipping from resin spilled on the floor. Risk of injury from resin spilled on the floor causing the operator to slip and fall. Remove resin spilled on the floor.

Fire hazard from inflammable materials getting in contact with hot melt. There is a fire hazard and a risk of personal injury from escaping melt getting in contact with inflammable materials and fluids. Never let melt get in contact with fluids and materials having an ignition temperature lower than that of the melt.

Fire hazard from inflammable materials getting in contact with hot machine surfaces. Inflammable materials and fluids may ignite themselves when getting in contact with hot machine surfaces and cause burns to the operator. Never let hot machine parts get in contact with materials and fluids having an ignition temperature lower than that of the hot machine parts.

2.2.4 Safety instructions on the winder

Risk of crushing and being caught in the nip of rotating rollers. There is a risk of crushing and getting caught in the nip of rotating rollers. Keep a safe distance from the rollers. Do not reach in-between rotating rollers.

Risk of being caught in the infeed nips of the rollers. There is a risk of getting caught or crushed in the infeed nips of the rollers (the place where the web is fed onto the rollers). Keep a safe distance from the rollers. Do not reach inbetween rotating rollers.

Risk of getting caught in the nip of the surface winder. There is a risk of getting caught and crushed between the film reel and the contact roller. Keep a safe distance. Do not reach into the machine.

Risk of being caught in the nip on the turret winder. There is a risk of getting caught and crushed between the film reel and the nip roll. Keep a safe distance. Do not reach into the machine.

Risk of crushing from the swiveling movements of the nip rolls. There is a risk of crushing when nip rolls are swung onto various rollers. Keep a safe distance from the rollers. Do not reach into the machine.

Risk of crushing on the surface winder during a roll change. There is a risk of crushing from the delivery arms swiveling out when performing a roll change. Wear industrial safety shoes. Pay attention to the "Risk of crushing" warning signs on the machine. Do not reach into the machine. Keep a safe distance from the delivery area.

Risk of crushing on the turret winder during a roll change. There is a risk of crushing from the turning station when swiveling out with the roll during a roll change. There is also a risk of crushing when setting down the film reel. Pay attention to the "Risk of crushing" warning signs on the machine. Do not reach into the machine. Wear industrial safety shoes. Keep a safe distance from the delivery area.

Risk of injury from electrostatic web charging. Strong electrostatic web charging may occur. Risk of personal injury from electric shocks caused by electrostatic charging which may result in uncontrolled reactions. Keep a safe distance. Deioniziing devices and suitable earthing must be provided, if necessary. Persons fitted with pacemakers or defibrillators are not allowed to operate or work on the machine unless they have consulted their doctor and/or the manufacturer of defibrillators/pacemakers.

2.2.5 Safety instructions on hot surfaces and materials

Risk of burns from hot machine parts and hot materials. The machine parts and used material may be heated up intensely and cause severe burns when touched. Dangerous areas where hot temperatures occur are marked with a "hot surface" danger sign. Be careful when handling hot fuels and auxiliary materials. Be careful when staying near heated machine parts. Wear personal protective equipment. Put up a sign at the installation site of the machine indicating the general risk of burns. Position the sign so that all persons with access to the machine can easily see the label.

In the event of a stop, supply system failure or Emergency-Stop, contact between hot surfaces and highly flammable material may cause a hazard. Provide suitable extinguishing media for fire fighting in accordance with local regulations in the machine area. Instruct the personnel on fire-fighting safety measures.

Risk of fire caused by open ignition sources, cable fire, short-circuit or overloading of electrical components.

Perform regular visual inspection of the electrical components.

Fires caused by a machine cause a risk of serious personal injury or property damage, as a fire can destroy not only the machine itself, but also surrounding equipment and buildings.

2.2.6 Hazardous substances

The term "Hazardous substances" covers the following substances:

- Oils
- Greases
- Chemical substances
- Gases
- Aggressive cleaning agents
- Nitrogen (with optional nitrogen feed)
- PIB (with optional PIB feed)
- Solvents
- Inks

- Always abide by the safety data sheets and safety instructions that apply to the product when handling hazardous substances. The safety data sheets can be requested from the supplies, if necessary. Wear personal safety equipment.
- Skin contact with gearbox oil and other lubricants is harmful to health, may cause removal of natural fat from the skin and skin inflammation. Wear suitable protective gloves when doing maintenance work. Prevent skin contact with lubricants. Use preventive skin protection.
- Hazardous substances escaping from leaky lines, hoses or threaded joints may cause air displacement, injuries or fires. Check all lines, hoses and threaded joints for leaks and damages periodically. Immediately replace hoses and lines when damaged.
- Dusts and gases can harm your respiratory tract. Wear an appropriate breathing mask when working on the machine. Wear suitable eye protection if there is a strong build up of dust. Make sure that the area for the operator to stay is sufficiently ventilated.
- The operating company must ensure that the inks, solvents and cleaning agents are disposed of correctly in order to protect the environment!

2.2.7 Longer shut-down periods

WARNING

Accidents and damages may occur from the machine if not monitored.

There is an increased risk of accidents and damages when switching on a machine that has not been monitored during longer shut-down periods.

Applies for longer shut-down periods:

Switch the control voltage off.

Disconnect the power supply. Switch the main switch in switch cabinet +E25 to "0".

Close the shut-off valve of the compressed air supply to the machine.

2.2.8 Residual risks

A residual risk is a hazard that despite meeting all measures, is an unavoidable non-apparent risk that results from using the machine.

BEWARE

Injuries can be caused by moving machine parts

Hands may be crushed or sheared off. Never reach between moving system parts. Keep a safe distance.

Injuries due to rotating chains, belts, gears and wheels

Parts of your body, clothes, hair and objects may be pulled in and crushed. Do not open or remove protective devices. Always wear tight-fitting clothes and a hair net. Work without industrial safety gloves Keep a safe distance.

Injuries can be caused by rotating rollers

Parts of your body, clothes, hair and objects may be pulled in and crushed. Keep your hands away from the roller gap! Keep a safe distance.

Injuries may be caused by the emerging webs

Parts of your body, clothes, hair and objects may be pulled in and crushed. Always wear tight-fitting clothes and a hair net. Work without industrial safety gloves Keep a safe distance.

Display safety information in places that have not been identified as sources of danger and ensure that special care is always taken.

Watch one of the locations identified by the safety information during production so that you can recognize a source of danger before touching the machine.

2.2.9 Equipotential bonding

WARNING

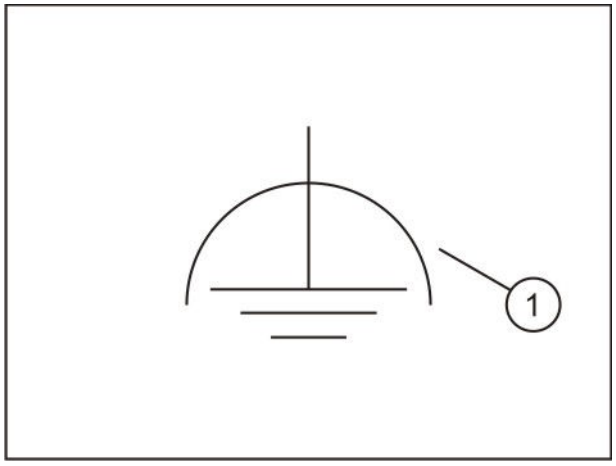
Risk of injuries from electric shocks if live components are touched

Equipotential bonding is required for the machine to prevent its components from having applied to them different electrical potentials. Whenever such unequal electrical potentials are connected with each other, e.g. through tools used on the machine, current flow is liable to take place and sparking may occur.

Make sure the local regulations for equipotential bonding are observed.

The electrical components are connected by the protective earth conductor as well as with all conductive elements forming part of the machine. Never use the protective earth conductor in the sense of the machine's equipotential bonding.

(1)	Connecting point marks for equipotential bonding



- All of the threaded holes marked with the symbol (1) are provided for connection of the equipotential bonding system on site by means of copper wire.

  Copper wire range: 16 mm^2
  The resistance to earth should be less than 10^6 Ohm.

  When using a different material, it is essential that its conductivity is equal to that of a 16 mm^2 copper wire.

  The symbol also appears in the installation drawing.

2.2.10 Internet Remote Service

Safety instructions

DANGER

Non-explosion proof equipment

Risk of explosion from non-explosion proof equipment.

Do not use the service notebook in the machine's explosion-risk area.

DANGER

Remote-controlled machinery

Using remote-controlled machinery can cause severe injuries. Using remote-controlled machinery can cause damage to the machine.

Keep continuous phone contact to the accessing person during remote access.

DANGER

Software or hardware changes on the machine

Defect software or hardware changes can injure persons and cause damage to the machine.

Test the software or hardware changes during phone contact.

Operating company's safety and supervisory obligations

The INTERNET REMOTE SERVICE mode serves as a service mode only and cannot be regarded as the normal production mode. It is imperative for the operating company to fulfill his safety and supervisory obligations.

- The operating company must fulfill his supervisory obligations.
- The operating company must ensure that specially qualified personnel take care of the machine.
- Before activating INTERNET REMOTE SERVICE mode, make sure that all safety devices are in an orderly and proper condition and that they are in a state for which they were designed.
- It is forbidden for persons to stay in the machine's danger zone while INTERNET REMOTE SERVICE mode is running.
- Any actions carried out in connection with the INTERNET REMOTE SERVICE mode generally require the momentary individual approval of the operating company.
- No other than reading activities are allowed when the machine is running (production run, maintenance run or similar). Generally, any altering access to the machine's control system is forbidden.
- It must be possible to shut down all machine/line activities by means of the EMERGENCY-STOP mode.
- Alterations to the control software or process interferences are allowed only under the supervision of the operating company and when the machine is at a standstill.

2.3 Workstations

2.3.1 Workstations....36
Polytex....36
Workstations....36

X000038391

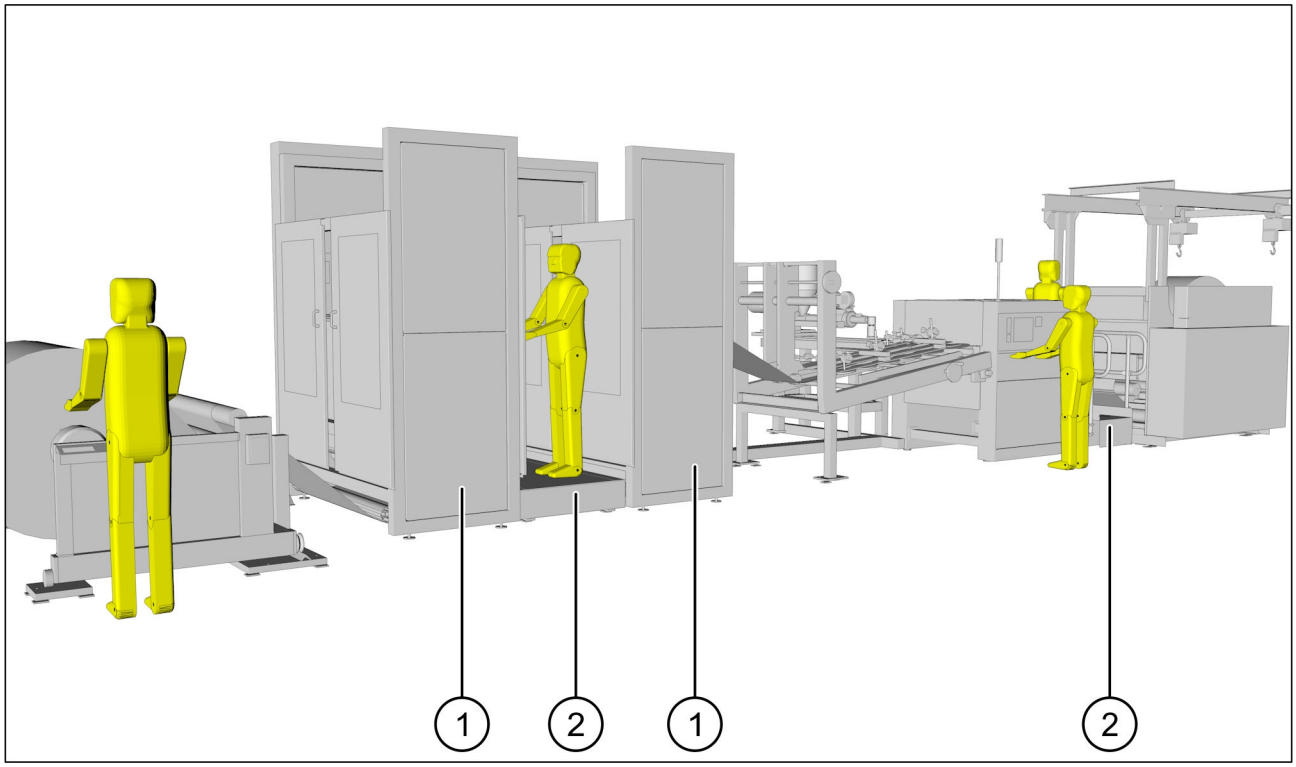
2.3.1 Workstations

Polytex

Adopt a position on the stepladder between the web processing stations for observing and adjusting web processing.

The web travel prevents the observation or adjustment of web processing from a position upstream of the web processing stations.

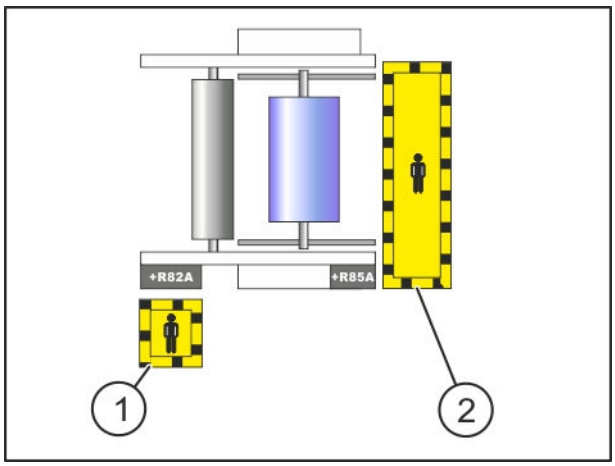
Adopt one of the illustrated positions for observing the web processing stations.



(1)	Web processing	(2)	Stepladder

Workstations

(1)	Workstation +R82A operator panel
(2)	Workstation Winding station



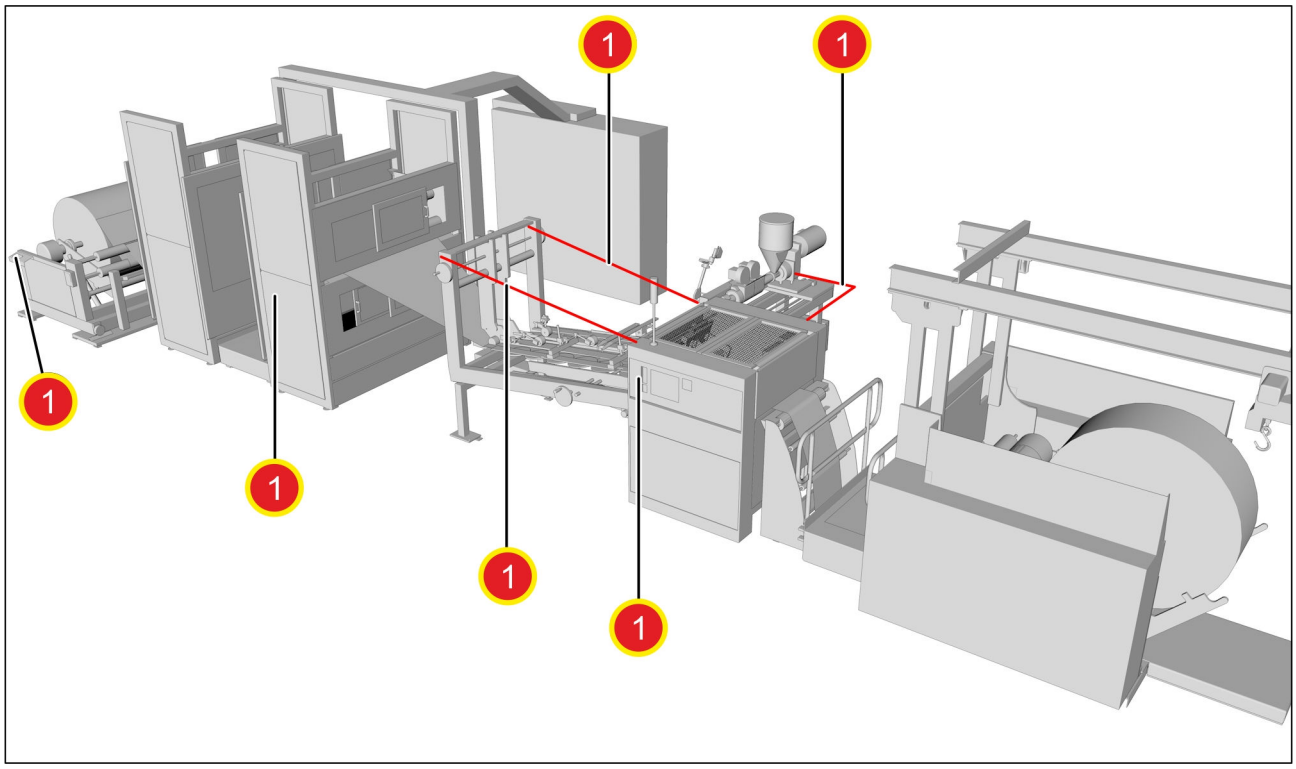
2.4 Safety devices: Overview

2.4.1 Polytex Emergency-Stop 39

2.4.2 Safety devices Polytex 40
Guard doors 40
Safety chains 40

2.4.3 Safety devices at the winder 42

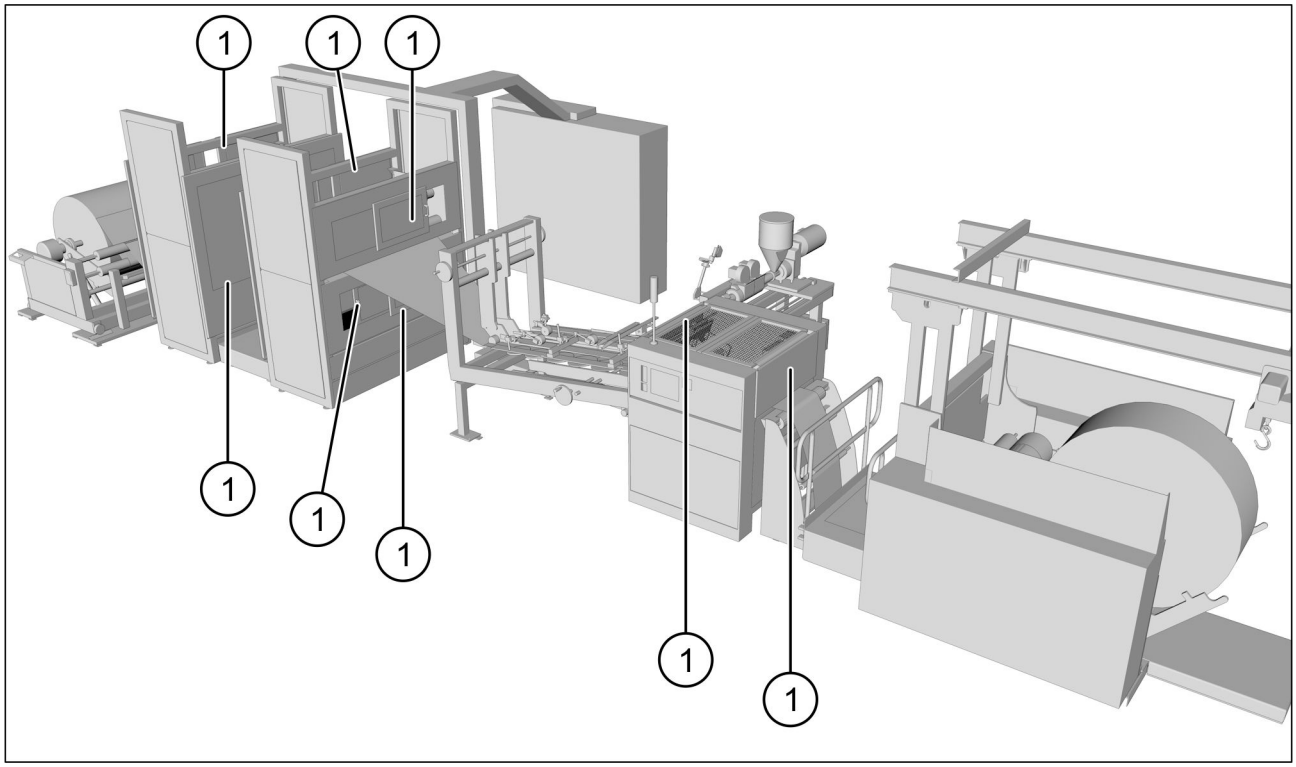
2.4.1 Polytex Emergency-Stop



(1)	EMERGENCY-STOP switch or trip wire or safety shutdown board with Emergency-Stop function		

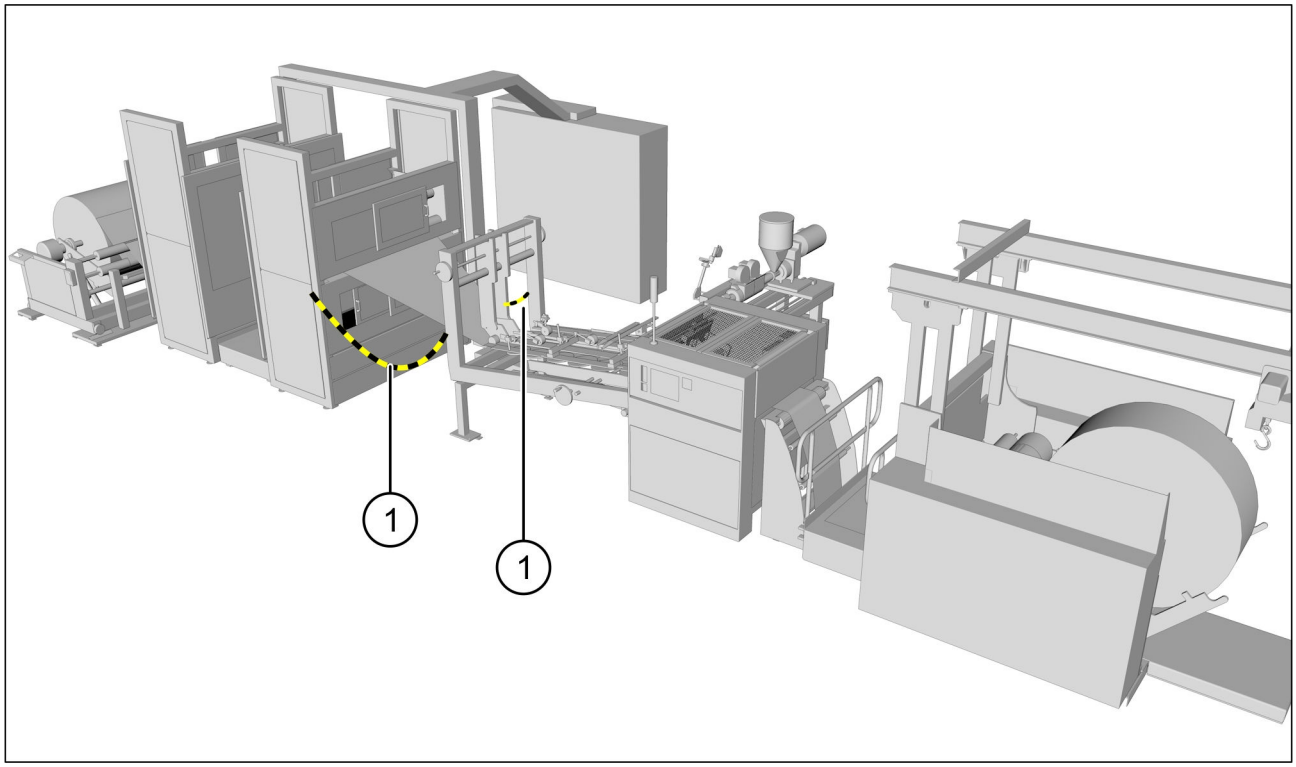
2.4.2 Safety devices Polytex

Guard doors



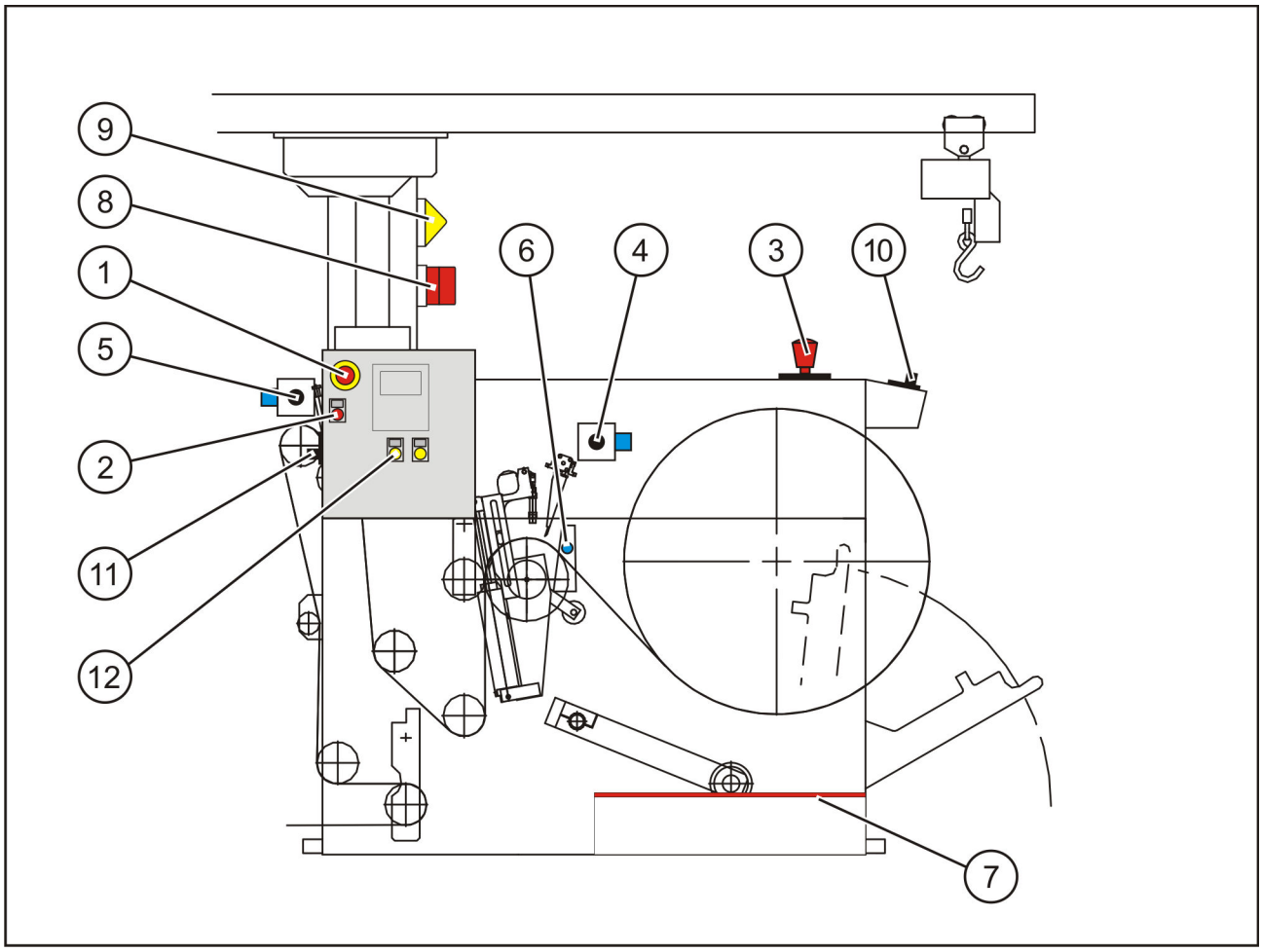
(1)	Sliding doors as guard doors with interlocking. The sliding doors cannot be opened while the machine is running.		

Safety chains



(1)	Safety chain		

2.4.3 Safety devices at the winder



(1)	EMERGENCY-STOP switch in the main operator panel +R82A/B	(2)	WINDING STATION STOP button in the main operator panel +R82A/B
(3)	WINDING STATION STOP button on the side frame	(4)	Emergency trip wire providing a 'Winding station Stop' function at the winding roll infeed nip
(5)	Emergency trip wire providing a 'Winding station Stop' function on the slitting assembly	(6)	Light barrier above the winding roll infeed nip between the contact roller and the film roll provided with the 'Winding station Stop' and the 'Open nip' function
(7)	Safety shutdown mat to safeguard the infeed nip within the contact roller area	(8)	Signal lamp, orange
(9)	Signal lamp, yellow	(10)	Key switch to select the 'Startup' or 'Production' mode in operator panel +R85A/B
(11)	Key switch to safeguard the winding station in the main operator panel +R82A/B	(12)	BYPASS LIGHT BARRIER button in the main operator panel +R82A/B

2.5 Safety devices: Description

2.5.1 EMERGENCY-STOP....0

2.5.2 Emergency stop trip wire....45
Safe behavior....45
Triggering an Emergency-Stop....45

2.5.3 Effective range of Emergency-Stop....46

2.5.4 Main switch....47

2.5.5 Signaling device....48

2.5.6 Guard door....48

2.5.7 Opening and closing the guard door....49
Unlocking and locking the guard door....49
Draw – Opening the guard door....49

2.5.8 Winder Stop....51

2.5.9 Winding station Stop....52

2.5.10 Light barrier....53

2.5.11 Warning lamp with horn....53

2.5.12 Strobe light....54

2.5.13 Key switch....55
Mode selector switch....55
Key interlock switch....55

2.5.14 Safety cut-out mat....56

2.5.1 EMERGENCY-STOP

When hitting the EMERGENCY-STOP button, the machine will be stopped and all machine movements will slow down in a controlled manner. Keep access areas to the Emergency-Stop devices free of obstacles.

DANGER

Risk of injury from the machine when starting.

Persons may well be injured when restarting the machine before the dangerous situation has been cleared.

Eliminate all danger sources and make sure that the machine's protected area is clear of persons before unlocking the button. You are only permitted to restart the machine after the dangerous situation has been eliminated.

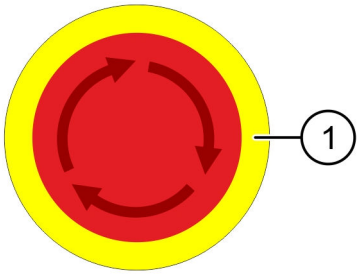
NOTE

Damage when misusing the EMERGENCY-STOP switch.

The machine will be exposed to great stress when actuating the EMERGENCY-STOP switch. Damage to the machine may occur if the machine is usually switched off via the EMERGENCY-STOP function although there is no threat of personal danger.

Only ever press the EMERGENCY-STOP switch if there is a threat of personal danger.

(1)	EMERGENCY-STOP



- ▶ Press the EMERGENCY-STOP button whenever a dangerous situation arises.

  The entire machine is set to a standstill in a controlled manner and depressurized. The button is locked.

- ▶ Clear the dangerous situation.
- ▶ Turn the switch in the direction of the arrow or pull it out.

  The switch is unlocked.

The machine is now ready for a restart.

2.5.2 Emergency stop trip wire

Safe behavior

DANGER

The Emergency-Stop function must always be operational.

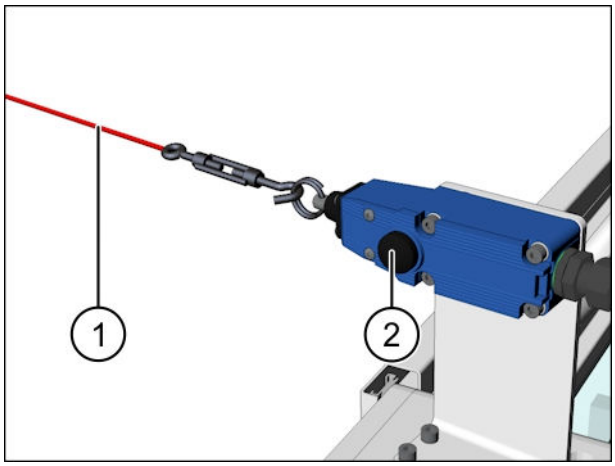
Never modify the Emergency-Stop function.

The machine must not be operated with a defective or modified Emergency-Stop function - mortal danger!

Triggering an Emergency-Stop

Emergency trip wires are fitted in every machine section where safety is crucial.

(1)	Emergency trip wire
(2)	Button, UNLATCH SWITCH



- Actuate the emergency trip wire immediately whenever a dangerous situation arises.

  The machine stops. Control voltage is cut off. The switch of the actuated emergency trip wire is latched.

- After triggering the Emergency-Stop, eliminate the danger source before turning the control voltage back on.
- To unlatch the actuated emergency trip wire switch, press button (2) at the enclosure of the emergency trip wire switch.

X000017813

2.5.3 Effective range of Emergency-Stop

THE EMERGENCY-STOP FUNCTION MUST ALWAYS BE OPERATIONAL.

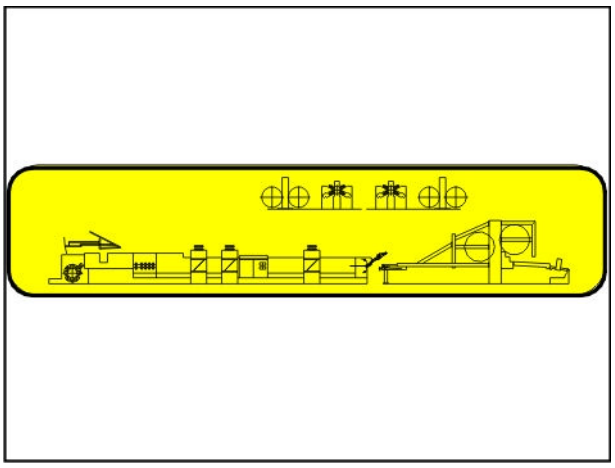
Never modify the Emergency-Stop function.

The machine must not be operated with a defective or modified Emergency-Stop function - mortal danger!

E-Stop buttons are fitted in all of the machine sections where safety is important.

Signs on the machine show the entire effective range of the E-Stop buttons.

All other areas are outside the effective range of the E-Stop buttons.



2.5.4 Main switch

DANGER

Risk of injury from electric shock if live components are touched

Disconnect the main power supply and secure it against unintentional reactivation. Verify the safe isolation from supply. Ground and short-circuit the line part. Cover any adjacent parts that are still live. Only skilled electricians are permitted to work on the unit. Keep a safe distance.

Circuits which have not been switched off by the mains disconnect (main switch) (1)) are still live! These components are scoop-proof, labeled with a danger sign (3) and identified by orange cables. An additional danger sign can be attached at the outside of the machine cabinet.

DANGER

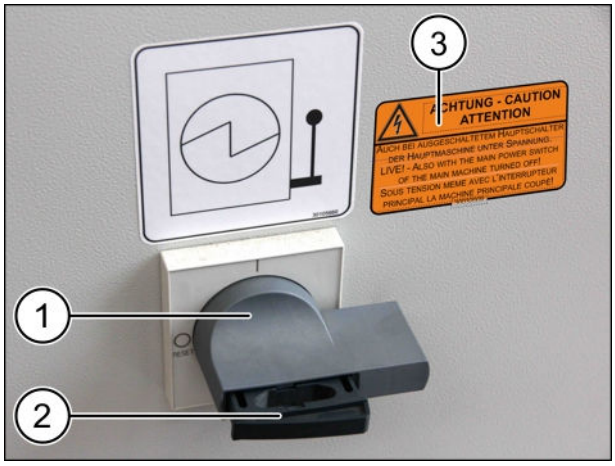
For service or repair works, turn the main switch to '0', pull out the locking bar (2) and secure it with a padlock against unauthorized reactivation.

Attach a clearly visible sign to the main switch, which has the name of the person who switched the machine off and the reason for doing it marked on it.

After completion of the service or repair works, remove the padlock, push in the locking bar and turn the main switch to '1'.

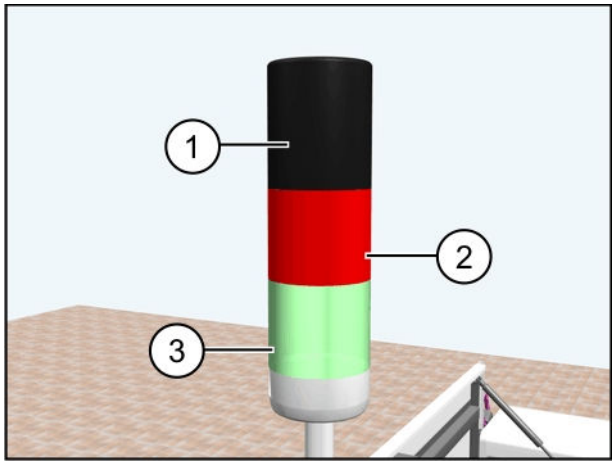
Only authorized personnel are permitted to switch the machine back on, as there is an increased risk of accidents and injuries caused by electric shock.

(1)	Main switch for electrical power supply to the entire machine and all its components. 0 - power supply switched off, I - power supply switched on
(2)	Locking bar
(3)	Danger sign - Live components



2.5.5 Signaling device

(1)	Horn
(2)	Red lamp
(3)	Green lamp



Red = Fault

Red, flashing = Data exchange in progress

Green = Production

Green, flashing = Machine ready to switch on

2.5.6 Guard door

There are different types of guard doors which may vary as per the machine layout:

- Pivotable guard door
- Stationary guard door
- Slidable guard door
- Protective flap

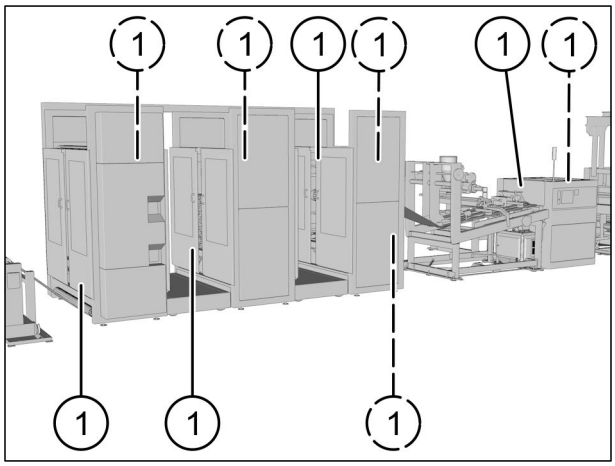
Guard doors prevent the operator from reaching into or entering dangerous zones. Guard doors feature electrical guard lockings to prevent opening of the doors when the machine is running.

2.5.7 Opening and closing the guard door

Unlocking and locking the guard door

Unlock and lock the guard doors with interlocking guard

(1)	Guard door



- ▶ Press the STOP button.

  The machine stops.
  The drives are not switched off. The web tension remains.

- ▶ Press the STOP button for 3 seconds.

  The drives are switched off. The web tension is lost.
  Guard doors (1) are being unlocked.

- ▶ Open the guard doors.
- ▶ Perform works on the machine parts that would otherwise be blocked by the guard doors.
- ▶ Close all guard doors.
- ▶ Press a button to trigger a machine movement.

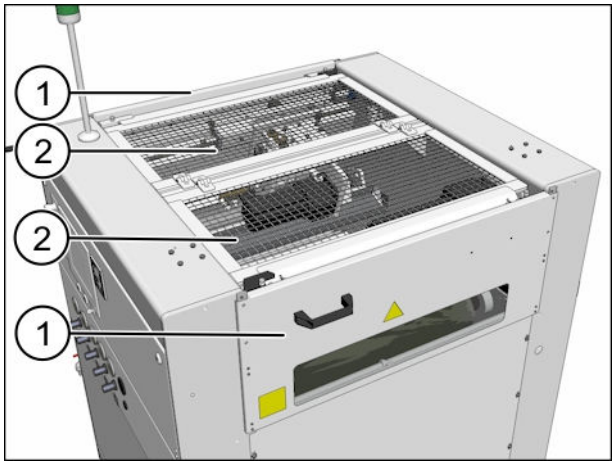
  All guard doors are being locked.

Draw – Opening the guard door

Requirements:

The guard doors are unlocked.

(1)	Guard door
(2)	Grid shutter



- ▶ Open the guard door (1) at the draw.

  The grid shutter (2) can be opened when opening the guard door.
- ▶ Open the grid shutter and put it down on the opposite grid shutter.

  The interior of the draw can now be accessed.
- ▶ Perform works in the interior of the draw.
- ▶ Close the grid shutter and the guard door.
- ▶ Press a button to trigger a machine movement.

  The guard door is locked.

2.5.8 Winder Stop

The WINDER STOP pushbutton serves to switch off the entire winder whenever a dangerous situation arises.

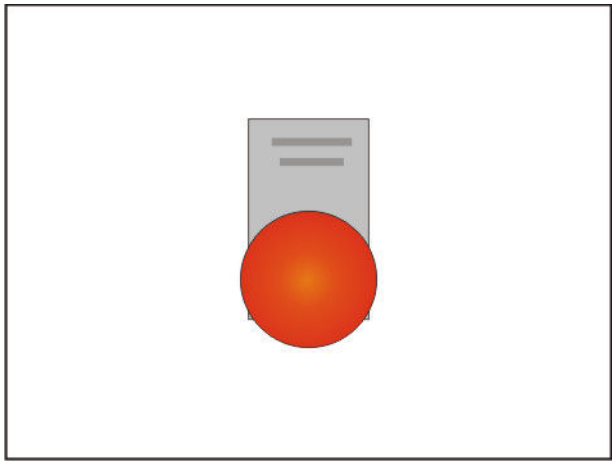
DANGER

Risk of injury from the machine when starting.

Persons may well be injured when restarting the machine before the dangerous situation has been eliminated.

You are only permitted to restart the machine after the dangerous situation has been eliminated.

(-)	WINDER STOP



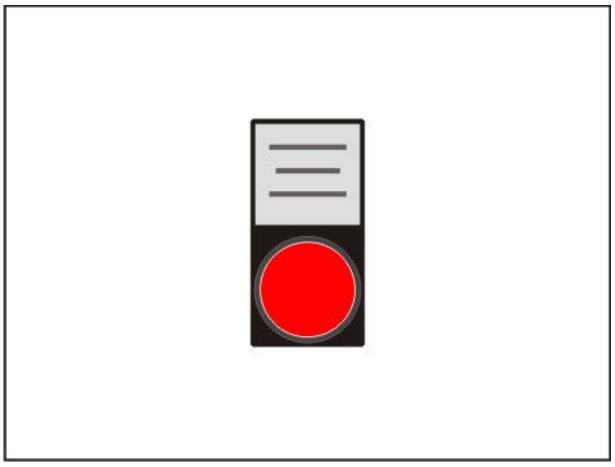
- ▶ Press the WINDER STOP pushbutton when a dangerous situation arises.
  - *All winder components (winding station and draw) will be switched off.*
  - *The extrusion line continues production.*
- ▶ Clear the potentially dangerous situation or movement.
- ▶ Turn the locked WINDER STOP pushbutton counter-clockwise until the pushbutton is unlocked.
- ▶ Press the RESET pushbutton on the control console.

The winder can now be restarted.

2.5.9 Winding station Stop

The WINDING STATION STOP pushbutton serves to switch the winding station off when a potentially dangerous situation arises.

	WINDING STATION STOP

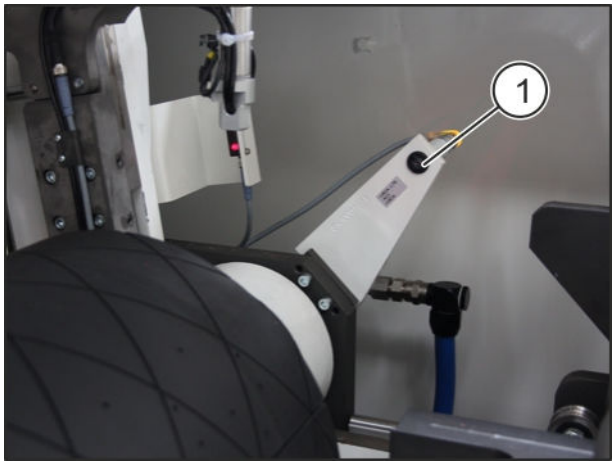


- ▶ Press the WINDING STATION STOP pushbutton when a dangerous situation arises.
  - *The winding station stops.*
  - *The other winding station and the film draw will continue running.*
  - *The extrusion line continues production.*
- ▶ Clear the potentially dangerous situation or movement.
- ▶ Press the RESET pushbutton on the control console.

The winding station can now be restarted.

2.5.10 Light barrier

(1)	Light barrier for gap monitoring



In underwind mode, the light barrier monitors the infeed nip between the contact roller and the film reel.

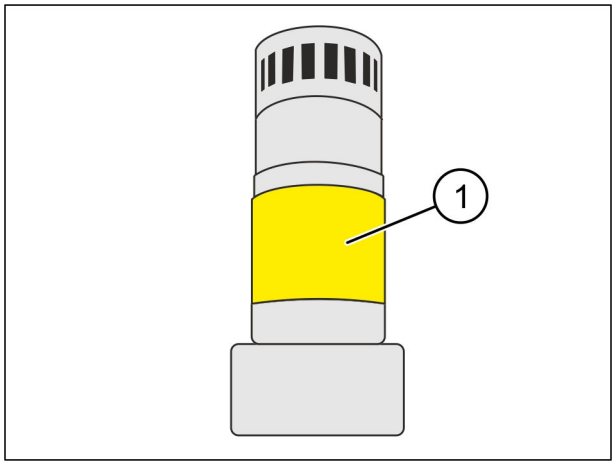
Entering the safety shutdown mat and interrupting the light barrier trips the 'Winding station STOP' function at the relevant winding station. The winder drives are stopped in a minimum of time and the film reel is retracted from the contact roller to a safe distance of at least 120 mm. The other winding station and the film draw continue running.As long as someone is present on the safety shutdown mat, a new roll change cannot be initiated. A already initiated film cut will be finished.

When interrupting the light barrier, e.g. by a film strip, without the safety shutdown mat being involved, the flash lamp flashes for 40 seconds. Simultaneously entering the safety shutdown mat triggers the 'Winder STOP' function as well.

The infeed nip monitor can be switched off for a limited period of time by pressing the yellow BYPASS SAFETY AREA button. A message box appears on the screen for 60 seconds. A horn sounds. After another 15 seconds, the winding station will be stopped.

2.5.11 Warning lamp with horn

(1)	Warning lamp with horn

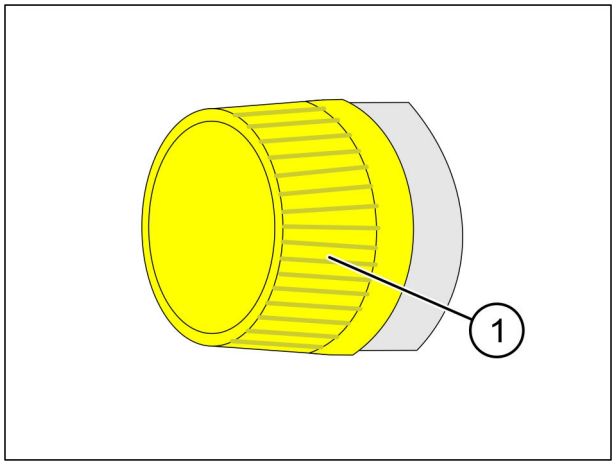


The warning lamp with horn (1) flashes during the entire roll change. The horn sounds for five seconds at the beginning of the roll change. The horn also informs you of alarm messages. It sounds until the alarm messages are acknowledged.

2.5.12 Strobe light

(1)	Strobe light

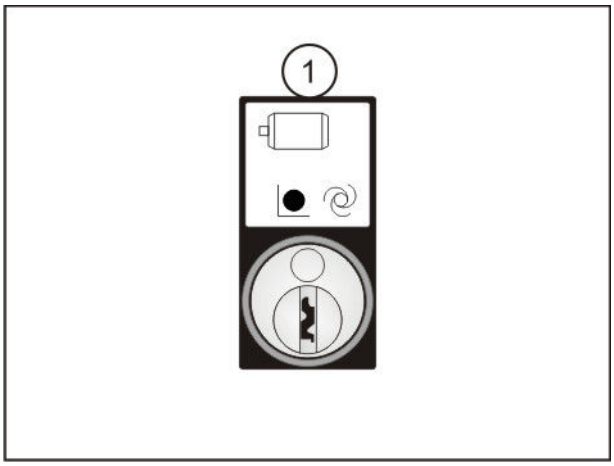
The strobe light warns the operator of dangerous situations on the machine.



2.5.13 Key switch

Mode selector switch

(1)	WINDING STATION key switch



The mode selector switch serves to select '**Start-up** ' or '**Production**' mode.

Key switch positions:

- Mid-position: 'Start-up mode'
- Right position: 'Production mode'

For safety reasons, the following restrictions apply when using 'Start-up' mode:

- Depending on the product, the winder should be started at the lowest possible thread-up speed.
- Due to the surface drive, the web tension is limited to 150 33.7[N] (33.7 lbf).
- The center drive cannot be switched on.

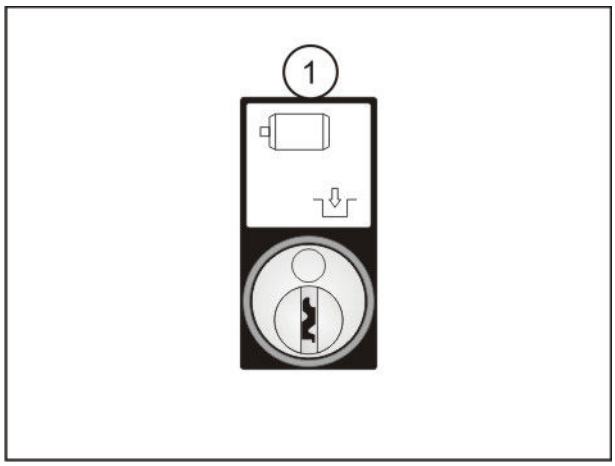
When in 'Start-up' mode, the safety cut-out mat within the winding area may be entered for threading up the film web and for initial winding. The light barrier in the infeed nip is bypassed. The yellow warning light flashes and urges you to take utmost care. Withdraw the key while using startup mode.

After initial winding of the tubular or flat film, the safety cut-out mat should be left immediately. Turn the mode selector switch to the right position. Only then can the line be run up to the web tension required for production. The center drive can be switched on.

When entering the safety cut-out mat the yellow warning light starts flashing and urges you to take utmost care. At the same time, the light barrier in the infeed nip is activated. When interrupting the light beam in the light barrier the WINDING STATION STOP function is triggered. The winding roll is retracted at least 120 mm (0.4 ft) from the contact roller and the roll drives are stopped as quickly as possible.

Key interlock switch

(1)	WINDING STATION SECURED



The key interlock switch is a key switch that is used to secure the winding station. Drives and pivoting arrangements are secured against unintentional activation.

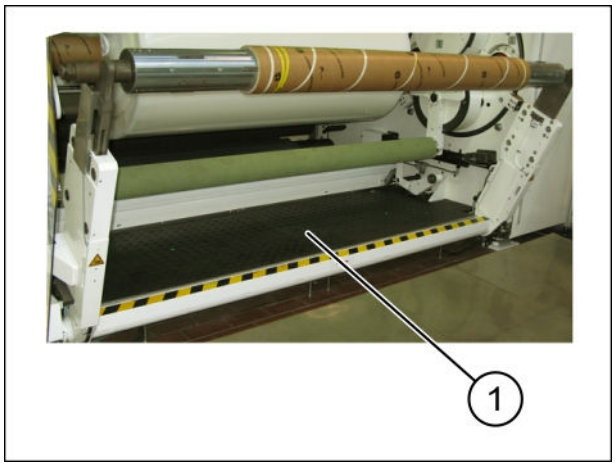
- mid-position: 'Winding station **not** secured'
- right position: 'Winding station secured'

Secure each winding station separately. The winding station is secured by turning the key interlock switch to the right position and then withdrawing it. The control voltage of the respective winding station is switched off.

For unlocking, turn the key interlock switch into the mid-position. Press the RESET button to release the winding station. All winding station drives can now be restarted.

2.5.14 Safety cut-out mat

(1)	Safety cut-out mat



The safety cut-out mat secures the winding station and must only be entered when

- threading the film into the winder: In STARTUP MODE, the safety cut-out mat may carefully be entered. Meanwhile, the flashing light is blinking.
- During film sampling: In PRODUCTION mode, the safety cut-out mat may carefully be entered when a second machine operator holds the SAFETY AREA BYPASS SAFETY CUT-OUT MAT I/II pushbuttons. Meanwhile, the flashing light is blinking.

When entering the safety cut-out mat in PRODUCTION mode, the following safety devices become active:

- The flashing light is blinking and urges you to take utmost care.
- The light barriers for gap monitoring are activated.

When the safety cut-out mat is active **and** the light barriers for gap monitoring are interrupted, the winding station will be stopped.

2.6 Danger signs

2.6.1 Information on the warning signs 0
2.6.2 Unwind 59
2.6.3 Draw and dancer roller 60
2.6.4 Web processing station with micro-perforator 61
2.6.5 Web processing and embossing unit 63
2.6.6 Gusseted tube forming unit 64
2.6.7 Longitudinal seam extruder 65
2.6.8 Main draw 67
2.6.9 Cut edge blocking 68
Cut edge blocking 68
Cut edge blocking, web travel 68
2.6.10 Switch boxes 69
2.6.11 Enclosure 0

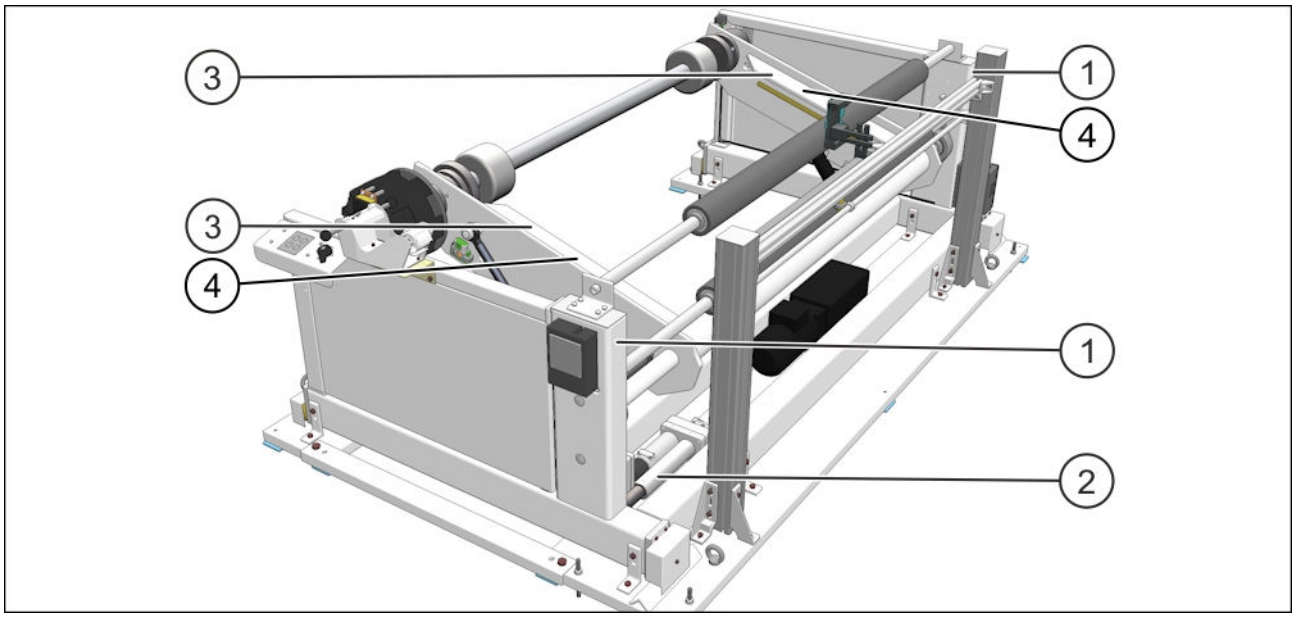
2.6.1 Information on the warning signs

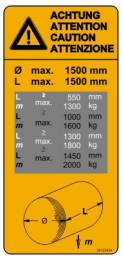
Warning signs on the machine alert to residual risks. The personnel must pay special attention in these areas.

- The warning signs must be kept clean and legible.
- Replace the warning signs immediately when damaged. Please place a spare parts order with us when new warning signs are required.

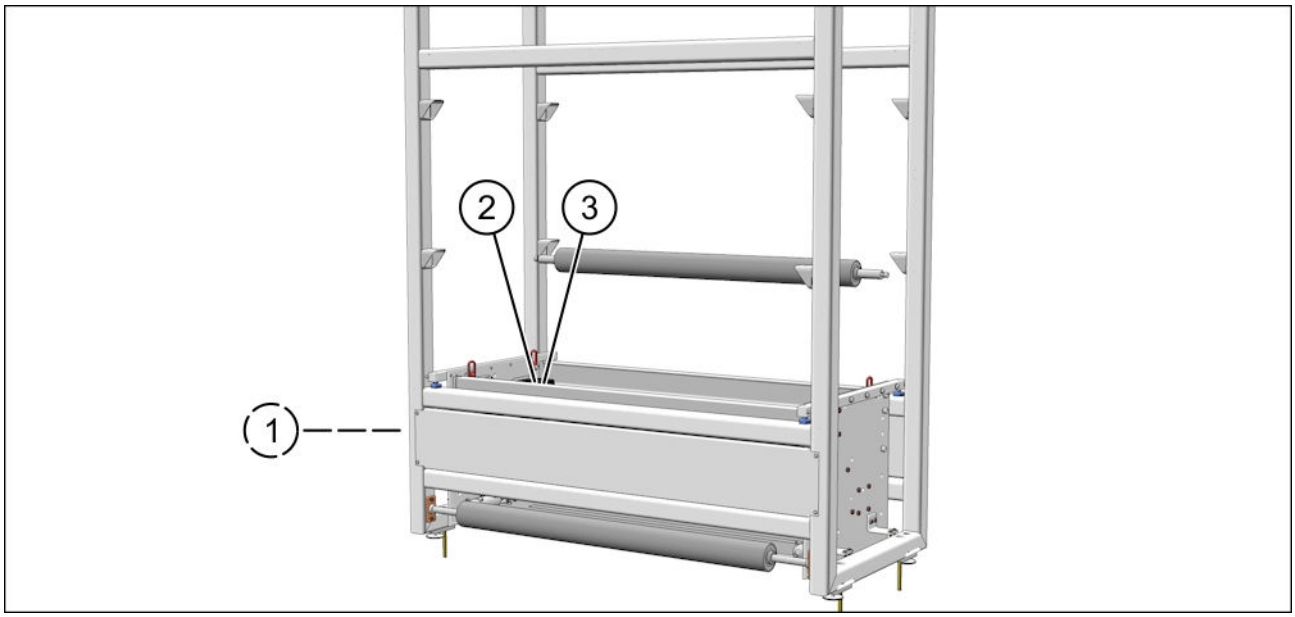
The following figures show parts of the machine including various options that may exist and the warning signs attached to them. The warning signs not listed here can be found in the parts lists.

2.6.2 Unwind



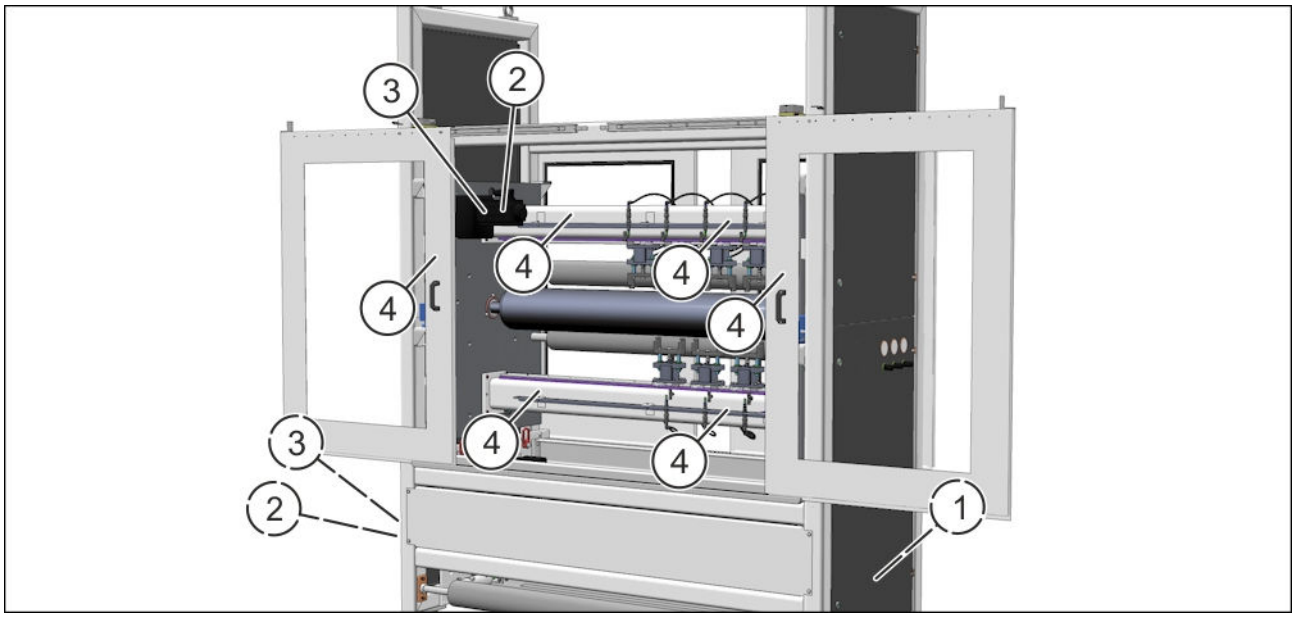
Pos.	Danger sign	Location	Warning sign
(1)		On both side frames.	Risk of injury from infeed nips. Your arms and hands might be caught and crushed. Do not reach into the machine. Keep a safe distance.
(2)		On the guide of the web guide control unit.	Risk of injury from moving machine parts Hands might be crushed or sheared off. Do not reach inbetween moving machine parts. Keep a safe distance.
(3)		On both delivery arms.	Risk of Injuries from moving machine parts Your feet might be crushed or sheared. Wear industrial safety shoes. Keep a safe distance.
(4)		On both delivery arms.	The permitted dimensions and roll weights must be heeded. See the specification on the rating plate.

2.6.3 Draw and dancer roller



Pos.	Danger sign	Location	Warning sign
(1)		On the switch cabinets.	Mortal danger from electric shockRisk of injury from electric shock when touching machine parts when live. Risk of serious or even fatal injury when working on machine parts when live. Be sure not to start working on the machine unless the relevant machine components have been de-energized. Secure against being switched back on. Verify the safe isolation from supply. Ground and short-circuit the machine part. Pay attention to possibly existing stored energy. Pay attention to separate source voltages provided from external sources. Make sure to isolate any adjacent machine parts if still live. Only electrically skilled personnel should carry out work on the unit.
(2)		On the motors.	Risk of burns from hot surfaces. Parts of your body might be burned by hot machine parts. Wear heat-resistant protective clothing. Keep a safe distance.
(3)		On the motors.	Risk of injury from moving machine parts Hands may be crushed or sheared off. Do not reach in between moving machine parts. Keep a safe distance.

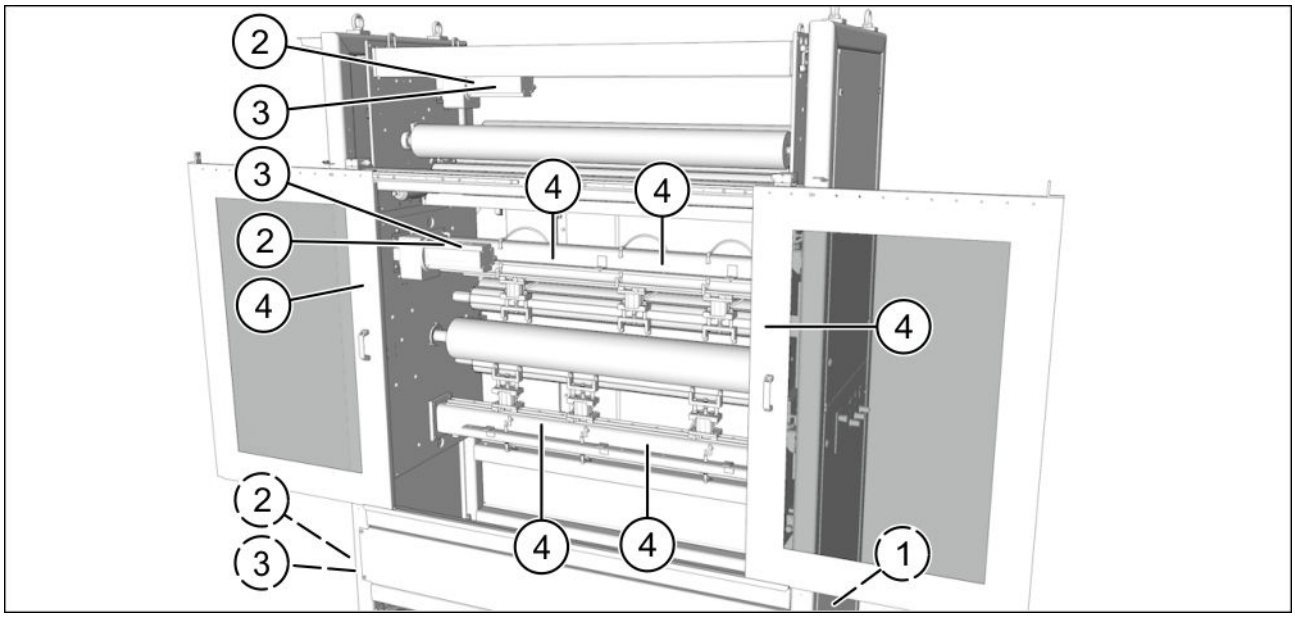
2.6.4 Web processing station with micro-perforator



Pos.	Warning label	Location	Warning sign
(1)		On the switch boxes.	Mortal danger from electric shockRisk of injury from electric shock when touching machine parts when live. Risk of serious or even fatal injury when working on machine parts when live. Be sure not to start working on the machine unless the relevant machine components have been de-energized. Secure against being switched back on. Verify the safe isolation from supply. Ground and short-circuit the machine part. Pay attention to possibly existing stored energy. Pay attention to separate source voltages provided from external sources. Make sure to isolate any adjacent machine parts if still live. Only electrically skilled personnel should carry out work on the unit.
(2)		On the motors.	Risk of burns from hot surfaces. Parts of your body might be burned by hot machine parts. Wear heat-resistant protective clothing. Keep a safe distance.
(3)		On the motors.	Risk of injury from moving parts Risk of crushing and shearing your hands. Do not reach in between moving components. Keep a safe distance.

Pos.	Warning label	Location	Warning sign
(4)		On both sides of the tool beams. On both sides of the guard doors.	Risk of injury from needle rollers. Needles may cause personal injury. Wear cut-resistant gloves.

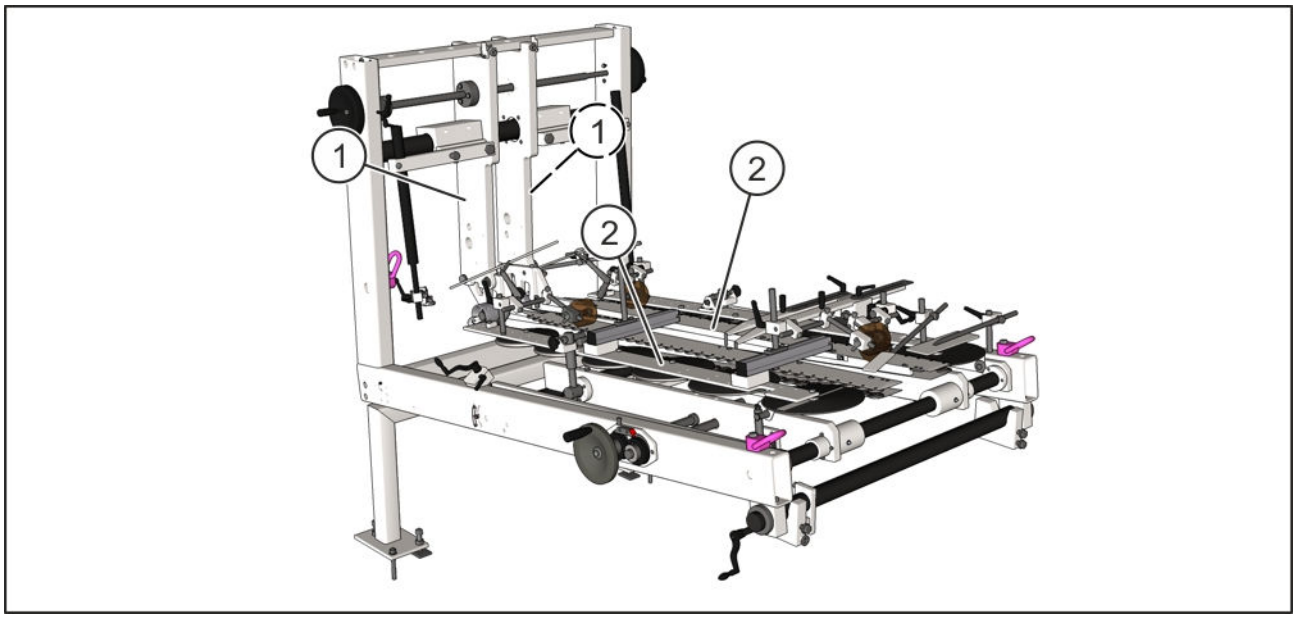
2.6.5 Web processing and embossing unit



Pos.	Warning label	Location	Warning sign
(1)		On the switch boxes.	Mortal danger from electric shockRisk of injury from electric shock when touching machine parts when live. Risk of serious or even fatal injury when working on machine parts when live. Be sure not to start working on the machine unless the relevant machine components have been de-energized. Secure against being switched back on. Verify the safe isolation from supply. Ground and short-circuit the machine part. Pay attention to possibly existing stored energy. Pay attention to separate source voltages provided from external sources. Make sure to isolate any adjacent machine parts if still live. Only electrically skilled personnel should carry out work on the unit.
(2)		On the motors.	Risk of burns from hot surfaces. Parts of your body might be burned by hot machine parts. Wear heat-resistant protective clothing. Keep a safe distance.
(3)		On the motors.	Risk of injury from moving parts Risk of crushing or shearing your hands. Do not reach in between moving components. Keep a safe distance.

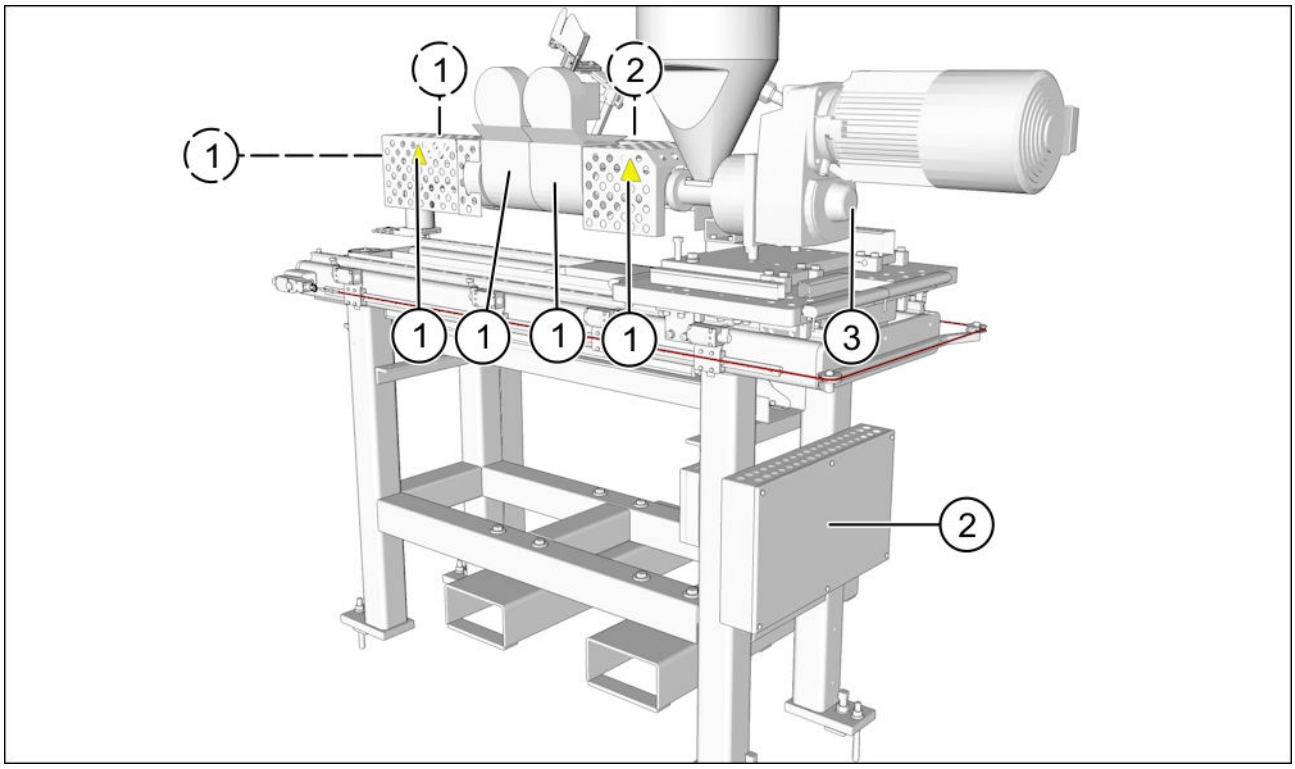
Pos.	Warning label	Location	Warning sign
(4)		On both sides of the tool beams. On both sides of the guard doors.	Risk of injury from needle rollers. Needles may cause personal injury. Wear cut-resistant gloves.

2.6.6 Gusseted tube forming unit



Item	Warning sign	Location	Warning label
(1)		At the infeed section of the inner side frames	Risk of injury from infeed nips. Your arms and hands might be caught and crushed. Do not reach into the machine. Keep a safe distance.
(2)		On the supports of the gusset forming unit.	Risk of injury from moving machine parts. Persons may get caught and crushed by moving machine parts. Never reach into the running machine. Keep a safe distance.

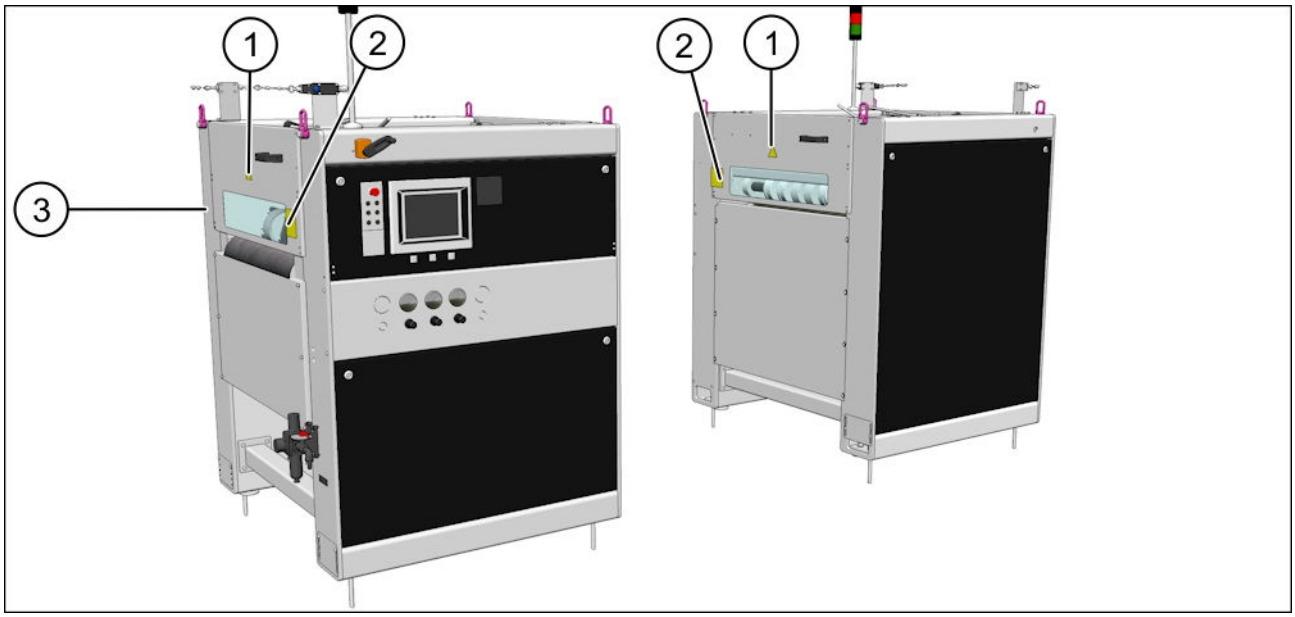
2.6.7 Longitudinal seam extruder



Pos.	Danger sign	Location	Warning sign
(1)		On all sides of the extruder guards. On the heater guards.	Risk of burns from hot surfaces. Parts of your body might be burned by hot machine parts. Wear heat-resistant protective clothing. Keep a safe distance.
(2)		On the switch cabinets.	Mortal danger from electric shockRisk of injury from electric shock when touching machine parts when live. Risk of serious or even fatal injury when working on machine parts when live. Be sure not to start working on the machine unless the relevant machine components have been de-energized. Secure against being switched back on. Verify the safe isolation from supply. Ground and short-circuit the machine part. Pay attention to possibly existing stored energy. Pay attention to separate source voltages provided from external sources. Make sure to isolate any adjacent machine parts if still live. Only electrically skilled personnel should carry out work on the unit.

Pos.	Danger sign	Location	Warning sign
(3)		On the cover of the screw.	Without the cover being mounted, rotating components are exposed. Risk of injury to your hands when rotating components are exposed. Do not operate the extruder without the cover. The cover must be mounted when operating the machine.

2.6.8 Main draw

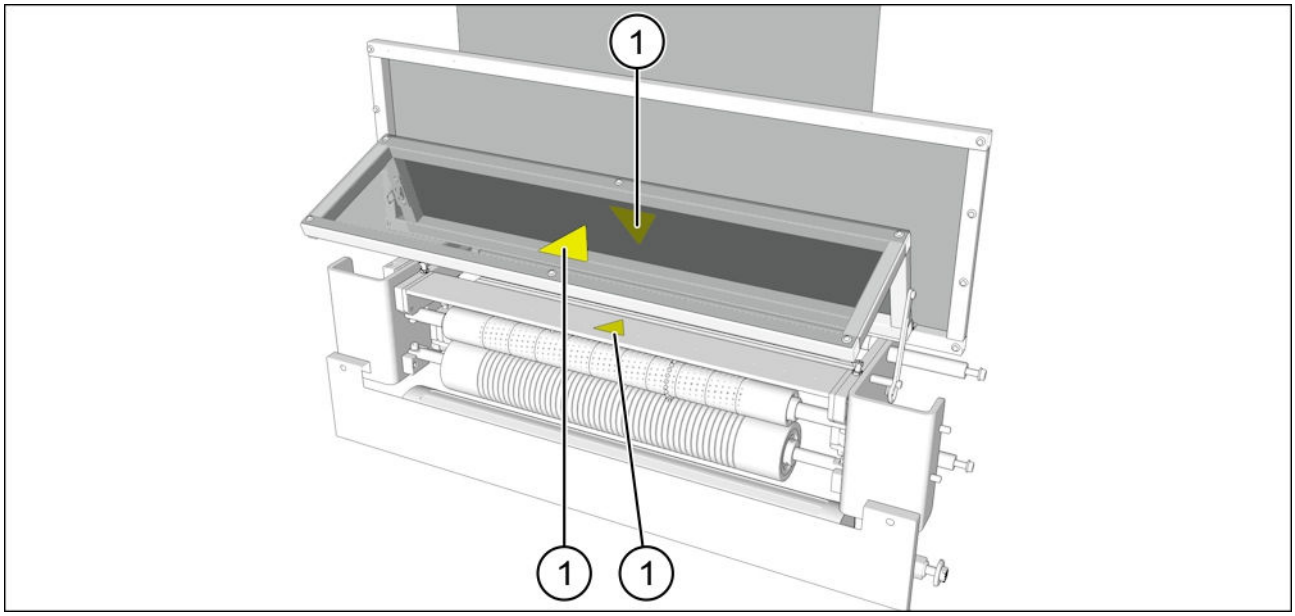


The figure shows the front and rear side of the main draw.

Pos.	Danger sign	Location	Warning sign
(1)		On the slide protections. On the frame of the web guide control unit.	Risk of injury from moving machine parts Hands might be crushed or sheared off. Do not reach inbetween moving machine parts. Keep a safe distance.
(2)		On the slide protection at the infeed. On the frame of the web guide control unit.	Risk of injury from moving machine parts. Persons may get caught and crushed by moving machine parts. Never reach into the running machine. Keep a safe distance.
(3)		On the frame of the main draw on the infeed.	Risk of crushing for persons Persons are at risk of being crushed between elevated devices. Do not stay between moving machine parts. Keep a safe distance.

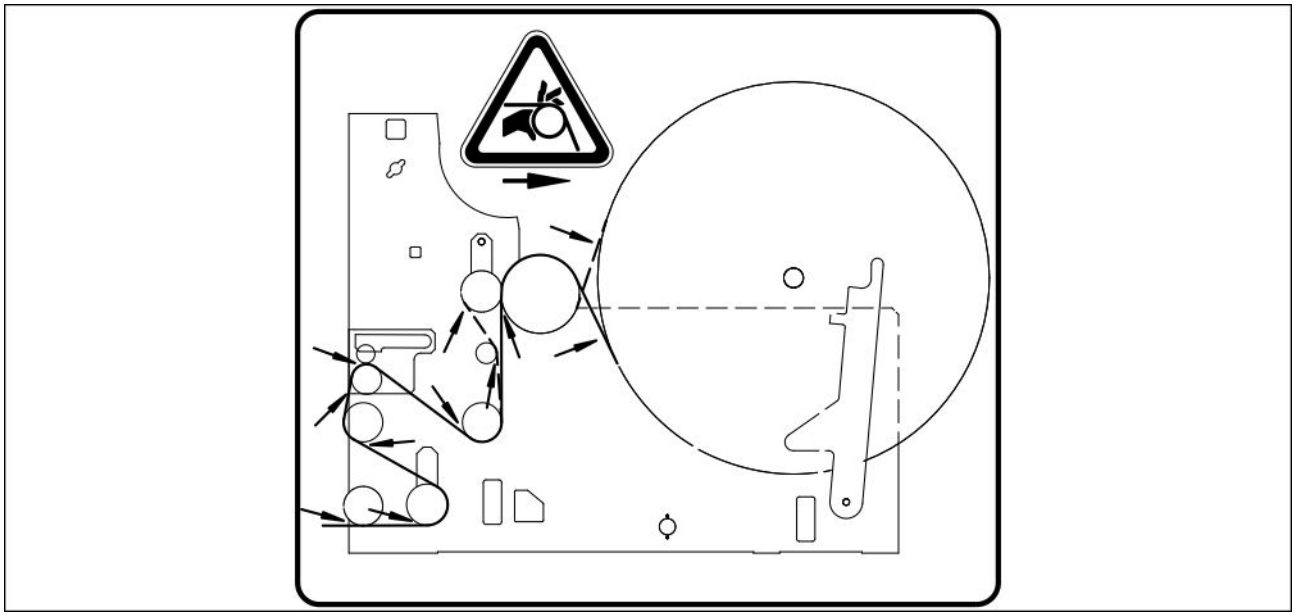
2.6.9 Cut edge blocking

Cut edge blocking



Pos.	Danger sign	Location	Warning sign
(1)		Warning signs on the guard. Warning sign on the cross-beam.	Needle roller Needles may cause personal injury. Wear cut-resistant gloves.
			The guard is provided with a lock. The guard cannot be opened while the machine is running.

Cut edge blocking, web travel



Pos.	Danger sign	Location	Warning sign
(1)		Information signs [arrows] referring to infeed nips and the web travel	Infeed nip Hands may be caught in the nip and crushed. Keep hands clear of the nip between the deflection pulley and the web.

2.6.10 Switch boxes

Pos.	Danger sign	Location	Warning sign
(-)		Switch cabinets, switch boxes, electric motors, electrical sub-assemblies	Risk of injuries from electric shocks if live components are touched Disconnect the main power supply and secure it against unintentional reactivation.. Verify the safe isolation from supply. Ground and short out the relevant parts. Any adjacent parts that are still live must be covered. Only electrically skilled personnel should carry out work on the unit.Keep a safe distance.

2.6.11 Enclosure

Pos.	Warning sign	Location	Warning instruction
(-)		Surfaces that must not be walked on	Do not walk on the surface Persons that walk on surfaces identified in this way may injure themselves or damage the machine.

2.7 Residual risks

2.7.1 Winder....72
Safety instructions....72
Spreader roller....72
Slitting assembly....72
Danger of being pulled in or being crushed....72
Vacuum cutover drum....72
Fault display from the vacuum cutover drum....72
Pulling in positions in the winding station....72

2.7.1 Winder

Safety instructions

Despite of all safety precautions that have been taken there still remain certain risks when operating the film winder. Especially when working near the infeed nip in-between the contact roller and the film roll. At any rate, great care must be taken when working in the danger area around the film winder.

CAUTION

The machine owner must train his operating personnel through regular, repeated instruction and inform them of any residual risks!

CAUTION

Extra care must be taken when winding the web onto the spliced winding core and during sampling.

Danger of being pulled into the winder's infeed nip! Never reach into the winder's infeed nip!

Spreader roller

Due to the adjustable roller sag an infeed guard cannot be installed between the film web and spreader roller.

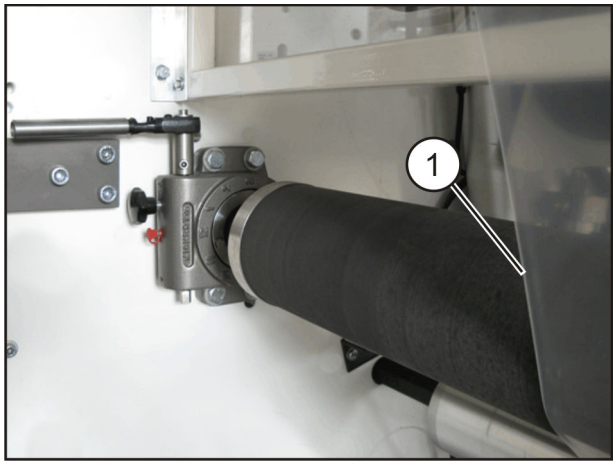
WARNING

You can be injured when the web is running

Parts of your body, clothes, hair and objects may be pulled in and crushed. Always wear tight-fitting clothes and a hair net. Work without industrial safety gloves Keep a safe distance.

(1)	Pulling in position on the spreader roller

Sufficient safety clearance must be maintained to the pull in gap on the spreader roller.



Slitting assembly

CAUTION

The machine owner must train his operating personnel through regular, repeated instruction and inform them of any residual risks!

WARNING

Injuries through cutting blades, circular knives or sharp-edged machine parts

Wear cut-resistant safety gloves.

DANGER

Risk of injuries due to hot surfaces, temperatures exceeding 60°C / 140°F

Your hands and other parts of the body can be burnt due to excessive heat. Wear safety gloves, safety shoes and long-sleeved clothing. Keep a safe distance.

CAUTION

If the slitting assembly is not used, it is essential to move the blade sliders in the 'rest position', especially when threading the film!

Danger of being pulled in or being crushed

Despite all safety devices installed, a residual risk remains when operating the film winder, especially at the different locations where you can be pulled in or crushed.

WARNING

You can be injured when the web is running

Parts of your body, clothes, hair and objects may be pulled in and crushed. Always wear tight-fitting clothes and a hair net. Work without industrial safety gloves Keep a safe distance.

DANGER

Injuries can be caused by rotating rollers

Parts of your body, clothes, hair and objects may be pulled in and crushed. Always wear tight-fitting clothes and a hair net. Work without industrial safety gloves Keep your hands away from the roller gap! Keep a safe distance.

WARNING

There is a danger of being crushed when lowering the film rolls.

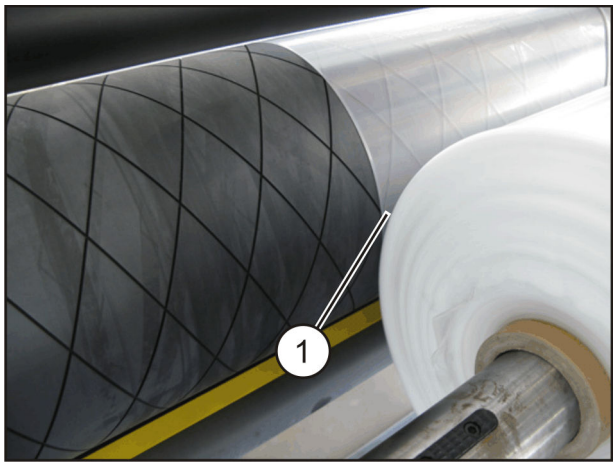
The lowering area for finished film rolls must be clear from personnel and objects. The system operator must make sure that the roll lowering area is secure before lowering the film roll. The system operator must be able to see the lowering area at all times during the lowering procedure.

Sufficient safety clearance must be maintained to the following dangerous areas when operating the film winder.

(1)	Pull-in gap between the contact roller and film winder

The danger exists of being caught in the winder pull-in gap between the film winder and the contact roller when pulling the film web in and when checking the film winding quality.

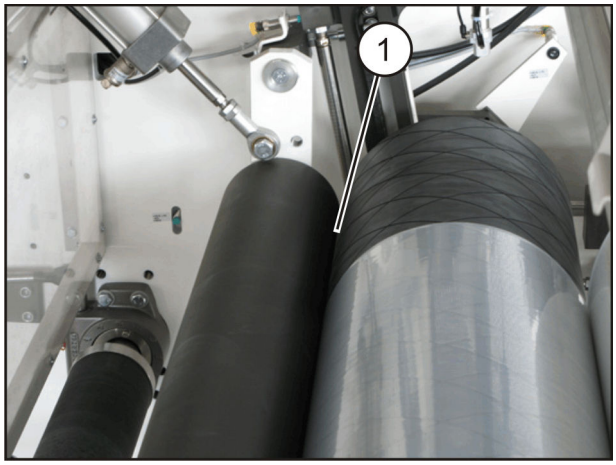
Ensure that you maintain an adequate safety distance. Never reach into the winder pull-in gap!



(1)	Pull-in gap between the contact roller and the lay-on roller

There is a risk that parts of the body might be pulled into the pull-in gap between the contact roller and the lay-on roller when in the 'Winding from above' operating mode.

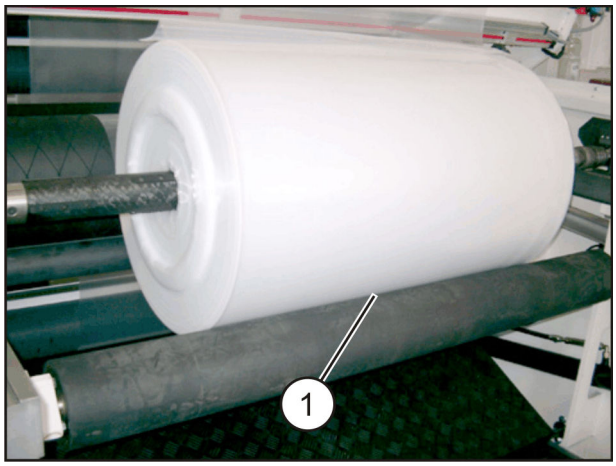
Ensure that you maintain an adequate safety distance. Never reach into the winder pull-in gap!



(1)	Pull-in gap between the film winder and the auxiliary winder drive **(Option)**

The auxiliary winder drive is used in the version 'without central drive'. When changing the roll, the auxiliary winder drive is swung up against the film roll from beneath. It continues to drive the film roll while it is pulled away from the contact roller.

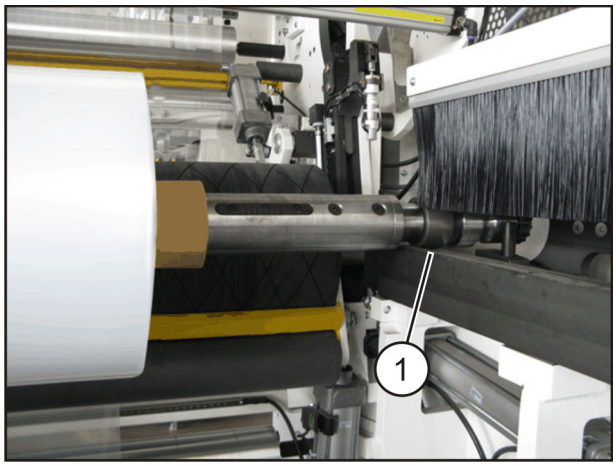
A flashing light and the horn provide acoustic and visual warnings that a roll change is taking place. The warning signals mean that extra care must be taken. If someone steps on the safety cut-out mat in the winding area the winding station will be switched to a safe state immediately.



(1)	Pull-in gap between the winding shaft and the parallel guide

The clearance to the mounting rails of the parallel guide is approx. 50 mm (2 inches) when winding up film with the maximum working width in the 'Winding from above' mode. There is a risk of being pulled in. The pulling in positions are protected against pulling in parts of a body by side-mounted guards.

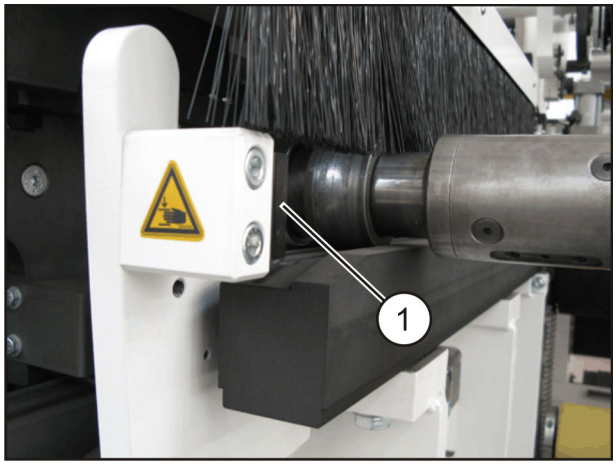
Ensure that you maintain an adequate safety distance. Never reach into the winder pull-in gap!



(1)	Rubber stop plates on the delivery arm

When changing the roll, the winding shaft moves on the mounting rails of the parallel guide in the direction of the delivery arms. When being transferred to the delivery arms' pick-up trough, the winding shaft is slowed down and braked by the rubber stop plates (1) on both sides. There is a risk of crushing between the winding shaft and the delivery arms.

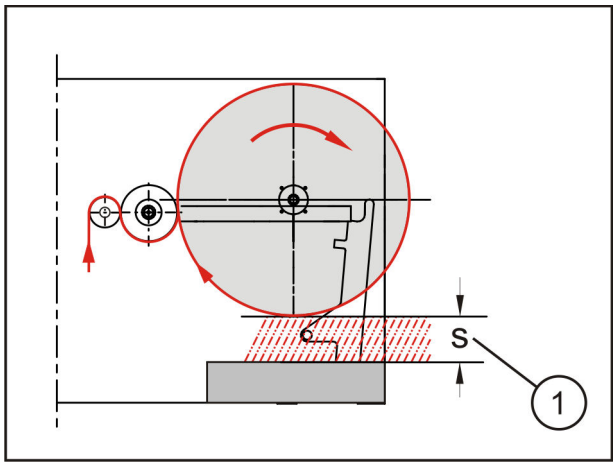
Make sure that you maintain an adequate safety distance. Never reach into the winder crushing section!



(1)	Distance s between the film winder and the safety cut-out mat

In the 'Winding from above' operating mode the distance s between the film winder and the safety cut-out mat is reduced to approx. 200 mm (7.8 inches) when working with a maximum roll diameter. It is strictly forbidden for the operator to crawl beneath the film winder due to the danger of being crushed.

There is no danger of your feet being pulled in.



Vacuum cutover drum

DANGER

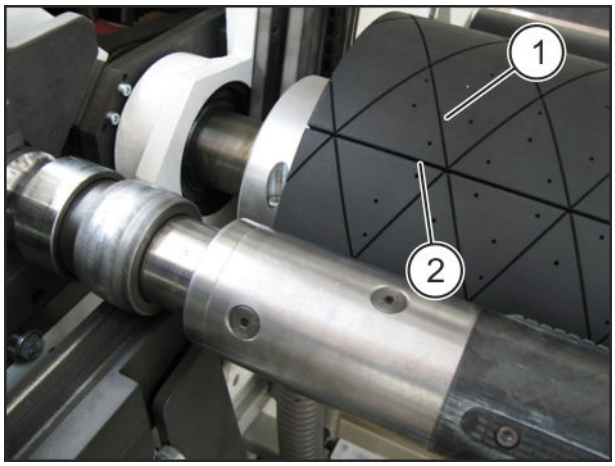
Risk of severe cut injuries from flicking out blades

Turn the WINDING STATION key switch to 'Secured' and withdraw the key before starting any maintenance work. The pneumatic system is depressurized. Delayed pressure reduction (residual pressure may cause the blade to flick out of the longitudinal slot in the drum shell.

Always turn the vacuum cutover drum so that the longitudinal slot points away from your body.

Never touch the longitudinal slot in the drum shell with your body.

(1)	Vacuum cutover drum
(2)	Longitudinal slot in the drum shell



X000175608

Fault display from the vacuum cutover drum

WARNING

Risk of injury from malfunctions on the vacuum cutover drum.

Defective components or dirty pneumatics may cause malfunctions of the vacuum cutover drum.

Check the vacuum cutover drum for defective components and dirty pneumatics if a fault message is displayed. Before starting any work on the unit, turn the vacuum cutover drum so that the blade in the roller faces away from the operator. Do not continue working on the machine or operating the machine unless it has been checked.

The button switches on the vacuum cutover drum might send a faulty signal. In such cases, a fault message informs you of the faulty signal:

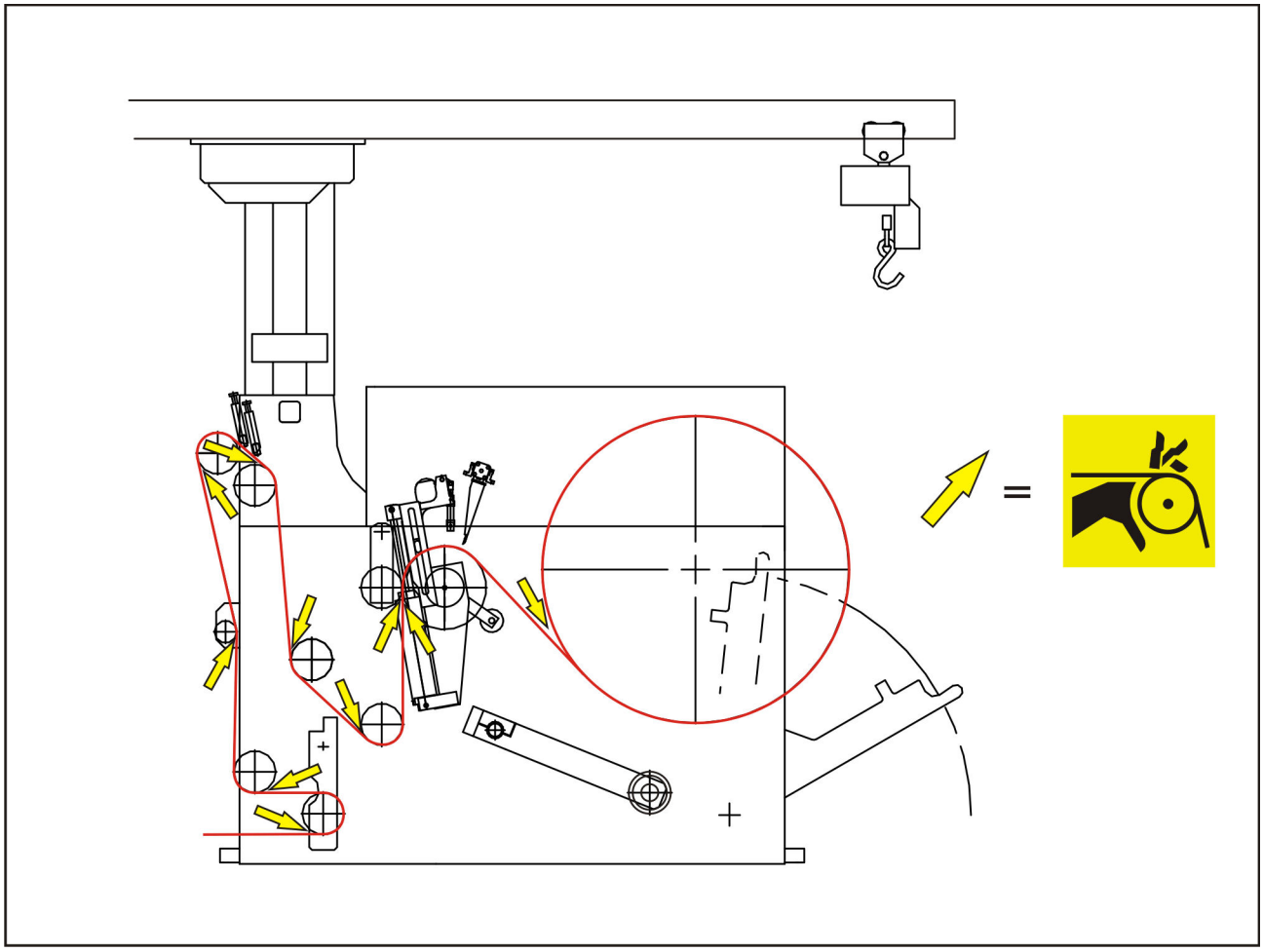
Caution! The button switches on the vacuum cutover drum send a faulty signal. This may result in faulty cuts affecting production and endanger persons when working on the machine. The vacuum cutover drum must be checked immediately for proper functioning of all components and possible contamination of the pneumatics.

After a malfunction of the vacuum cutover drum, the following works should be carried out to prevent dangerous situations from occurring:

- Check the machine and particularly the vacuum cutover drum for defective components.
- Check the machine and particularly the vacuum cutover drum for dirty pneumatics.
- Check the shock absorbers on the vacuum cutover drum for proper air permeability.

Replace contaminated or defective components whenever necessary.

Pulling in positions in the winding station



BEWARE

Injuries resulting from the pulling in positions

Parts of your body, clothes, hair and objects may be pulled in and crushed. Always wear tight-fitting clothes and a hair net. Work without industrial safety gloves Keep a safe distance.

The danger of being pulled into the wedge-shaped gap between the film web and the surface of the roller exists at the pulling in positions marked by an arrow. Not all of the pulling in positions can be protected by a run-on guard. Warning signs have therefore been fitted on the side frame to mark the danger spots. Extra care must be taken when working on the winding stations. Always wear safe, tight-fitting clothes.

2.8 Intended use

2.8.1 Intended use....81

2.8.2 Intended use....82

2.8.3 Intended use – Winding shaft....83
Danger notes....83
Safety instructions....83
Functional elements....83
Improper use....83
Intended use....83

2.8.4 Reasonably foreseeable misuse....85

2.8.1 Intended use

- The machine is intended for the production of a one-layer tube made from one film web . Any other use or use going beyond the listed ones are considered to be unintended use.

The intended use includes:

- the standard mode
- the startup mode
- the trial mode
- the controlled shutdown of the machine
- the inspection, maintenance and repair work
- putting the machine out of operation

Abiding by the instruction manual and the sub-vendor documentation and adhering to the manufacturer's regular cleaning, maintenance and lubricating intervals also count as part of the intended use.

2.8.2 Intended use

The machine is a film winder. The film winder is intended to wind up flexible film webs on reels.

The intended use includes:

- the standard mode
- the startup mode
- the trial mode
- the controlled shutdown of the machine
- the inspection, maintenance and repair work
- putting the machine out of operation

The following resins are to be used on the machine:

- Polypropylene (PP)
- Polyethylene (PE)
- Polyamide (PA)
- Barrier layer material (e.g. EVOH)
- Polystyrene (PS)
- Bonding agent (HV)

Only films collapsed symmetrically across the entire machine width are allowed to be wound up in the machine.

The following resins must **not** be processed on the machine:

- chloric plastics, e.g. polyvinyl chloride (PVC)
- toxic raw materials
- explosive raw materials

The machine is provided to be operated only when functioning properly. The processing of raw materials depends on various factors and must be agreed upon beforehand between the manufacturer and the operating company.

Only qualified personnel are allowed to operate and maintain the machine (Qualification of the personnel).

Abiding by the operating instructions and the manufactrurer's instruction manual and adhering to the manufactrurer's specified regular cleaning, maintenance and lubricating intervals also count as part of the intended use.

Another use or a use in excess thereof, e.g. the processing of raw materials not provided for this purpose are to be regarded as unintended use.

2.8.3 Intended use – Winding shaft

Danger notes

WARNING

There is a risk of severe injuries and damage to the machine.

Prevent damages to the winding shaft from shocks, impacts or drops. Under operating conditions, a damaged winding shaft is at risk of breakage.

Have the winding shaft checked by qualified personnel after improper stress has occurred or on suspicion of damage.

Immediately replace the winding shaft when damaged!

CAUTION

Risk of injury from pinch points, high weight and suspended loads when handling a winding shaft.

Works on the winding shaft and the chuck cones should be carried out only when at a standstill.

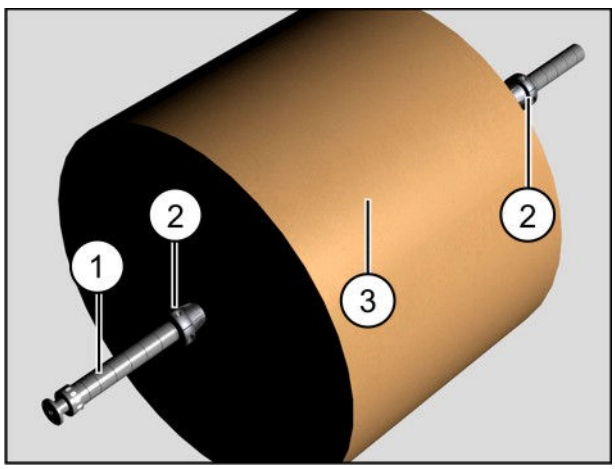
Omit any working manners that may affect safety.

Safety instructions

- A winding shaft must only ever be used as intended and in the immediately vicinity of the machine.
- Pay attention to the given specifications of the roll with regard to its diameter, width and weight.
- The roll must be wound up in a faultless manner. The roll must be perfectly round without showing any imbalance.
- Follow the instructions and notes concerning the winding shaft given in the instruction manual and on the machine.
- Pay attention to the startup, operating and maintenance conditions.
- Never change the winding shaft and the chuck cones.

Functional elements

(1)	Winding shaft
(2)	Chuck cone
(3)	Roll



Improper use

Using a winding shaft for the transportation of rolls is strictly forbidden!

Using a winding shaft for transporting rolls between the deposit and the machine is considered improper use.

A roll handling attachment must be used for roll transport between the deposit and the machine.

Use a roll handling attachment as per Directive 2006/42/EC of the European Parliament or a corresponding one pertaining to the local laws and regulations.

Intended use

Using a winding shaft is considered intended use when used

1. as a roll bearing in the winding station when operating the machine considering the given specification of the roll and
2. for lifting the roll into the winding station in the immediate vicinity of the machine (as a machine component) considering the given specifications of the load handling attachment used (crane assembly) as well as the safety regulations and instructions for the transportation of suspended loads.

2.8.4 Reasonably foreseeable misuse

Any use other than or beyond the use intended by the manufacturer is considered to be misuse. The manufacturer/supplier cannot be held liable for any damage resulting from such use. The responsibility rests solely with the operating company.

The following is considered to be unintended use:

- Use under difficult conditions, e.g. in corrosive atmospheres.
- Feeding material which does not meet the requirements as regards the quality and properties or which deviates from the specified standards.
- The manufacture of products which is considered unintended use.
- Loading foreign software or modifying software by persons which have not been trained for this by the manufacturer.
- Unauthorized modifications to the machine such as mechanical reconstructions, modifications to the electrical installation or the controller.
- Feeding or removing material in other places than those intended for this purpose except for removing material or clearing jams.
- Transportation of persons (except lifting platforms).
- Operation without wearing protective clothing or protective devices.
- Non-observance of the safety arrangement, the safety instructions and the pertaining instructions for the prevention of accidents.
- Transportation of rolls with a winding shaft.
- The maximum permissible winding shaft speed specified by the manufacturer must not be exceeded.
- Operating the entire machine in explosive atmospheres.
- The data and the quality properties of the material to be processed as specified in the chapter under the heading "Technical specifications" must not be exceeded or fallen below.
- Operating the machine if not in proper condition.

3 Machine description

3.1 Polytex

3.1.1 Machine overview
3.1.2 Unwind
3.1.3 Web processing station, outer web – Draw with dancer roller
3.1.4 Micro-perforator, outer web
3.1.5 Continuous embossing unit
3.1.6 Side gusseted tube forming station
3.1.7 Seam extruder
3.1.8 Main draw
3.1.9 Cooling zone
3.1.10 Cut edge blocking
3.1.11 Oscillation

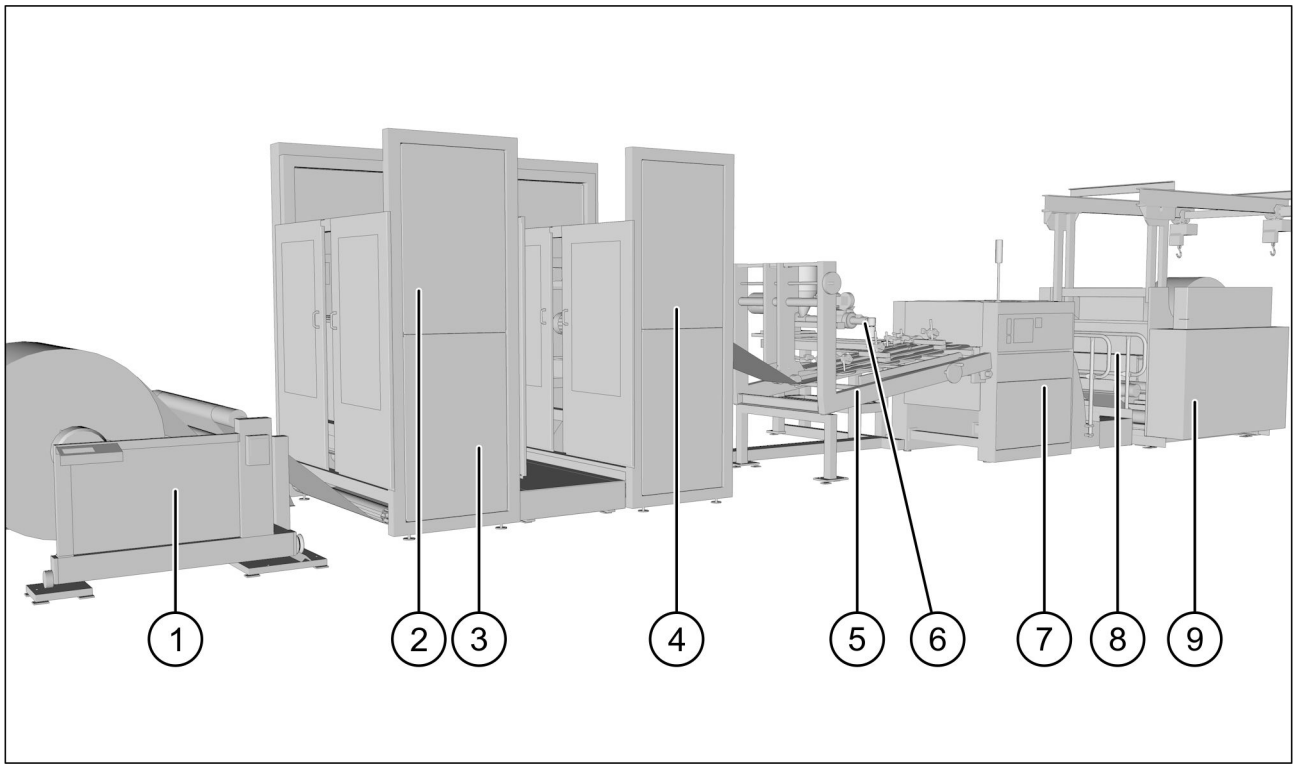
3.2 Winder

3.2.1 Winding principles
3.2.2 Overall view
3.2.3 Functional groups
3.2.4 Vacuum cutover drum
3.2.5 Cut edge blocking
3.2.6 Winding shaft loading system
3.2.7 Center drive
3.2.8 Web guide control

3.1 Polytex

3.1.1 Machine overview....88
3.1.2 Unwind....88
3.1.3 Web processing station, outer web – Draw with dancer roller....89
3.1.4 Micro-perforator, outer web....89
3.1.5 Continuous embossing unit....90
3.1.6 Side gusseted tube forming station....90
3.1.7 Seam extruder....91
3.1.8 Main draw....91
3.1.9 Cooling zone....92
3.1.10 Cut edge blocking....92
3.1.11 Oscillation....93

3.1.1 Machine overview

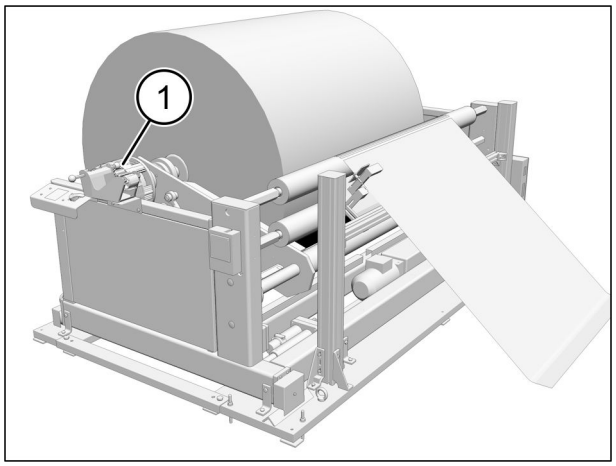


(1)	Unwinding unit	(2)	Venthole perforation
(3)	Draw with dancer roller	(4)	Continuous embossing unit
(5)	Side gusseted tube forming unit	(6)	Longitudinal seam extruder
(7)	Main draw with chill roll	(8)	Cut edge blocking
(9)	Winder with stand displacement		

3.1.2 Unwind

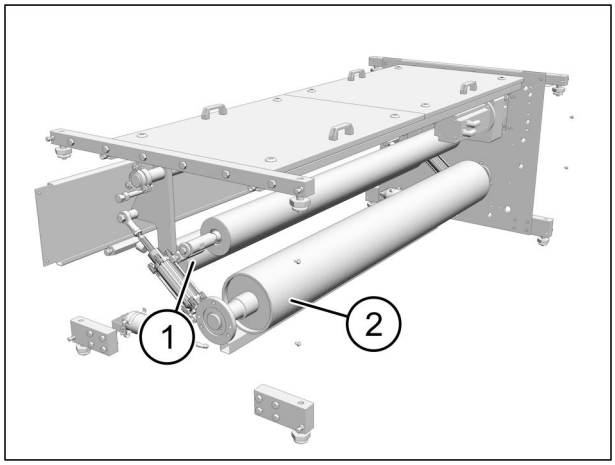
(1)	Roll brake

The unwind station is equipped with a roll brake (1).



3.1.3 Web processing station, outer web – Draw with dancer roller

(1)	Dancer roller
(2)	Draw roller

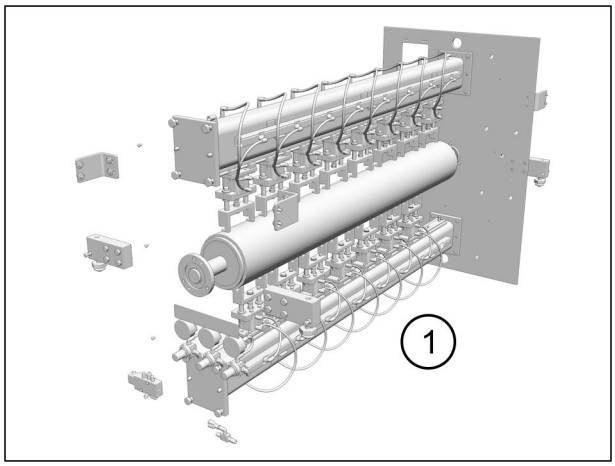


The dancer roller (1) controls the web tension. It keeps the web tension constant within defined limits. The web is wrapped around the dancer roller by 180°. Pneumatic cylinders at the swiveling levers generate the counter force to the web tensile strength and thus the web tension.

If tensile strength increases or decreases, the dancer roller leaves its mid-position to keep the web tension constant or evens it out. The sensor transmits a tension value analog to the dancer roller position to the controller of the drive of the draw roller (2) wrapped around by the web. This has an impact on the draw roller speed and the dancer roller is kept in its mid-position.

3.1.4 Micro-perforator, outer web

(1)	Micro-perforator



The micro-perforator (1) perforates the web.

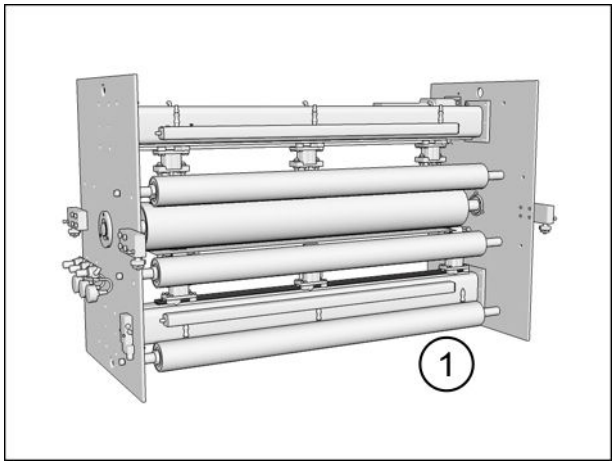
The air can later escape from the perforated sack while bagged.

3.1.5 Continuous embossing unit

(1)	Continuous embossing unit

The embossing unit (1) is provided with blunt needles which are pressed into the material. Doing so, the material gets a texture.

The texture prevents slippage when placing the sacks on top of each other.



3.1.6 Side gusseted tube forming station

(1)	Side gusseted tube forming station

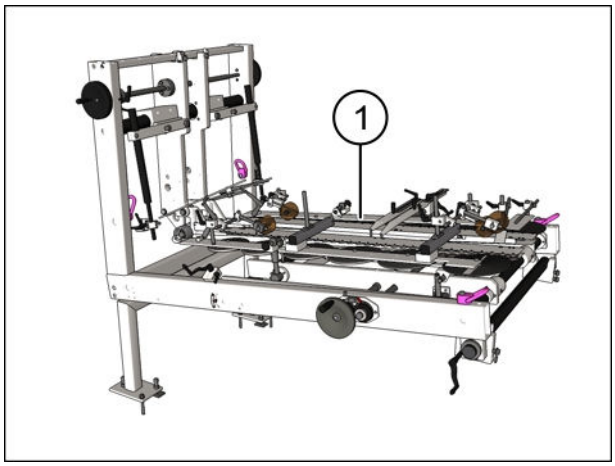
The material web is formed to a tube by former plates, rollers and guide rods in the tube forming station and overlapped in such a way that its edges can be glued together.

The web infeed section contains adjustable tooth rollers intended as deviating rolls.

In the infeed section only the center part of the material web is tensioned and the edges of the material webs can align themselves.

The edges of the material web are glued.

The tube is additionally provided with side gussets, e.g. for sewn or pinch-bottom bags.

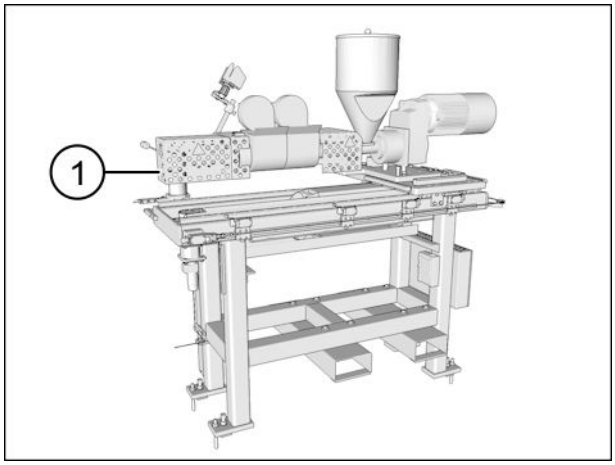


3.1.7 Seam extruder

(1)	Seam extruder

The seam extruder serves to produce plastic melt for web pasting.

The screw feeds in the resin-shaped raw material (thermoplastic polymers), to plasticize and homogenize it and then force the resultant plastic melt through the applicator nozzles under high pressure.



3.1.8 Main draw

(1)	Main draw
(2)	Seam cooling roller

The draw roller in the main draw (1) is separated in segments. The draw roller segments may be adjusted in the transverse direction to comply with the production conditions.

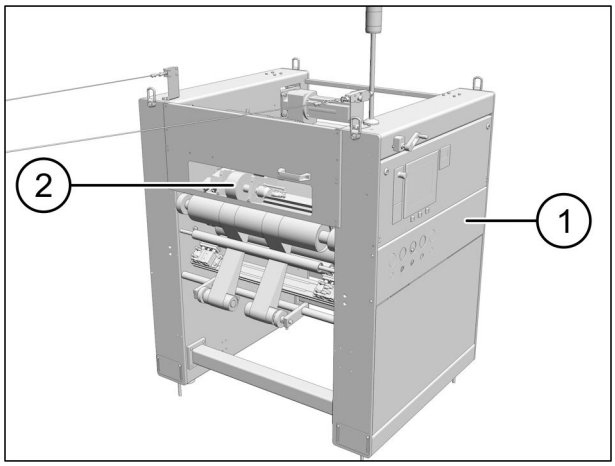
The cooling roller (2) cools and smoothens the longitudinal seam of the tube.

Cooling accelerates the solidification of the plastic melt from the longitudinal seam extruder. The joint between the folded ends of the web reaches the required strength faster.

The cooling roller can be moved transversely to position it at the center of the seam.

The cooling roller can be swiveled horizontally and vertically to influence the position of the plastic melt in the longitudinal seam.

The counter roller of the cooling roller is also cooled to accelerate the solidification of the plastic melt.



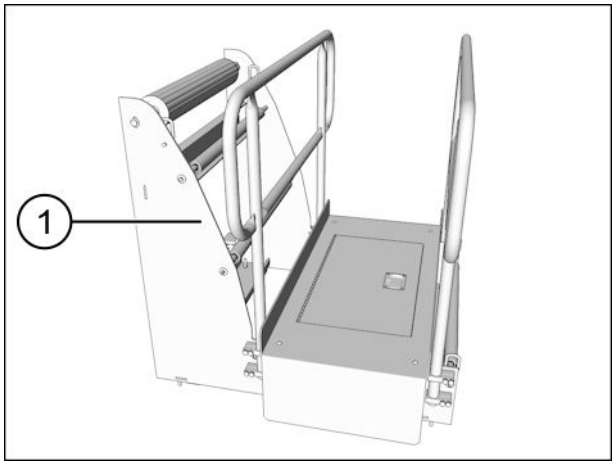
3.1.9 Cooling zone

(1)	Cooling zone

In the cooling zone, the longitudinal seam cools further down.

The rollers in the cooling zone support the web and feed it towards the rewind.

The shape of the cooling zone prevents strong web creasing.

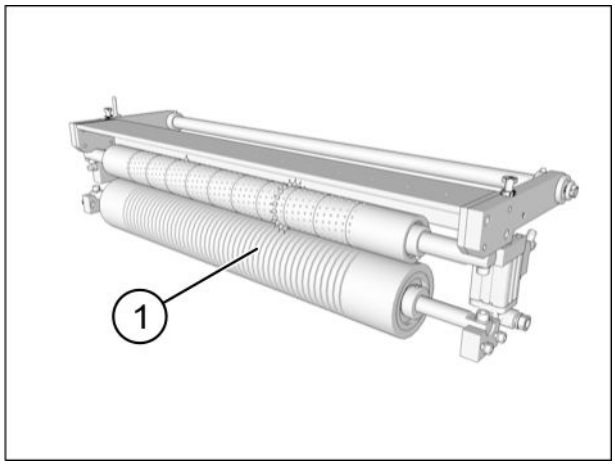


3.1.10 Cut edge blocking

(1)	Cut edge blocking

The needle roller in the cut edge blocking unit pierces through the film tube.

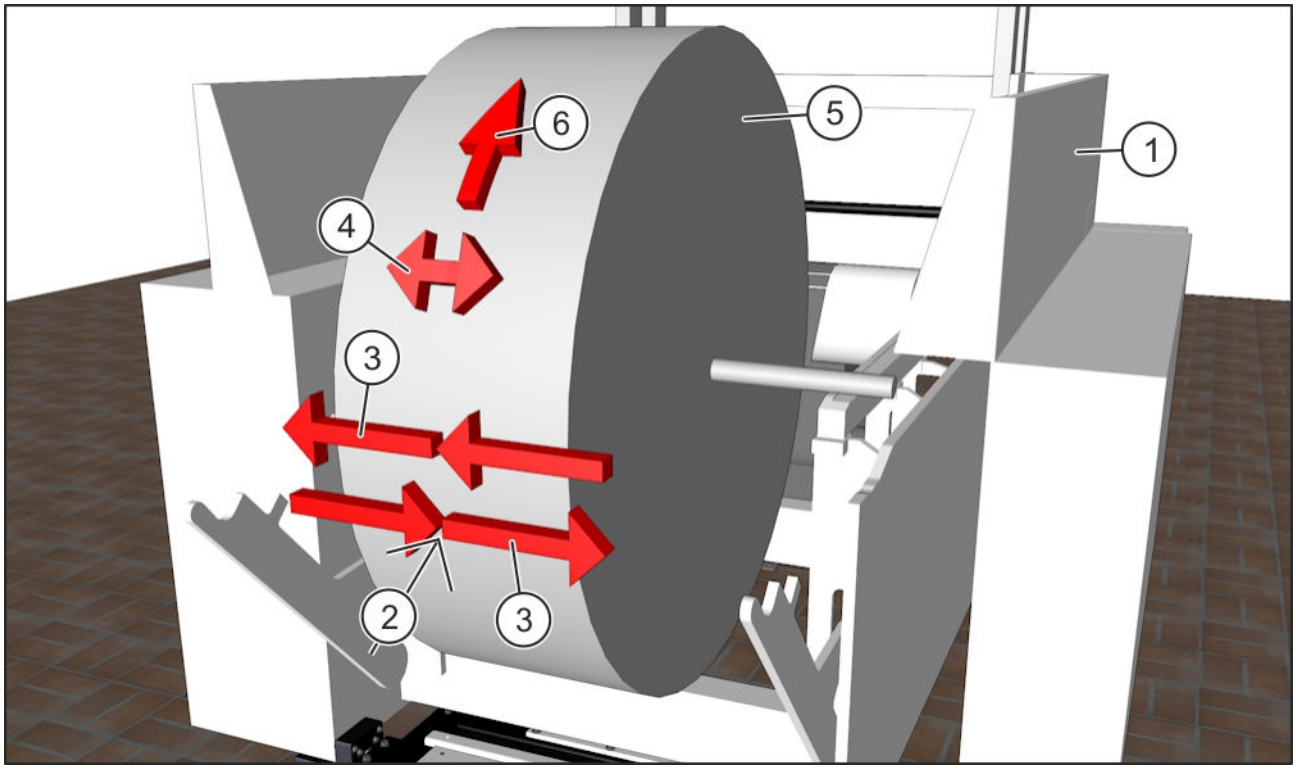
The holes caused by the needles connect the upper and the lower tube side. By connecting the tube sides the film tube remains together when cut off for the roll change in the winder.



3.1.11 Oscillation

The oscillating assembly continually offsets the web within the limited traversing ranges to keep the formation of thick spots in the winding roll to a minimum.

The entire winder can be moved. A drive slowly traverses the entire winder.



Pos.	Designation	Description
(1)	Winder	Winder with oscillating assembly
(2)	Zero postition	The winder traverses from its zero position.
(3)	Traversing range	The traversing range of the winder from its zero position.
(4)	Lateral offset	Traversing range of the winder per one rotation of the winding roll.
(5)	Winding roll	Wound up web
(6)	Rotational direction	Rotational direction of the winding roll

3.2 Winder

3.2.1 Winding principles 95

3.2.2 Overall view 96

3.2.3 Functional groups 97
- Contact roller with press roll 97
- Roll change system 97
- Cut-off knife assembly 97
- Winding shaft parallel guide 97
- Auxiliary roll drive 97
- Hydraulic roll delivery unit 97
- Hydraulic unit 97
- Web tension control 97
- Winding shaft loading system 97
- Deionization 97
- Web guide control unit 97
- Hoisting gears 97

3.2.4 Vacuum cutover drum 107

3.2.5 Cut edge blocking 108

3.2.6 Winding shaft loading system 109

3.2.7 Center drive 109

3.2.8 Web guide control 0

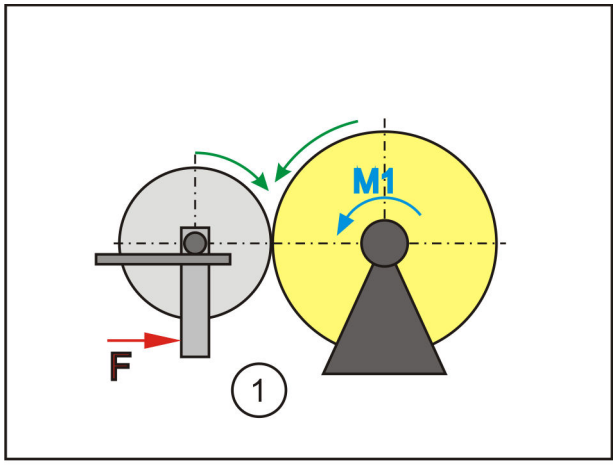
3.2.1 Winding principles

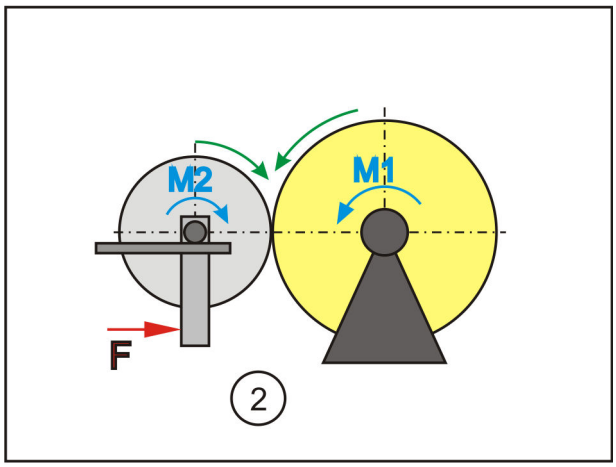
(1)	Surface winding
(2)	Surface/Center winding
(3)	Gap winding

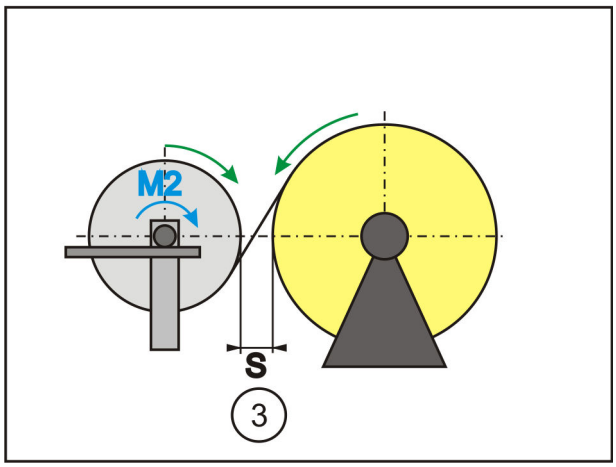
With the **surface winding** (1) alone the film roll is advanced against the driven contact roller using the contact force F. The center drive is uncoupled. The torque M1 of the speed controlled contact roller is transferred to the film roll from outside in a force-fitting manner and causes it to turn. To influence the roll hardness the contact force F can be changed linearly between a selectable initial and end diameter within a range from -50 and 300%.

With the combined **surface/center winding** (2) the film roll is driven from the outside via the contact roller with the torque M1 and centrally via the coupled current controlled center drive with the drive torque M2. To influence the roll hardness the contact force F can continually be changed between a selectable initial and end diameter within a range from -50 and 300%. Using the current controlled center drive, the web tension ratio can be tapered linearly between a selectable initial and end diameter by 0 to 50%.

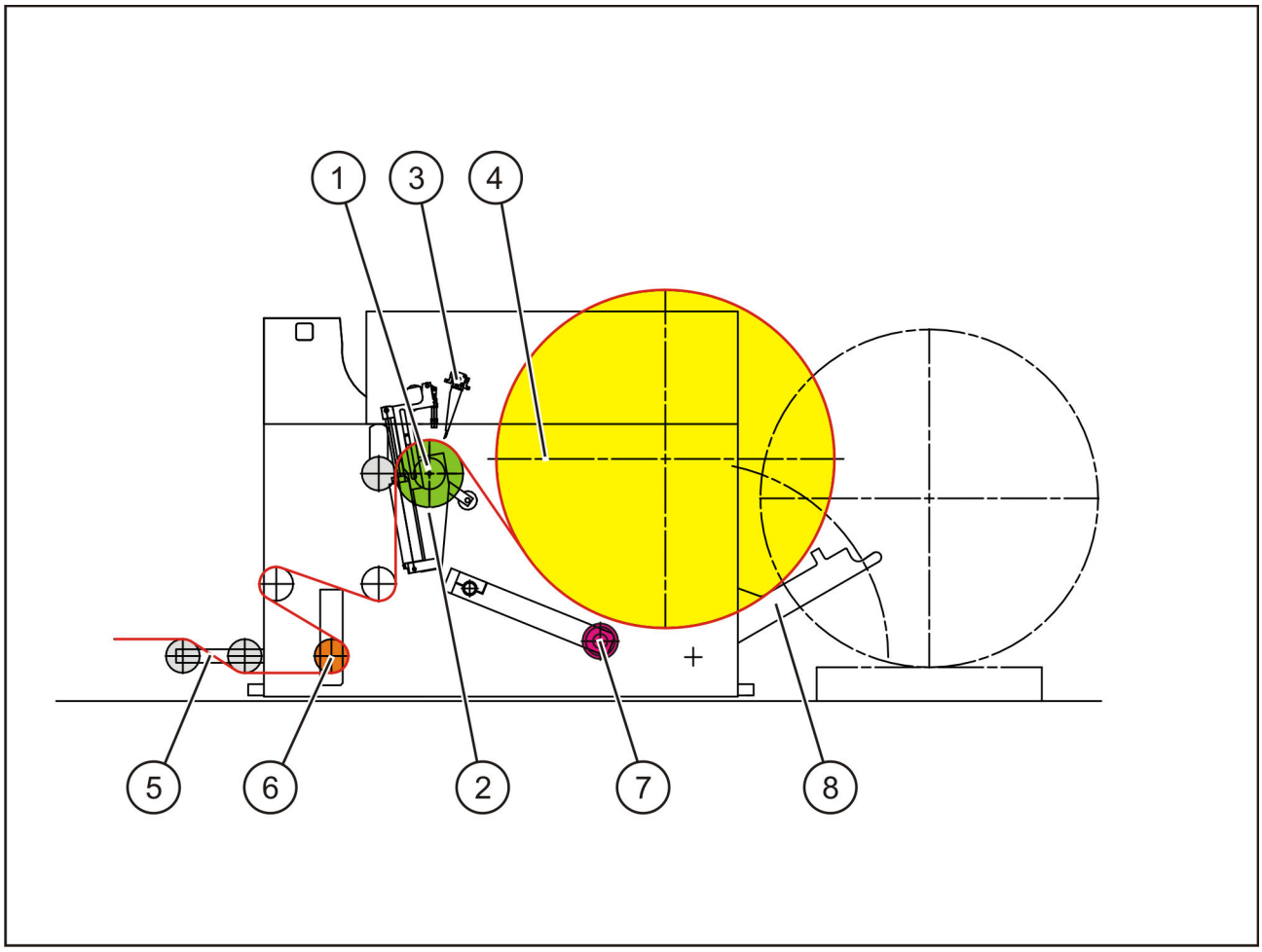
When **gap winding** (3) is used, the film roll never comes into contact with the contact roller. The nip between the film roll and the contact roller remains constant irrespective of the roll weight and roll diameter. The winding torque M2 is generated by a current controlled center drive. To influence the roll hardness the web tension ratio can be tapered linearly between a selectable initial and end diameter by 0 to 50%.







3.2.2 Overall view

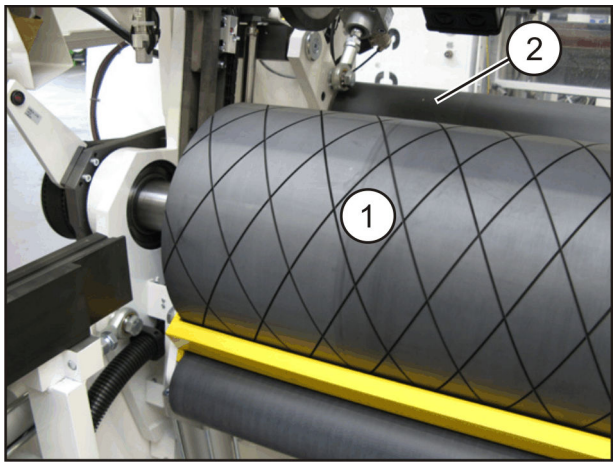


(1)	Contact roller with lay-on roller	(2)	Roll change system
(3)	Cross cutting system with flying knife	(4)	Winding shaft parallel guide
(5)	Web guide control unit	(6)	Web tension control
(7)	Auxiliary roll drive	(8)	Delivery arms

3.2.3 Functional groups

Contact roller with press roll

(1)	Contact roller
(2)	Press roll



During the contact winding, the contact roller (1) transfers the winding torque to the film reel with a defined contact force. A three-phase drive moves the stationary contact roller into position. The rubber coating of the roller can be smooth or cross-grooved. The air in front of the winder gap escapes via the grooves in the contact roller.

The contact roller uses friction to drive the pneumatically pressed down film reel via the circumference and transports the film up to the winding point. The contact force has a decisive effect on the winding strength of the film reel.

The pneumatically swiveling lay-on roller (2) reduces the air pulled in between the film and the contact roller and thus ensures that the film remains flat.

Roll change system

(1)	Roll change system with take-up fork and deflection roller

In the take-up position the new winding shaft is either taken out of the winding shaft magazine or directly inserted into the take-up fork of the roll change system (1) and locked.

As the roll change starts the winding shaft is pneumatically forced onto the contact roller carrying the web. The deflection roller swivels underneath the web and lifts it off the contact roller surface. Separating and initial winding of the web takes place in the initial winding position. After the separating cut, the roll change system carrying the new film reel swivels in its transfer position and places the reel down on the mounting rail of the parallel guide.

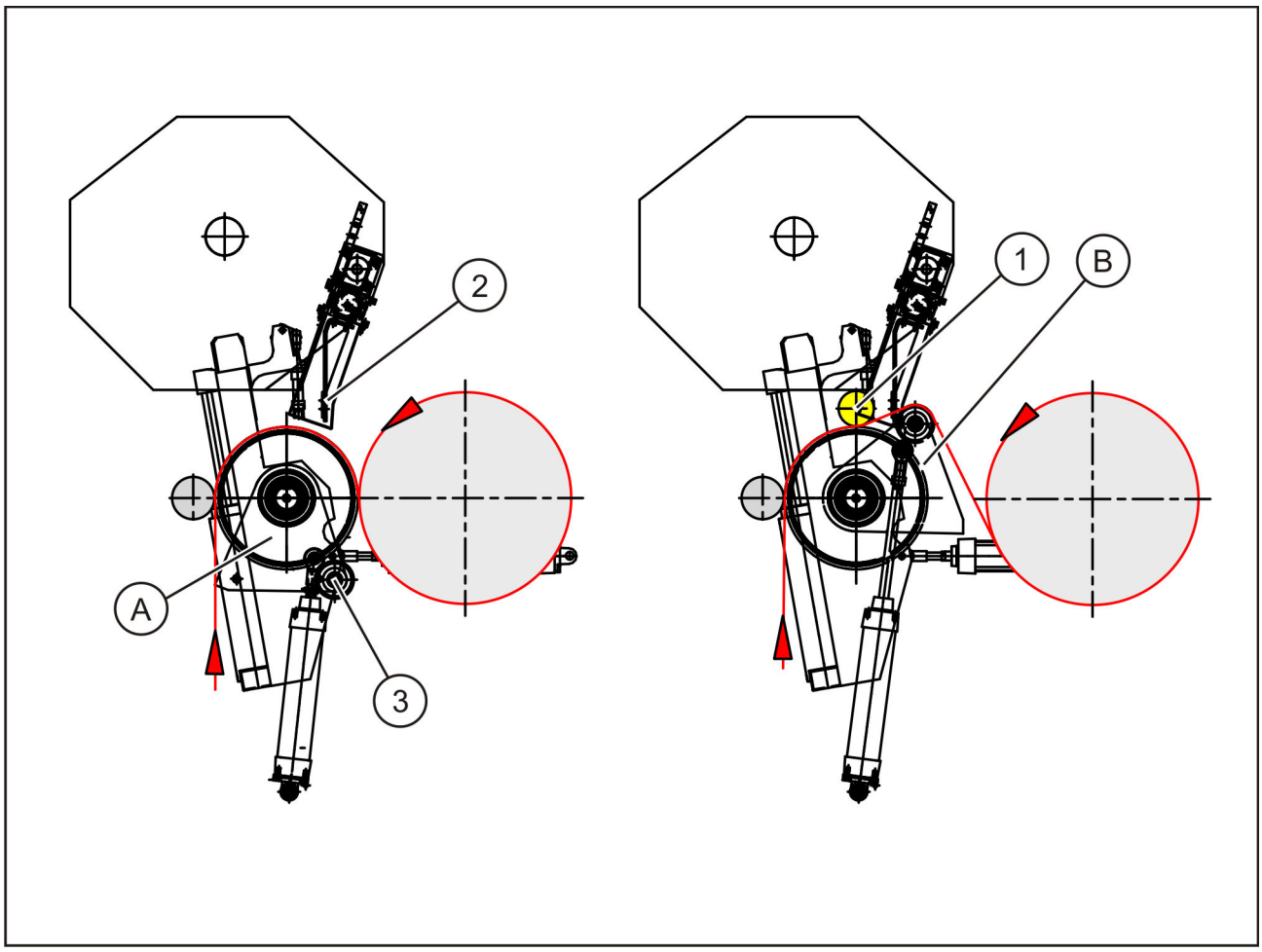
Cut-off knife assembly

Function

(1)	Draw blade

The cut-off assembly separates the web during reel change using a pneumatically driven draw blade (1) (diagonal cut).

Cross cutting sequence



(1)	Winding shaft	(2)	Draw blade
(3)	Deflection roller for cross cutting	(A)	Deflection roller in home position
(B)	Deflection roller in cutting position		

After the winding roll has been retracted from the contact roller the deflection roller swivels from its home position (A) into its cutting position (B). At the same time, it moves the web into the cutting position. The pneumatically driven flying knife (2) cuts the web between the newly inserted winding shaft (1) and the deflection roller. Afterwards, the deflection roller returns to its home position.

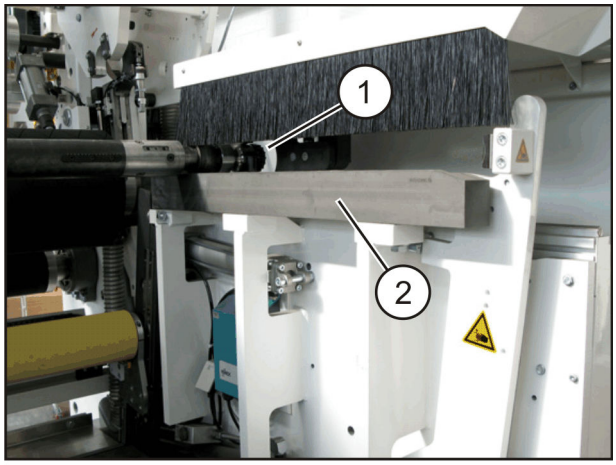
Winding shaft parallel guide

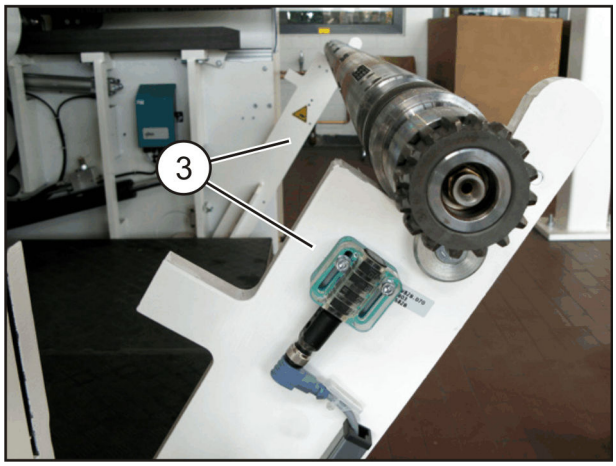
NOTE

The electromagnetic brake is a component important to safety and for this reason it has to be checked for correct functioning at regular intervals. It is imperative to keep the friction surfaces free of grease.

The electromagnetic brake of the parallel guide disposes of a function monitoring system. If faulty appears the message 'STOP BRAKE PARALLEL GUIDE DEFECTIVE'. For safety reasons, the stop brake must be repaired immediately in case of malfunction.

(1)	Jacks
(2)	Mounting rails
(3)	Delivery arm





Synchronously movable jacks (1) on both sides take up the winding shaft with the film reel and displace it on the mounting rails (2) towards the delivery arms (3) according to the increasing reel diameter parallel to the contact roller. Pneumatically operated linear cylinders displace the film reel. An electromagnetic brake decelerates the film reel and keeps it in position. In the 'Surface winding' mode, the pneumatic cylinders force the film reel against the contact roller at a pre-selected contact pressure.

In the Low Pressure Mode (LPM) the pneumatic cylinders are de-pressurized. The film winder automatically returns to the 'Center winding' mode. The web tension force slightly forces the film reel up against the contact roller. Sensitive films can be wound accordingly.

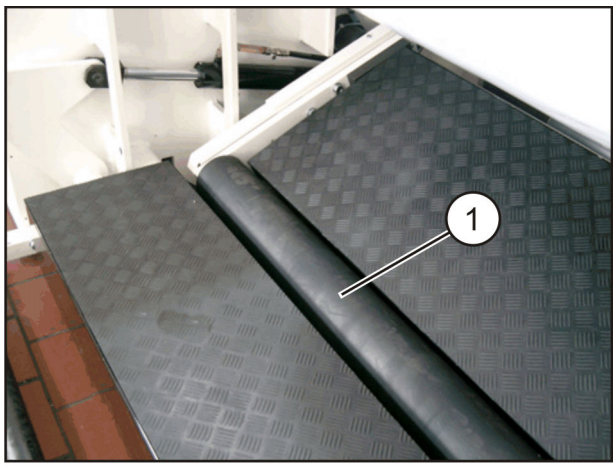
During reel change the film reel is retracted from the contact roller. The electromagnetic brake keeps the film reel in this very position.

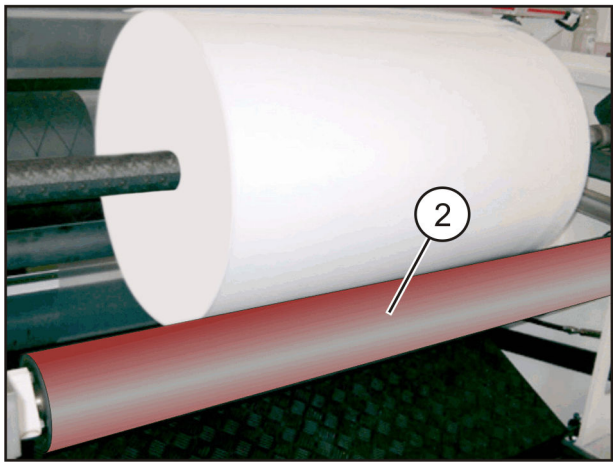
In the 'Gap winding' mode the film reel is incrementally retracted from the contact roller according to its increase in diameter. The nip between the film reel and the contact roller is between 5 and 10 mm. The electromagnetic brake keeps the film reel in this very position.

Auxiliary roll drive

(1)	Auxiliary roll drive
(2)	Auxiliary roll drive, loaded

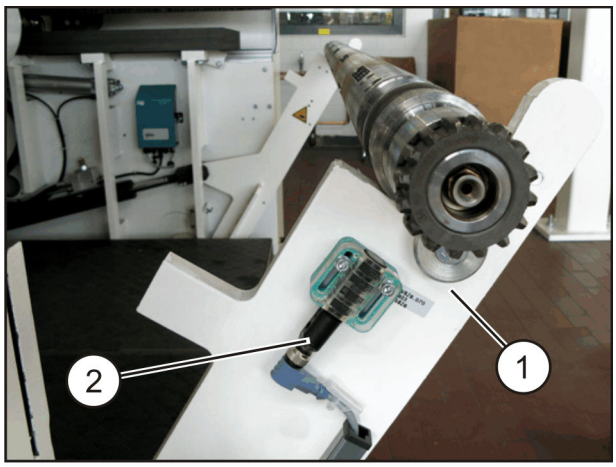
The pivotable auxiliary roll drive (1) is used during roll changes. It consists of a driven roller or driven rolls. The loaded auxiliary roll drive (2) drives the film roll when retracted from the contact roller and keeps the web tension constant. Therefore, the auxiliary roll drive is loaded to the film roll from below using adjustable pressure.

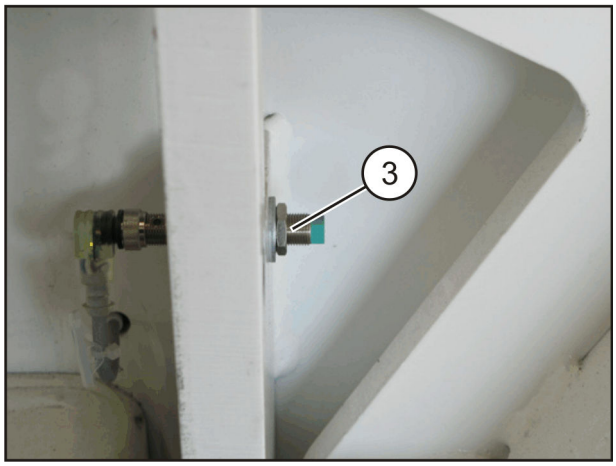




Hydraulic roll delivery unit

(1)	Delivery arms
(2)	Sensor for monitoring the winding shaft arrangement
(3)	Sensor for monitoring the delivery arm positioning





At the end of the mounting rails the jacks of the parallel guide separate from the winding shaft. Hydraulically pivotable delivery arms (1) which are interconnected in the center of rotation take the winding shaft together with the film reel and lower it in inching mode onto an underlay.

The sensors (2) monitor the winding shaft arrangement on the delivery arms. The sensors (3) monitor the delivery arm position.

The sensors dispose of an automatic function monitoring system. In case of malfunction appears the message 'PROXIMITY SENSOR AT LOWERING ARM DEFECTIVE'. For safety reasons, the sensors must be repaired immediately in case of malfunction.

Hydraulic unit

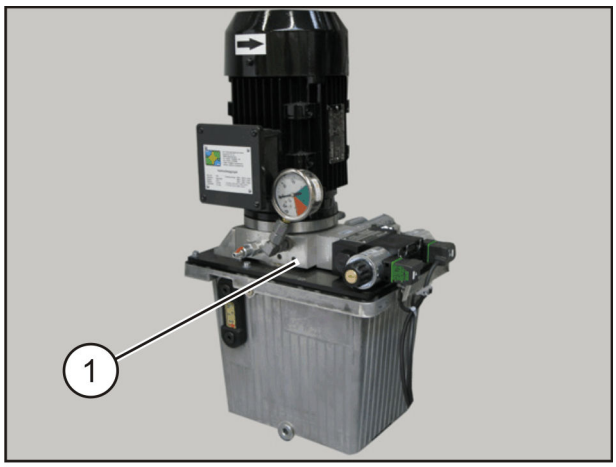
WARNING

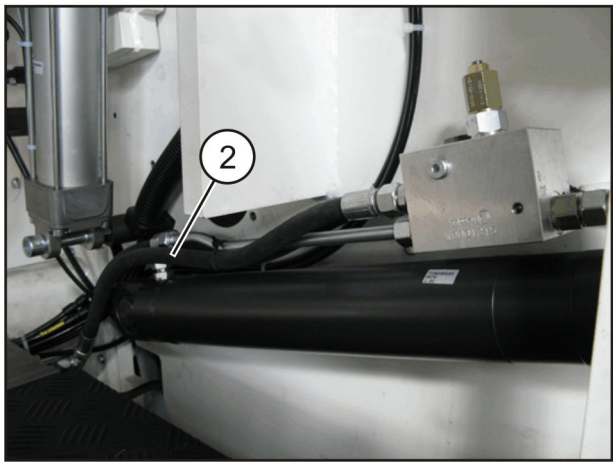
Check all lines, hoses and screw connections for leaks and externally visible damage at regular intervals! Repair any damage immediately. Spraying oil can cause injury and fires.

For safety reasons, the screwed hose connections, connectors and devices must not be loosened if the line is still pressurized. Reduce the load beforehand and switch off the pump.

Replace the hydraulic hoses every 4 years at the latest irrespective of their condition.

(1)	Hydraulic unit
(2)	Hydraulic hoses





The hydraulic unit (1) generates the oil pressure and the volumetric flow rate for swiveling the delivery arms of the reel delivery unit. The hydraulic cylinders are connected to the hydraulic unit via pipes and hoses (2).

The maximum operating pressure is 150 bar.

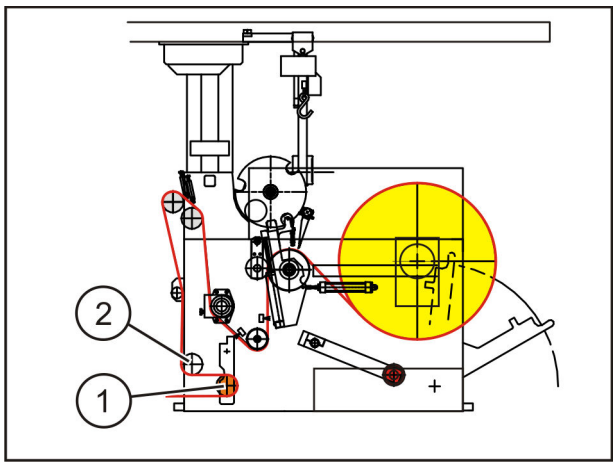
Depending on the control valve's position the hydraulic cylinders swivel the lowering arms downwards or upwards. An excess flow cut-off valve is provided to decelerate the lowering movement in case of hose breakage.

The hydraulic unit essentially consists of:

- an oil container
- an electric motor, a coupling and a pump
- a return filter
- a pressure control device
- a control valve unit
- an oil level indicator
- an oil drain plug
- an air filter

Web tension control

(1)	Dancer roller
(2)	Load-cell roller



The web tension control serves to keep the web tension constant within defined limits. This can be realized either via the dancer roller (1) or via the load-cell roller (2).

The web is guided around the dancer roller (1) via idler rollers. Pneumatic cylinders at the cam followers generate the counter-force to the web tensile force and thus the web tension. As the web tensile force increases or decreases the dancer roller deviates from its vertical center position. A tension value analog to the dancer roller position is transmitted to the contact roller drive control by means of a position proximity switch. Thereby, the speed of the contact roller drive is affected and the actual value of the web tension is adapted to the set web tension setpoint.

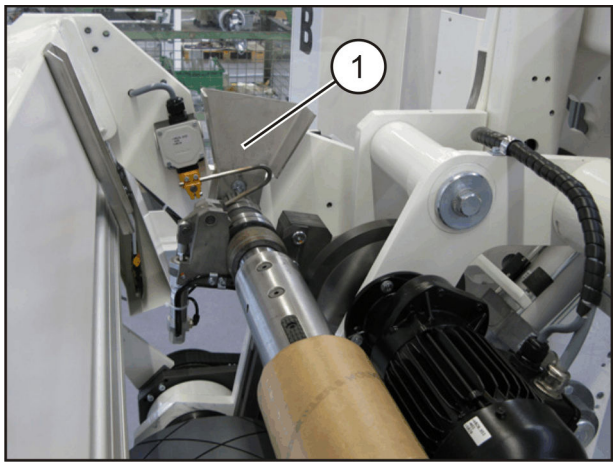
Winding shaft loading system

NOTE

The winding shaft must only be inserted in the extended delivery arms for fitting the winding cores. Then, use a hoisting gear or take the winding shaft by hand to insert it in the winding shaft loading system.

It is forbidden to insert the winding shaft from any other position as this will cause damage to the winder.

(1)	Winding shaft loading system



The winding shaft loading system serves to guide the winding shaft when it is inserted into the take-up fork of the roll change system. The limit switch indicates the loading of the take-up fork with a winding shaft.

The winding shaft magazine is loaded either manually or by using a hoisting gear.

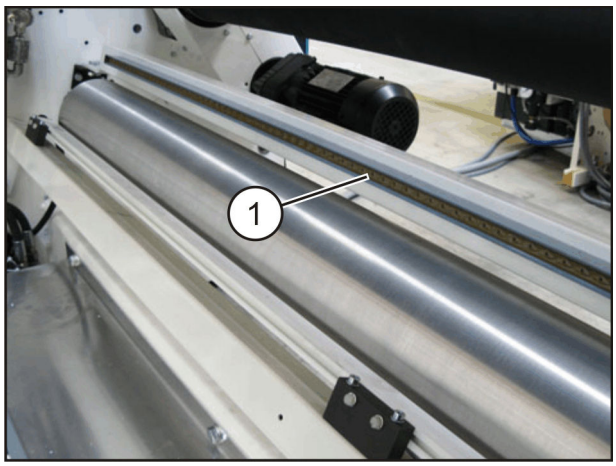
The winding shaft always runs through the sections of the winding station in the following sequence:

- Winding shaft magazine
- Roll change system
- Parallel guide
- Delivery arms

Deionization

(1)	Anti-static bar

The anti-static bar (1) removes static charges from the electro-statically charged film surface.The high-voltage applied to the anti-static bar causes artificial discharges at its tips. Positive and negative ions induce a balance of the charges on the web surface.



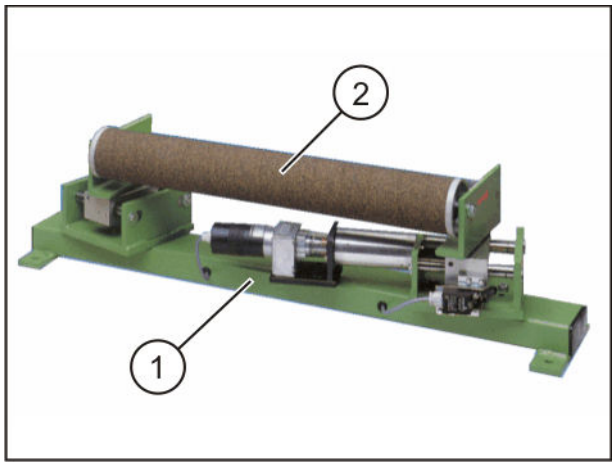
Web guide control unit

(1)	Web guide control unit
(2)	Swivel roller

The web guide control unit (1) provides a straight-edged web infeed into the pretreater and onto the winding roll. The aligning of the film web takes place according to an imaginary center line or the left or right web edge resp..

The web guide control (1) consists of a swivel roller (2) equipped with a servo drive and a control unit.

Moving the swivel roller causes the web to be correctively positioned in the desired direction. In case of an oncoming web wander the positioning unit moves to the side and guides the exiting web through a scanning system in a defined position on the next deflection roller.



Hoisting gears

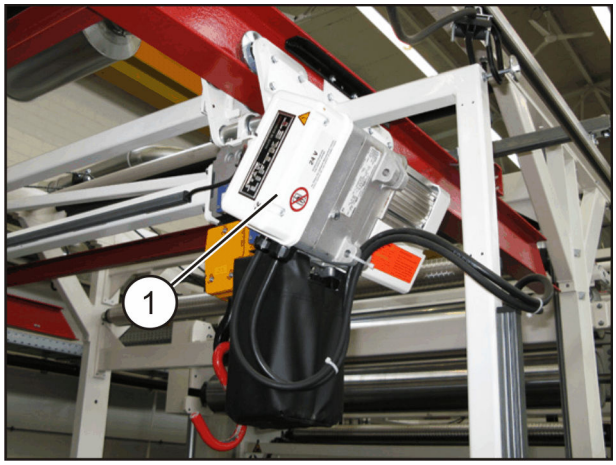
NOTE

The hoisting gears must only be used to transport empty winding shafts.
The weight of a winding shaft must not exceed 300 kg (67.5 lbf).

For further information on the hoisting gears please refer to the sub-vendor documentation.
For the sling points (position, load bearing capacity) please refer to the transport and handling plan.

(1)	Crane trolley with 2 electric chain hoists

The crane trolley featuring two electric chain hoists is used to take the winding shafts back out of the delivery arms into the winding shaft magazine's inserter or directly into the roll change system. The crane trolley can be moved from the outside without having to go into the interior of the winder. The chain hoists are power-driven.



3.2.4 Vacuum cutover drum

When using the vacuum cutover drum the maximum permissible web speed during roll change depends on the air pressure in the cutting cylinders and on the working width of the winder. The programmable logic controller (PLC) monitors the maximum permissible web speed.

Option 1: System pressure < 4 bar (58 psi)	
Compressed-air supply pressure	< 4 bar (58 psi)
Cutting pressure	< 4 bar (58 psi)
Pressure switch setting	< 4 bar (58 psi)

Working width of film winder [mm] ([inch])	**Max. web speed [m/min] ([ft/min])**
900 - 2,200 (35 - 87)	200 (656)

Option 2: System pressure 4 - < 6 bar (58 - < 87 psi), equipped with pressure transmission	
Pneumatic system pressure	4 < 6 bar (58 - < 87 psi)
Cutting pressure	< 6 bar (87 psi)
Pressure switch setting	< 6 bar (87 psi)

Working width of film winder [mm] ([inch])	**Max. web speed [m/min] ([ft/min])**
2,200 - 3,600 (87 - 142)	200 (656)

At a cutting pressure between 4 and 6 bar (58 - 87 psi) a roll change can be performed at a maximum web speed of 120 m/min (394 ft/min).

If dropping below a mininum pressure of 6 bar the alarm message 'Pressure short fall on vacuum cutover drum' appears. At a web speed exceeding120 m/min (394 ft/min) the roll change will not be performed.

Option 3: System pressure < 6 bar (< 87 psi)	
Pneumatic system pressure	< 6 bar (< 87 psi)
Cutting pressure	< 6 bar (< 87 psi)
Pressure switch setting	< 6 bar (87 psi)

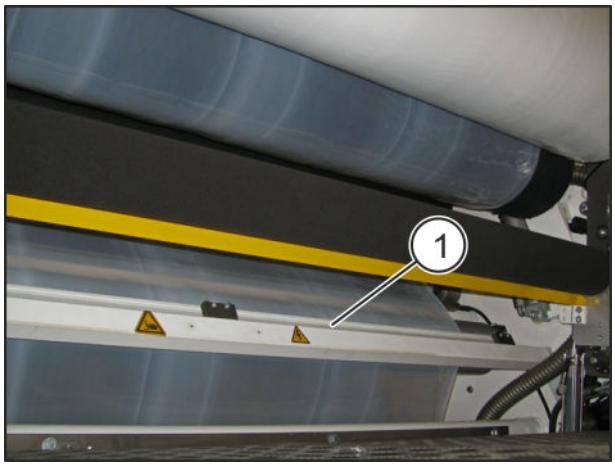
Working width of film winder [mm] ([inch])	**Max. web speed [m/min] ([ft/min])**
2,200 - 3,600 (87 - 142)	200 (656)

At a cutting pressure between 4 and 6 bar (58 - 87 psi) a roll change can be performed at a maximum web speed of 120 m/min (394 ft/min).

If dropping below a mininum pressure of 6 bar the alarm message 'Pressure short fall on vacuum cutover drum' appears. At a web speed exceeding120 m/min (394 ft/min) the roll change will not be performed.

3.2.5 Cut edge blocking

(1)	Charging electrode



During roll change, the cut edge blocking system prevents the tubular film from opening immediately after the cross cut. The cut edge blocking system consists of the charging electrode (1) and a high-voltage generator.

The high-voltage generator is powered up when pressing the START ROLL CHANGE pushbutton. The passing tubular film is statically charged by the charging electrode. The film halves adhere to each other and thus prevent the tubular film from opening.

For threading the tubular film into the winder, the charging electrode can be swung away.

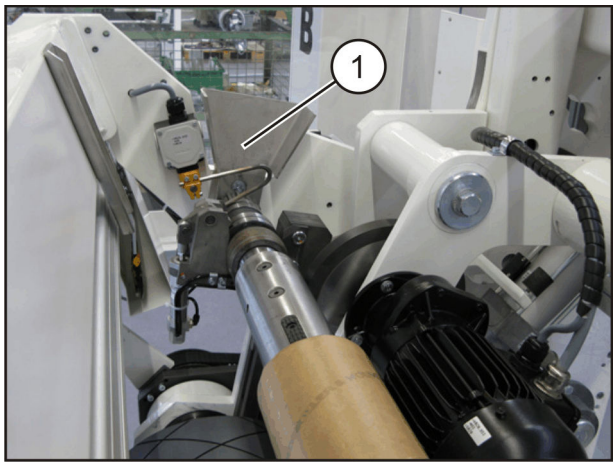
3.2.6 Winding shaft loading system

NOTE

The winding shaft must only be inserted into the extended delivery arms for fitting the winding cores. Then, use a hoisting gear or take the winding shaft by hand to insert it into the winding shaft loading system.

It is forbidden to insert the winding shaft from any other position as this will cause damage to the winder.

(1)	Winding shaft loading system



The winding shaft loading system serves to guide the winding shaft when it is inserted into the take-up fork of the roll change system. The limit switch indicates the loading of the take-up fork with a winding shaft.

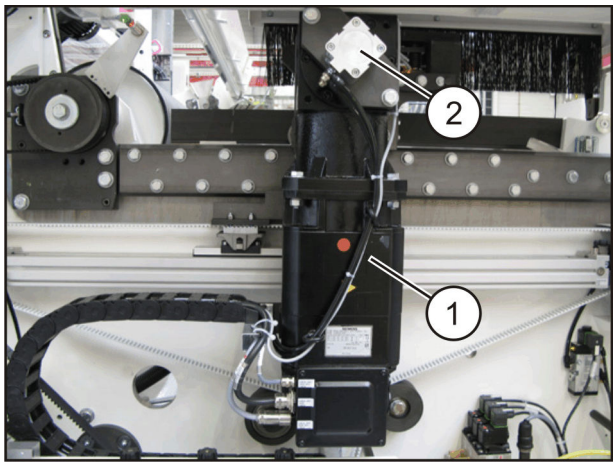
The winding shaft magazine is loaded either manually or by using a hoisting gear.

The winding shaft always runs through the sections of the winding station in the following sequence:

- Winding shaft magazine
- Roll change system
- Parallel guide
- Delivery arms

3.2.7 Center drive

(1)	Center drive
(2)	Bevel gear drive

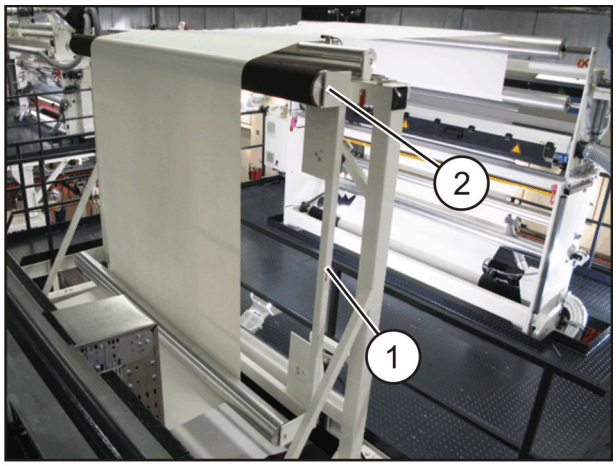


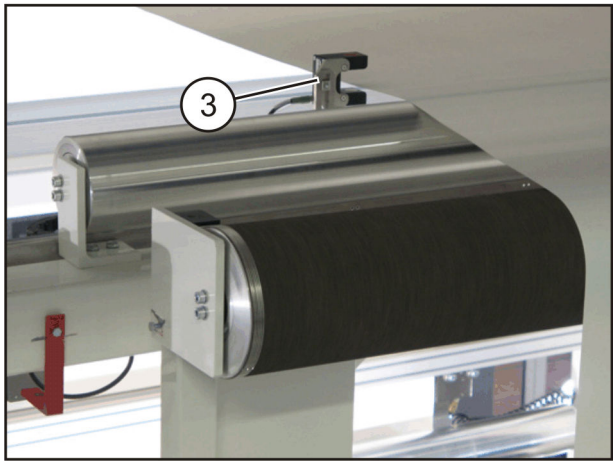
When in the 'surface/center winding' or 'gap winding' mode, the center drive (1) drives the winding shaft via the bevel gear drive (2). A coupling interconnects the bevel gear drive and the winding shaft.

3.2.8 Web guide control

For further information please refer to the chapter under the heading Sub-vendor documentation.

(1)	Web guide control
(2)	Control roller system
(3)	Sensor heads with ultrasonic sensors





The web guide control unit (1) provides a straight-edged web feed into the treater, into the slitting assembly and onto the winding roll. The aligning of the film web takes place according to an imaginary center line or the left or right web edge resp..

The web guide control unit features:

- a control roller system (2) with electro-motorized servo drive
- non-contact edge detection by two sensor heads with ultrasonic sensors (3)
- and a control unit for signal conversion.

The web is repositioned correctly as required by the movement of the control roller system laid out as a pivot-mounted base frame with idler rollers. An infeed web wander causes the positioning unit of the control roller system to turn around a pivot point on the infeed side and transfers the web through the detector system in a defined position onto the following deflection roller.

The web edges can be oscillatingly positioned by means of the oscillating sensor heads. Continuous repositioning of the web edges reduces strong edge thickening in the folded area of the lay-flat tube during winding.

4 Control elements and screen pages

4.1 Operator panels: Overview
4.1.1 Control elements
4.1.2 Polytex Operator panels

4.2 Operator panels: Description
4.2.1 Unwind
4.2.2 Web processing station with micro-perforator and embossing unit
4.2.3 Main draw

4.3 Screen pages
4.3.1 Information on screen pages
4.3.2 Home screen page
4.3.3 Production screen page
4.3.4 Display screen page
4.3.5 Parameters/settings screen page
4.3.6 I/O peripherals screen page
4.3.7 ACOPOS screen page
4.3.8 Touch screen screen page
4.3.9 Module overview screen page
4.3.10 Unwind screen page
4.3.11 Web processing screen page
4.3.12 Longitudinal seam screen page
4.3.13 Heating zones screen page
4.3.14 Temperature profile screen page
4.3.15 Longitudinal seam monitoring screen page
4.3.16 Draw screen page
4.3.17 Oscillation screen page
4.3.18 Cut edge blocking screen page
4.3.19 Alarm list screen page - Log
4.3.20 Enable screen page
4.3.21 Contact screen page

4.4 Screen pages: Operation
4.4.1 Operating concept
4.4.2 Header
4.4.3 Information line
4.4.4 Footer
4.4.5 Entering your password
4.4.6 Touch screen screen page
4.4.7 Setting up the web processing station
4.4.8 Setting up longitudinal seam monitoring
4.4.9 Enabling the software

4.5 Winder

4.5.1 Main control console +R82A for the winder
4.5.2 Screen in control console +R82A/B
4.5.3 Control elements on operator panel +R82
4.5.4 Operator panel +R85A
4.5.5 Control consoles +R84A/B
4.5.6 Pneumatic control elements in the winding station

4.1 Operator panels: Overview

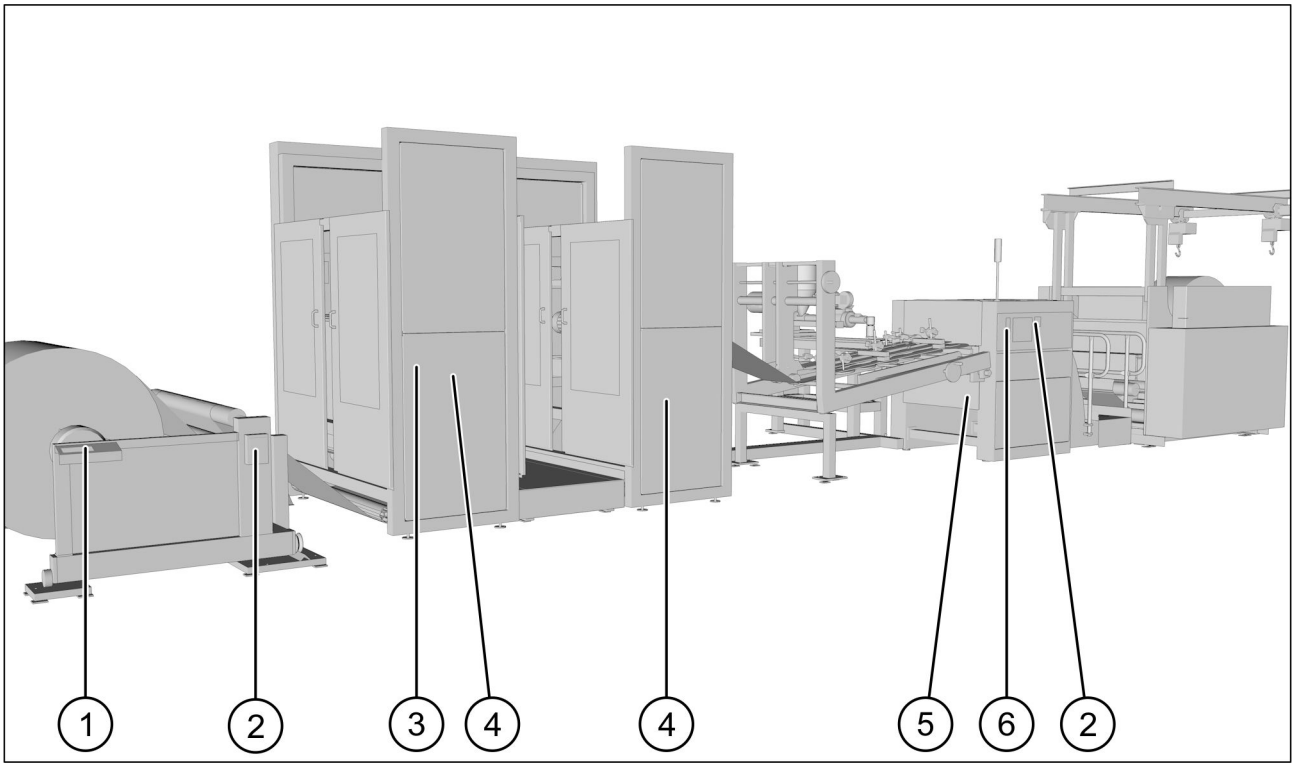
4.1.1 Control elements....114

4.1.2 Polytex Operator panels....114

4.1.1 Control elements

- In order for any function to work the control voltage must be on.
- An illuminated switch or button indicates that the function has been switched on.
- The switches and buttons are listed in accordance with their location seen in running direction of the machine.
- Depending on the machine layout, the operator panel controls might vary.

4.1.2 Polytex Operator panels

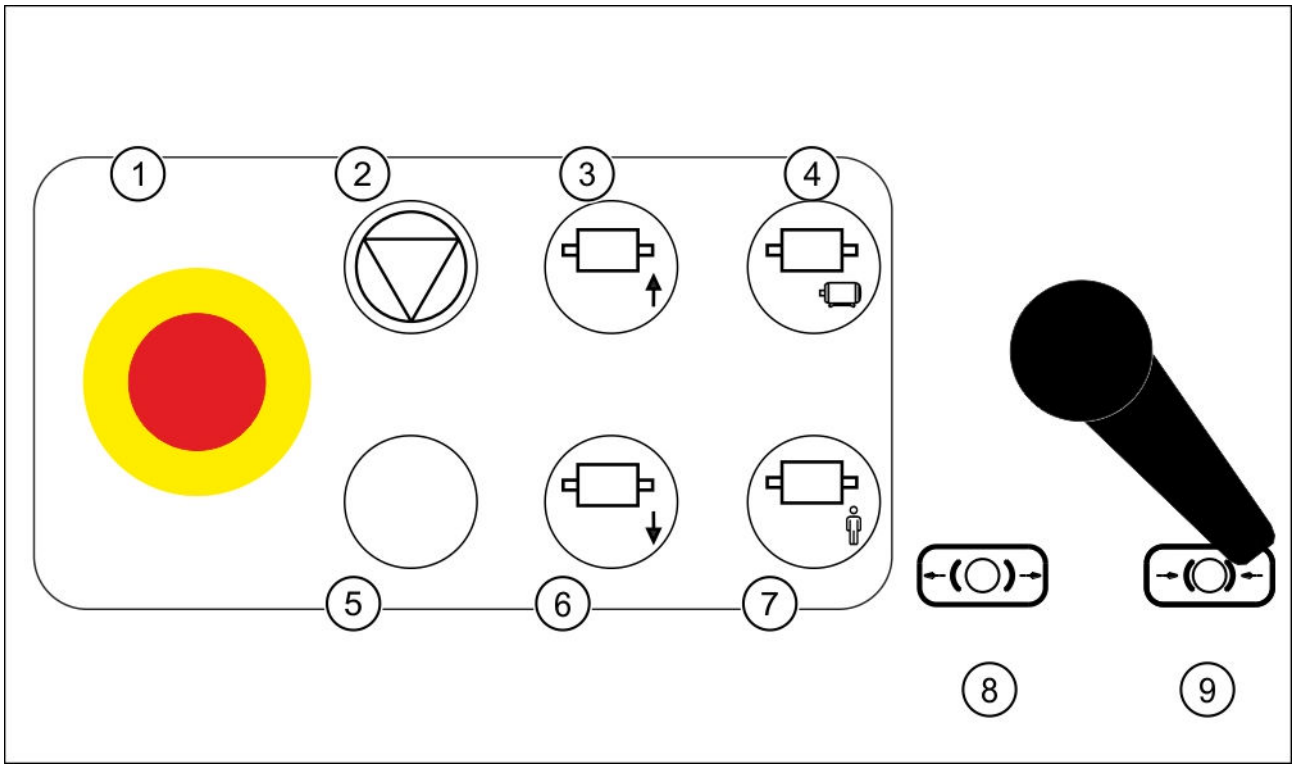


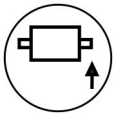
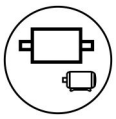
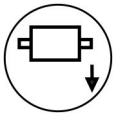
(1)	Operator panel Unwind, +R10	(2)	Operator panel Web guide control
(3)	Operator panel Continuous venthole perforation and embossing unit, +R15	(4)	Pneumatic setting elements
(5)	Compressed air conditioner, pneumatics	(6)	Main draw Operator panel +R20

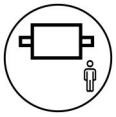
4.2 Operator panels: Description

4.2.1 Unwind....116
4.2.2 Web processing station with micro-perforator and embossing unit....118
4.2.3 Main draw....120

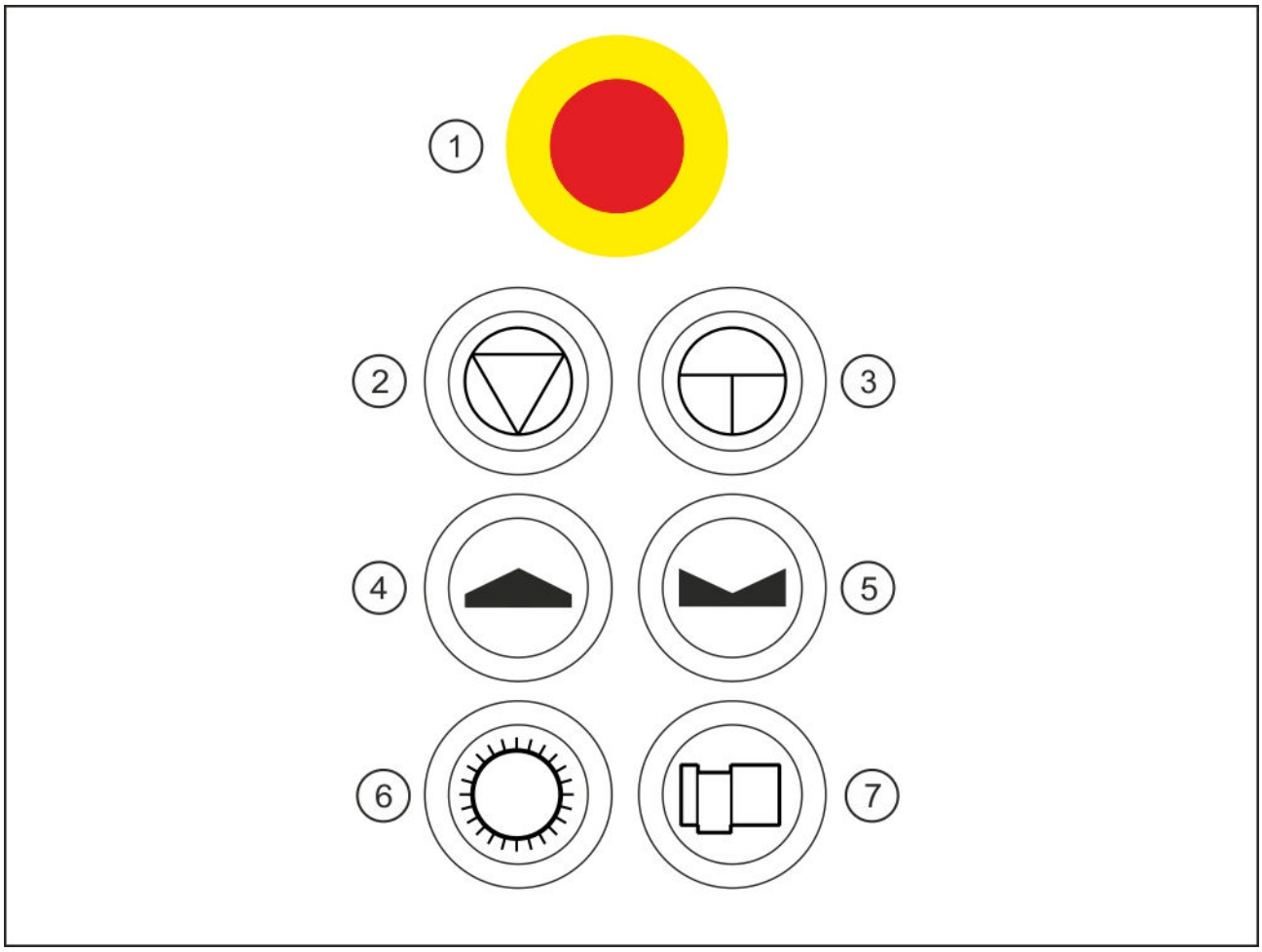
4.2.1 Unwind



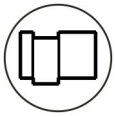
Pos.	Icon	Designation	Description
(1)		EMERGENCY-STOP	Press the Emergency-Stop switch whenever a dangerous situation arises. The machine is shut down in a controlled manner. The switch is locked.
(2)		STOP	Press the button once. After pressing the button the machine is set to a standstill in a controlled manner. Press the button twice. The machine is stopped quickly. Press the button for at least three seconds. The guard lockings of the guard doors open.
(3)		ROLL UP	Hold down the button to raise the roll.
(4)		MOVE ROLL TOWARDS THE DRIVE SIDE	Hold down the button to move the roll towards the drive side.
(5)	–	–	–
(6)		ROLL DOWN	Hold down the button to lower the roll.

Pos.	Icon	Designation	Description
(7)		MOVE ROLL TOWARDS THE OPERATOR SIDE	Hold down the button to move the roll towards the operator side.
(8)		POSITION: BRAKE OFF	Set the switch to this position to switch off the brake. There will be no controlled web tension between the unwind and the first driven roller or the draw. The roll unwinds in an uncontrolled manner.
(9)		POSITION: BRAKE ON	Set the switch to this illustrated position to switch on the brake. The brake keeps the set web tension constant.

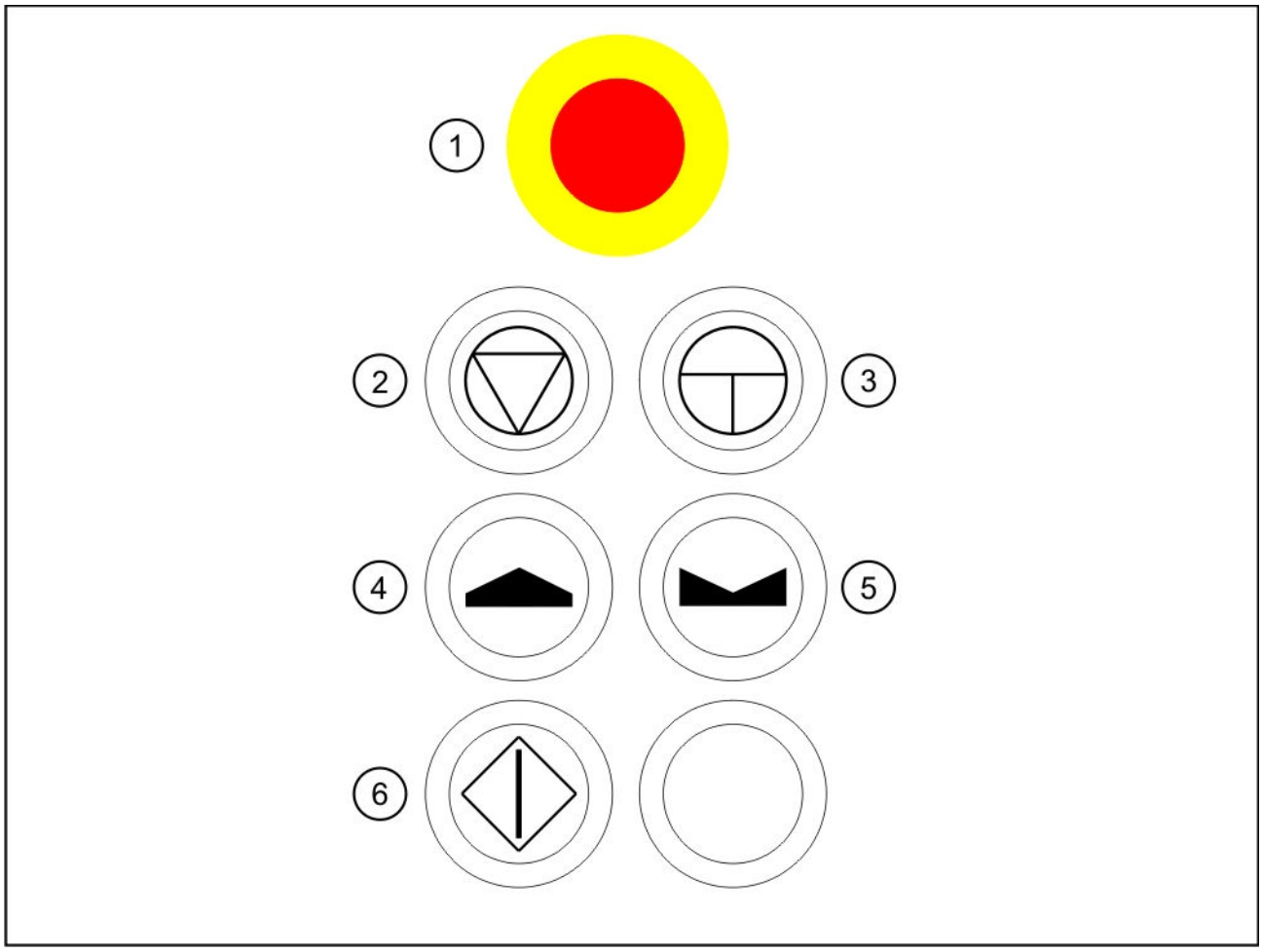
4.2.2 Web processing station with micro-perforator and embossing unit



Pos.	Icon	Designation	Description
(1)		EMERGENCY-STOP	Press the Emergency-Stop switch whenever a dangerous situation arises. The machine is shut down in a controlled manner. The switch is locked.
(2)		STOP	Press the button once. After pressing the button the machine is set to a standstill in a controlled manner. Press the button twice. The machine is stopped quickly. Press the button for at least three seconds. The guard lockings of the guard doors open.

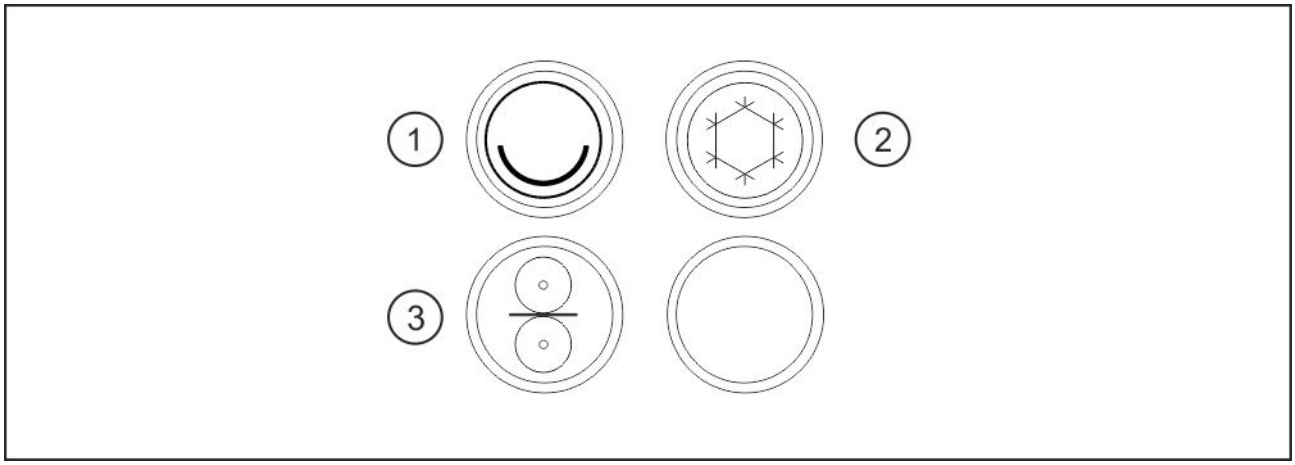
Pos.	Icon	Designation	Description
(3)		JOG	Press the button. The guard lockings of the guard doors close. A signal sounds for one second. Let go of the button. After two seconds, the machine is ready to be switched on. The button lights up. Press and hold down the button again within the next 10 seconds. The relevant drive is rotating and stops when letting go of the button. The jog mode drops out and the button goes dark unless the machine is started within the next 10 seconds. After dropping out, jog mode must be built up again.
(4)		FAST	Requirements: • The “Machine On” procedure has been passed. Press the button to increase the machine speed. The machine is accelerated while holding the button. The machine is accelerated to maximum production speed.
(5)		SLOW	Requirements: • Went through the “Machine On” and “Fast” procedure. Press the button to reduce the machine speed. The machine speed is reduced while holding the button.The machine speed drops to base speed.
(6)		VENT HOLE PERFORATION	Switches the venthole perforation on and off, the pushbutton lights up: the venthole perforation is switched on
(7)		EMBOSSING UNIT	Switches the embossing unit on and off, the button lights up: the embossing unit is switched on

4.2.3 Main draw

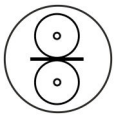


Pos.	Icon	Designation	Description
(1)		EMERGENCY-STOP	Press the Emergency-Stop switch whenever a dangerous situation arises. The machine is shut down in a controlled manner. The switch is locked.
(2)		STOP	Press the button once. After pressing the button the machine is set to a standstill in a controlled manner. Press the button twice. The machine is stopped quickly. Press the button for at least three seconds. The guard lockings of the guard doors open.

Pos.	Icon	Designation	Description
(3)		JOG	Press the button. The guard lockings of the guard doors close. A signal sounds for one second. Let go of the button. After two seconds, the machine is ready to be switched on. The button lights up. Press and hold down the button again within the next 10 seconds. The relevant drive is rotating and stops when letting go of the button. The jog mode drops out and the button goes dark unless the machine is started within the next 10 seconds. After dropping out, jog mode must be built up again.
(4)		FAST	Requirements: • The “Machine On” procedure has been passed. Press the button to increase the machine speed. The machine is accelerated while holding the button. The machine is accelerated to maximum production speed.
(5)		SLOW	Requirements: • Went through the “Machine On” and “Fast” procedure. Press the button to reduce the machine speed. The machine speed is reduced while holding the button.The machine speed drops to base speed.
		MACHINE ON	Press the button. The guard lockings of the guard doors close. The work flow 'Machine On' starts.



Pos.	Icon	Designation	Description
(1)		EDGE PRESSING UNIT/SQUEEZE ROLL	Lowers and raises the edge pressing unit/squeeze roll activated on the 'Draw' screen page.

Pos.	Icon	Designation	Description
(2)		SEAM COOLING ROLLER	Lowers and lifts the seam cooling roller in the infeed section of the main draw down on and off the longitudinal seam.
(3)		DRAW	Lowers and lifts the upper draw roller in the main draw onto and off the lower draw roller.

4.3 Screen pages

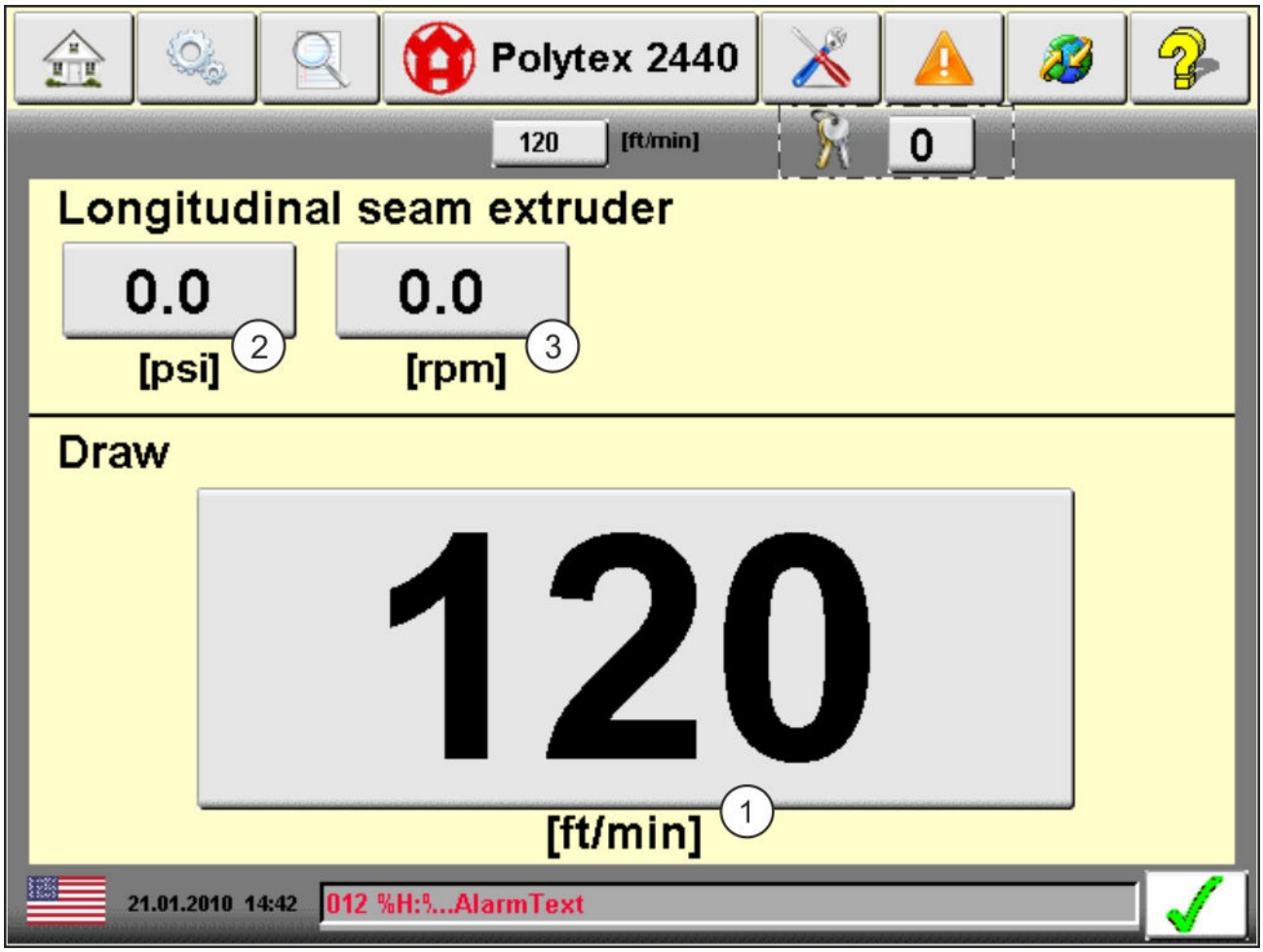
4.3.1 Information on screen pages....124
4.3.2 Home screen page....124
4.3.3 Production screen page....125
4.3.4 Display screen page....126
4.3.5 Parameters/settings screen page....127
4.3.6 I/O peripherals screen page....129
4.3.7 ACOPOS screen page....130
4.3.8 Touch screen screen page....131
4.3.9 Module overview screen page....132
4.3.10 Unwind screen page....133
4.3.11 Web processing screen page....134
4.3.12 Longitudinal seam screen page....136
4.3.13 Heating zones screen page....138
4.3.14 Temperature profile screen page....140
4.3.15 Longitudinal seam monitoring screen page....141
4.3.16 Draw screen page....142
4.3.17 Oscillation screen page....145
4.3.18 Cut edge blocking screen page....146
4.3.19 Alarm list screen page - Log....148
4.3.20 Enable screen page....149
4.3.21 Contact screen page....150

4.3.1 Information on screen pages

The illustrated screen pages show all possible options the machine can be equipped with.

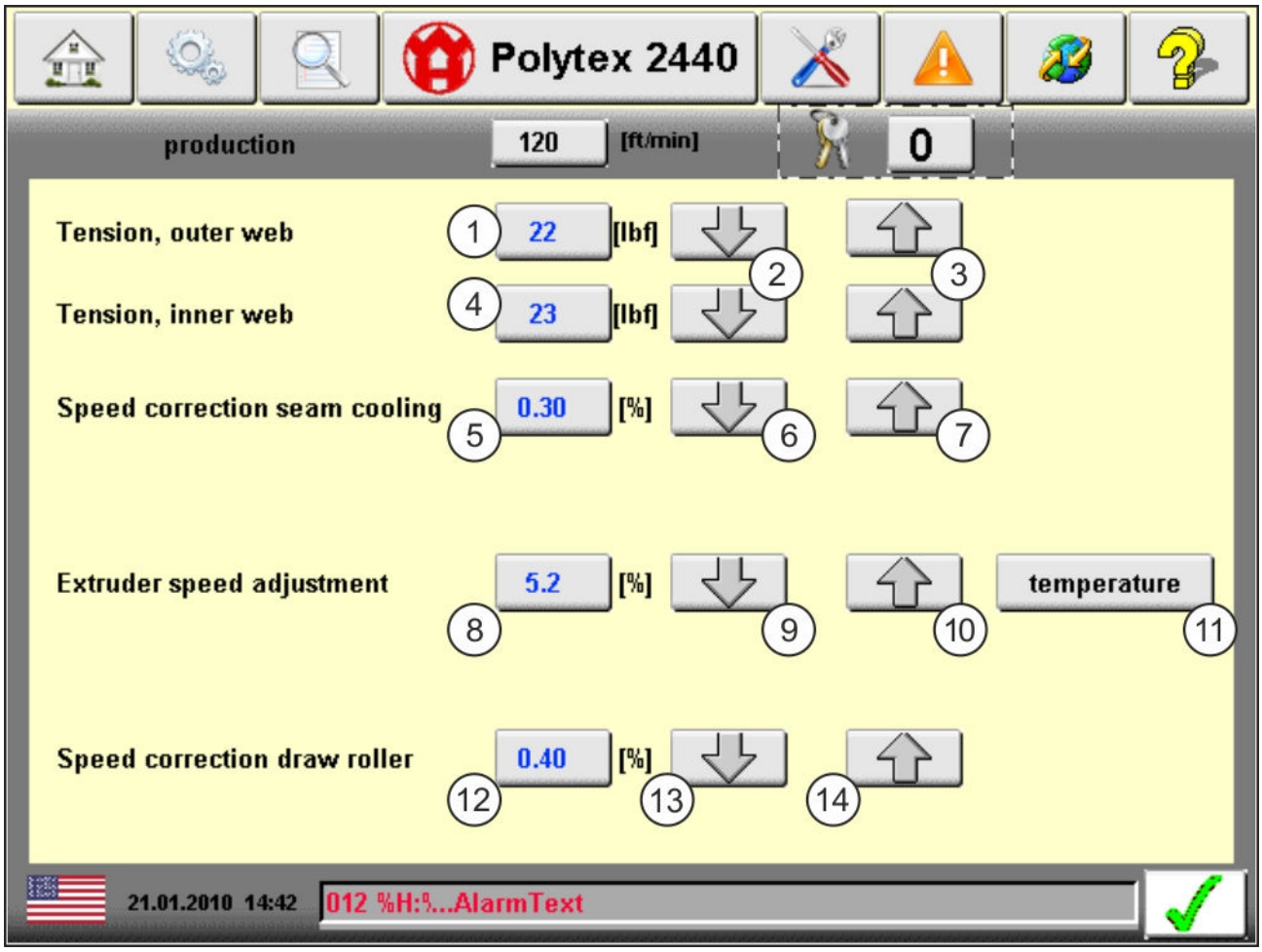
Your machine may not be equipped with all of the optional equipment dealt with in this instruction manual.

4.3.2 Home screen page



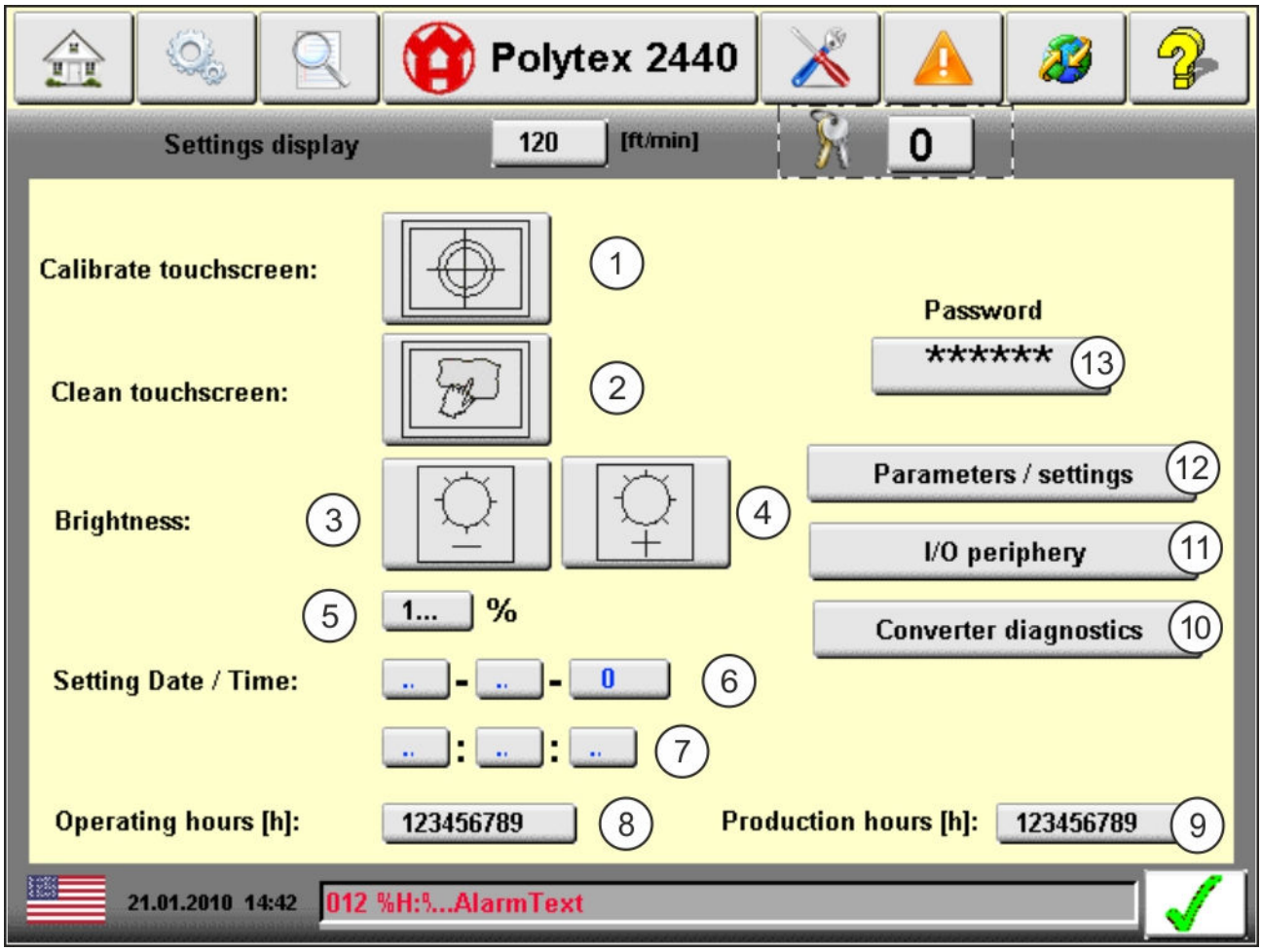
(1)	Draw, web speed	(2)	Longitudinal seam extruder, melt pressure in the extruder
(3)	Longitudinal seam extruder, screw speed per minute		

4.3.3 Production screen page



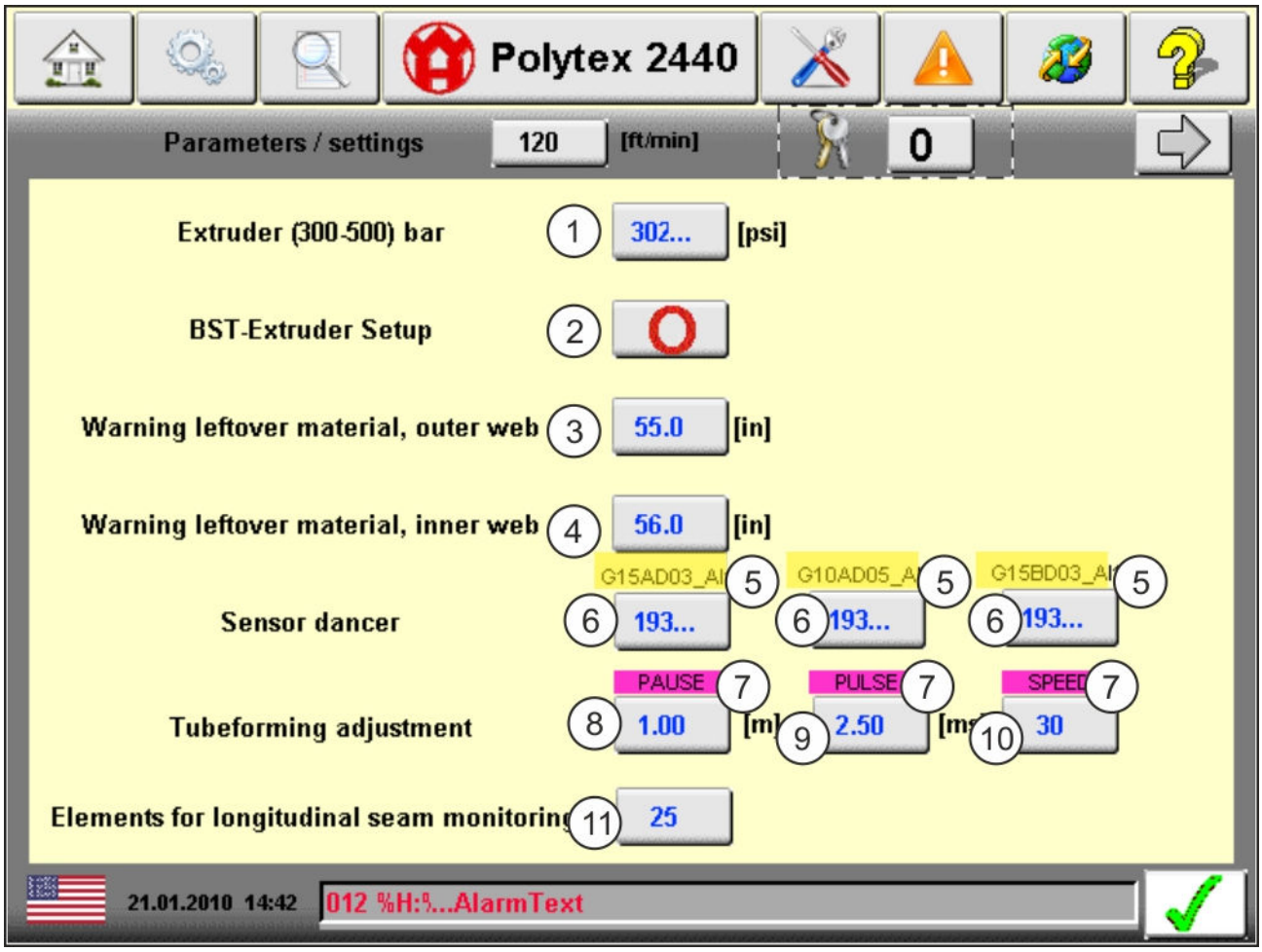
(1)	Set web tension, outer web	(2)	TAPER WEB TENSION
(3)	INCREASE WEB TENSION	(4)	Set web tension, inner web
(5)	Correction value for seam cooling speed	(6)	REDUCE CORRECTION VALUE
(7)	INCREASE CORRECTION VALUE	(8)	Correction value for screw speed
(9)	REDUCE QUANTITY TO BE APPLIED	(10)	INCREASE QUANTITY TO BE APPLIED
(11)	TEMPERATURE, OPENS THE EXTRUDER HEATING ZONES SCREEN PAGE	(12)	Correction value for draw roller speed
(13)	REDUCE CORRECTION VALUE	(14)	INCREASE CORRECTION VALUE

4.3.4 Display screen page



(1)	Open the screen page CALIBRATE TOUCH SCREEN	(2)	Open the screen page CLEAN TOUCH SCREEN
(3)	TOUCH SCREEN DARKER	(4)	TOUCH SCREEN BRIGHTER
(5)	Touch screen brilliance	(6)	Set up the date
(7)	Set up the time	(8)	Running time meter
(9)	Production hour counter	(10)	Open the screen page ACOPOS DIAGNOSIS
(11)	Open the screen page I/O PERIPHERALS	(12)	Open the screen page PARAMETERS/ SETTINGS, only possible when entering the password
(13)	ENTER YOUR PASSWORD		

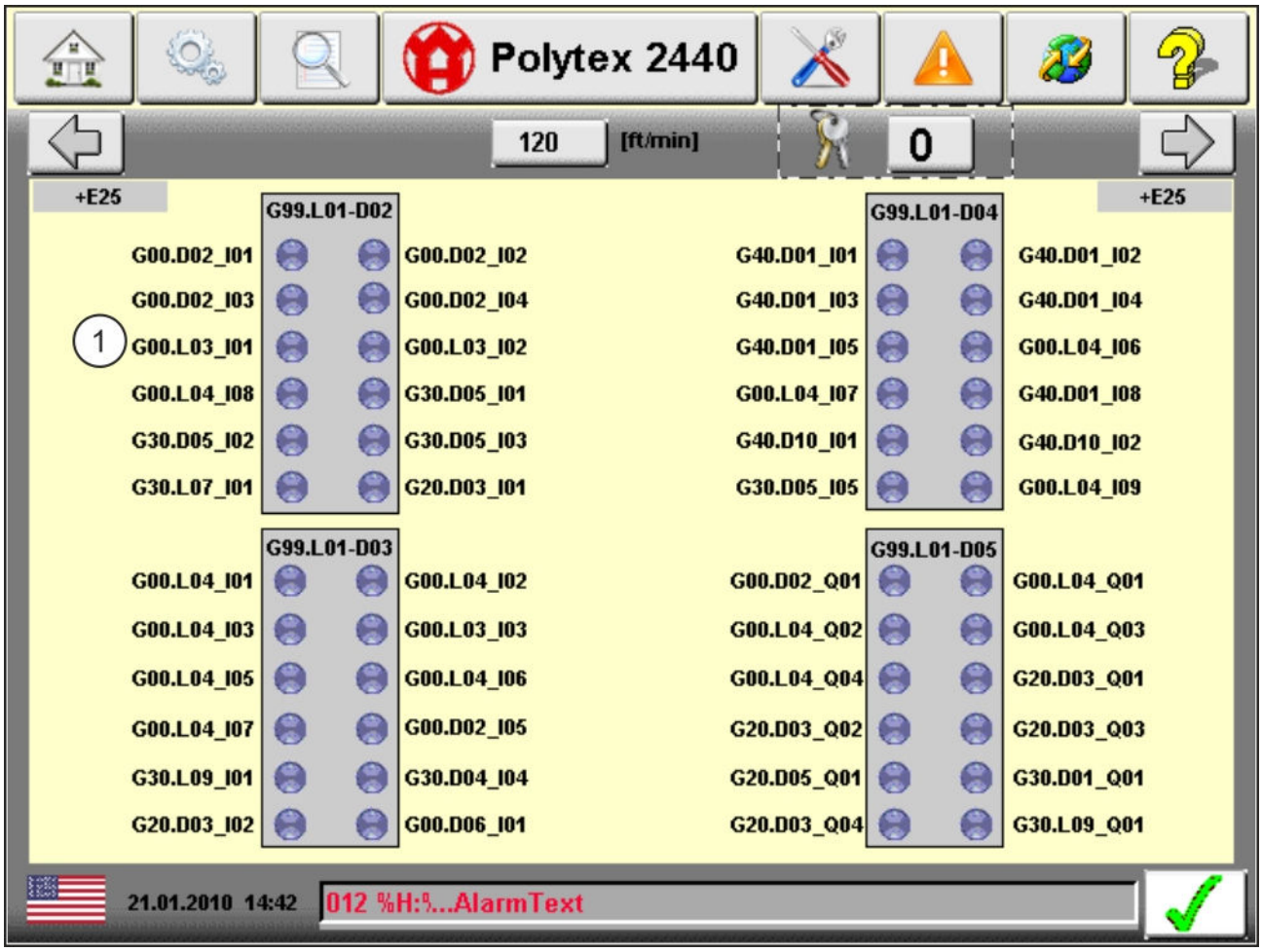
4.3.5 Parameters/settings screen page



Pos.	Designation	Description
(1)	Extruder switch-off	Set up the max. melt pressure for the deactivation of the extruder. The machine stops when reaching the set value.
(2)	BST extruder setup ON/OFF	Activate the operator panel for web guide control on the extruder. Web guide control can be adjusted via the operator panel.
(3)	Warning Leftover material Outer web	Define the material height on the roll for the outer web at which an acoustic signal shall be given. The signal indicates the end of the roll.
(4)	Warning Leftover material Inner web	Define the material height on the roll for the inner web at which an acoustic signal shall be given. The signal indicates the end of the roll.
(5)	EC-ID of the angle sensor	Identifies the sensor on the dancer roller by means of the electrical component identifier (2).
(6)	Angle sensor Dancer roller	Value of the dancer roller position. If necessary, set up the sensor on the dancer roller.

Pos.	Designation	Description
(7)	Parameters	The sensor in the web guide control unit detects the web edge in the tube forming unit and the web guide control unit re-positions the tube forming unit accordingly. Activate the mode on the 'Draw' screen page. The parameters specified below are for orientation purposes. The actual parameters should be determined under production conditions! Read the designations of the parameters in the relevant fields for tube forming adjustment. Pause: Control Off Pulse: Control On Speed: Production speed
(8)	Length of the web path without using web guide control mode	Enter the value for the length of the web path without using the web guide control mode. Enter the length in meters. Parameter value: 2.5 meters (8 13/64 in.) The web guide control unit operates in manual mode over the specified length.
(9)	Web guide control period	Enter the web guide control period. Enter the time in seconds. Parameter value: 0.5 seconds. During the time specified, web guide control is running in automatic mode.
(10)	Production speed: Web guide control On	Enter the production speed from which web guide control should be active. Web guide control starts on reaching the set speed. For controlling, the web guide control unit operates in automatic mode and manual mode, in turns. The time intervals for automatic mode and manual mode operation is determined by the length parameters for the web path. If the production speed drops below the specified value, there will be no web guide control. Enter the production speed in meters/min.. Parameter value: 0 meters.
(11)	Elements for seam monitoring	Enter the number of measurements for the determination of the temperature. The current glue line temperature is calculated from the average non-contact measurement values.

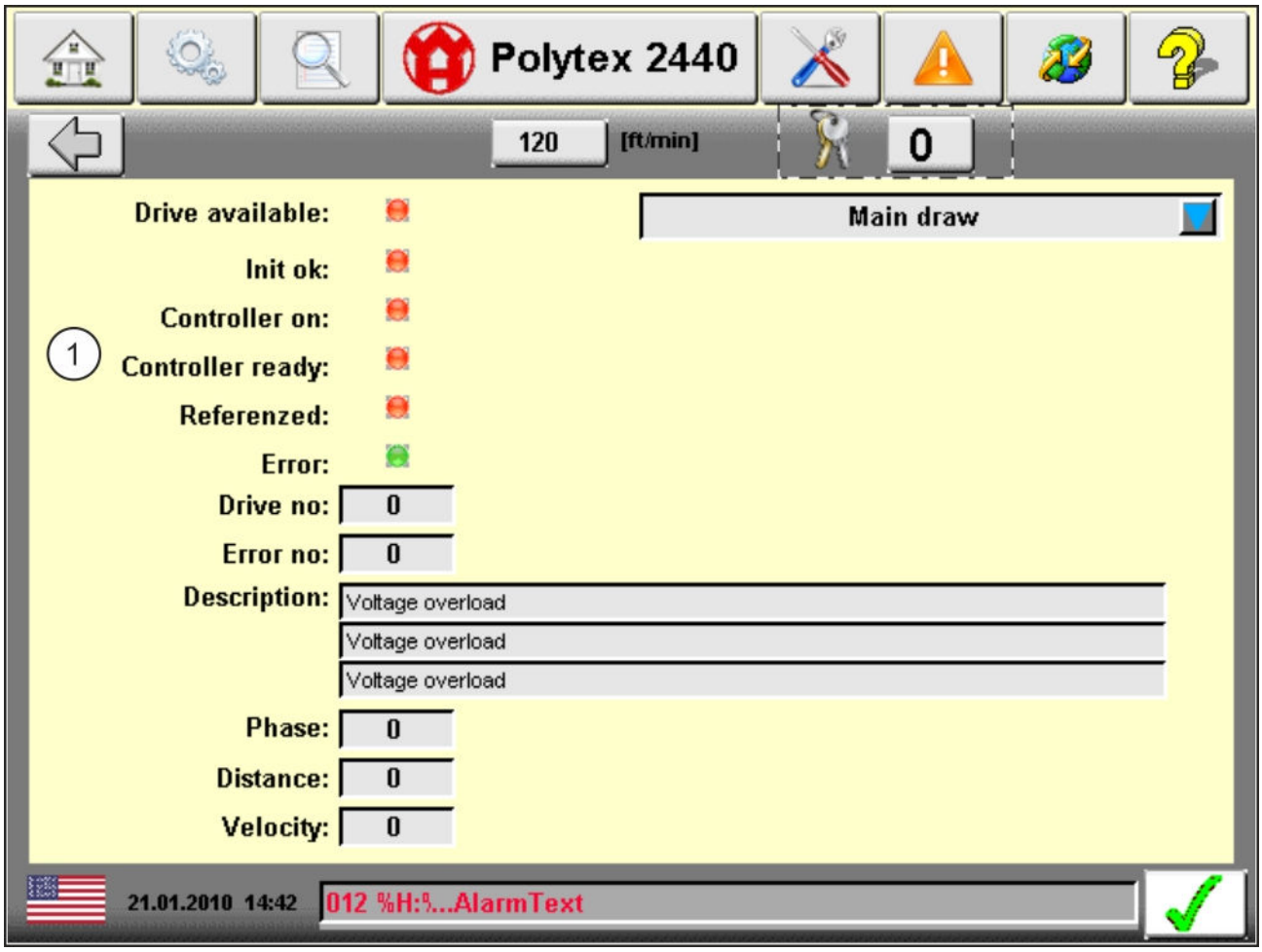
4.3.6 I/O peripherals screen page



(1)	Signal status of the individual inputs and outputs, signal lamp green: Input/ Output active, the signal state is displayed on multiple screen pages		

4.3.7 ACOPOS screen page

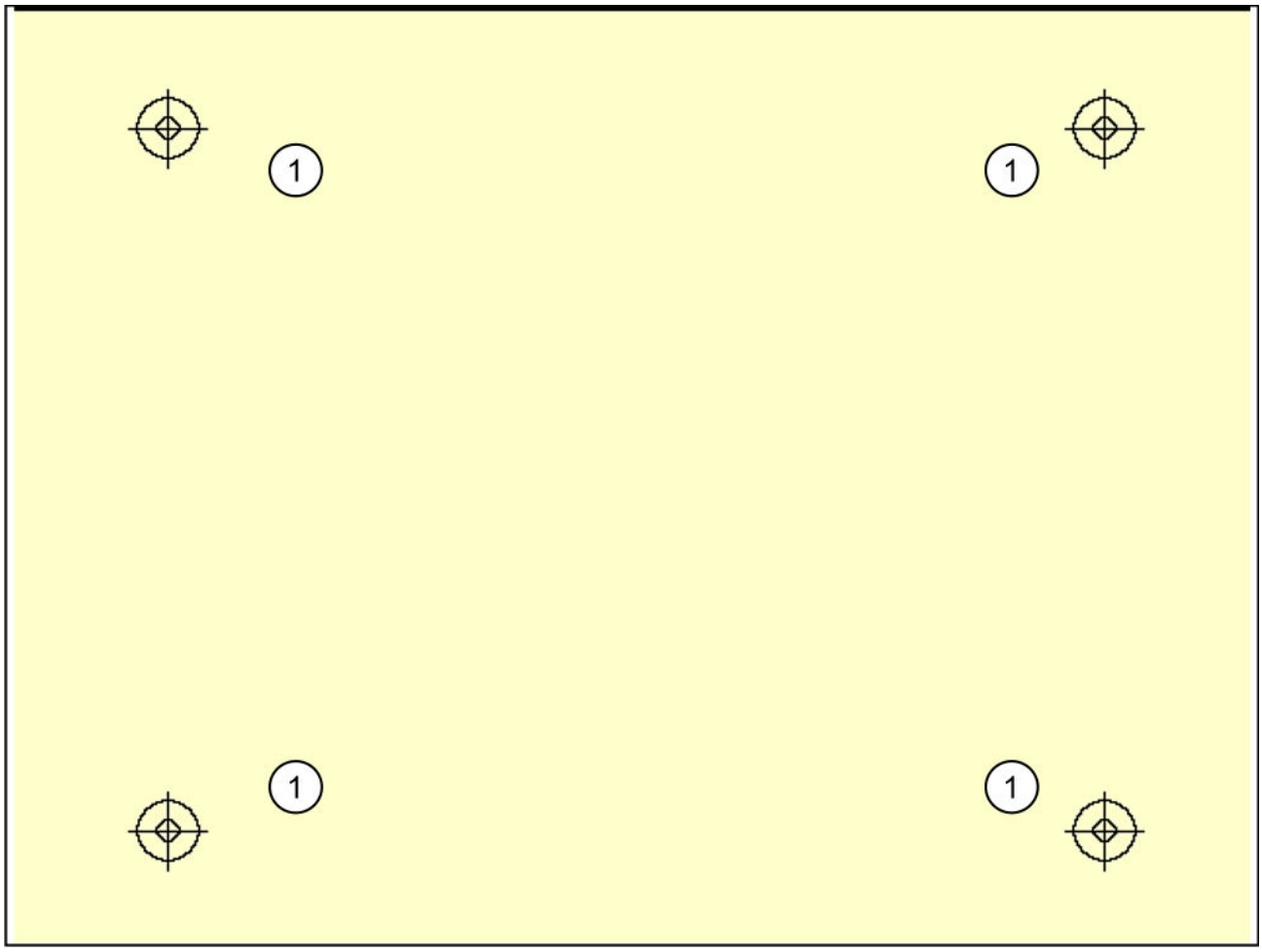
Converter screen page



1	Converter condition screen page		

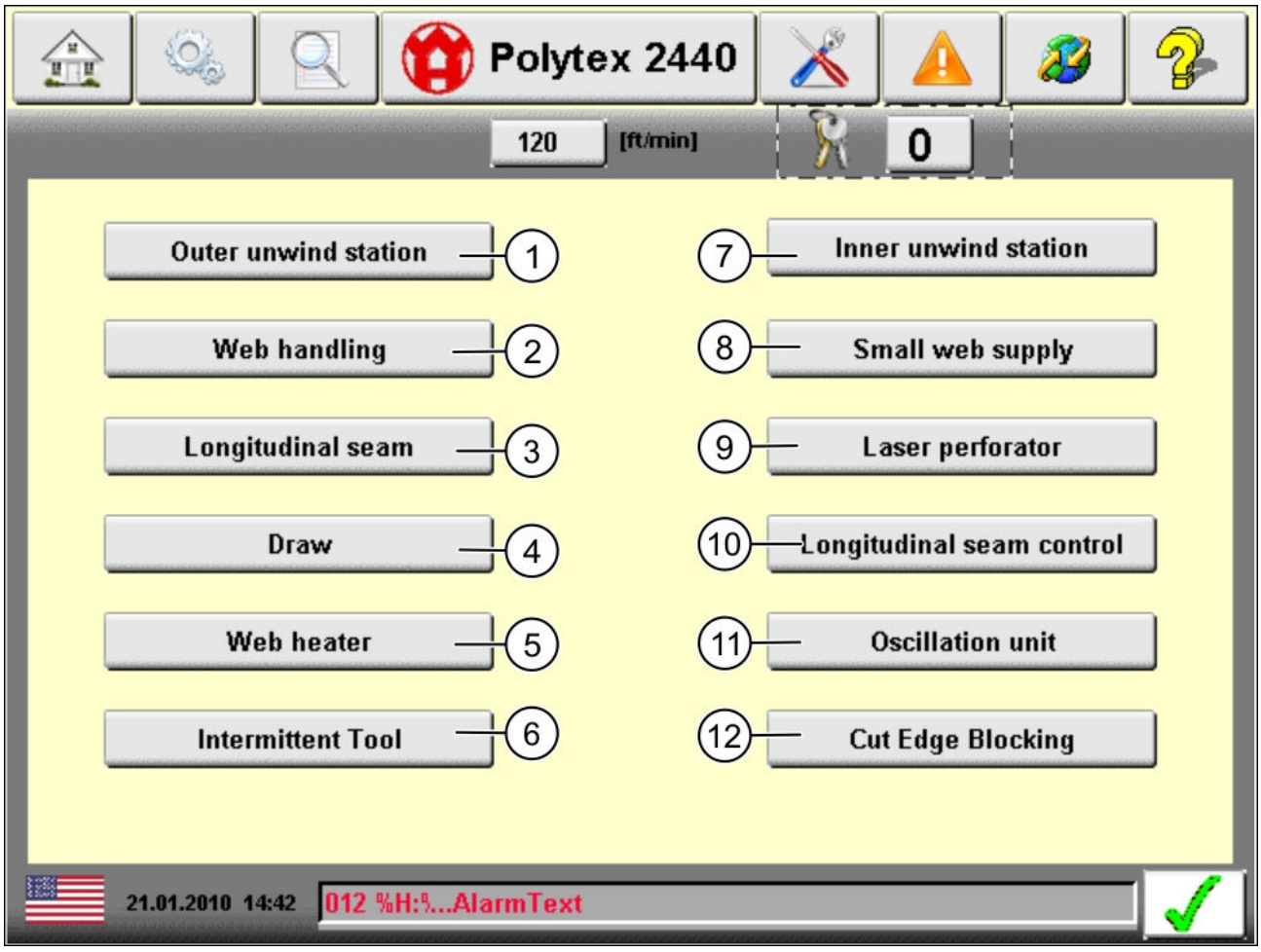
For the operating instructions for the converter and explanatory information on the screen page, see Sub-vendor documentation.

4.3.8 Touch screen screen page



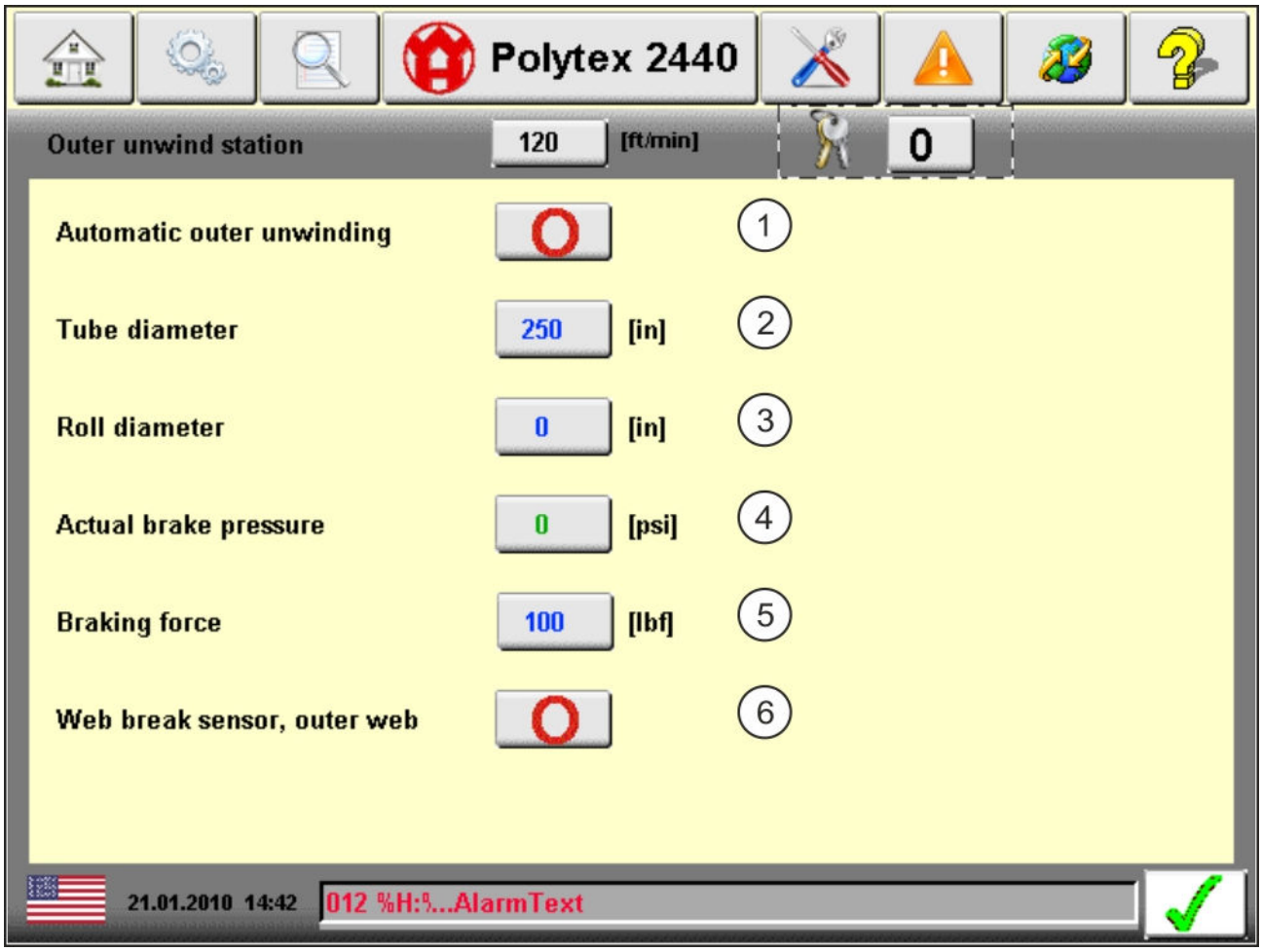
(1)	Touch point		

4.3.9 Module overview screen page



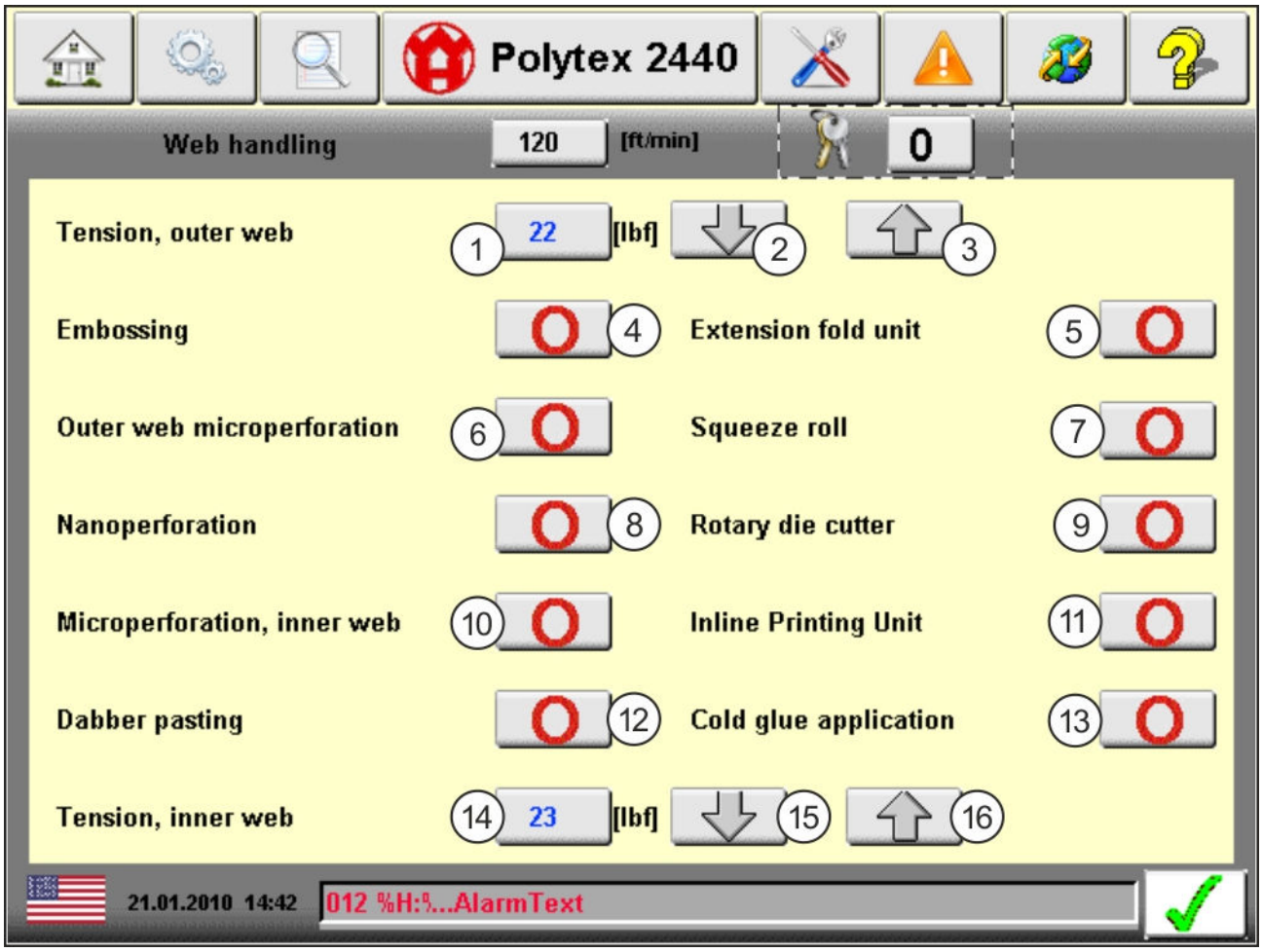
(1)	UNWIND UNIT, OUTER PLY	(2)	WEB PROCESSING STATION
(3)	SEAM EQUIPMENT	(4)	DRAW
(5)	WEB HEATER	(6)	INTERMITTENT TOOL
(7)	UNWIND UNIT, INNER PLY	(8)	INFEED STRIP
(9)	LASER PERFORATOR	(10)	SEAM MONITORING
(11)	OSCILLATION	(12)	CUT EDGE BLOCKING

4.3.10 Unwind screen page



Pos.	Designation	Description
(1)	Unwind, inner [outer], Automatic mode ON/OFF	Activate the brake at the unwind.
(2)	Core diameter, in millimeters	Enter the core diameter.
(3)	Roll diameter, in millimeters	Enter the roll diameter after the roll change.
(4)	Display field: Braking pressure, actual value, bar	Braking pressure value. The braking pressure depends on the braking force as adjusted.
(5)	Braking force, N	Adjust the braking force.
(6)	Outer web break On/Off	Activate the web break detector at the unwind.

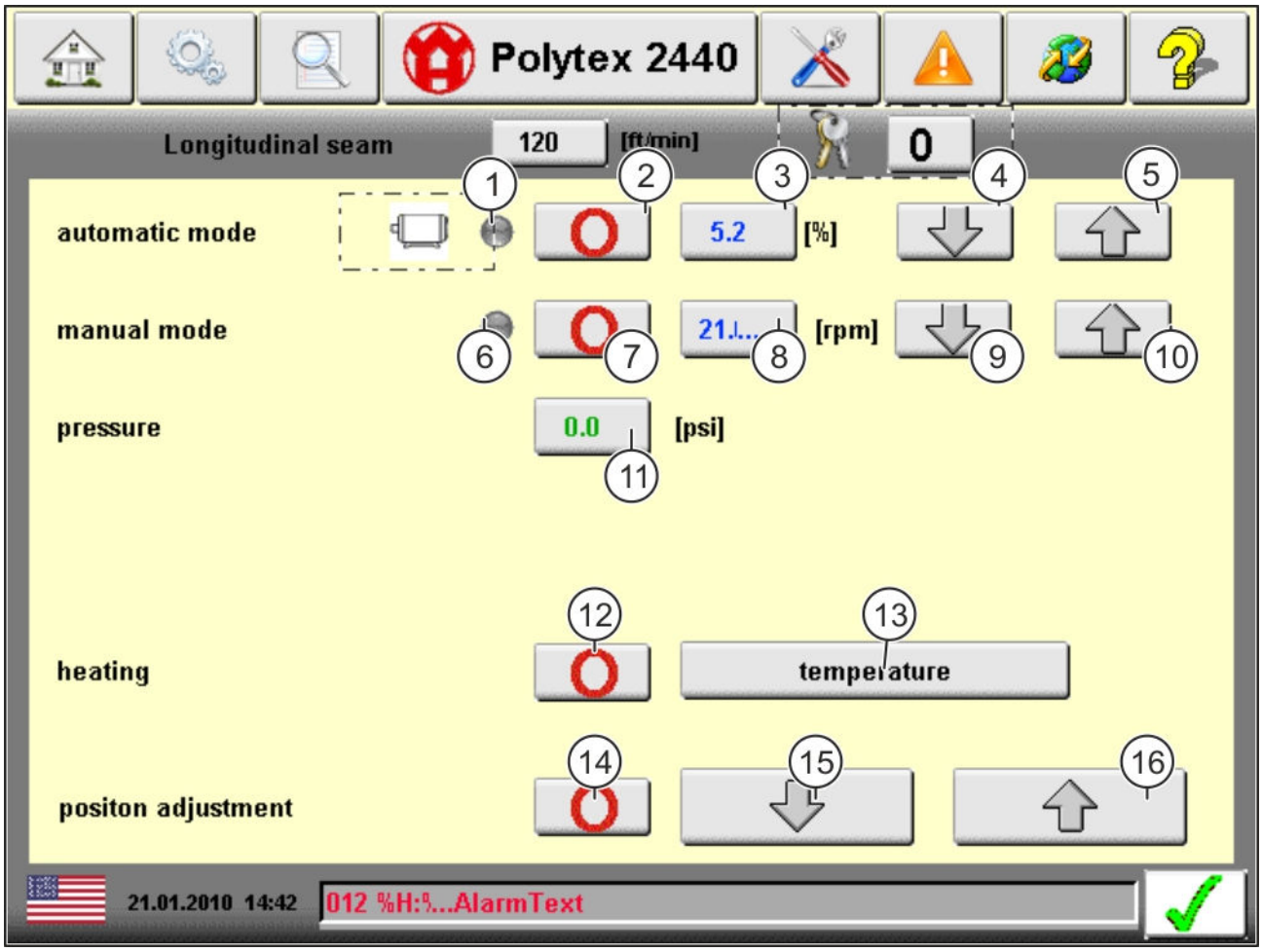
4.3.11 Web processing screen page



Pos.	Designation	Description
(1)	Tension, outer web, in N	Read the currently set web tension value and modify, whenever necessary. Setting up the web tension has an impact on the pressure which the dancer roller applies to the web wrapped around it. Set up the web tension so that dancer roller moves around its center position.
(2)	REDUCE WEB TENSION	Reduces the web tension. The value can be reduced within the limits displayed.
(3)	INCREASE WEB TENSION	Increases the web tension. The value can be increased within the limits displayed.

Pos.	Designation	Description
(4)	Embossing ON/OFF	Activates the tool. Press the button and select OFF. The tool is switched off. The tool can be switched on and off at production speeds of max. 35 meters (115 ft) per minute. If switched on, the tool will be activated when starting the machine. Switch off the tool at the machine. The tool will become inactive.
(5)	Gusseting ON/OFF	
(6)	Micro-perforation, outer web ON/OFF	
(7)	Squeeze roll ON/OFF	
(8)	Nano perforation ON/OFF	
(9)	Rotary die cutter ON/OFF	
(10)	Micro-perforation, inner web ON/OFF	
(11)	Inline printing unit ON/OFF	Activate the 'Inline printing unit' mode. The tail-end printer follows the tube former. Switch off the mode. The tail-end printer does not follow the tube former. The tube former will work as if no tail-end printer was installed.
(12)	Dabber pasting [hot glue application] ON/OFF	Activates the tool. Press the button and select OFF. The tool is switched off. The tool can be switched on and off at production speeds of max. 35 meters (115 ft) per minute. If switched on, the tool will be activated when starting the machine. Switch off the tool at the machine. The tool will become inactive.
(13)	Cold glue application ON/OFF	
(14)	Tension, inner web, in N	Read the currently set web tension value and modify, whenever necessary. Setting up the web tension has an impact on the pressure which the dancer roller applies to the web wrapped around it. Set up the web tension so that the dancer roller moves around its center position.
(15)	REDUCE WEB TENSION	Reduces the web tension. The value can be reduced within the limits displayed.
(16)	INCREASE WEB TENSION	Increases the web tension. The value can be increased within the limits displayed.

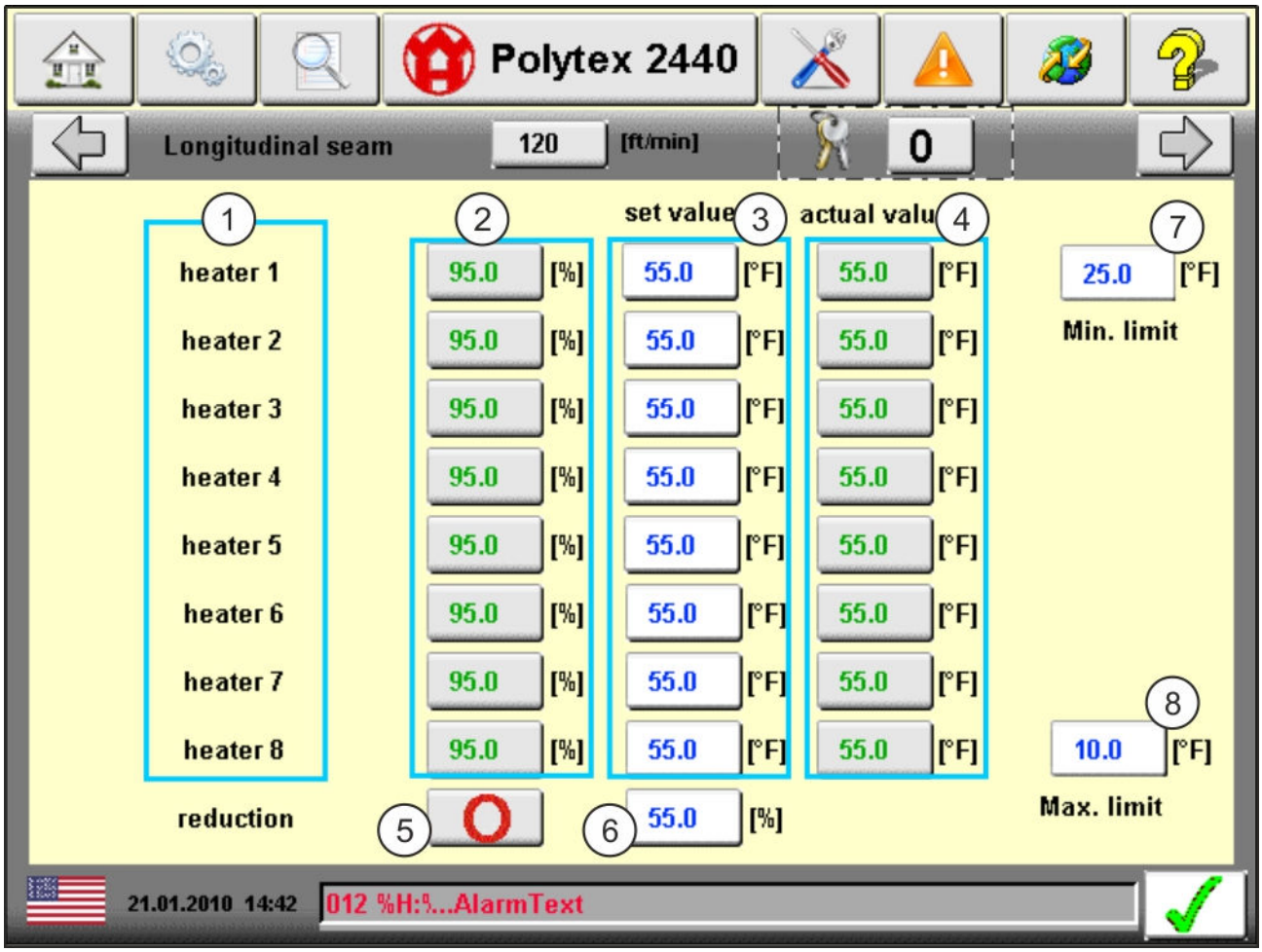
4.3.12 Longitudinal seam screen page



Pos.	Designation	Description
(1)	Signal lamp: the screw is rotating [automatic mode]	Signal lamp green: the screw is rotating Signal lamp red: the screw is at a standstill
(2)	ON/OFF	Switches 'automatic mode' On/Off Automatic mode is active. The extruder starts the production of plastic melt. When starting the machine, the applicator moves to its production position. Depending on the production speed, the applied plastic melt quantity will be increased or reduced.
(3)	Display field: quantity to be applied	Read the value for the quantity of applied plastic melt by which this quantity differs from the automatically determined quantity.
(4)	REDUCE QUANTITY TO BE APPLIED	Reduces the quantity that differs from the quantity determined automatically.
(5)	INCREASE QUANTITY TO BE APPLIED	Increases the quantity that differ from the quantity determined automatically.
(6)	Signal lamp: the screw is rotating [manual mode]	Signal lamp green: the screw is rotating Signal lamp red: the screw is at a standstill
(7)	ON/OFF	Switches 'Manual' mode On/Off

Pos.	Designation	Description
(8)	Display field: Revolutions	Set revolution speed of the screw Revolutions per minute
(9)	REDUCE RPM	–
(10)	INCREASE RPM	–
(11)	Display field: Pressure	The display shows the current material pressure in the extruder.
(12)	ON/OFF	Switches the heaters of the extruder On/Off.
(13)	TEMPERATURE	Screen page: Open 'Extruder's heating zones'.
(14)	ON/OFF	Used for manual adjustment of the extruder position.
(15)	ARROW DOWN	Moves the extruder die into the longitudinal seam.
(16)	ARROW UP	Retract extruder die from longitudinal seam.

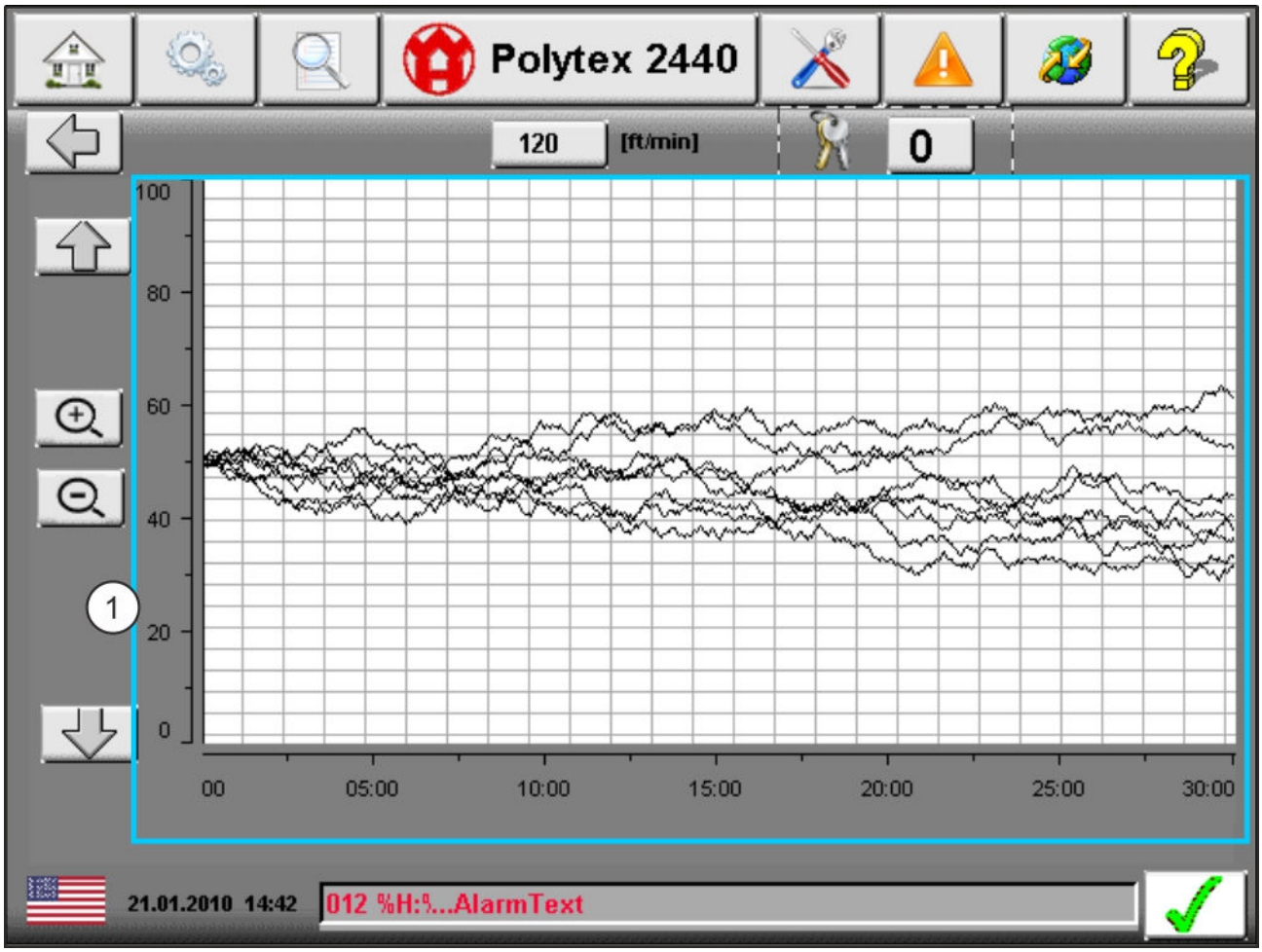
4.3.13 Heating zones screen page



Pos.	Designation	Description
(1)	Heating zone	–
(2)	Display field: Heating capacity	Read off the switch-on time for each of the heating zones in the display fields. The switch-on time in percent describes the time which a heating unit is switched on in order to reach and maintain the given setpoint temperature. Example: a switch-on time of 100 percent results if a heating unit is heated from the room temperature to the setpoint temperature.
(3)	Text field: Set temperature value	Set up the temperature setpoint for this heating zone.
(4)	Display field: Actual temperature value	Read off the current heating zone temperature.
(5)	Reduction ON/OFF	Switches the mode for temperature setpoint reduction On/Off. The activation of the mode is signified by a message.

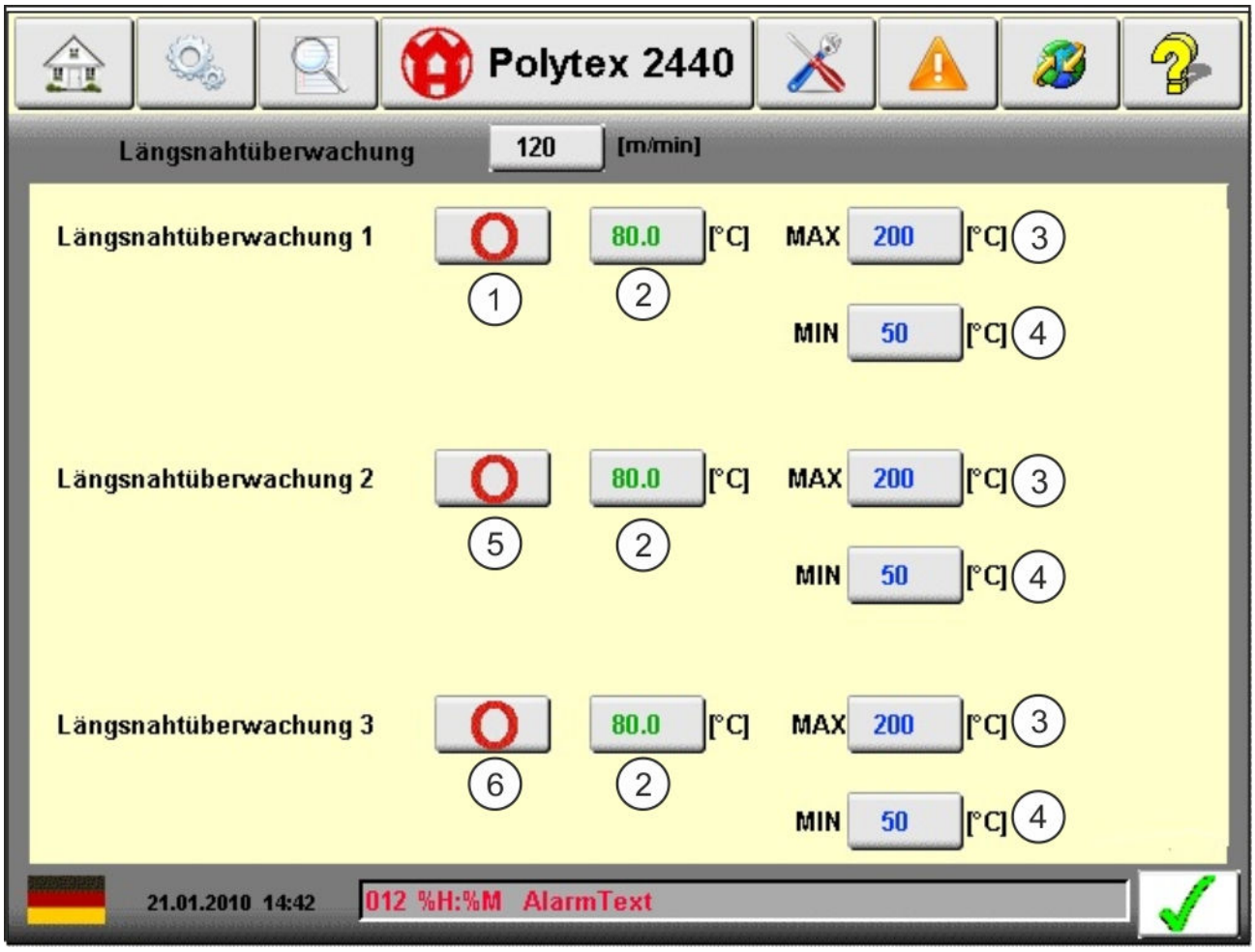
Pos.	Designation	Description
(6)	Reduction, in percent	Enter a percentage value for the setpoint temperatures to drop to this level. Example: If a temperature is set to 120 degrees Celsius (248 degrees Fahrenheit) and should fall to a value of 55 percent, then the set temperature will fall to 66 degrees Celsius (151 degrees Fahrenheit). Reduction mode active: the set temperatures will be permanently adjusted to the reduced temperatures based on the percentage value. After switching off the reduction mode, the extruder's heating zones are heated up again to 100 percent of the set temperatures.
(7)	Excess temperature, temperature above set value	Enter the excess temperature value. Enter a value in degrees Celsius by which the adjusted setpoint temperature may be exceeded. If the temperature reaches the value of the setpoint temperature plus the value of the over-temperature, production is aborted! Automatic mode ends. A message informs you that automatic mode is not possible.
(8)	Low temperature, temperature below set value	Enter the low temperature value. Enter a value in degrees Celsius by which the temperature may fall below the setpoint temperature. If the temperature reaches the value of the setpoint temperature minus the value of the under-temperature, production is aborted! Automatic mode ends. A message informs you that automatic mode is not possible.
–	NEXT	Open the 'Temperature curves' screen page.

4.3.14 Temperature profile screen page



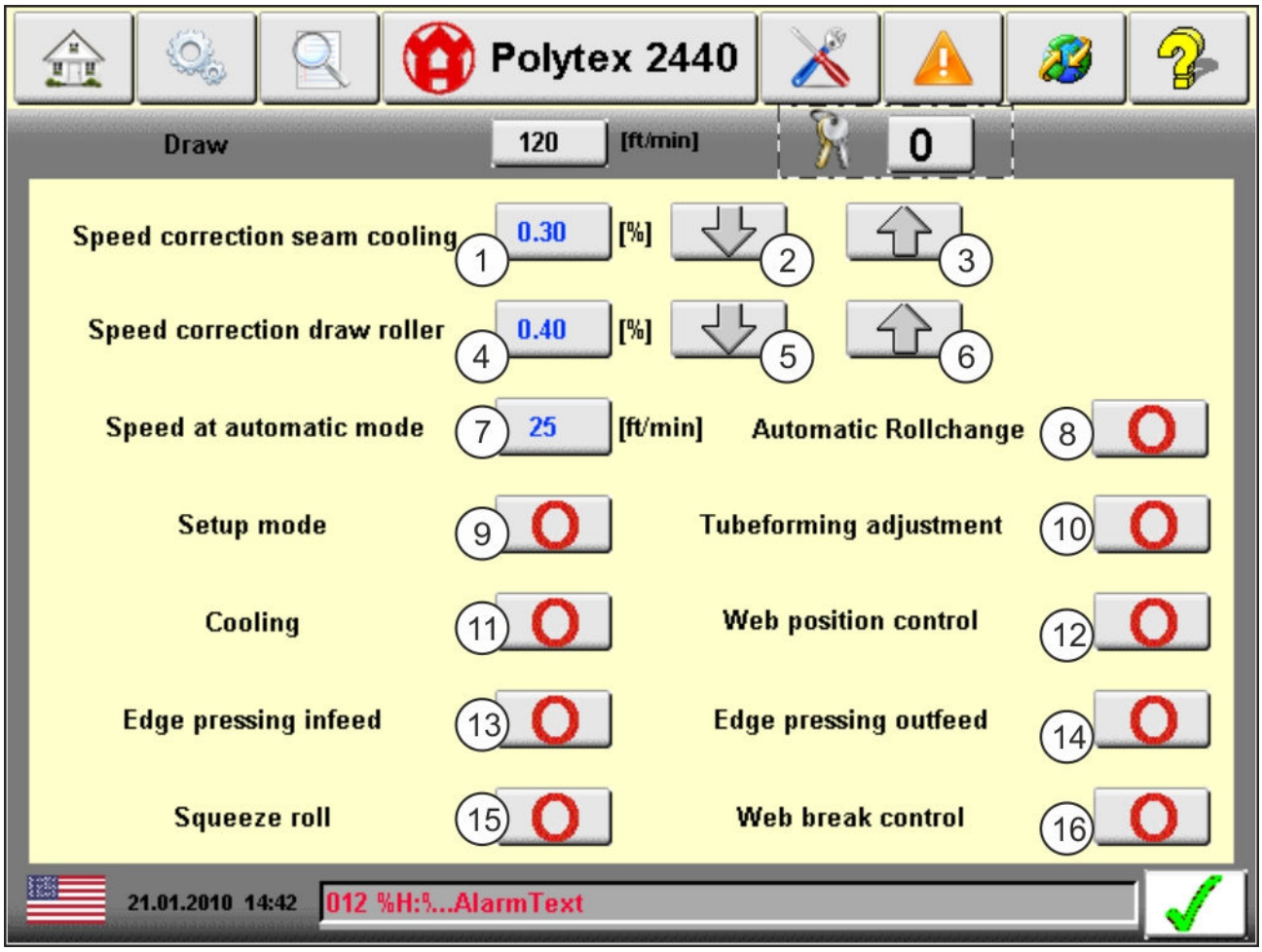
(1)	Graphical overview of the temperature range in the heating zones		

4.3.15 Longitudinal seam monitoring screen page



(1)	Longitudinal seam monitoring 1 ON/OFF	Switches monitoring for the 1st glue line on/off
(2)	Display field: Current temperature	
(3)	Text field: Maximum temperature	Enter the temperature
(4)	Text field: Minimum temperature	Enter the temperature
(5)	Longitudinal seam monitoring 2 ON/OFF	Switches monitoring for the 2nd glue line on and off
(6)	Longitudinal seam monitoring 3 ON/OFF	Switches monitoring for the 3rd glue line on and off

4.3.16 Draw screen page



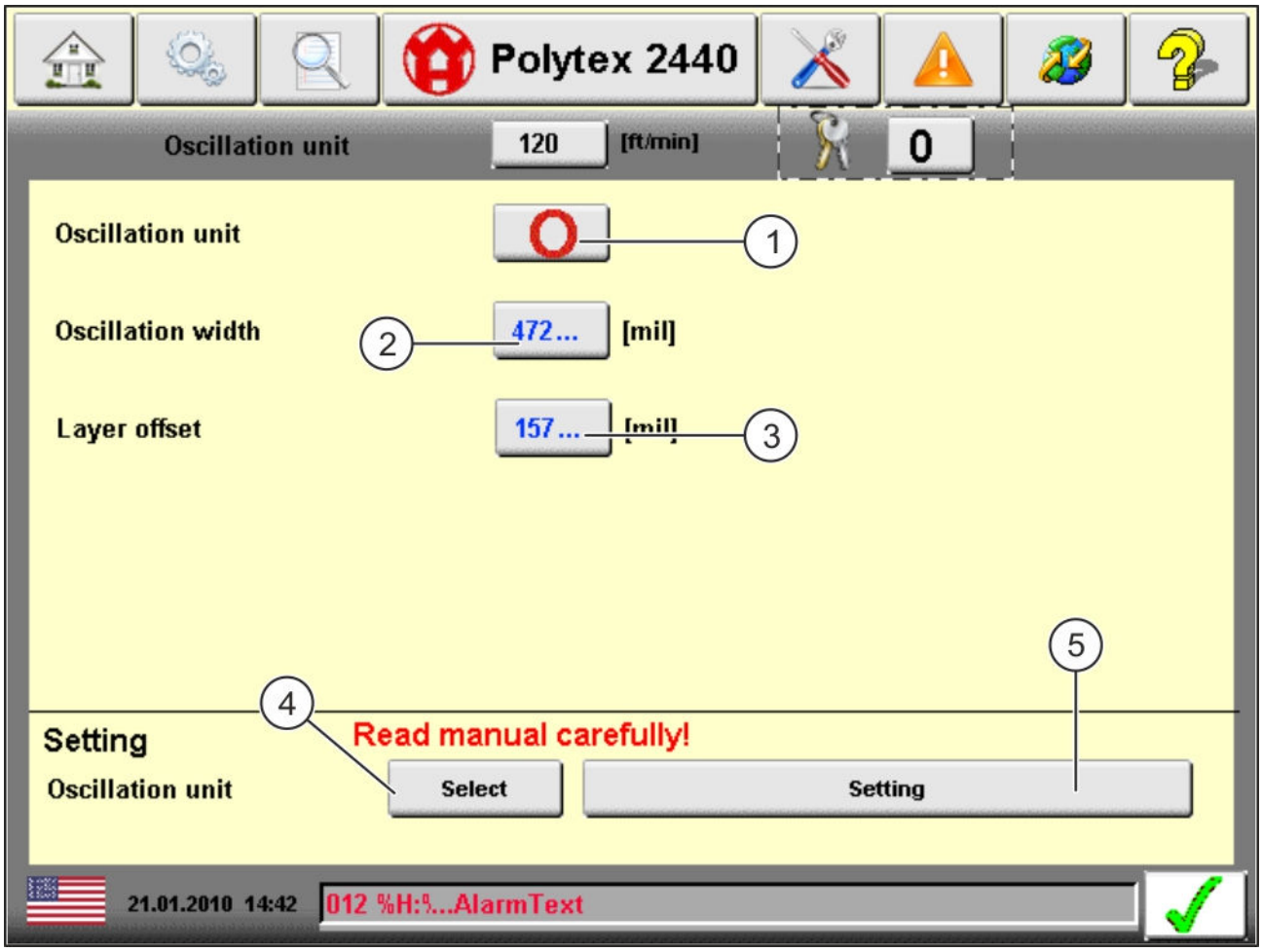
Pos.	Designation	Description
(1)	Correction value for longitudinal seam cooling speed	Read and modify the speed correction value in the display and input field.
(2)	REDUCE correction value	Reduces the roller speed. The value can be reduced within the limits displayed.
(3)	INCREASE correction value	Increases the roller speed. The value can be increased within the limits displayed. The maximum value which can be set is restricted by the maximum possible production speed. The rpm cannot increase beyond this speed.
(4)	Correction value for draw roller speed	Read and modify the speed correction value in the display and input field.
(5)	REDUCE correction value	Reduces the roller speed. The value can be reduced within the limits displayed.
(6)	INCREASE correction value	Increases the roller speed. The value can be increased within the limits displayed. The maximum value which can be set is restricted by the maximum possible production speed. The rpm cannot increase beyond this speed.

Pos.	Designation	Description
(7)	Set up speed for automatic mode	Used to set up the base speed for automatic mode. The window for entering numerical values pops up. Enter and confirm the base speed value. Possible base speed settings: from 5 to 150 meters (from 17 to 492 feet) per minute. The window disappears after confirmation.
(8)	Automatic roll change mode ON/OFF	For tube formers with turret winder [Unwind] and winder [Rewind] Switch 'Automatic roll change' mode on and off. With the 'Automatic roll change' mode activated, the tube former initiates a roll change on the winder due to the relevant information received from the turret winder. Mode of operation: • The turret winder performs a roll change. • The winder also performs a roll change to prevent the splice from being wrapped into the new material roll by the roll change at the turret winder. • The splice from the roll change at the turret winder is wound up at the end of the old material roll and the roll change is carried out at the winder. • Cut off the splice from the end of the old material roll.
(9)	Setup mode ON/OFF	Stop the machine! Activate the setup mode. With the setup mode deactivated, the web cannot be pulled through the processing station by hand. The draw roller drive prevents the draw roller from rotating. The draw roller drive is enabled by activating the setup mode. The draw roller can be rotated by hand. The web can be pulled through the processing station by hand.
(10)	Adjustment Tube forming station ON/OFF	Before activating the setup mode for the tube forming station, see the chapter under the heading 'Setting up the tube former including the web guide control'. Switch tube forming adjustment on and off.
(11)	Cooling ON/OFF	Switches the blower of the cooling tower on and off. The blower starts running only when activating the machine.
(12)	Web edge inspection ON/OFF	Switches the mode on and off. The sensor of the web edge inspection unit observes the lower inside running web edge in the longitudinal seam. The sensor recognizes when this web edge deviates into or out of the tube. The machine stops if the web edge inspection unit detects an improperly positioned lower inside running web edge.

Pos.	Designation	Description
(13)	Edge pressing Infeed ON/OFF	Activate this mode for advancing and retracting the edge pressing unit at the main draw infeed. The edge pressing unit can be switched on and off only via the EDGE PRESSING INFEED button on the operator panel, if the mode was activated. Button to ON: The button is used for advancing and retracting the edge pressing unit. Button to OFF: The edge pressing unit cannot be advanced and retracted.
(14)	Edge pressing Outfeed ON/OFF	Activate this mode for advancing and retracting the edge pressing unit at the main draw outfeed. The edge pressing unit can be switched on and off only via the EDGE PRESSING OUTFEED button on the operator panel, if the mode was activated. Button to ON: The button is used for advancing and retracting the edge pressing unit. Button to OFF: The edge pressing unit cannot be advanced and retracted.
(15)	Squeeze roll ON/OFF	Activate this mode for advancing and retracting the squeeze roll. The squeeze roll can be switched on and off only via the SQUEEZE ROLL button on the operator panel, if the mode was activated. Button to ON: The squeeze roll can be advanced and retracted by pressing the button. Button to OFF: The squeeze roll cannot be advanced and retracted.
(16)	Web break detector ON/OFF	Switches the mode on and off. The sensor of the web break detector monitors the web travel through the main draw. If a web break occurs between the web processing station and the main draw, the activated detector recognizes the web break and the machine stops in a minimum of time.

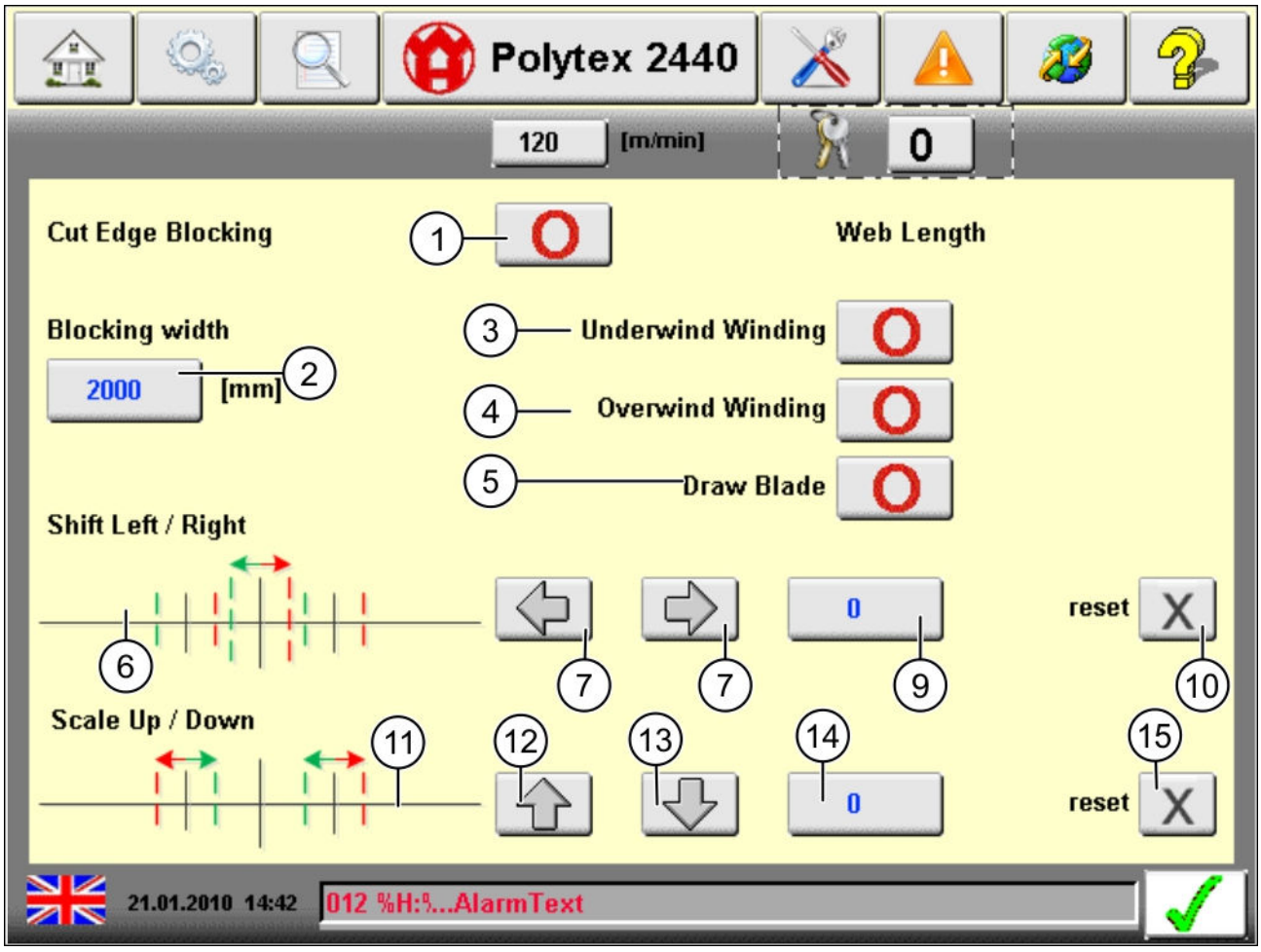
Production at base speed for automatic mode	
Requirements: The current production speed is lower than the set base speed for automatic mode	
Hold down the JOG and FAST buttons.	The machine accelerates to the set base speed for automatic mode.
Hold down the JOG and SLOW buttons.	No function. The current production speed does not change.
Requirements: The current production speed is higher than the set base speed for automatic mode	
Hold down the JOG and FAST buttons.	No function. The current production speed does not change.
Hold down the JOG and SLOW buttons.	The machine slows down to the set base speed for automatic mode.

4.3.17 Oscillation screen page



Pos.	Designation	Description
(1)	Oscillation ON/OFF	Switches oscillation on/off.
(2)	Oscillating range	In the text field, enter the traversing range for the winding roll to be formed (the wound up web). From the zero position, the winder traverses within the range entered.
(3)	Lateral offset	In the text field, enter the lateral offset for the web to be wound up. The winder traverses within the given range per one rotation of the winding roll.
Referencing the drive Switch off the machine. Switch the machine back on. Leave the control voltage switched off. When switching on control voltage the buttons will be suppressed.		
(4)	Set up oscillation, SELECT	Press the button. The drive of the oscillating assembly is selected.
(5)	Set up oscillation, SETTING	Press the button. The current drive position data of the oscillating assembly are transmitted to the machine controller.

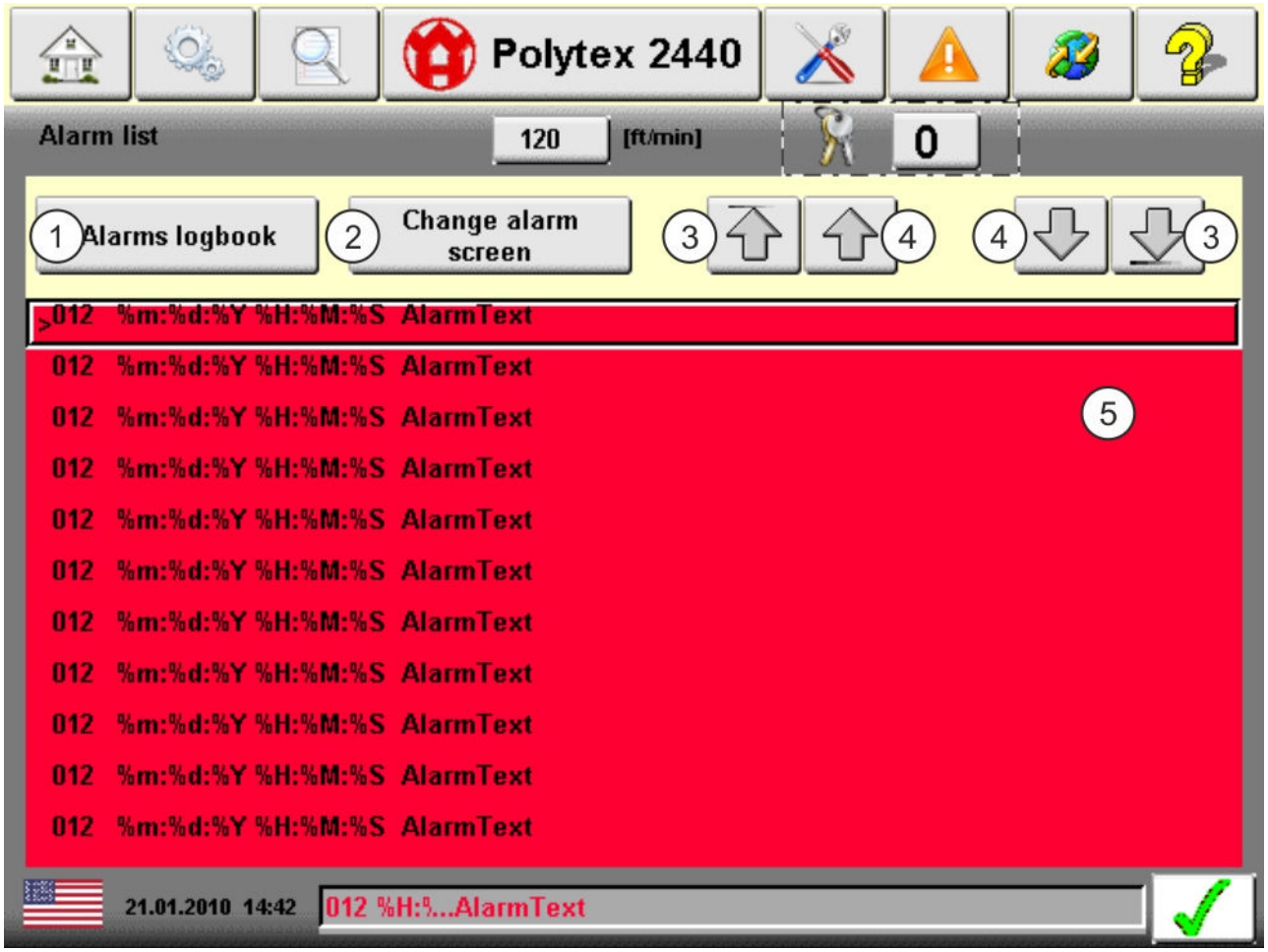
4.3.18 Cut edge blocking screen page



Pos.	Designation	Description
(1)	Cut edge blocking ON/OFF	Switches cut edge blocking on and off.
(2)	'Blocking width' display and input field	The length of the path for the tube to be needled [blocked]. Enter the path length in millimeters.
The 'Winding mode check box' is displayed according to the design of the winder.		
(3)	Underwind mode ON/OFF	To be activated when the tube is wound up from below and cross cutting is carried out by a vacuum cutover drum.
(4)	Overwind mode ON/OFF	To be activated when the tube is wound up from above and cross cutting is carried out by a vacuum cutover drum.
(5)	Draw blade ON/OFF	To be activated when the tube is wound up from below and cross cutting is carried out by a flying knife.
(6)	Figure: Shift Left/Right	–
(7)	TO THE LEFT	Retracts the blocking assembly against the web flow direction.
(8)	TO THE RIGHT	Advances the blocking assembly in the web flow direction.

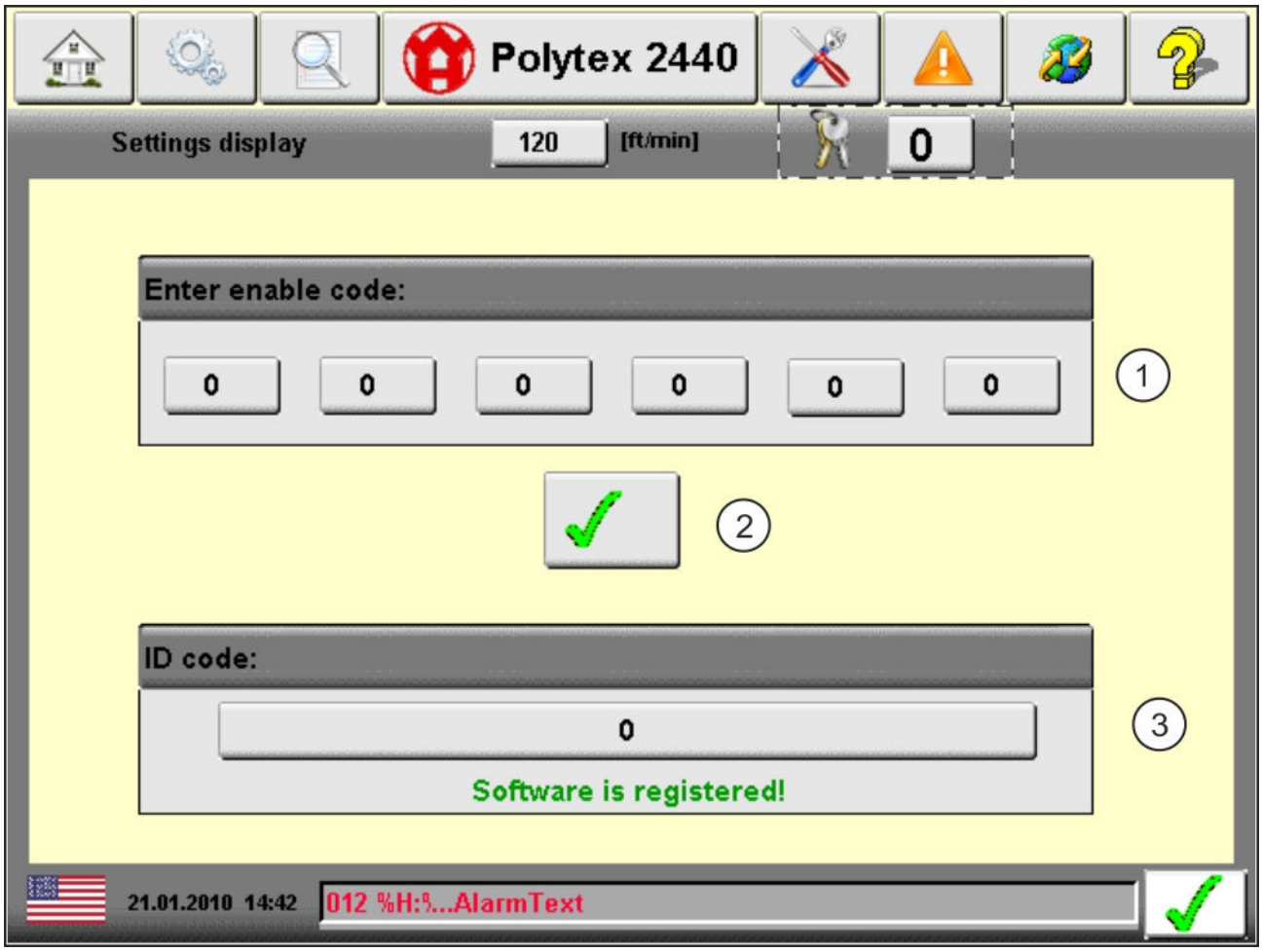
Pos.	Designation	Description
(9)	'Shift' display and input field	Enter the length of the path for the blocking assembly to be shifted. The display shows the length of the path by which the blocking assembly was shifted.
(10)	RESET	Reset the change of position to zero.
(11)	Figure: Scale up/Down	–
(12)	SCALE UP	Increases the blocked tube length.
(13)	SCALE DOWN	Reduces the blocked tube length.
(14)	'Scale Up/Down' input and display field	Enter the length of the blocked tube. The display shows the length of the blocked tube.
(15)	RESET	Resets the change of length to zero.

4.3.19 Alarm list screen page - Log



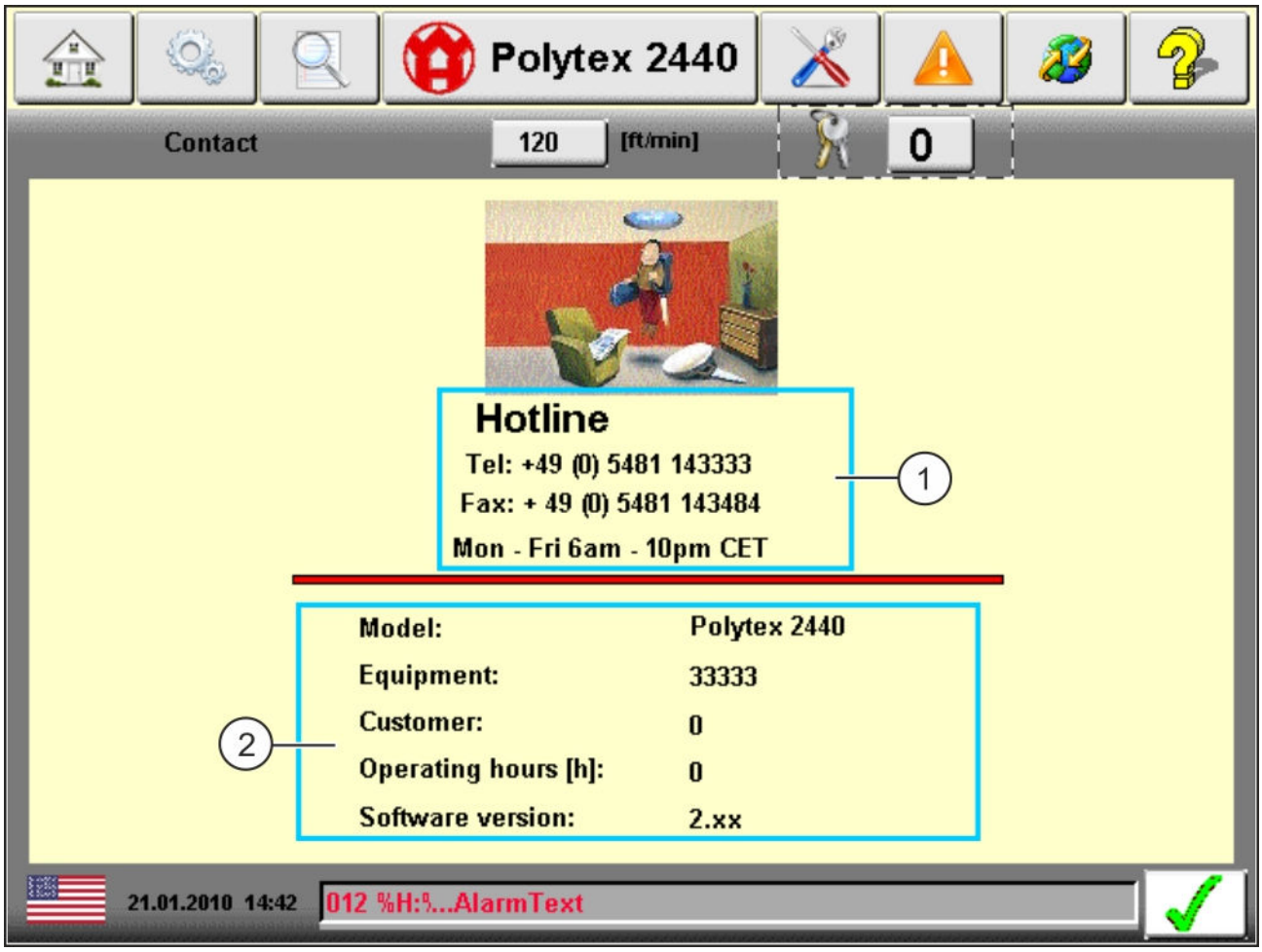
(1)	LOG ALARMS, call the log alarms screen page. The list contains a maximum of 500 messages, sorted by date.	(2)	ALARM SCREEN CHANGE, when the alarm screen change mode is switched on, the alarm list screen page will be opened for each message.
(3)	GO TO START/END of the list	(4)	SCROLL UP/DOWN in the list
(5)	List of current alarms		

4.3.20 Enable screen page



(1)	Registration number sequences	(2)	CONFIRM ENTRY
(3)	Identification code		

4.3.21 Contact screen page



(1)	W&H hotline hours	(2)	Machine and software information

4.4 Screen pages: Operation

4.4.1 Operating concept 152
General 152
Preventing damage from occurring 152
Instruction manual notes 152

4.4.2 Header 152

4.4.3 Information line 153

4.4.4 Footer 153

4.4.5 Entering your password 154

4.4.6 Touch screen screen page 154

4.4.7 Setting up the web processing station 155
Setting up the web tension 155
Switching on the micro-perforator 155
Switching on the embossing unit 155

4.4.8 Setting up longitudinal seam monitoring 157

4.4.9 Enabling the software 158
Activating the software 158
Software not yet activated 158

4.4.1 Operating concept

General

Depending on the machine configuration, the design of the screen pages may vary.

Preventing damage from occurring

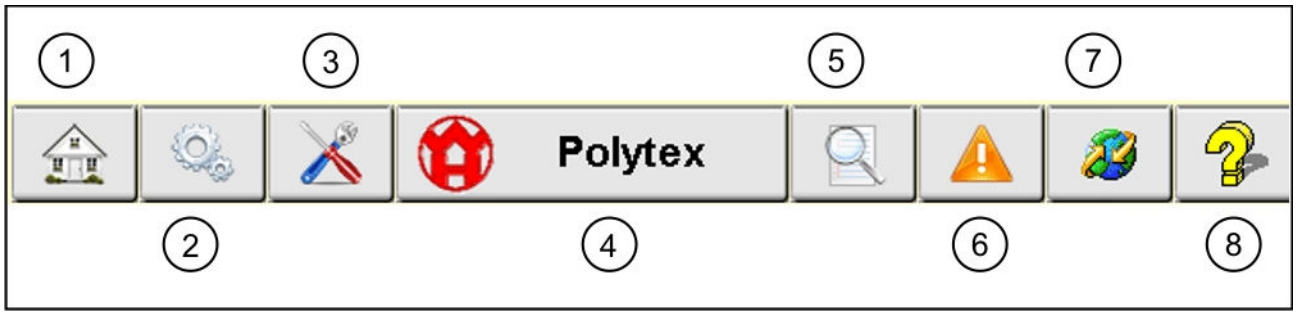
- Never use a hard object to touch the touchscreen.
- The touchscreen is fitted with protective film.. Replace if damaged.
- Only use your finger to touch the selection, switching and input fields that the program displays on the screen to call up other screen pages, start processes and to enter specific values.
- Whether a number pad or a keypad is opened depends entirely on the inputting field that you press. The data is entered using the number pad or keypad.

Instruction manual notes

Text displayed in bold capital letters indicates a pushbutton or a key.

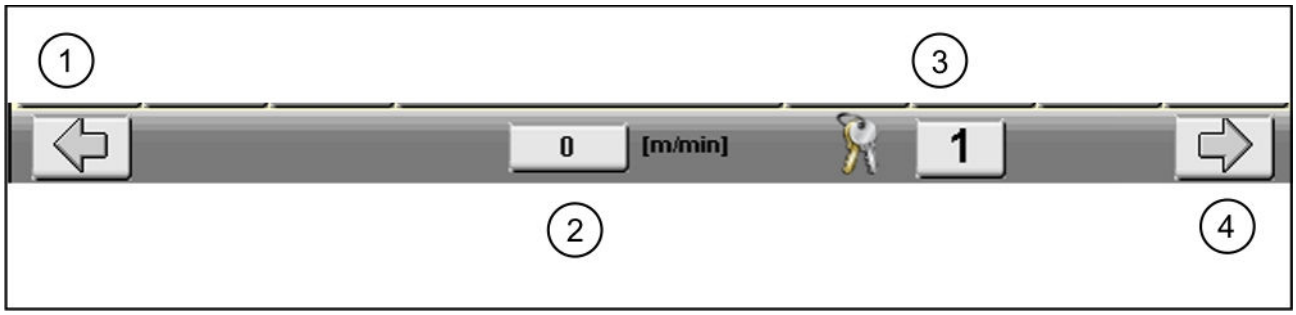
Screenshots might still be in German. The item number in the screenshot refers to the picture caption and the accompanying handling text or the description.

4.4.2 Header



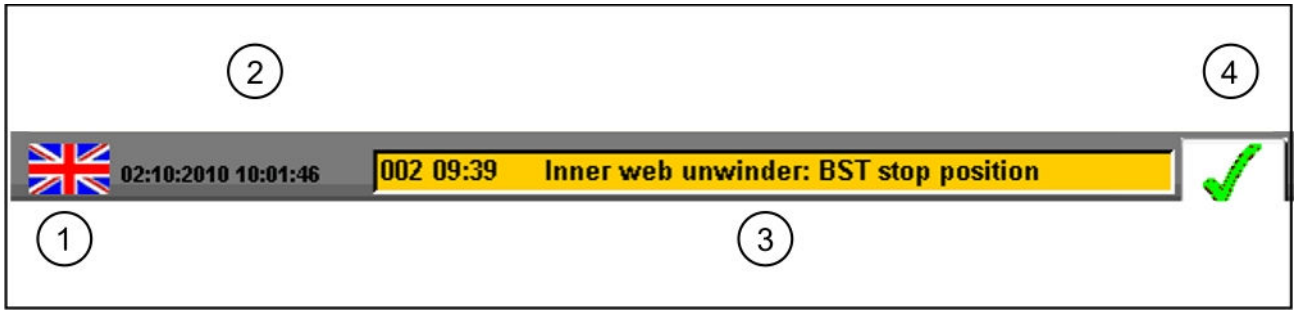
(1)	Open the start page	(2)	Change settings.
(3)	Set basic settings	(4)	Call up information
(5)	Open actuators overview	(6)	Open alarm list
(7)	Open display language	(8)	Open help

4.4.3 Information line



(1)	Back one screen page	(2)	Web speed
(3)	Enter password, protected screen pages can be opened, "Service level 1" activated	(4)	Forward one screen page

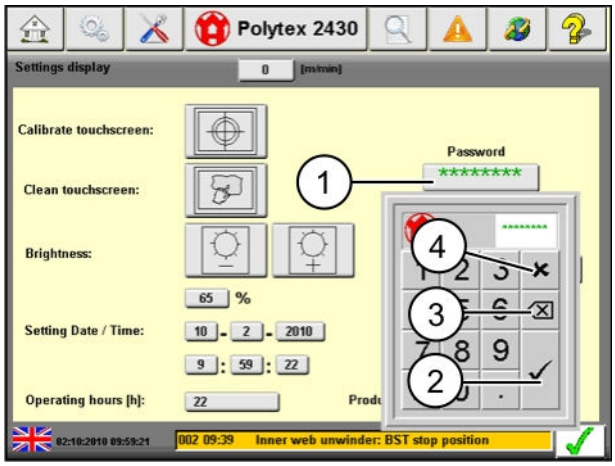
4.4.4 Footer



(1)	National flag - Display language	(2)	Date and Time
(3)	Message: Message number, time of fault, reason for the message	(4)	Confirm message

4.4.5 Entering your password

(1)	PASSWORD
(2)	CONFIRM
(3)	DELETE
(4)	CLOSE WINDOW



- ▶ Press the pushbutton (1) in order to enter the password.

  Enter a registered password.

  The numerical keypad window opens.

- ▶ Enter a password and press the pushbutton (2) in order to confirm the password.

  The window closes.

- ▶ Press the pushbutton (3) in order to reset incorrect entries.

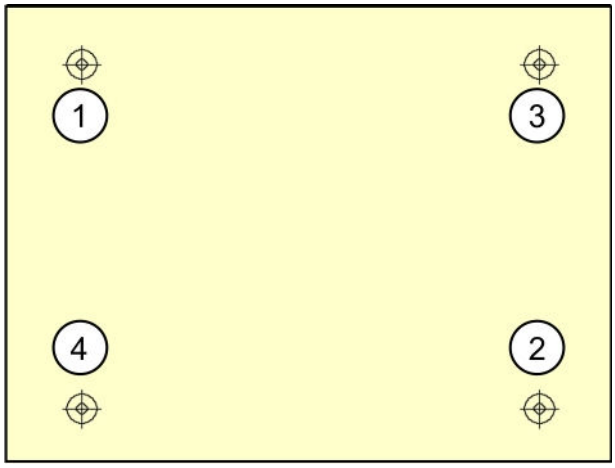
  Press the pushbutton briefly in order to delete one digit.
  Press and hold the pushbutton in order to delete all of the digits.

- ▶ Press the pushbutton (4) in order to open the numerical keypad window.

  Changes made in the window are not saved.

4.4.6 Touch screen screen page

(1)	Touch point
(2)	Touch point
(3)	Touch point
(4)	Touch point



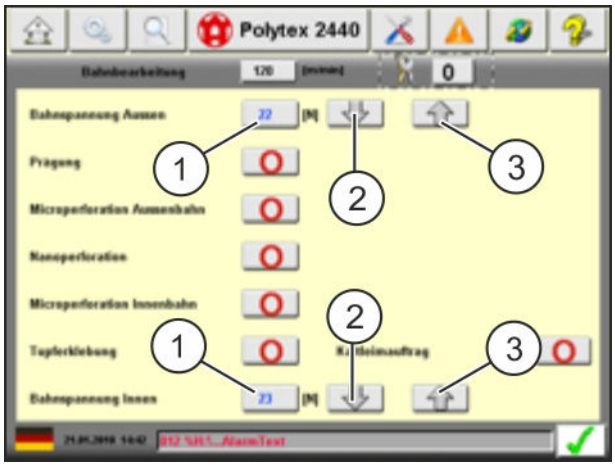
- ▶ Touch all of the points precisely and in the specified sequence (1) to (4).

  The "Basic settings" screen page opens as soon as all of the points have been touched.

4.4.7 Setting up the web processing station

Setting up the web tension

(1)	Set web tension
(2)	TAPER WEB TENSION
(3)	INCREASE WEB TENSION



- Read off the currently set web tension value in the display and entry field (1) and change whenever necessary.

  Setting up the web tension has an impact on the pressure which the dancer roller applies to the web wrapped around it.
  Set up the web tension until the dancer roller moves around its center position.

- Press the button (2) to taper the web tension.

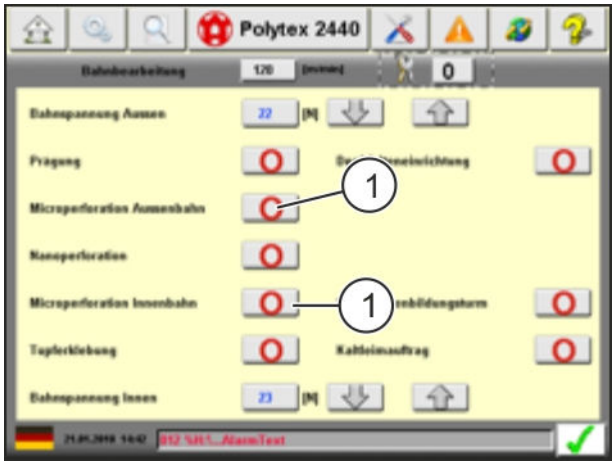
  The value can be reduced between the displayed limits.

- Press the button (3) to increase the web tension.

  The value can be increased between the displayed limits.

Switching on the micro-perforator

(1)	Micro-perforator ON/OFF



- Press the button (1) to switch on the tool.
- Press the button and select OFF to switch off the tool.

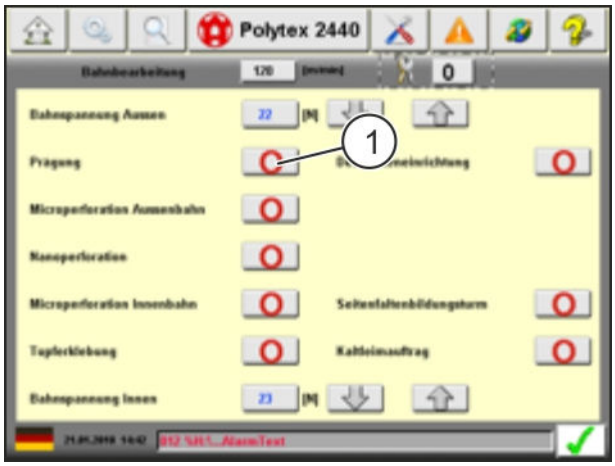
  The tool can be switched on and off at production speeds of max. 35 meters (115 ft) per second.
  If switched on, the tool is activated when starting the machine.

- Switch off the tool at the machine.

  The tool becomes inactive.

Switching on the embossing unit

(1)	Embossing ON/OFF



- ▶ Press the button (1) to switch on the tool.
- ▶ Press the button and select OFF to switch off the tool.

  The tool can be switched on and off at production speeds of max. 35 meters (115 ft) per second.
  If switched on, the tool is activated when starting the machine.
- ▶ Switch off the tool at the machine.

  The tool becomes inactive.

4.4.8 Setting up longitudinal seam monitoring

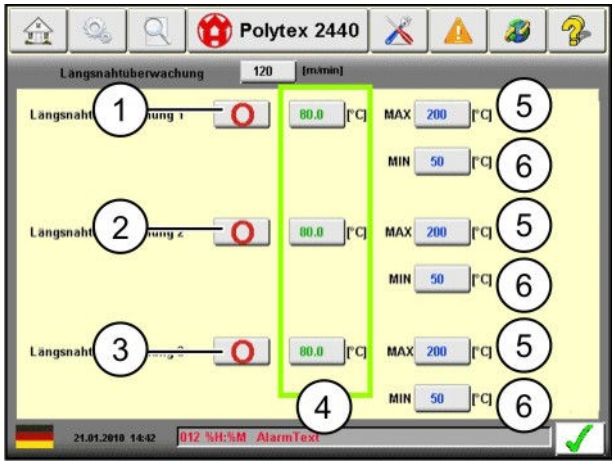
Arrangement of the sensors for longitudinal seam monitoring

The sensor for longitudinal seam monitoring 1 is located on the extruder side.

The sensor for longitudinal seam monitoring 2 is located in the middle between the two other sensors.

The sensor for longitudinal seam monitoring 3 is located on the operator terminal side.

(1)	Longitudinal seam monitoring 1 ON/OFF
(2)	Longitudinal seam monitoring 2 ON/OFF
(3)	Longitudinal seam monitoring 3 ON/OFF
(4)	Display field: Current temperature
(5)	Text field: Maximum temperature
(6)	Text field: Minimum temperature



- ▶ Press the button (1) to switch on longitudinal seam monitoring for the 1st glue line.
- ▶ If necessary, press button (2) and (3) to switch on longitudinal seam monitoring for the 2nd and 3rd glue line.
- ▶ Read the current temperature of the relevant glue line from the display fields (4).
- ▶ Open text field (5) and enter the maximum temperature for the 1st glue line.
- ▶ Open text field (6) and enter the minimum temperature for the 1st glue line.

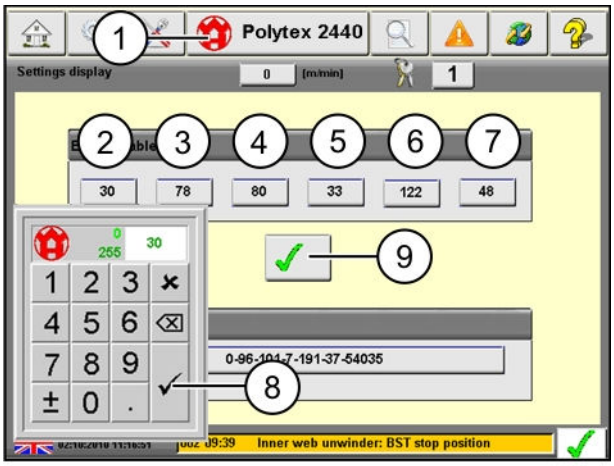
  When falling below or exceeding the set temperature for the 1st glue line, a message appears and the signaling device flashes red. Eliminate the cause for the wrong temperature and confirm the message.
- ▶ If necessary, set up the temperatures for the 2nd and 3rd glue line.

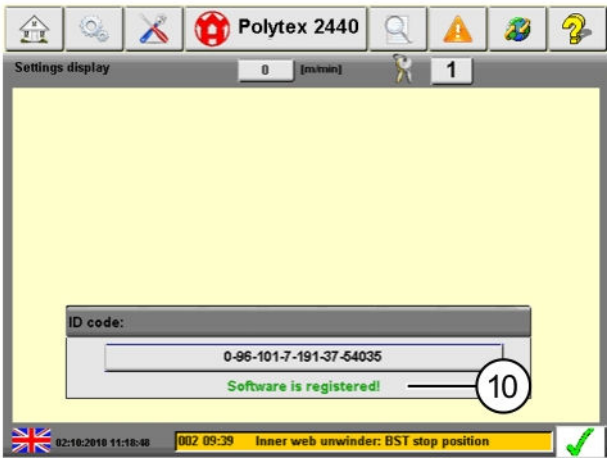
4.4.9 Enabling the software

Activating the software

Register the software.

(1)	CALL UP INFORMATION
(2-7)	Registration number sequences
(8)	CONFIRM
(9)	CONFIRM ENTRY, Confirm the number sequences entered
(10)	Message





- ▶ Press the pushbutton (1) in order to call up the "Call up information" screen page.
- ▶ Press the six pushbuttons (2) to (7) in sequence.
- ▶ The numbers displayed on the pushbuttons are entered into the window which opens.
- ▶ Press the pushbutton (8) after entering the complete number sequence.

  The window closes.
- ▶ Press the pushbutton (9) in order to confirm the entries.

  The pushbuttons displaying the numbers are not longer displayed.
  The following message (10) appears beneath the identification code: The software is registered.

Software not yet activated

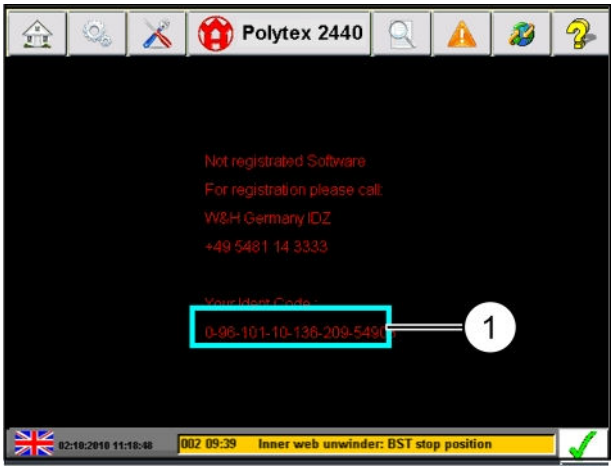
NOTE

The machine control software is not yet activated.

The registration message is displayed until the software has been registered.

The machine can still be operated.

(1)	Identification code



- Inform W&H of the identification code (1) in order to complete the registration.

  The message is no longer displayed once the registration has been completed.

4.5 Winder

4.5.1 Main control console +R82A for the winder....0

4.5.2 Screen in control console +R82A/B....0
Overall view....163
Cursor keypad....163
Numeric keypad....163
Control keypad....163
Service screens....163
Web pulling forces and web tensions....163
Contact forces....163
Low Pressure Mode (LPM)....163
Production data....163
Counter parameters....163

4.5.3 Control elements on operator panel +R82....173
Control panel - Part 1....173
Control panel - Part 2....173

4.5.4 Operator panel +R85A....176

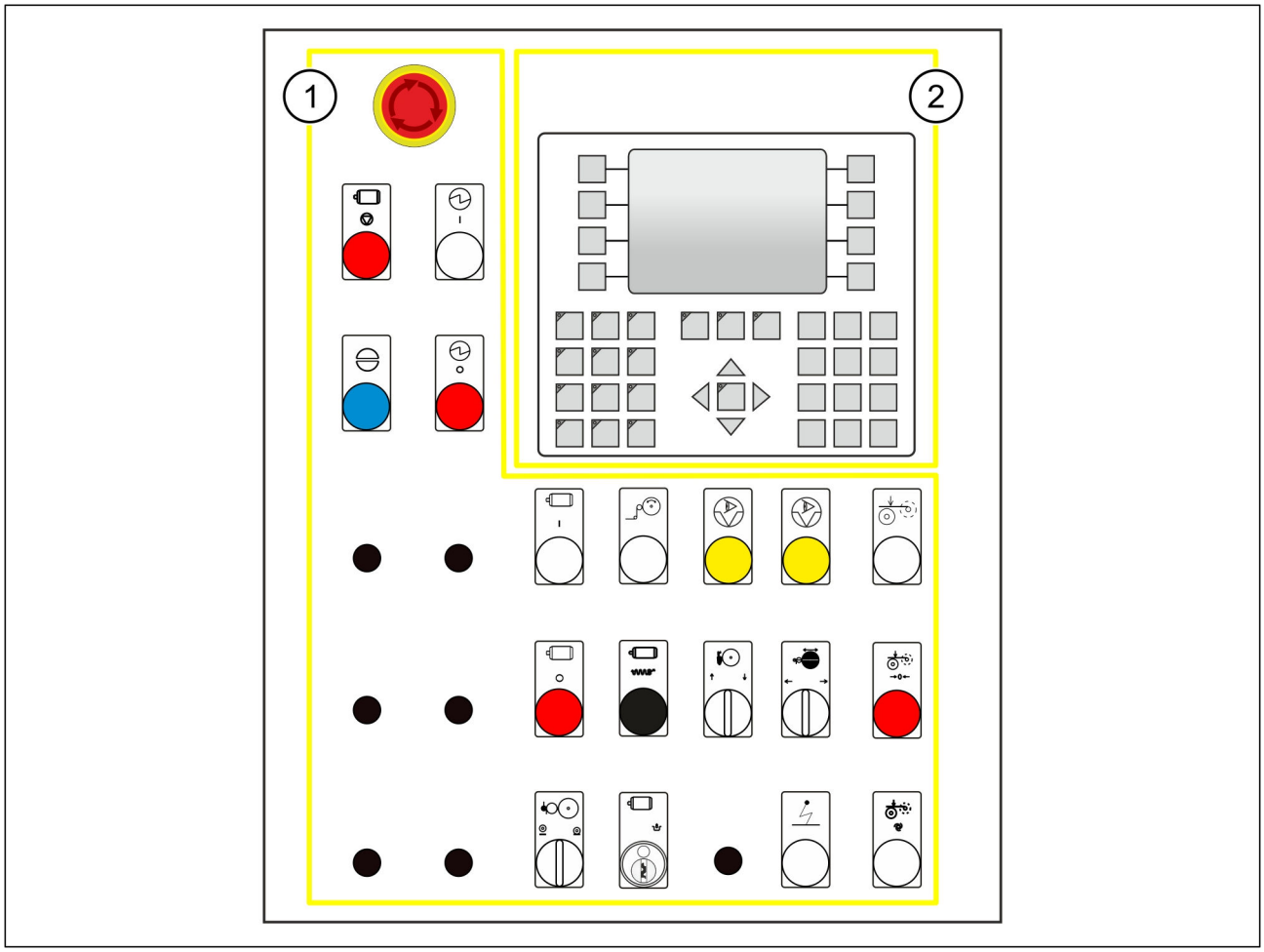
4.5.5 Control consoles +R84A/B....177

4.5.6 Pneumatic control elements in the winding station....178
Maintenance unit for the compressed air supply....178
Pressure settings....178

4.5.1 Main control console +R82A for the winder

The winder is operated from the main control console +R82A.

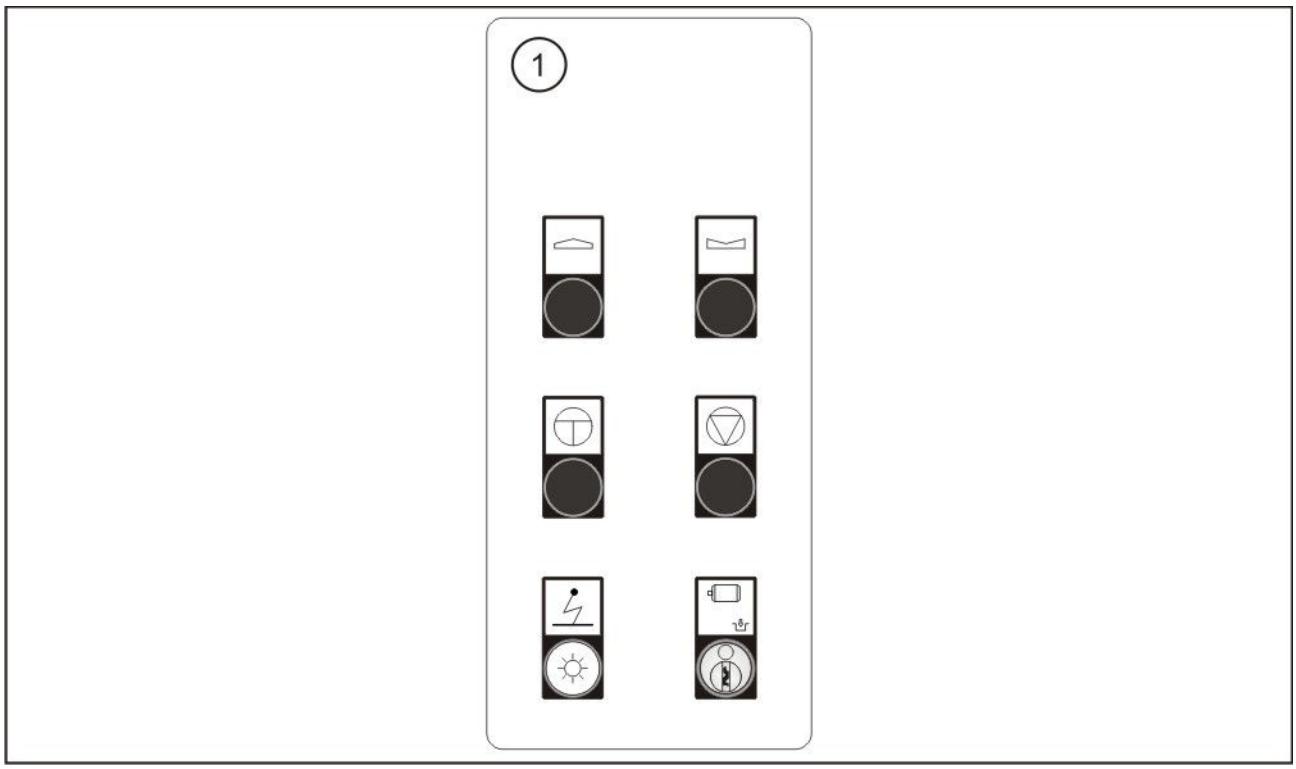
Overall view



(1)	Control panel including control elements	(2)	Display screen with cursor, numeric and functional keys

The +R82A operator panel comprises one control panel each including the control elements (1) and the visual display unit including the cursor, numeric and function keys (2). The functions of individual sub-assemblies are switched on or off using the control elements. The screen pages serve to display and enter process and production data and to show alarm messages. The service screen pages which can be accessed only by entering a password support the service personnel when troubleshooting and calibration becomes necessary.

Side view



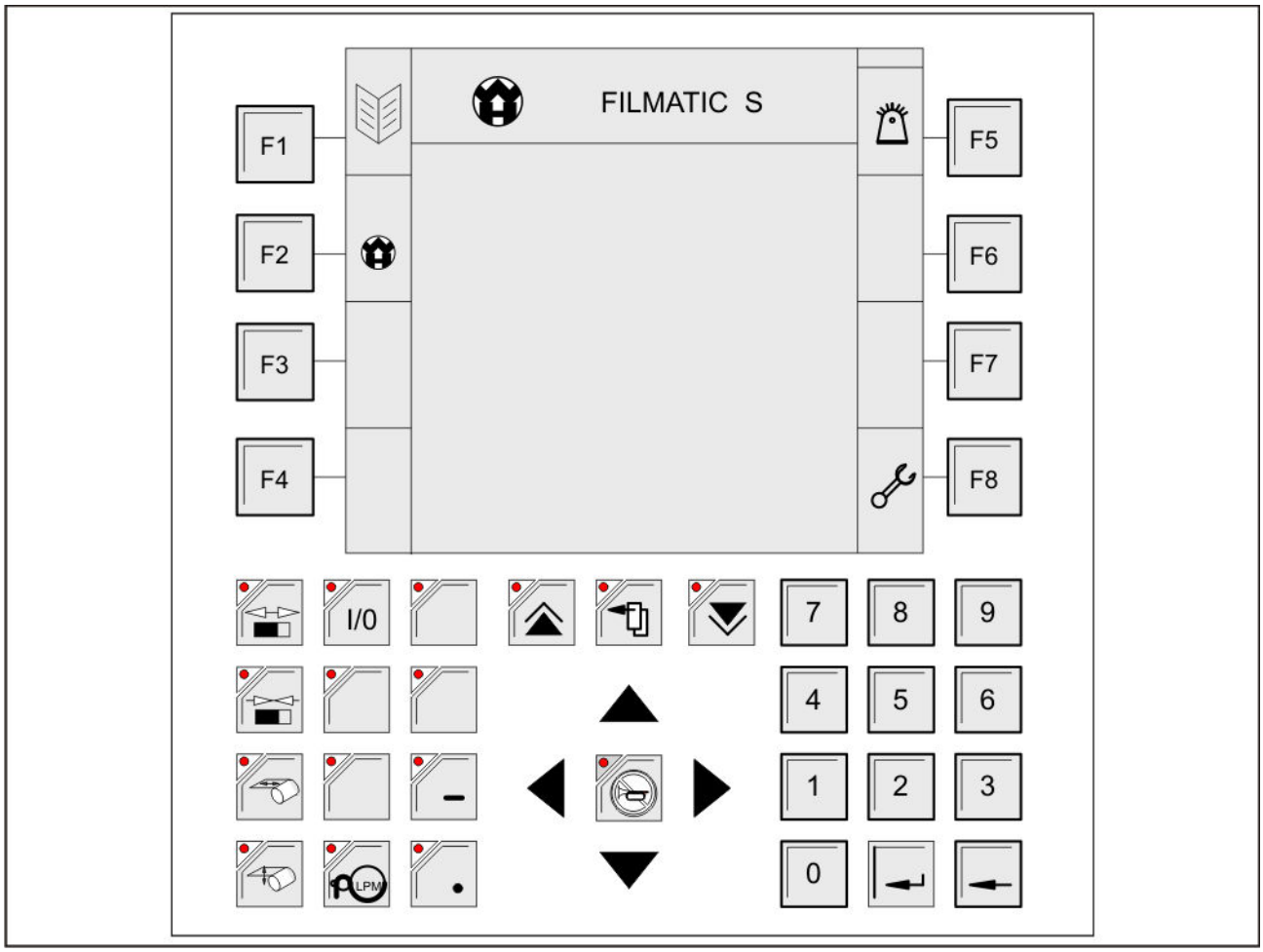
(1)	Control panel with control elements		

Seen from the side view, the control console +R82A contains the control panel with the control elements (1). The functions of individual sub-assemblies are switched on or off using the control elements.

4.5.2 Screen in control console +R82A/B

The screen integrated in control console +R82A/B serves to display processing and production parameters as well as fault messages.

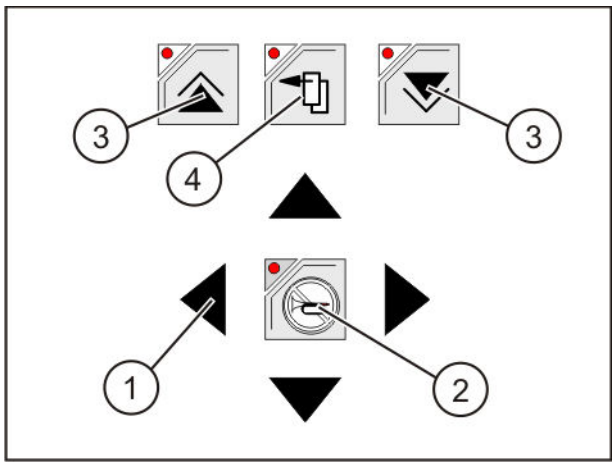
Overall view



The service screens are intended to support the service personnel in troubleshooting and calibrating procedures. Modifications on the service level mean that machine settings, calibration factors and other parameters are changed. Only specialists or service personnel are permitted to undertake this. Basic settings, calibrations and parameterizing will be undertaken by W&H service personnel during the commissioning of the line. The service and parameter level is protected and can only be enabled after a password has been entered.

Cursor keypad

(1)	Direction arrows for the field cursor
(2)	Pushbutton for acknowledging
(3)	Direction arrows for the line cursor
(4)	Pushbutton for paging



The pushbuttons (1) are used to move the field cursor within the screen page.

All alarm reports in the "Alarms" screen page can be acknowledged together and the horn will be switched off if the button (2) in the centre is pressed. Individual alarms can be acknowledged via the selection key [F5]. The ‘Q’ or ‘G’ acknowledged sign is displayed in the right column alongside the alarm reports.

Acknowledge symbol Q:

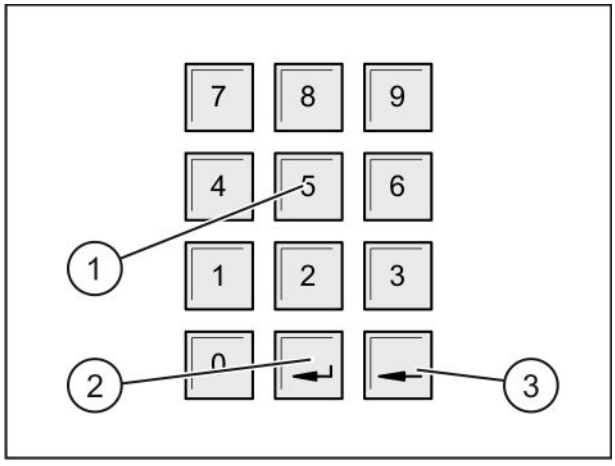
The acknowledgement is carried out before ending the error status. The alarm will be deleted from the alarm screen when the alarms are exited.

Acknowledge symbol G:

The acknowledgement is carried out after ending the error status. The alarm will be deleted immediately from the alarm screen upon acknowledgement.

Numeric keypad

(1)	Numeric keypad
(2)	ENTER key
(3)	DELETE key



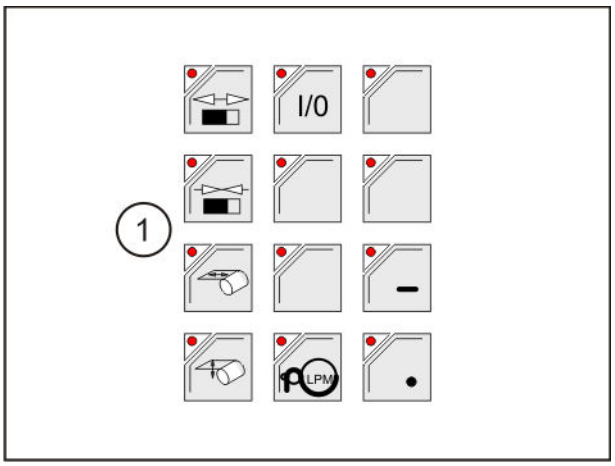
The individual numbers can be entered in the input fields via the numeric keypad (1).

The entry of a data set is confirmed with ENTER (2).

The characters on the left hand side of the cursor in the entry line are deleted by pressing the DELETE key (3).

Control keypad

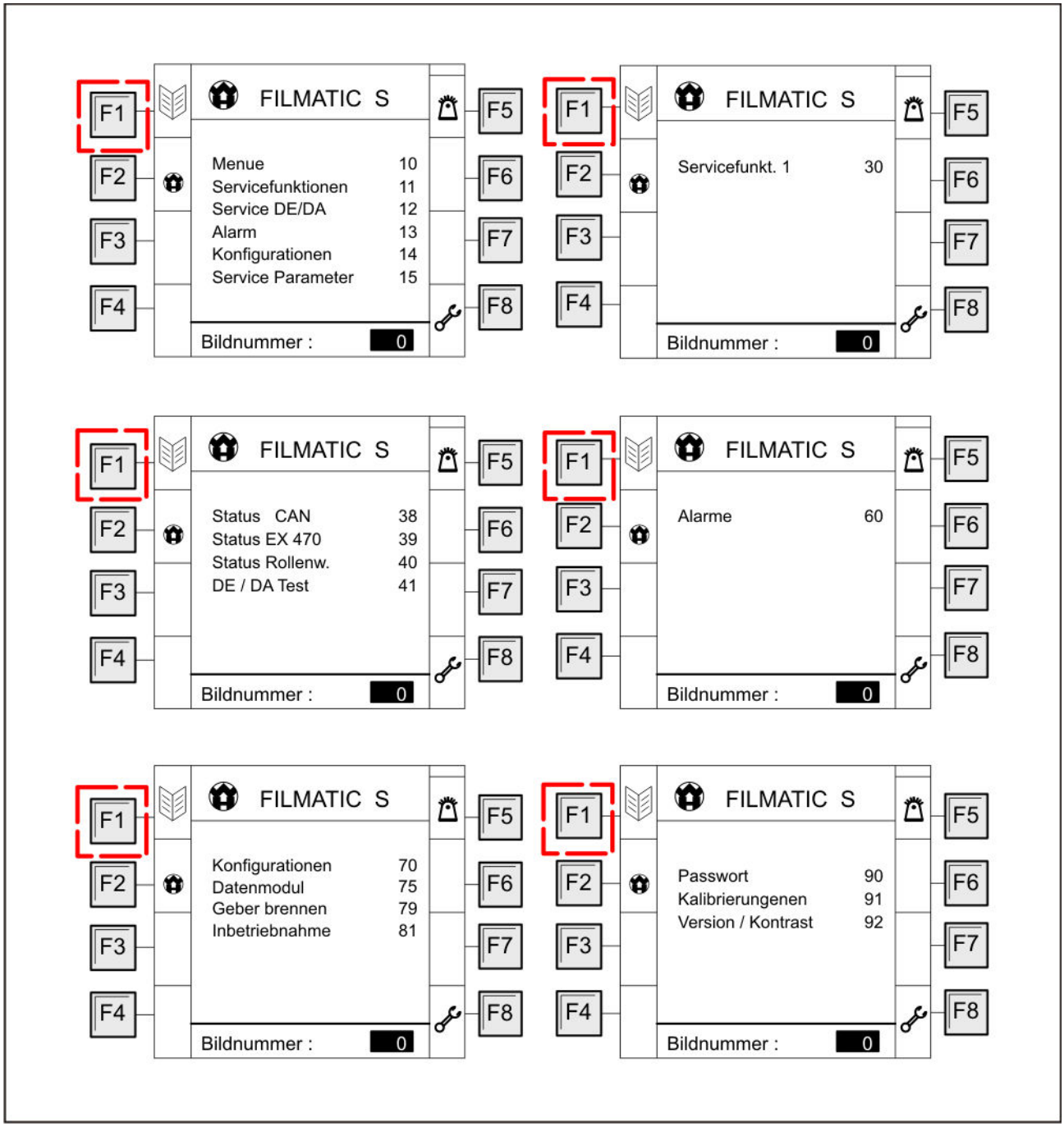
(1)	Control keypad



The following screen pages can be called up via the control keypad (1):

- Displaying and entering the web pulling forces and web tensions (setpoint and actual settings)
- Displaying and entering the contact forces (setpoint and actual settings)
- Displaying and entering the production data (setpoint and actual settings)
- Displaying and entering the counter data (set-point and actual settings)

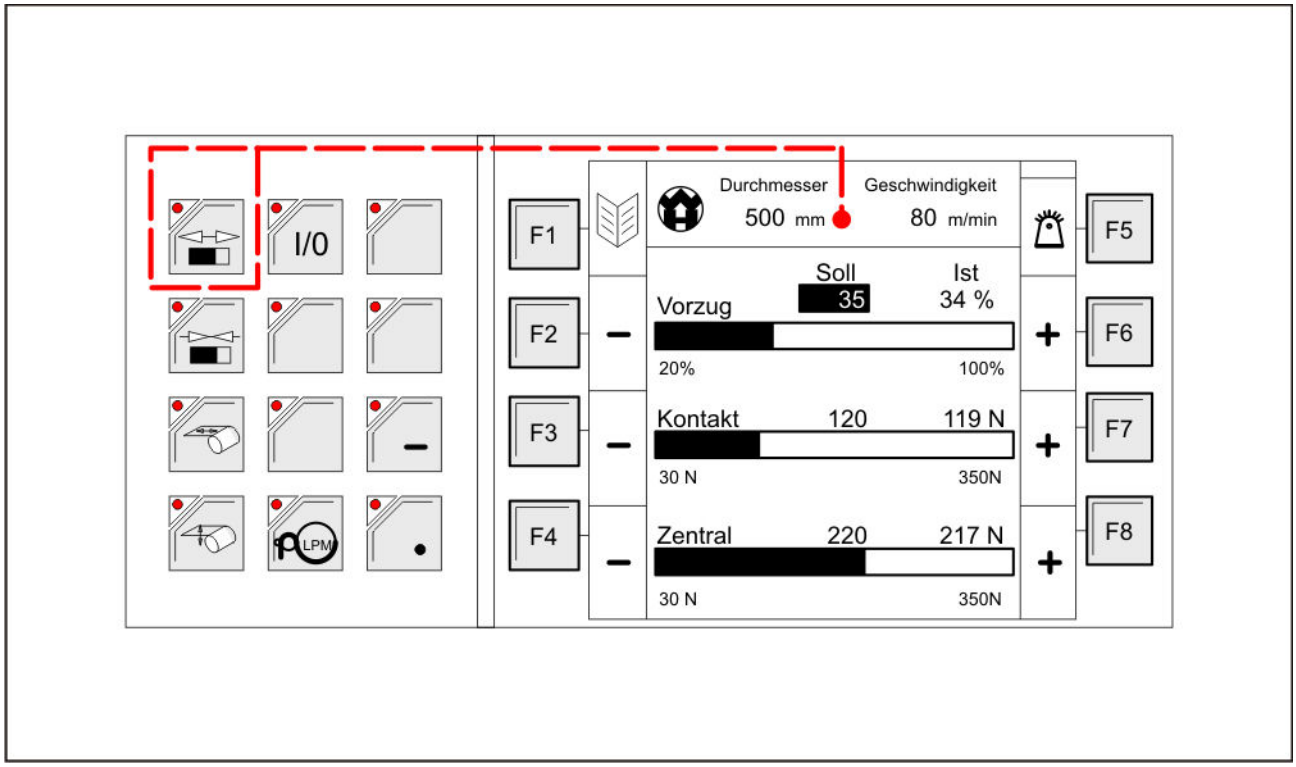
Service screens



Press the [F1] function key several times to call up the individual service pages. The graphical display gives a clear summary of the service level menu and sub-menu pages that can be displayed on the screen.

The service pages are only enabled after a password has been entered.

Web pulling forces and web tensions

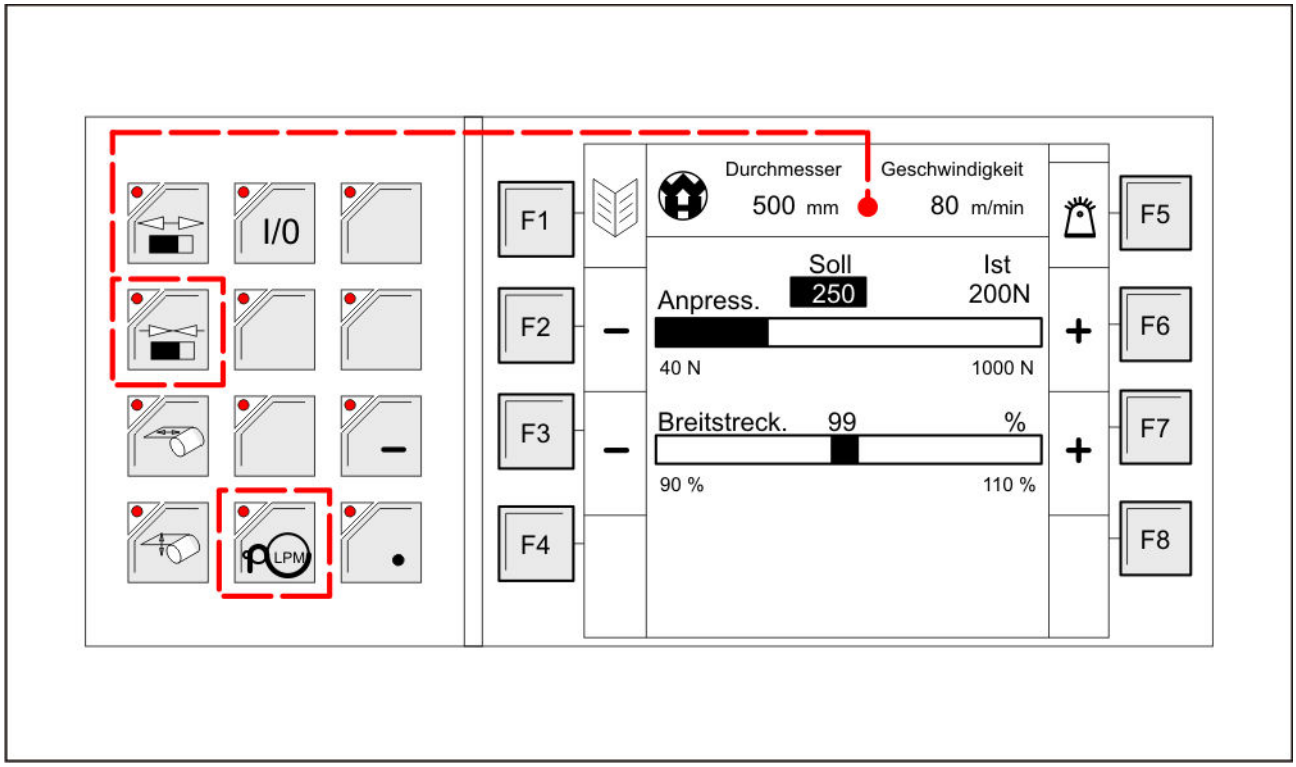


The screen page displays the set values and actual values that affect the web tension. The bar charts display the set values.

The set values are entered numerically via the inputting box. In addition to this, the set value can be modified using the ± keys on the left and right of the bar charts. Press the relevant F2, F3, F5, F6, etc., function keys for this purpose.

Draw	Motor load set value and actual value in [%] of film draw.
Contact	Set value and actual value for the web tension in [N], which is generated by the driven contact roller.
Central	Set value and actual value for the web tension in [N], which is generated by the centrally driven winding shaft.

Contact forces



The screen page displays the contact force set value and actual value between the contact roller and the film roll as well as the driven spreader roller's advance. The bar charts display the set value.

The set value is entered numerically via the inputting box. In addition to this, the set value can be modified via the ± keys on the left and right of the bar chart. Press the relevant F2, F3, F5, F6, etc., function key for this purpose.

Contact force	Contact force set value and actual value in [N] between the contact roller and the film roll
Spreader roller	Set value of the driven spreader roller in [%] of the web speed. The spreader roller can advance by a maximum of ± 10% of the film web speed.

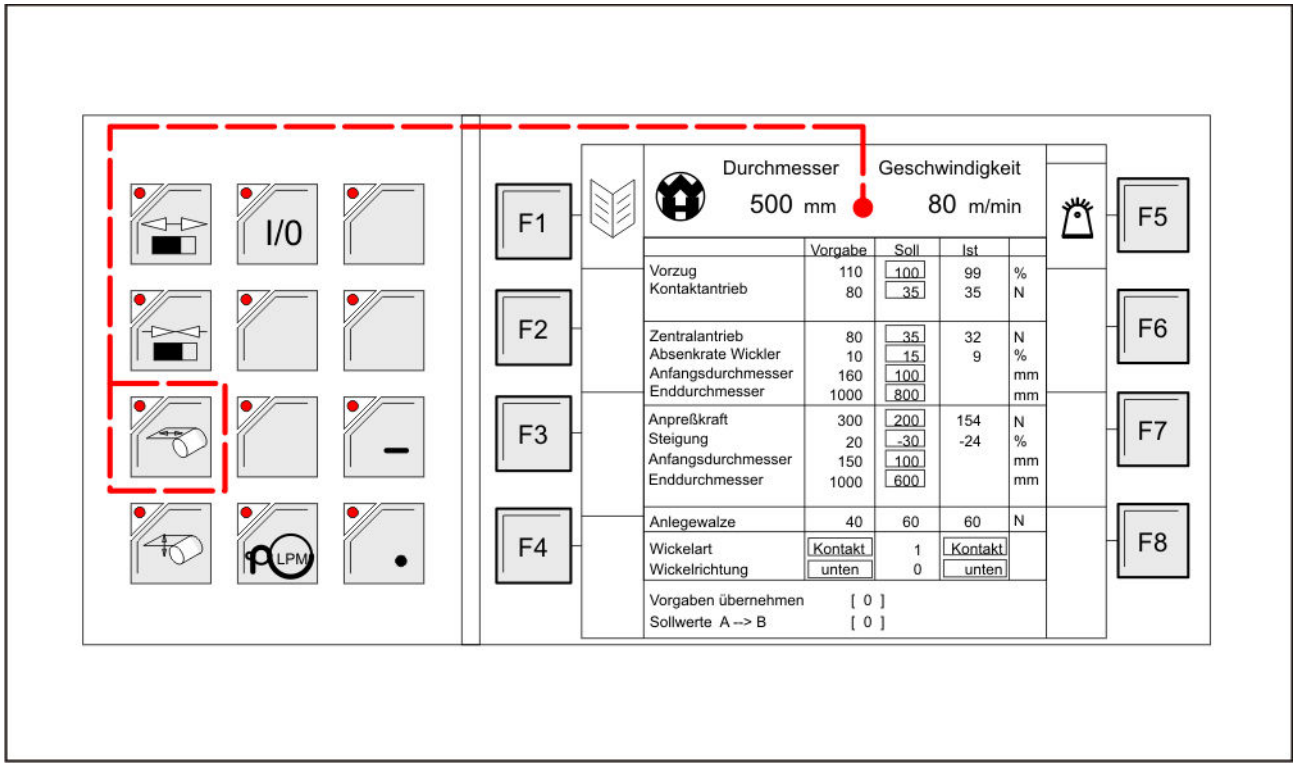
Low Pressure Mode (LPM)

With winders that have a central drive, in Low Pressure Mode sensitive films are wound between the contact roller and the film roll without additional contact force. The web pulling force pulls the film roll only lightly against the contact roller.

By pressing the LPM pushbutton on the control panel, the pneumatic cylinders for pressing the film roll against the contact roller are depressurized. The winder automatically switches to 'central winding' operation mode. The central drive begins to run, accelerates to synchronous speed and then couples.

Pressing the LPM pushbutton again switches the Low Pressure Mode (LPM) back off. The pneumatic cylinders will be charged with pressure again. The central drive continues to operate. It can be switch back off by pressing the CENTRAL WINDING pushbutton.

Production data



The screen page contains the following information:

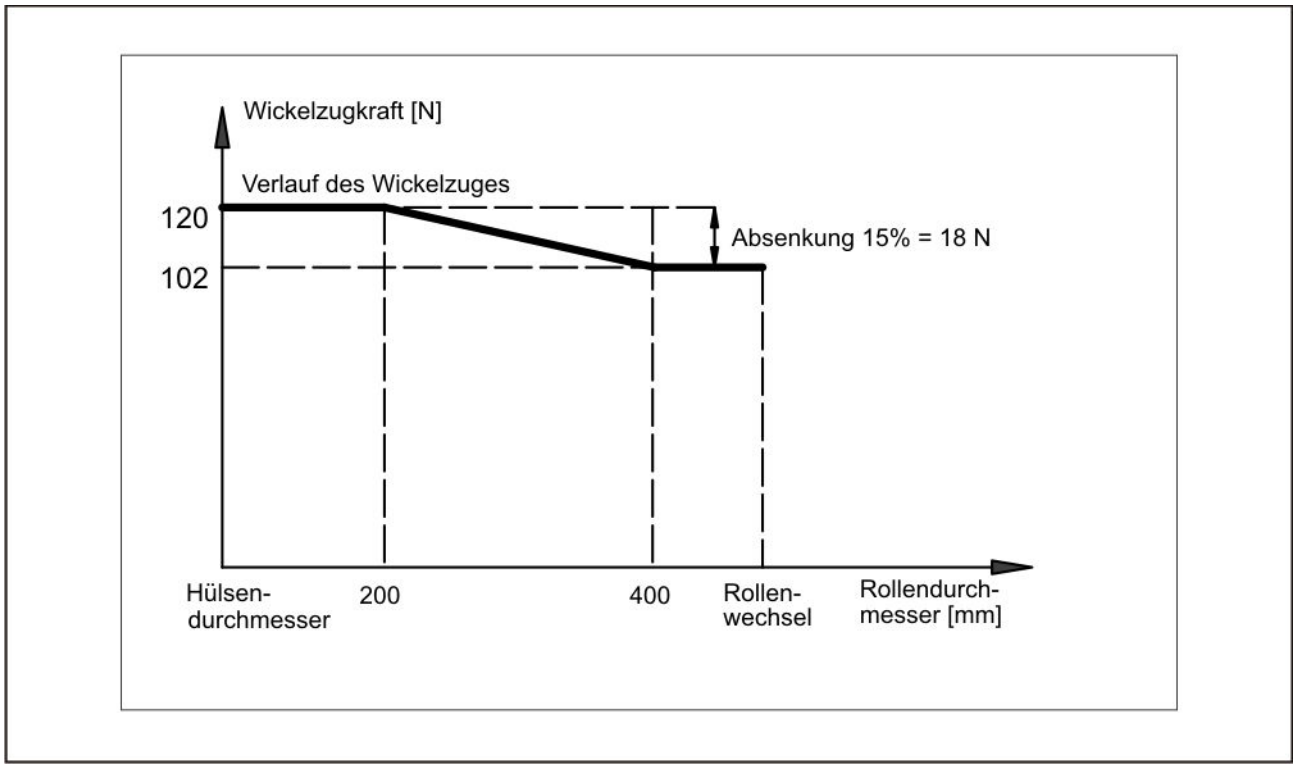
- Film draw motor load in [%] of maximum value
- Web pulling force in [N] before the driven contact roller
- Winding in force in [N], which is generated by the central drive system
- Winding in force reduction rate in [%]
- Start and final diameters for restricting the range over which the winding in force reduction rate is effective.
- Contact force in [N] between the film roll and the contact roller
- [%] increase or decrease in the contact force
- Start and final diameters to restrict the range in which the increase or decrease in the contact force between the film roll and contact roller is effective.
- Selecting the winding modes 'CONTACT', 'CENTRAL CONTACT, 'GAP'
- Winding direction with web guide from 'above' or from 'below'

The production data can be entered or modified within the permitted limits from the extruder system PROCONTROL control column and in conjunction with the W2 module. The production data will be transferred to the winder as the default settings.

A '1' must be entered in the inputting box and confirmed by pressing 'ENTER' in order to transfer the settings from the 'DEFAULT' column to the 'SETTINGS' column. The input limits are checked. The 'COMMAND NOT EXECUTED' report will be displayed if the inputting limits are infringed. The set values will be activated immediately after they have been imported.

Copy winding data (settings). The production data will be copied from one winding page over to the other (A -> B, B -> A). A '1' must be entered in the inputting box and confirmed by pressing 'ENTER' for this purpose.

Reducing the winding in force



The reduction of he winding in force affects the hardness of the film roll. The start and final reduction diameters must be determined during production depending on the material.

Example:

Winder starting tension	120 [N]
Reduction	15 [%]
Start diameter	200 [mm]
Final diameter	400 [mm]

If the start diameter is set to "0", the reduction starts at the current outer diameter of the winding core.

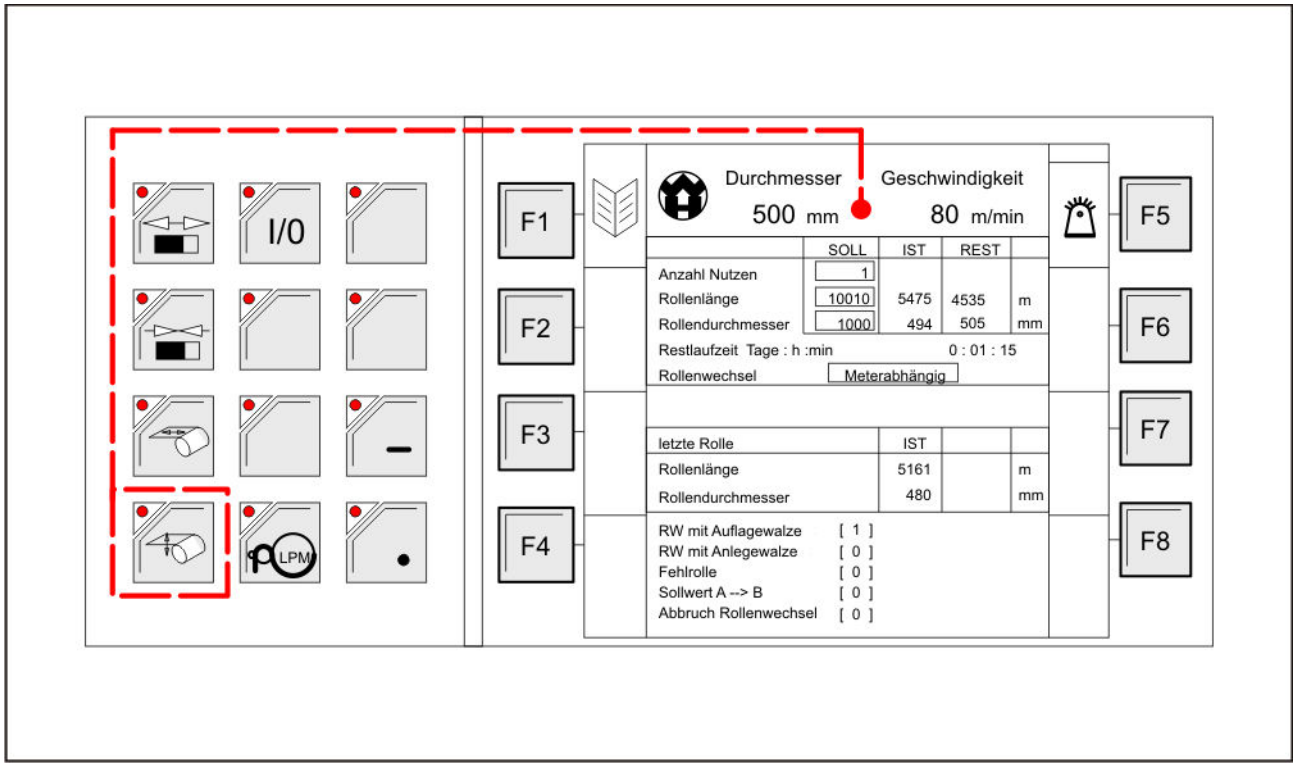
The final diameter must always be entered when entering the winding in force reduction!

Contact force reduction

The contact force between the film roll and the contact roller will be displayed as well as any changes. To increase or decrease the winding contact force the start and final diameters can be freely selected within the predefined range.

If the contact force exceeds the upper limit (650 N) because the selected increase rate was too high, winding will continue at this maximal setting until the final winding diameter is realized.

Counter parameters



The screen page includes the details of the film roll that is running and the job details for winding station A or B:

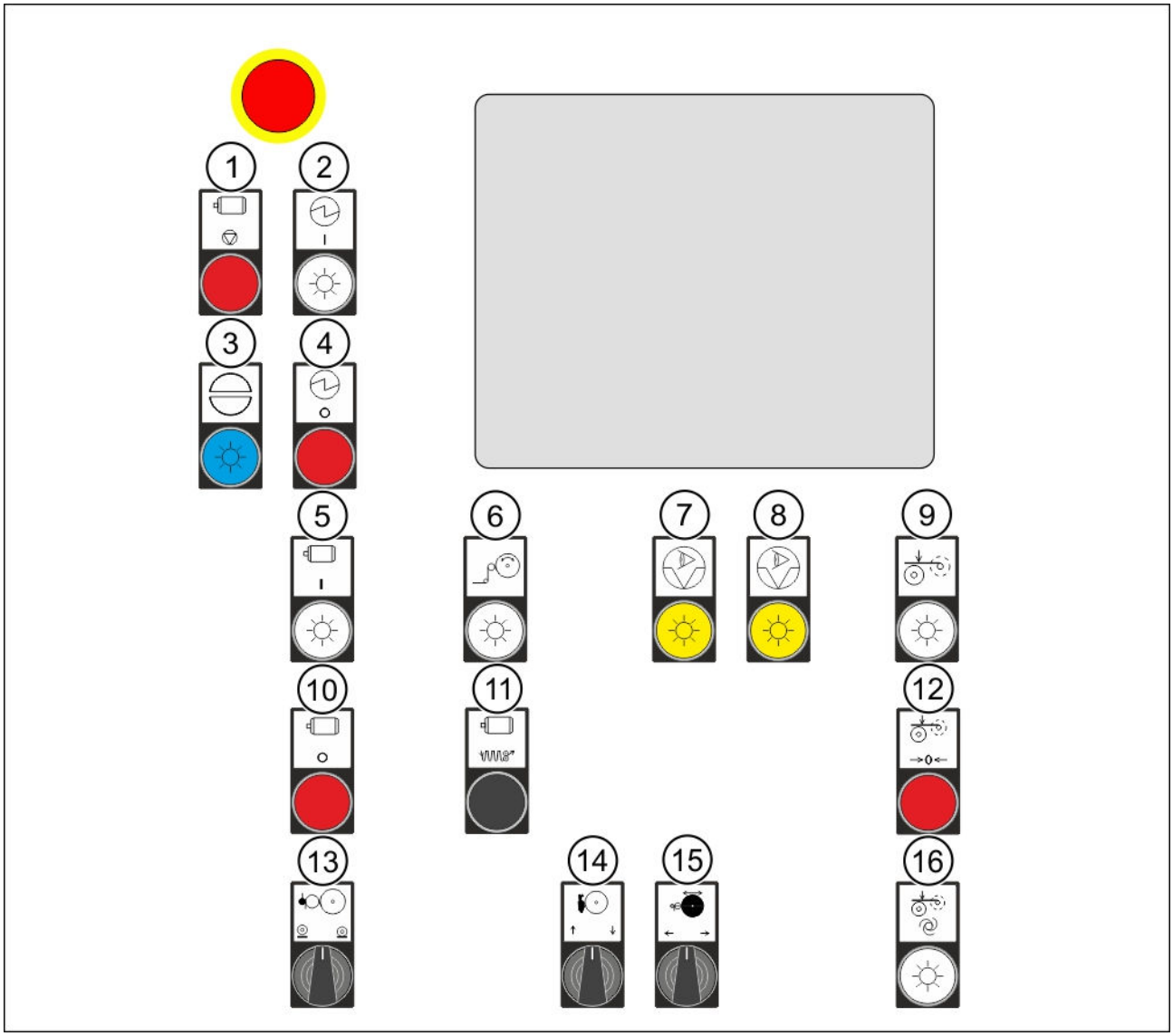
- Set number of webs in the winding station
- Roll length of the running film roll in [m] (set, actual and remaining value)
- Roll diameter of the running film roll in [m] (set, actual and remaining value)
- Running time left for the film roll [days : h : min]
- 'Meter dependent' or 'diameter dependent' roll change
- Actual roll length of the last roll in [m]
- Actual roll diameter of the last roll in [mm]
- Roll change with press roll up against the contact roller. A '1' must be entered in the input field and confirmed by pressing ENTER.
- Roll change with lay-on roller from below when using the vacuum cutover drum (option). A '1' must be entered in the input field and confirmed by pressing ENTER.
- Roll change with lay-on roller from below. The auxiliary winder drive is used as lay-on roller (option). A '1' must be entered in the input field and confirmed by pressing ENTER.
- Declare a waste roll. If production deems the quality of the running film roll as poor, then this film roll can be declared a waste roll. A '1' must be entered in the input field and confirmed by pressing ENTER. Start a roll change afterwards. For job balancing, the waste roll length, consisting of

the number of rolls x roll length will not be taken into consideration. In addition, the roll length and the roll diameter of the last roll are displayed. If no 'W2' interconnection to the extrusion line is available, the job length and the running time left for the job will also be displayed.

- Copy the set winding parameters. The counter parameters will be copied from one winder page to the other winder page (A -> B, B -> A). A '1' must be entered in the input field and confirmed by pressing ENTER.
- Abort roll change. Use this to abort the roll change, if a fault occurs. A '1' must be entered in the input field and confirmed by pressing ENTER. The roll change system must be returned to the start position afterwards using the PIVOT RCS swivel switch.

4.5.3 Control elements on operator panel +R82

Control panel - Part 1

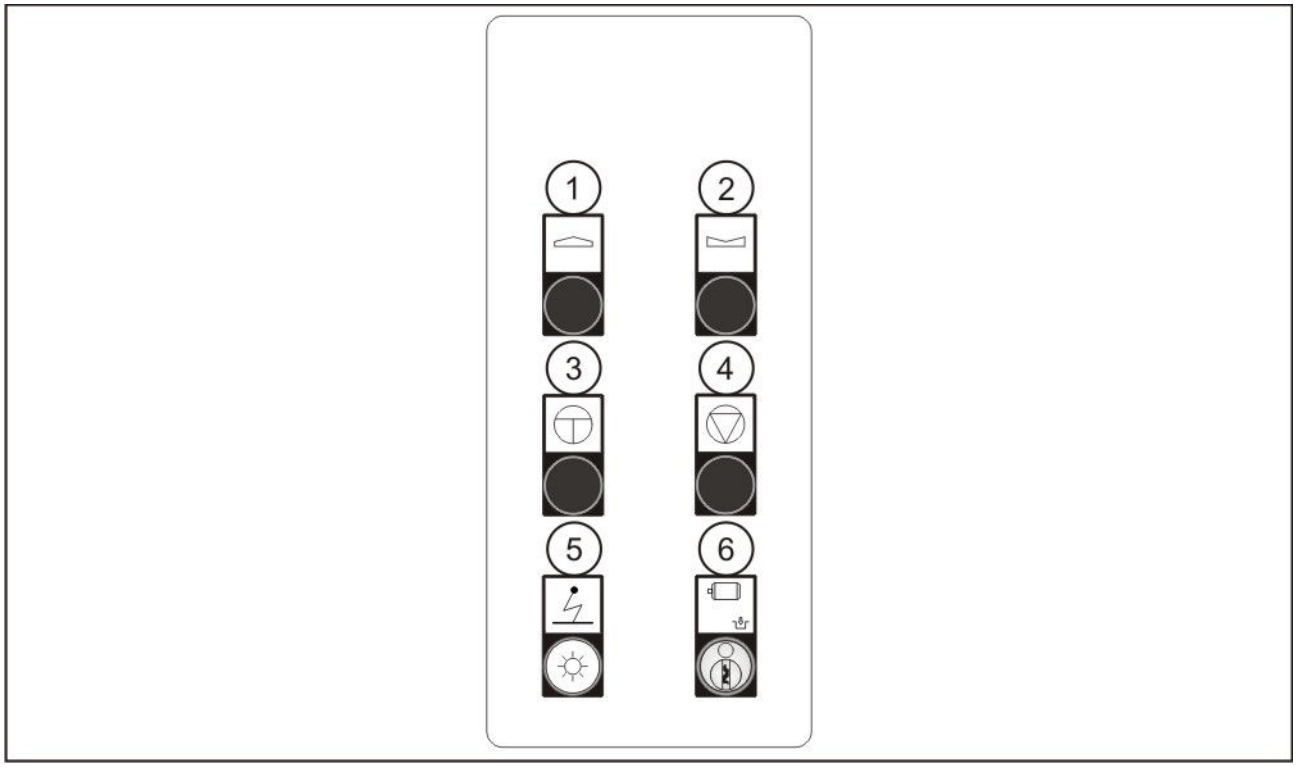


Pos.	Designation	Description
(1)	WINDING STATION STOP	Switches off the winding station.
(2)	CONTROL VOLTAGE ON	Switches on control voltage.
(3)	RESET	Enables the drives in the winding station to be switched back on.
(4)	CONTROL VOLTAGE OFF	Switches off control voltage.
(5)	WINDER ON	Used to switch on the drives in the winding station.
(6)	CENTER WINDING	Used to switch on 'Center winding' mode. The center drive to drive the winding shaft is started when disengaged and reengaged when reaching the synchronous torque.

Pos.	Designation	Description
(7/8)	BYPASS LIGHT BARRIER	Used to bypass the light barrier in the infeed nip for film sampling by simultaneously pressing both the buttons.
(9)	START ROLL CHANGE	Used to start a roll change irrespective of the selected 'Automatic roll change' or 'Manual roll change' mode.A roll change can be started at any time.
(10)	WINDER OFF	Used to switch off the drives in the winding station.
(11)	SPEED UP WEB	Used to even out the web sag (slack web) between the draw and the winding station.
(12)	ROLL CHANGE STOP	Stops all movement during a roll change. Rewinding of the web is continued. A roll change can be started again at any time.
(13)	LAY-ON ROLLER ON/OFF	Used to swivel the lay-on roller onto or away from the contact roller. Left-hand position = swivel away, right-hand position = swivel in.
(14)	SWIVEL ROLL CHANGE SYSTEM (RCS)	Used to pivot the roll change system manually. '↓': Used to swivel the roll change system including the winding shaft from the take-over position into the transfer position. '↑': Used to swivel the roll change system without the winding shaft back from the transfer position into the take-over position.
(15)	PARALLEL GUIDE	Shifts the winding shaft including the roll on the mounting rails of the parallel guide.
(16)	ROLL CHANGE AUTO / MANUAL	Select 'Automatic roll change' or 'Manual roll change' mode. The illuminated button indicates that 'Automatic' mode is active.

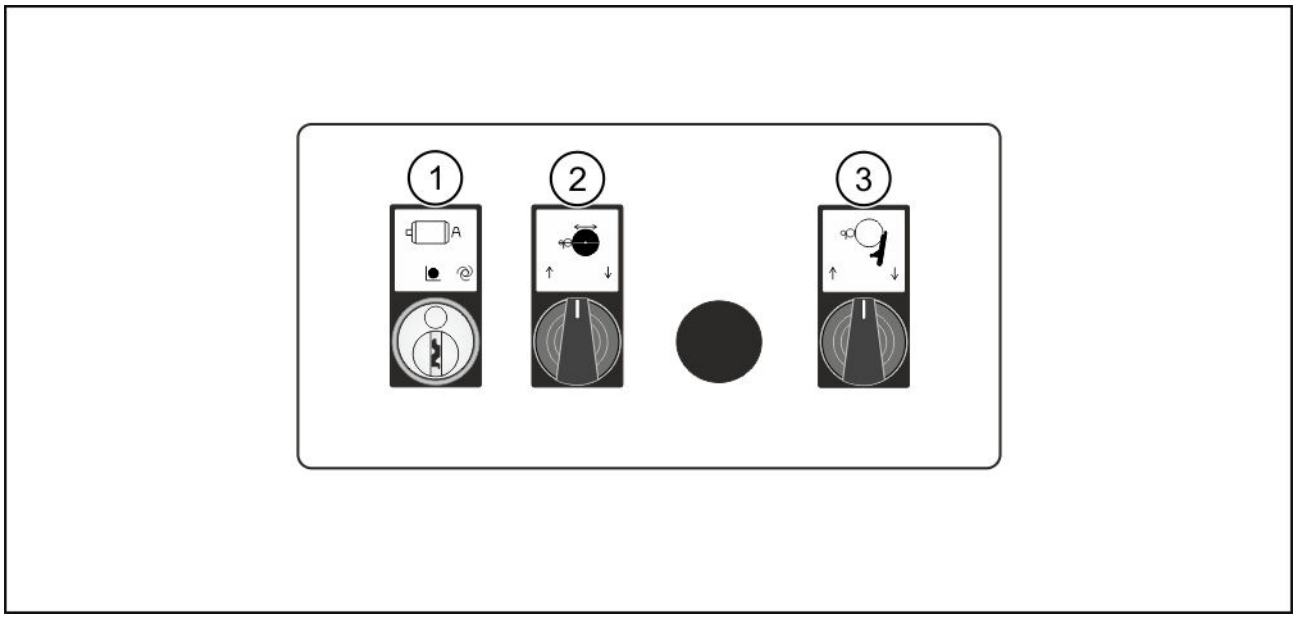
Control panel - Part 2

The control elements are provided on the side where the main control console is installed.



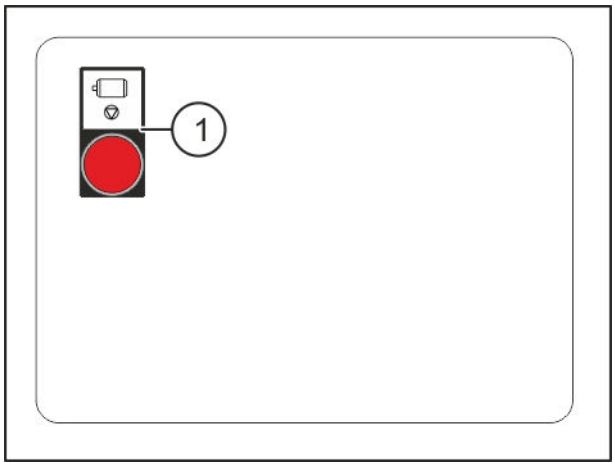
Pos.	Designation	Description
(1)	FAST	Press and hold down the button to run up the machine to maximum speed.
(2)	SLOW	Press and hold down the button to slow the current machine speed down to base speed.
(3)	JOG	Switch on the printing unit drive by pressing the button twice. At the first time, hold it down for 1 to 3 seconds. A warning signal sounds when the printing unit is started.
(4)	STOP	Stops the printing unit drive.
(5)	DEIONIZATION	Switches deionization mode on and off to remove static charge from the film surface.
(6)	WINDING STATION SECURED	Key switch to secure the winding station against unintentional switching on during maintenance works, **Mid-position:** Winding station not secured, **right-hand position:** Winding station secured.

4.5.4 Operator panel +R85A



Pos.	Designation	Description
(1)	MODE SELECTOR SWITCH	Select mode 0 = 'Startup mode' or I = 'Production mode' .
(2)	PARALLEL GUIDE	Shifts the nip roll and the web cut-off assembly with the roll on the mounting rails of the parallel guide. '↓': Retracts the roll from the contact roller'←': Moves the roll towards the contact roller.
(3)	LOWERING ARMS	Raises and lowers the lowering arms for roll handling. The hydraulic unit starts running when actuating the rotary switch.

4.5.5 Control consoles +R84A/B



(1)	WINDING STATION STOP	Stop the winding station whenever a dangerous situation arises. The other winding station and the extrusion line will continue running.

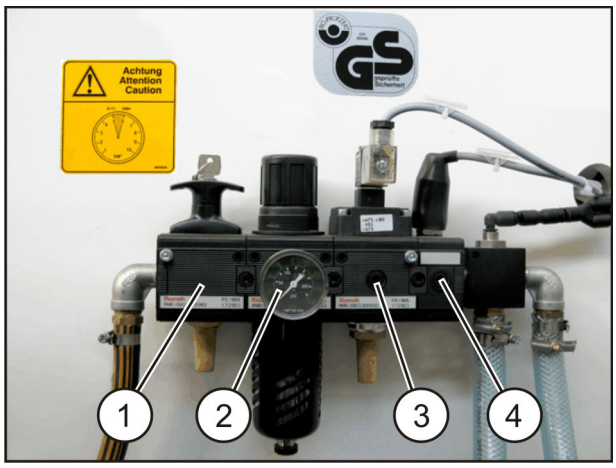
4.5.6 Pneumatic control elements in the winding station

Maintenance unit for the compressed air supply

NOTE

Pneumatic valves and cylinders must be supplied with compressed air free of oil.

(1)	Main shut-off valve
(2)	Filter pressure regulator with manometer
(3)	Pilot valve
(4)	Distributor block with pressure switch



The maintenance unit is equipped with a main shut-off valve (1) fitted in the compressed air line immediately in front of the filter pressure regulator. To operate the unit the valve must be opened.

The filter pressure regulator (2), consisting of a filter with water trap and pressure control valve with manometer, has the task of cleaning and removing liquid and solid constituents from the compressed air (the resource) as well as controlling the air pressure.

Any condensation or impurities collected in the filter can be removed by opening the drainage screw (must be checked at regular intervals!).

The pressure regulator reduces the network pressure to the required operating pressure and maintains this value as a constant setting regardless of the amount of air consumed or pressure fluctuations. The network pressure must be 6 bar and the operating pressure, at least 5 bar.

Increasing the pressure: Turn hand wheel clockwise

Decreasing the pressure: Turn hand wheel counter-clockwise

The pneumatic system can be depressurized via the pilot valve (3).

The pressure switch (4) monitors the minimum pressure in the pneumatic system.

Pressure settings

Setting the contact force between the press roll and the contact roller

The precision control valve is used to set the contact pressure between the lay-on roller and the contact roller. The manometer indicates the set pressure.

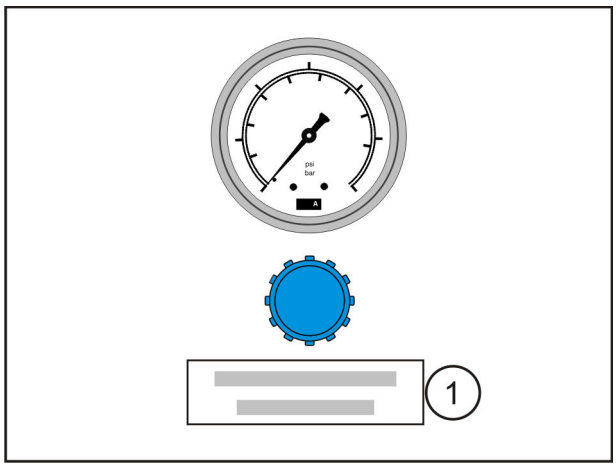
Loosen the set collar to adjust the precision control valve.

CAUTION

Screw-in throttles serve to set the piston speed within the pneumatic cylinders. After setting the piston speed, use a thread locker to ensure that the screw-in throttles do not come loose.

(1)	LAY-ON ROLLER pressure control valve

- Turn the hand wheel for the pressure control valve (1) clockwise to increase the pressure.
- Turn the hand wheel for the pressure control valve (1) counter-clockwise to reduce the pressure.



X000050624

Setting the air pressure at the roll change system

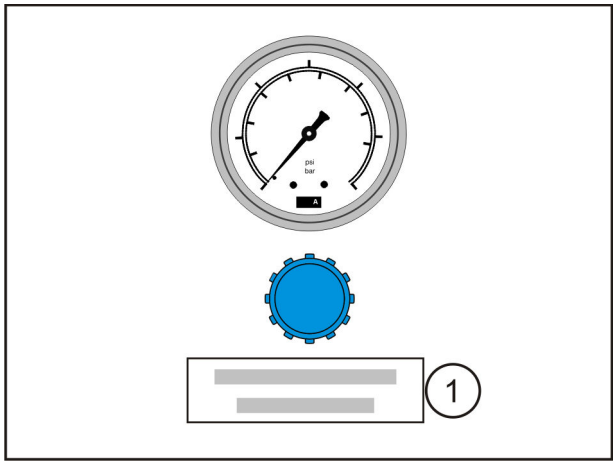
Use the precision control valve to set the air pressure for pivoting the roll change system. The manometer indicates the set pressure.

Loosen the set collar to adjust the precision control valve.

CAUTION

Screw-in throttles serve to set the piston speed within the pneumatic cylinders. After setting the piston speed, use a thread locker to ensure that the screw-in throttles do not come loose.

(1)	PIVOT ROLL CHANGE SYSTEM precision control valve



- Turn the hand wheel for the pressure control valve (1) clockwise to increase the pressure.

  The minimum operating pressure should be approx. 3.0 bar (43.5 psi).

- Turn the hand wheel for the pressure control valve (1) counter-clockwise to reduce the pressure.

Setting the pneumatic pressure for retracting the film roll

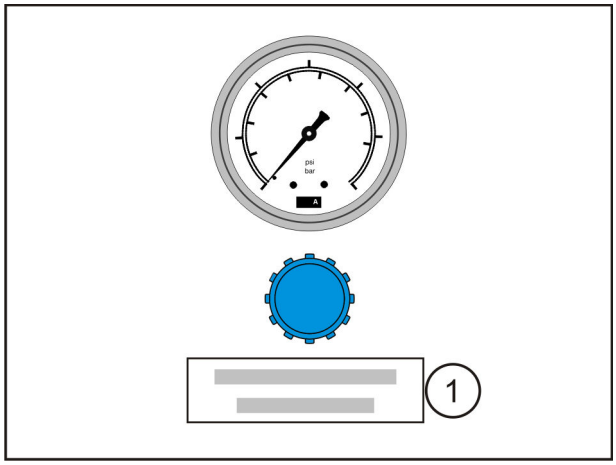
Use the precision control valve to set the air pressure for retracting the film roll from the contact roller (parallel guide). The manometer indicates the set pressure.

Loosen the set collar to adjust the precision control valve.

CAUTION

Screw-in throttles serve to set the piston speed within the pneumatic cylinders. After setting the piston speed, use a thread locker to ensure that the screw-in throttles do not come loose.

(1)	RETRACT PARALLEL GUIDE pressure control valve



- Turn the hand wheel for the pressure control valve (1) clockwise to increase the pressure.

  The minimum operating pressure should be approx. 4.0 bar (58 psi).

- Turn the hand wheel for the pressure control valve (1) counter-clockwise to reduce the pressure.

X000050624

Setting the contact pressure between the lay-on roller and the film roll

The precision control valve is used to set the contact pressure between the lay-on roller and the film roll during roll change. The manometer indicates the set pressure.

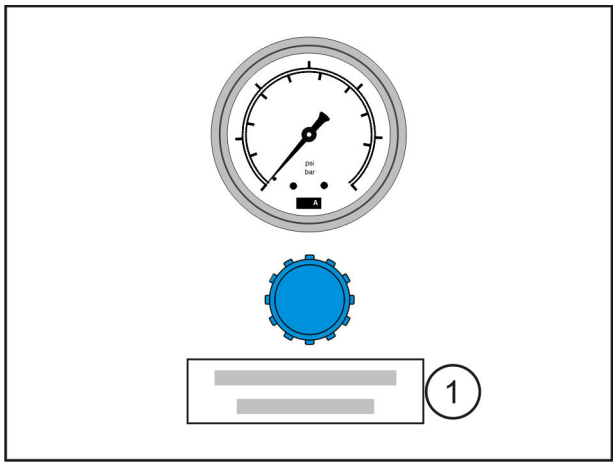
Loosen the set collar to adjust the precision control valve.

NOTE

Screw-in throttles serve to set the piston speed within the pneumatic cylinders. After setting the piston speed, use a thread locker to ensure that the screw-in throttles do not come loose.

(1)	LAY-ON ROLLER pressure control valve

- Turn the hand wheel for the pressure control valve (1) clockwise to increase the pressure.
- Turn the hand wheel for the pressure control valve (1) counter-clockwise to reduce the pressure.



Setting up the contact pressure between the peel-off rollers

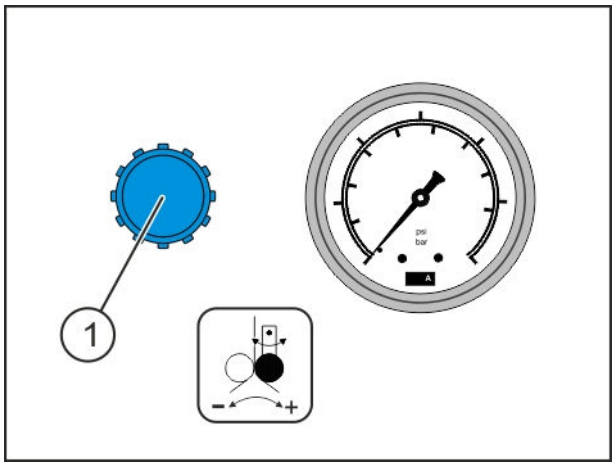
The pressure control valve is used to set up the contact pressure of the pivotable peel-off roller applied to the driven peel-off roller. The pressure gauge indicates the set pressure.

Loosen the set collar to adjust the pressure control valve.

NOTE

Screw-in throttles serve to set the piston speed within the pneumatic cylinders. After setting the piston speed, use a thread locker to ensure that the screw-in throttles do not come loose.

(1)	Pressure control valve PEEL-OFF ROLLER



- Turn the handwheel for the pressure control valve (1) clockwise to increase the pressure.
- Turn the handwheel for the pressure control valve (1) counter-clockwise to reduce the pressure.

5 Operating the machine

5.1 Unwind
5.1.1 Unwinder

5.2 Web processing station
5.2.1 Dancer roller
5.2.2 Micro-perforator
5.2.3 Embossing

5.3 Tube forming station
5.3.1 Web edge - Safe behavior
5.3.2 Tensioning the emergency stop trip wire
5.3.3 Guide elements at the infeed section
5.3.4 Side gusset tube forming unit - Infeed rollers
5.3.5 Adjusting the side gusseted tube forming
5.3.6 Adjusting the side gusset depth
5.3.7 Side gusset tube forming unit - Former plates

5.4 Seam extruder
5.4.1 Longitudinal seam extruder
5.4.2 Transporting the frame
5.4.3 Switching on Automatic mode
5.4.4 Refilling resin – Safety instructions
5.4.5 Resin monitoring
5.4.6 Thermocouples – Overview
5.4.7 Thermocouples
5.4.8 Heating elements
5.4.9 Melt pressure sensor
5.4.10 Longitudinal seam monitoring – Drag rope
5.4.11 Longitudinal seam extruder - Collecting tank

5.5 Main draw
5.5.1 Control voltage
5.5.2 Key switch
5.5.3 Machine ON
5.5.4 Jog
5.5.5 Switching on Automatic mode
5.5.6 Setting up the contact pressure
5.5.7 Main draw
5.5.8 Main draw – Compressed air supply
5.5.9 Longitudinal seam monitoring
5.5.10 Seam cooling roller
5.5.11 Replacing the seam cooling roller
5.5.12 Supplying coolant
5.5.13 Nip roll
5.5.14 Cut edge blocking

5.6 Web guide control
5.6.1 Web guide control, ekr
5.6.2 Web guide control - Material setup

5.7 Winder
5.7.1 Control voltage
5.7.2 Change winding shaft
5.7.3 Fixing the winding core
5.7.4 Starting the winder
5.7.5 Web sag on startup
5.7.6 Automatic step sequence during a roll change
5.7.7 Moving the film reel
5.7.8 Film roll delivery
5.7.9 Shut-down sequence when producing tubular film

5.1 Unwind

5.1.1 Unwinder 187
Safety instructions 187
Changing the roll 187
Setting up the web guide controller 187

5.1.1 Unwinder

Safety instructions

Electromagnetic fields

⚠ DANGER

It is forbidden for people fitted with a pacemaker to work on or operate the machine

Electromagnetic fields on the film winder, caused by weather-dependent static charging of the film, can cause pacemakers to malfunction. This is why individuals with pacemakers are prohibited from operating the film winder or being in its immediate vicinity.

⚠ DANGER

Electrical discharges

High voltages are present on the charge and discharge electrodes.

Contact with the electrodes can cause subsequent accidents as a consequence of reflexive reactions to the electric shock. Persons in the vicinity of rotating rollers and drives are in mortal danger! Persons with pacemakers are at severe risk!

Observe the operating instructions from the component manufacturer.

⚠ WARNING

Electrostatic charges

Electrostatic charges are liable to occur between the film and the rollers when separating the film from rollers.

Therefore you must ensure that these components are always properly grounded and that conductive drive materials are always used.

Roll dimensions

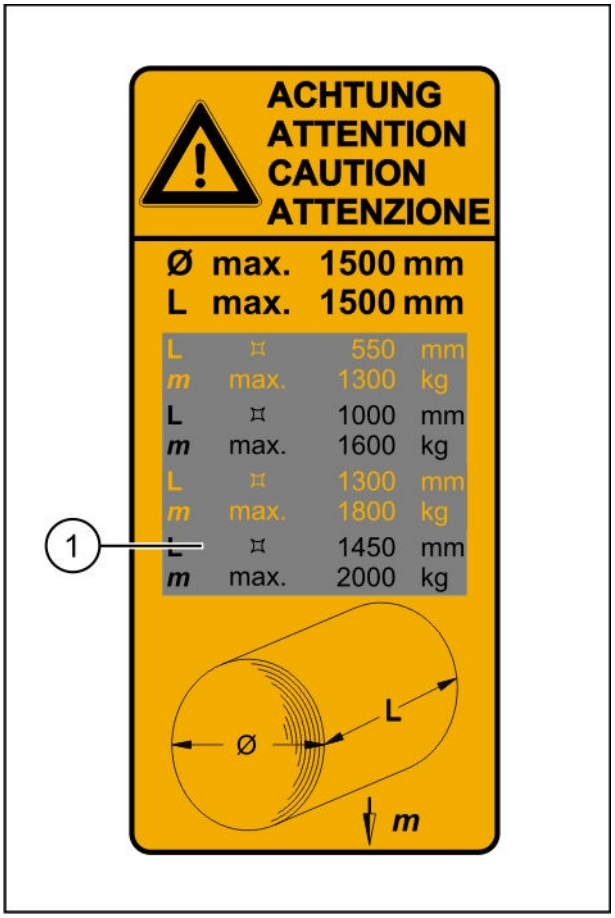
⚠ WARNING

Risk of injury due to winding shaft breakage and the roll rolling out of the unwind stand after that.

Do not exceed the specified weight, width and diameter of the roll.

Exceeding the specified dimensions and weights causes winding shaft to break.

(1)	Label: Dimensions and weights to be adhered to



Safe behavior

DANGER

Injuries due to falling load

Do not walk below suspended loads.

Wear a hardhat and industrial safety shoes.

Observe the permisslble roll weight!

When lift trucks and crane system are to be used the manufacturer's instructions must be adhered to.

WARNING

Automatically moving machine parts

Risk of crushing and shearing your hands and feet.

Wear protective footwear.

Before initiating and performing machine movements, the operator must ensure that the danger zone is clear of persons. He must permanently view into the danger zone to be able to stop machine movements immediately whenever a dangerous situation arises.

Pay attention to the lowering roll and the swivel arms. Keep a safe distance between the roll and the swivel arms.

Risk of injury from the roll rolling out of the unwind stand.

Never switch on the machine with the lock open.

Close the bearings before starting the machine.

CAUTION

Rotating rollers and wheels, circulating chains and belts

Parts of your body, clothes, hair and objects can be caught in the nip and crushed.

Do not open or remove protective devices. Keep hands clear of the infeed nip. Always wear tight-fitting clothes and a hairnet. Work without industrial safety gloves.

CAUTION

Web infeed nips

Parts of your body, clothes, hair and objects can be caught in the nip and crushed.

Do not reach into the running machine. Always wear tight-fitting clothes and a hairnet. Work without industrial safety gloves.

CAUTION

Free-running webl

A free-running web may cut into parts of your body.

Wear cut-proof protective gloves.

NOTE

Carefully transport the rolls onto and insert them into the unwind to make sure not to damage the material.

Always have a ready-made roll at hand to allow for a roll change without delay.

Insert the roll into the ready-made unwind unit so that the spacing of the lateral edge of the new roll and the spacing of the lateral edge of the expiring roll are the same.

When using machines equipped with a web guide control unit, be sure to splice the web edges with utmost care to allow for the web guide control sensor to reliably scan the web edges.

X000047219

Changing the roll

Attaching the roll

Use chucks to secure the roll on the winding shaft in place.

CAUTION

Wear safety goggles when working with compressed air!

CAUTION

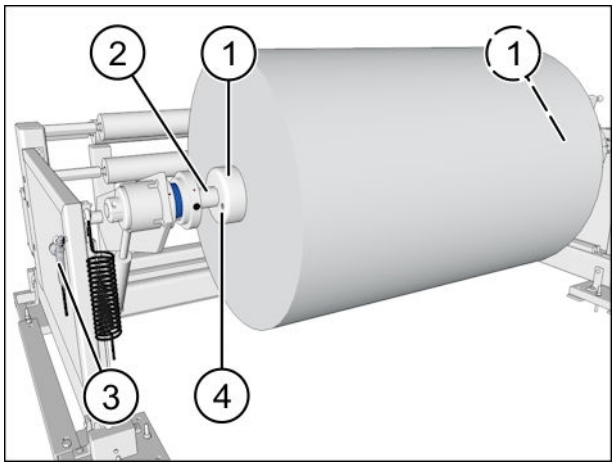
Bursting bellows

Never inflate the chucks without winding sleeves.

Secure entrainment of the winding shaft.

Always establish a fixed joint between the winding shaft and the roll.

(1)	Chuck
(2)	Winding shaft
(3)	Air gun
(4)	Valve



- Feed the winding shaft through the core into the roll.
- Push the left and right chucks (1) onto the winding shaft (2) and into the roll core.
- Align the winding shaft until its center aligns with the the machine center.
- Screw the chucks into place in the determined position on the winding shaft.
- Place the air gun (3) against the valve (4) on the chuck and inflate the chuck.

  Inflate both chucks until the roll is securely clamped in place.

  The roll is securely attached to the winding shaft.

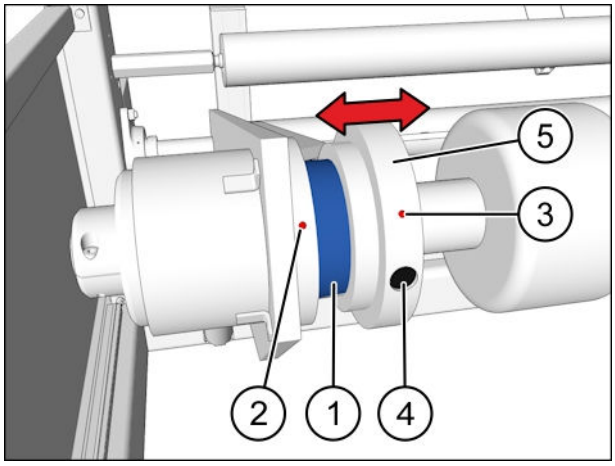
Place the winding shaft together with the roll into the unwind unit.

Load the winding shaft into the unwind for underwind winding.

Inserting a material roll

Open and close the bearings

(1)	Sliding bearings
(2)	Mark
(3)	Mark
(4)	Button
(5)	Lock



- ▶ Turn the sliding bearings (1) until the marks (2) and (3) are aligned with each other.
- ▶ Press the pushbutton (4) in order to open the sliding bearing lock (5).

  Press the pushbutton and simultaneously push the lock towards the lifting gear arm. Open both sliding bearings.
- ▶ Press and hold the LOWER MATERIAL ROLL pushbutton in order to lower the lifting gear.
- ▶ Roll the material roll with the fitted winding shaft over the bearings opened upward.
- ▶ If necessary, press and hold the MATERIAL ROLL TOWARDS DRIVE SIDE or MATERIAL ROLL TOWARDS OPERATOR SIDE push-button in order to move the bearings into the pick-up position.
- ▶ Press and hold the RAISE MATERIAL ROLL pushbutton in order to swivel the lifting gear upwards.

  Swivel the lifting gear upwards until the winding shaft journals are located in the bearing positions.
- ▶ Push the sliding bearing locks towards the material roll in order to close them.

Splicing the web

Butt splicing

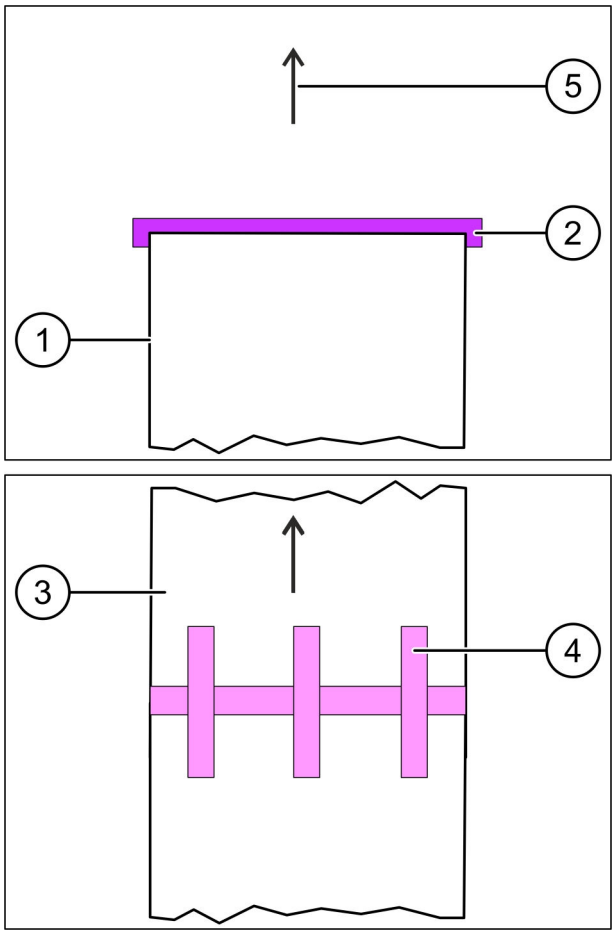
CAUTION

Risk of injury from blades, knives or sharp-edged machine parts

Wear cut-resistant gloves.

Splice printed webs so that the spacing between the print marks remains equal.

(1)	New web
(2)	Adhesive tapes
(3)	Old web
(4)	Adhesive tapes
(5)	Material flow direction



- ▶ Place the beginning of the new web (1) onto the roll with the adhesive tapes reversed (2).
- ▶ Place the end of the old web (3) end to end or slightly overlapping on the beginning of the new web and connect both webs with several adhesive tapes (4).
- ▶ Tighten the roll.

Setting up the web guide controller

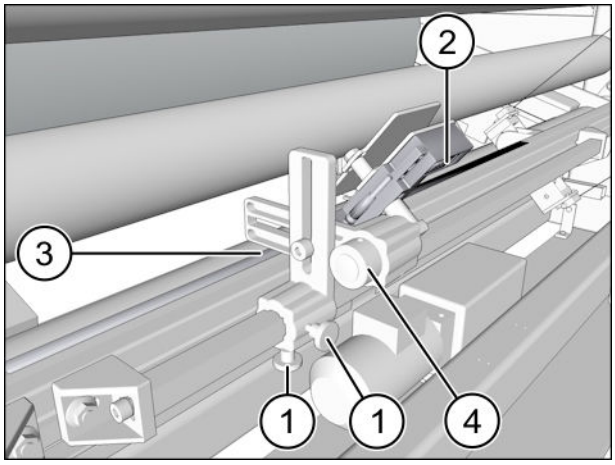
CAUTION

Risk of injury from rotating rollers

Parts of your body, clothes, hair and objects can be caught and crushed.

Do not reach into the nip. Keep a safe distance.

(1)	Screw
(2)	Sensor
(3)	Scale, memory scale: Half the web width
(4)	Handwheel



- ▶ Slacken the screws (1).
- ▶ Push the sensor (2) to the web edge.
- ▶ Set up the sensor to the web width using the scale (3).
- ▶ Tighten the screws (1).
- ▶ If necessary, turn the handwheel (4) for fine adjustment of the sensor position.

The end position is visually indicated at the web guide controller and during production additionally by a separate acoustic signal.

Set up the sensor to the web.

5.2 Web processing station

5.2.1 Dancer roller....195
Safe behavior....195
Adjusting the sensor....195

5.2.2 Micro-perforator....196
Safe behavior....196
Switching the tool on and off....196
Adjusting the tool....196
Dismantling and reinstalling the needle roller....196

5.2.3 Embossing....201
Safe behavior....201
Switching the tool on and off....201
Adjusting the tool....201
Dismantling and reinstalling of the embossing roller....201

5.2.1 Dancer roller

Safe behavior

CAUTION

Injuries can be caused by moving parts

Hands might be crushed or sheared off. Do not reach in between moving components. Keep a safe distance.

Risk of crushing during the swiveling of the dancer roller.

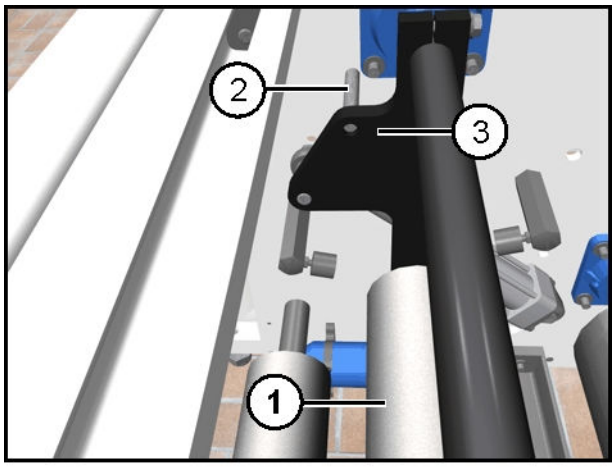
Take great care during the swiveling of the dancer roller, hands and fingers can be crushed or sheared off.

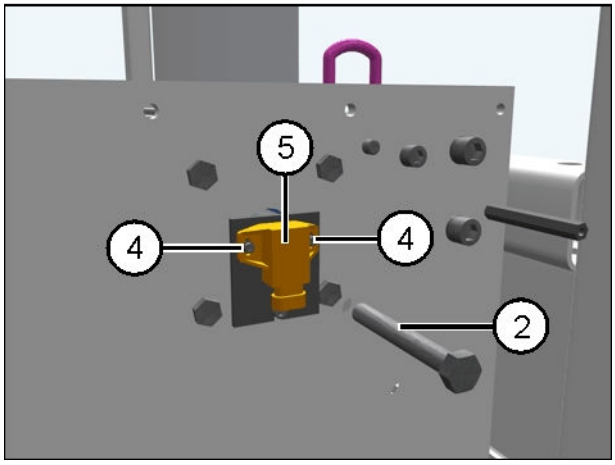
When the dancer roller is swiveling do not grasp the dancer roller lever and its stop.

Adjusting the sensor

Adjust the angle sensor on the dancer roller.

(1)	Dancer roller
(2)	Screw
(3)	Lever
(4)	Screw
(5)	Sensor





- Fix the dancer roller (1) in its center position.

  In the center position the dancer roller hangs down completely vertically.
- Push the screw (2) through the hole in the frame, screw and tighten into the lever (3).

  Depending on where the pneumatic cylinder was fixed onto the handle, push the screw through the upper or lower hole in the frame.
- Undo the screws (4) on the sensor (5).
- Turn the sensor until the value "19600" appears on the operating terminal.

  The sensor is adjusted to the center position of the dancer roller.
- Tighten the screws (4) in order to clamp the sensor in place.
- Unscrew the screw (2) from the lever and pull out of the frame.

5.2.2 Micro-perforator

Safe behavior

CAUTION

Rotating rollers

Parts of your body, clothes, hair and objects may be caught and crushed.

Wear tight fitting clothes and a hairnet. Work without industrial safety gloves. Do not reach into the nip. Keep a safe distance.

CAUTION

Risk of injury from cutting blades, knives or sharp-edged machine parts

Wear cut-resistant gloves.

CAUTION

Injuries can be caused by moving parts

Hands might be crushed or sheared off. Do not reach in between moving components. Keep a safe distance.

Risk of crushing during the advancing and retracting of the tools.

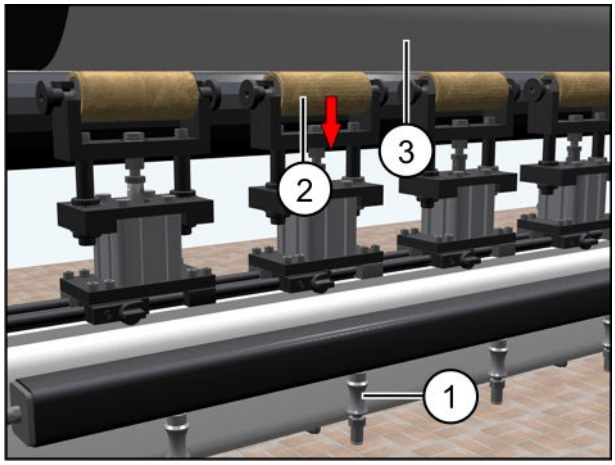
Take great care during the advancing and retracting of the tools, hands and fingers can be crushed or sheared off.

Do not reach for the tools or between the tools and the pneumatic cylinder when the tools are advancing or retracting.

Switching the tool on and off

Advance and retract an individual tool.

(1)	Pusher
(2)	Tool
(3)	Web



- ▶ More the pusher (1) to advance the tool (2) towards the web (3) or to retract it.

  Move the pusher away from the web:; the upper pusher upwards and the lower pusher downwards: Advance the tool towards the web, thus switching on the tool
  Move the pusher towards the web; the upper pusher downwards and the lower pusher upwards: Retract the tool from the web, thus switching off the tool

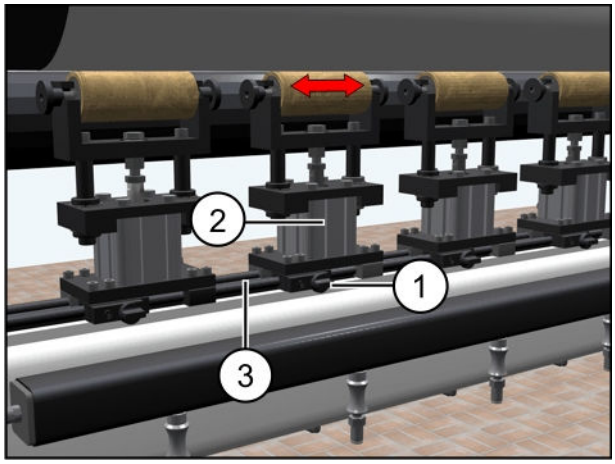
Adjusting the tool

Positioning the tool

Adjust the tool in the transverse direction.

Do not let the tools impact against each other when setting up.
If the tools impact against each other they do not work reliably.

(1)	Handle
(2)	Tool
(3)	Rail

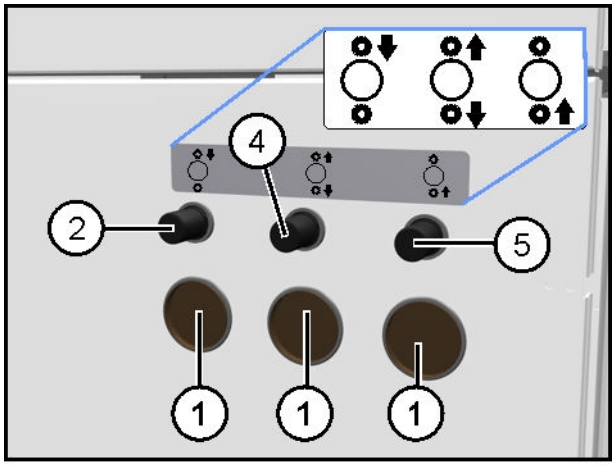


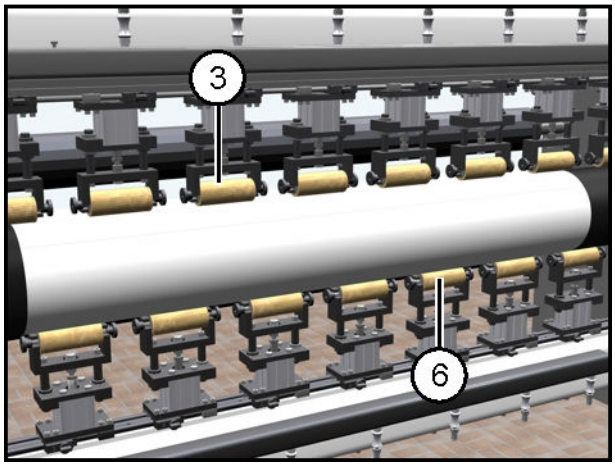
- Turn and loosen the handle (1) to loosen the tool (2).
- Slide the tool onto the rail (3) into the desired position.

  If the upper and lower tools are correctly adjusted, an overall needle pattern will appear on the material web.
- Turn and tighten the handle in order to fix the tool in position.

Setting up the contact pressure

(1)	Pressure gauge
(2)	Controller, setting up the contact pressure at the top
(3)	Upper tools
(4)	Controller, setting up the counterpressure
(5)	Controller, setting up the contact pressure at the bottom
(6)	Lower tools





- ▶ Read off the set up air pressure on the respective pressure gauge (1).
- ▶ Turn the controller (2) to set up the contact pressure for the upper tools (3).

  Air pressure:
- ▶ Turn the controller (4) to set up the air pressure for retracting the tools.

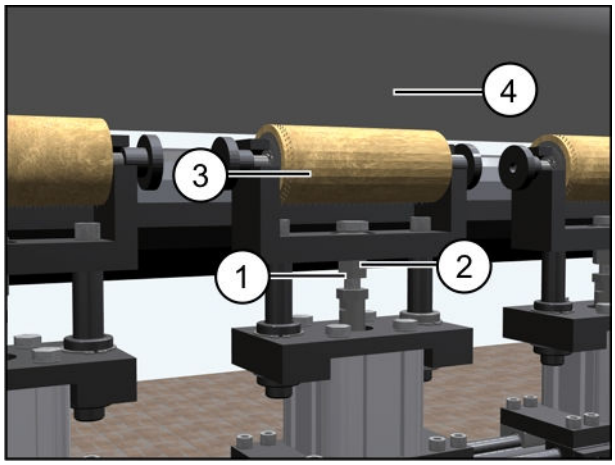
  Air pressure:
- ▶ Turn the controller (5) to set up the contact pressure for the lower tools (6).

  Air pressure:

Setting the insertion depth

Adjust the insertion depth of the tool until an even needle pattern appears.

(1)	Nut
(2)	Nut
(3)	Tool
(4)	Web



- ▶ Loosen the nut (1) and turn by a few revolutions away from the nut (2).
- ▶ Turn the nut (2) to raise and lower the tool (3).

  By raising and lowering the tool the diameter of the needle holes will be increased or decreased.
  Retracting the tool from the material web (4) reduces the diameter of the needle holes.
  Advancing the tool towards the material web increases the diameter of the needle holes.
  Turn the nut clockwise: The diameter of the needle holes increases.
  Turn the nut counter-clockwise: The diameter of the needle holes reduces.
- ▶ Turn the nut (2) until the desired needle hole diameter is set.
- ▶ Turn and tighten the nut (1) against the nut (2).

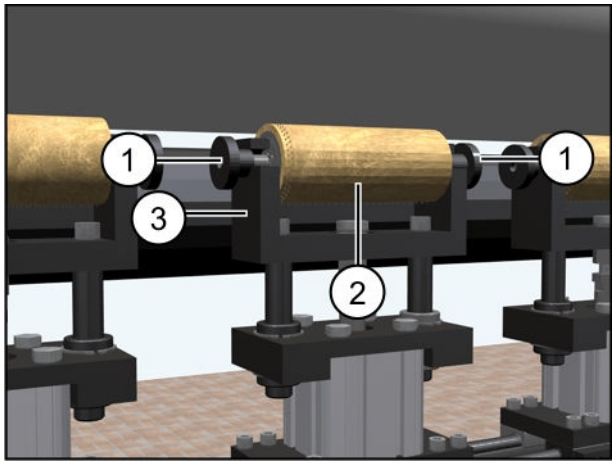
Adjust all other tools to have the same distance from the material web.

X000072555

Dismantling and reinstalling the needle roller

Before dismantling or reinstalling the needle roller the tool should be retracted from the web completely.

(1)	Nut
(2)	Needle roller
(3)	Holder



- Switch off the micro-perforator to retract the tool from the web completely.
- Slacken the nut (1).
- Pull the needle roller (2) out of the holder (3).
- Remove the nuts from the needle roller and screw them onto a new or cleaned needle roller.
- Insert and push the needle roller into the holder until it hits the stop.
- Tighten the nut (1).

5.2.3 Embossing

Safe behavior

CAUTION

Rotating rollers

Parts of your body, clothes, hair and objects may be caught and crushed.

Wear tight fitting clothes and a hairnet. Work without industrial safety gloves. Do not reach into the nip. Keep a safe distance.

CAUTION

Risk of injury from cutting blades, knives or sharp-edged machine parts

Wear cut-resistant gloves.

CAUTION

Injuries can be caused by moving parts

Hands might be crushed or sheared off. Do not reach in between moving components. Keep a safe distance.

Risk of crushing during the advancing and retracting of the tools.

Take great care during the advancing and retracting of the tools, hands and fingers can be crushed or sheared off.

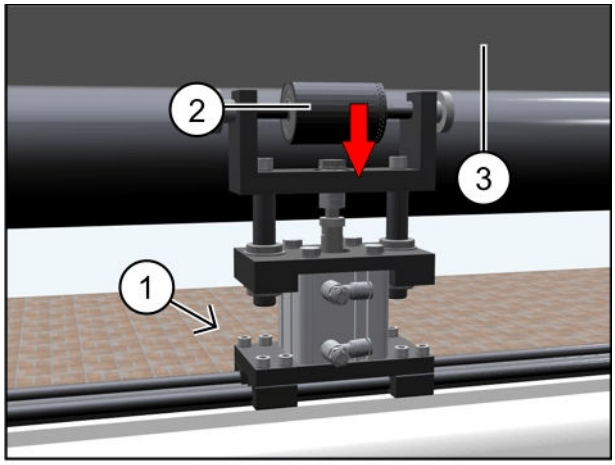
Do not reach for the tools or between the tools and the pneumatic cylinder when the tools are advancing or retracting.

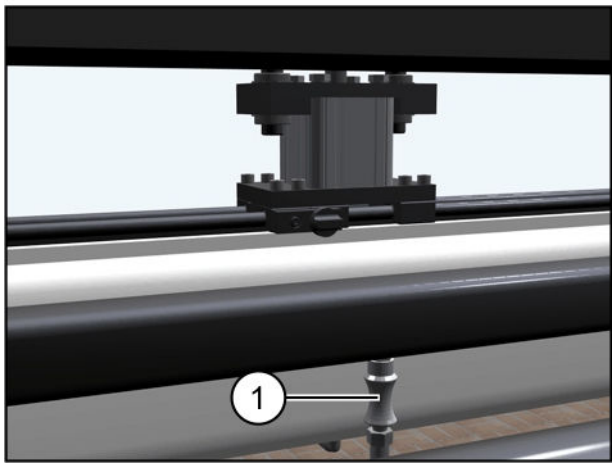
X000072557

Switching the tool on and off

Advance and retract an individual tool.

(1)	Pusher
(2)	Tool
(3)	Web





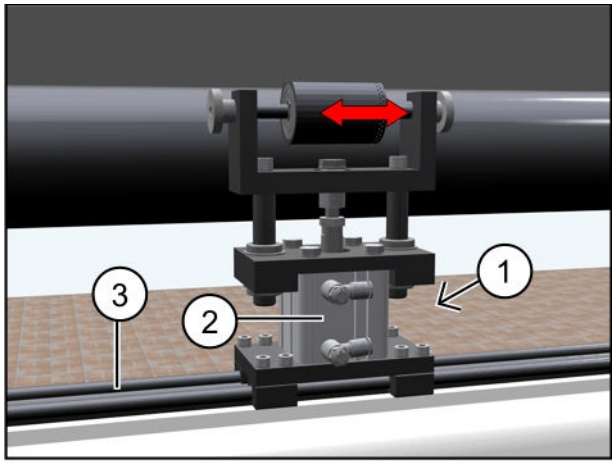
Adjusting the tool

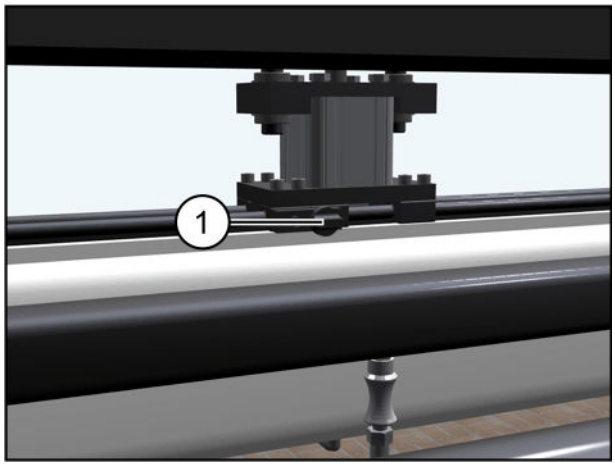
Positioning the tool

Adjust the tool in the transverse direction.

Do not let the tools impact against each other when setting up.
If the tools impact against each other they do not work reliably.

(1)	Handle
(2)	Tool
(3)	Rail

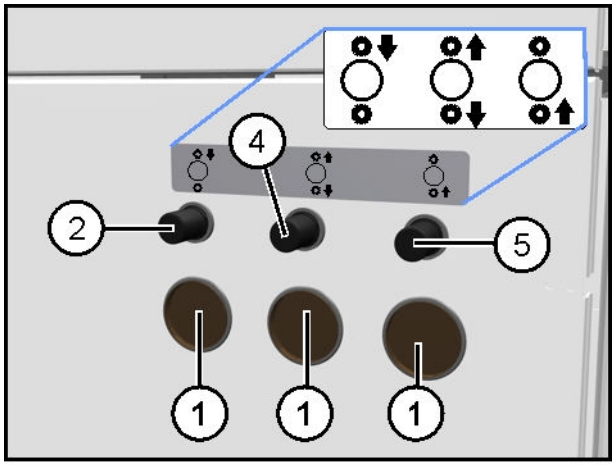


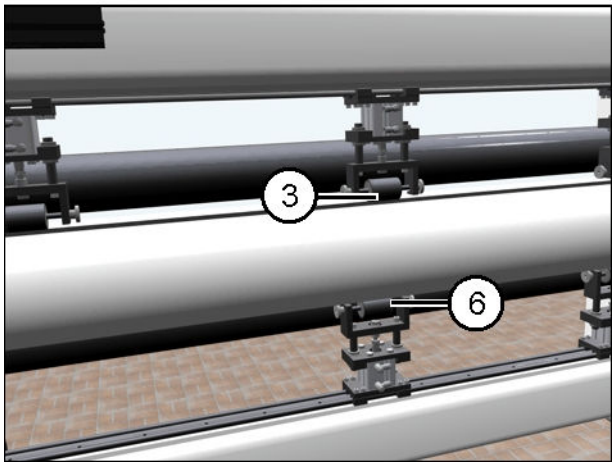


- ▶ Turn and loosen the handle (1) to loosen the tool (2).
- ▶ Slide the tool onto the rail (3) into the desired position.
- ▶ Turn and tighten the handle in order to fix the tool in position.

Setting up the contact pressure

(1)	Pressure gauge
(2)	Controller, setting up the contact pressure at the top
(3)	Upper tools
(4)	Controller, setting up the counterpressure
(5)	Controller, setting up the contact pressure at the bottom
(6)	Lower tools





- Read off the set up air pressure on the respective pressure gauge (1).
- Turn the controller (2) to set up the contact pressure for the upper tools (3).

  Air pressure:
- Turn the controller (4) to set up the air pressure for retracting the tools.

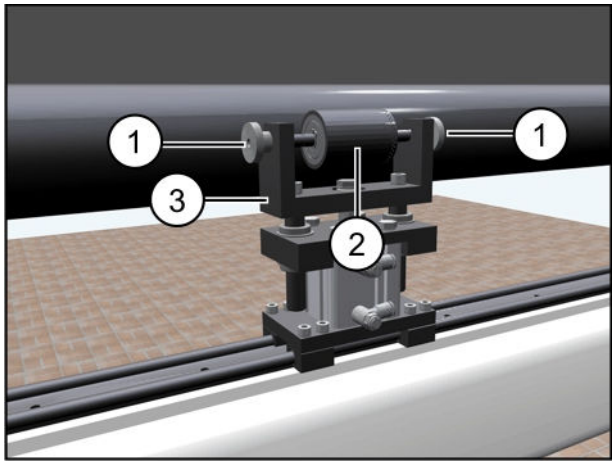
  Air pressure:
- Turn the controller (5) to set up the contact pressure for the lower tools (6).

  Air pressure:

Dismantling and reinstalling of the embossing roller

Before dismantling or reinstalling the embossing roller the tool should be retracted from the web completely.

(1)	Nut
(2)	Embossing roller
(3)	Holder



- Switch off embossing to retract the tool from the web completely.
- Slacken the nut (1).
- Pull the embossing roller (2) out of the holder (3).
- Turn the nuts out of the embossing roller and into a new embossing roller.
- Insert and push the embossing roller into the holder until it hits the stop.
- Tighten the nut (1).

5.3 Tube forming station

5.3.1 Web edge - Safe behavior....207
Free running webs....207
Danger zone....207

5.3.2 Tensioning the emergency stop trip wire....208
Unhooking and hooking the emergency trip wire....208
Adjusting the emergency trip wire....208

5.3.3 Guide elements at the infeed section....209
Setting up the guide elements....209
Aligning the guide element....209

5.3.4 Side gusset tube forming unit - Infeed rollers....210
General information....210
Removing the infeed roller....210
Installing the infeed roller....210
Preparing the scale....210
Adjusting the tube width....210

5.3.5 Adjusting the side gusseted tube forming....213
Functional elements....213
Adjusting the side gusset depth....213
Safe behavior....213
Adjusting the folding elements....213
Setting up the tube forming unit laterally....213
Raising the holder....213

5.3.6 Adjusting the side gusset depth....218
Graph caption....218
Scale at the adjusting device....218
Adjusting the side gusset depth....218

5.3.7 Side gusset tube forming unit - Former plates....221
Install the former plate holder....221
Changing the former plate....221

5.3.1 Web edge - Safe behavior

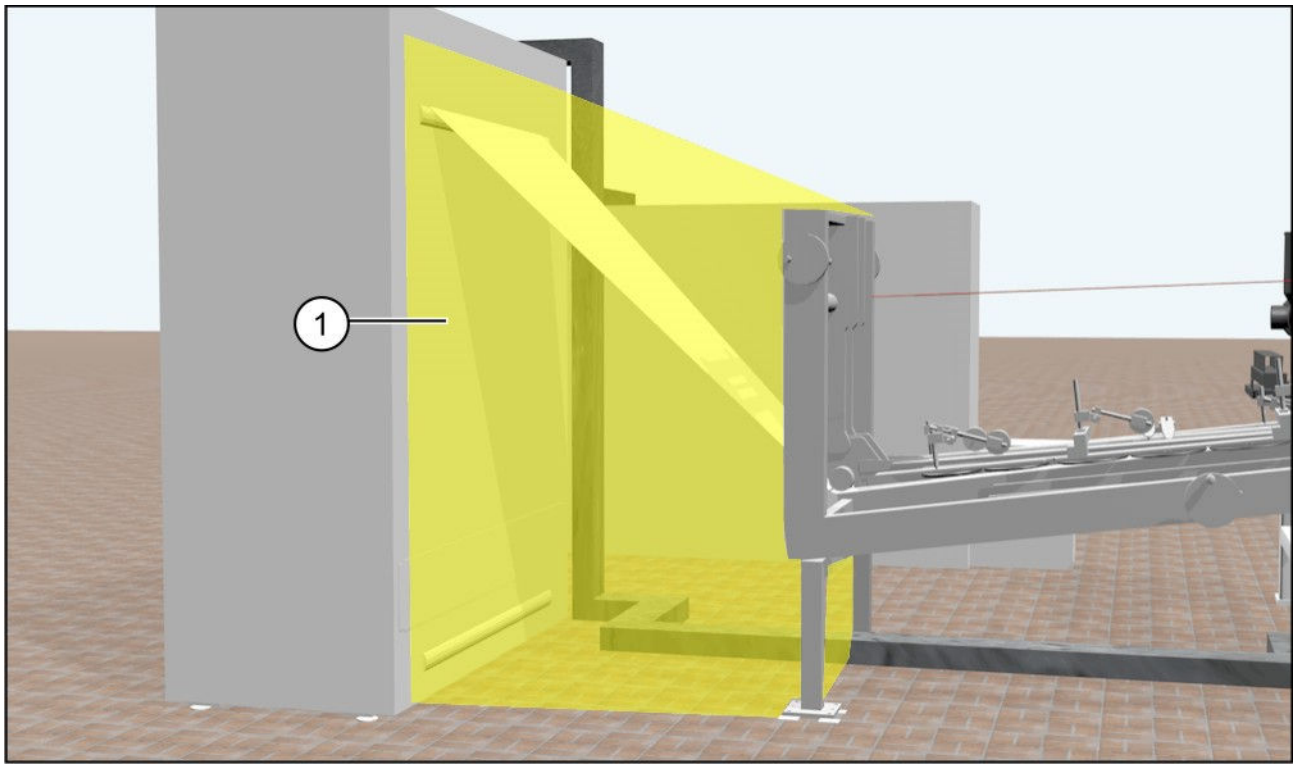
Free running webs

CAUTION

Sharp edges of free running webs

Risk of gashes in unprotected parts of your body from sharp edges of free running webs.
It is forbidden to enter the depicted danger zone when the machine is running!

Danger zone



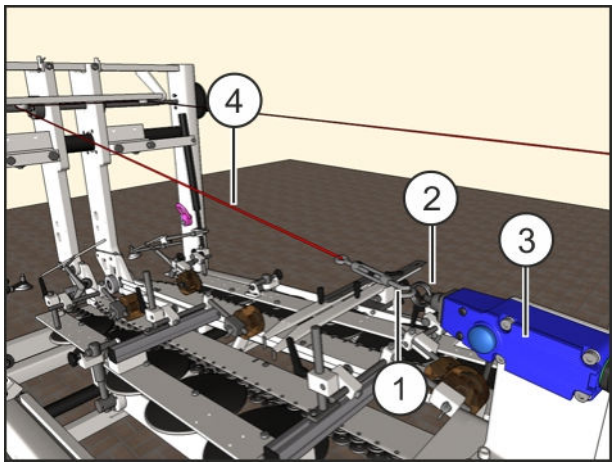
(1)	Zone in which the edges of a free running web may pose a health risk to the operator		

5.3.2 Tensioning the emergency stop trip wire

Unhooking and hooking the emergency trip wire

For dismantling the tube forming unit, unhook the emergency trip wire at the place provided for this purpose.

(1)	Hook
(2)	Ring
(3)	Switch
(4)	Emergency trip wire

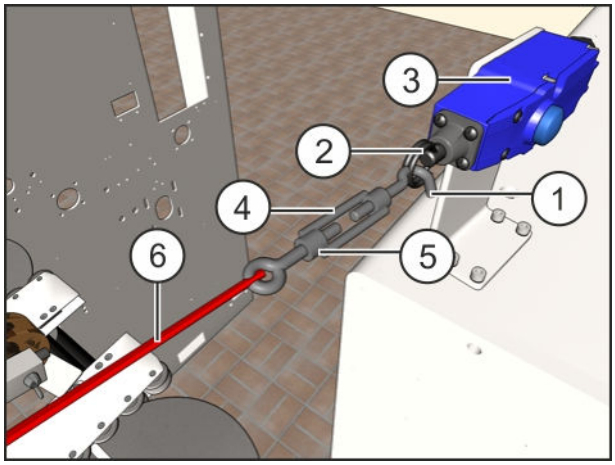


- ▶ Remove the hook (1) from the ring (2) on the switch (3).
- ▶ Deposit the emergency trip wire (4).
- ▶ Replace the tube forming station.
- ▶ Fit the hook into the ring on the switch.
- ▶ Check the emergency trip wire for proper functioning. Readjust the emergency trip wire whenever necessary.

Adjusting the emergency trip wire

Use a tension jack to tension or relieve the emergency trip wire until the Emergency-Stop function can reliably be triggered.

(1)	Hook
(2)	Ring
(3)	Switch
(4)	Center section
(5)	Tension jack
(6)	Emergency trip wire



- ▶ Fit the hook (1) into the ring (2) on the switch (3).
- ▶ Turn the center section (4) of the tension jack (5) to lengthen or shorten the emergency trip wire.
- ▶ Test the Emergency-Stop function

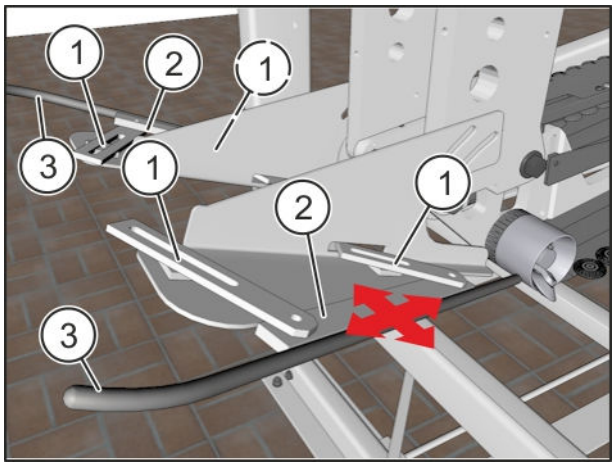
  Retension the emergency trip wire whenever necessary.

5.3.3 Guide elements at the infeed section

Setting up the guide elements

The guide elements at the infeed section of the tube former smooth the wrinkles out of the web and tauten the web in the transverse direction.

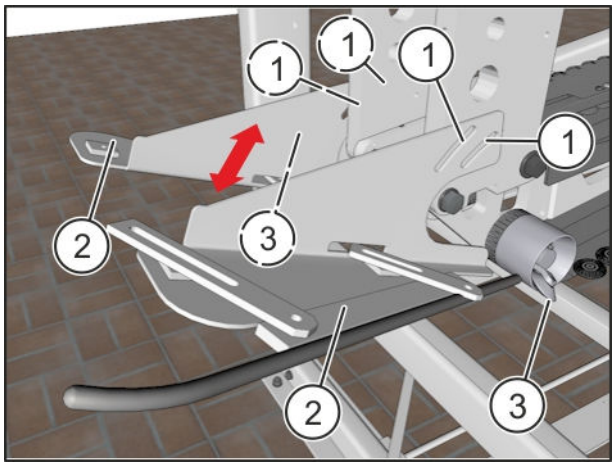
(1)	Screw
(2)	Guide element
(3)	Guide rod



- ▶ Slacken the screws (1).
- ▶ Shift the guide elements (2) until the guide rods (3) tauten the web.
- ▶ Tighten the screws.

Aligning the guide element

(1)	Screw
(2)	Guide element
(3)	Roller



- ▶ Slacken the screws (1).
- ▶ Shift the guide elements (2) until the web runs towards the rollers (3) .
- ▶ Tighten the screws.

5.3.4 Side gusset tube forming unit - Infeed rollers

General information

BEWARE

You can be injured when the web is running

Parts of your body, clothes, hair and objects may be pulled in and crushed. Always wear tight-fitting clothes and a hair net. Work without industrial safety gloves Keep a safe distance.

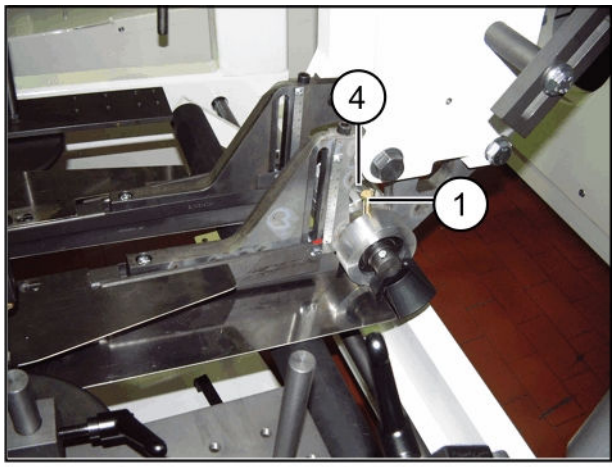
BEWARE

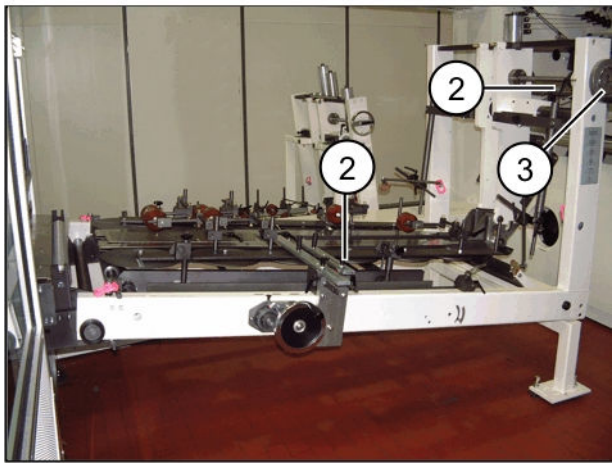
Injuries can be caused by moving machine parts

Hands may be crushed or sheared off. Never reach between moving system parts. Keep a safe distance.

Removing the infeed roller

(1)	Screw
(2)	Lever
(3)	Handwheel
(4)	Infeed roller





- ▶ Loosen the screws (1).
- ▶ Loosen the lever (2) and move holders apart using handwheel (3).
- ▶ Remove the infeed roller (4).

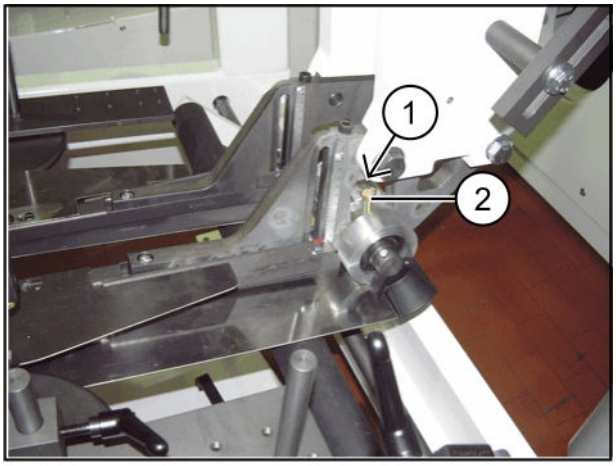
Installing the infeed roller

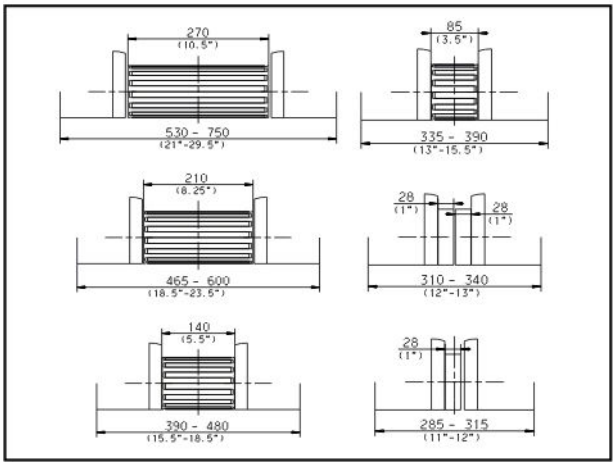
NOTE

The teeth of the geared infeed roller must remain intermeshed.

The distance between two smooth infeed rollers must not exceed a maximum of 20 mm (3/4 inch).

(1)	Infeed roller
(2)	Screw





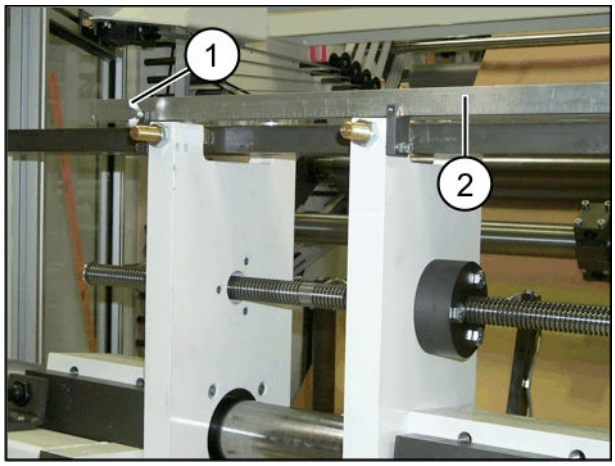
- Select infeed roller (1) depending on tube width.

  For extremely small tube widths only select a smooth infeed roller or do not use an infeed roller at all.

- Move holders together until infeed roller can be tightened with screws (2).

Preparing the scale

(1)	Wing screw
(2)	Scale

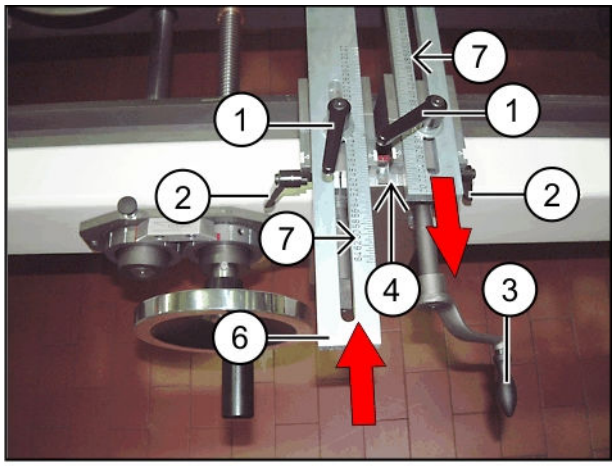


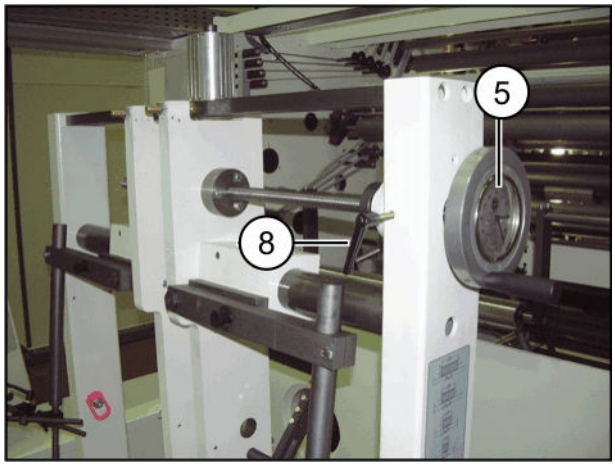
- ▶ Loosen the wing screw (1).
- ▶ Reposition of scale (2) in accordance with the tube width.

  Inner borehole = up to 56 centimeters (22 inches)
  Middle borehole = up to 50 centimeters (19 3/4 inches)
  Outer borehole = up to 44 centimeters (17 1/4 inches)
- ▶ Retighten the wing screw.

Adjusting the tube width

(1)	Lever
(2)	Lever
(3)	Crank handle
(4)	Scale
(5)	Handwheel
(6)	Former plate holder
(7)	Scale
(8)	Lever

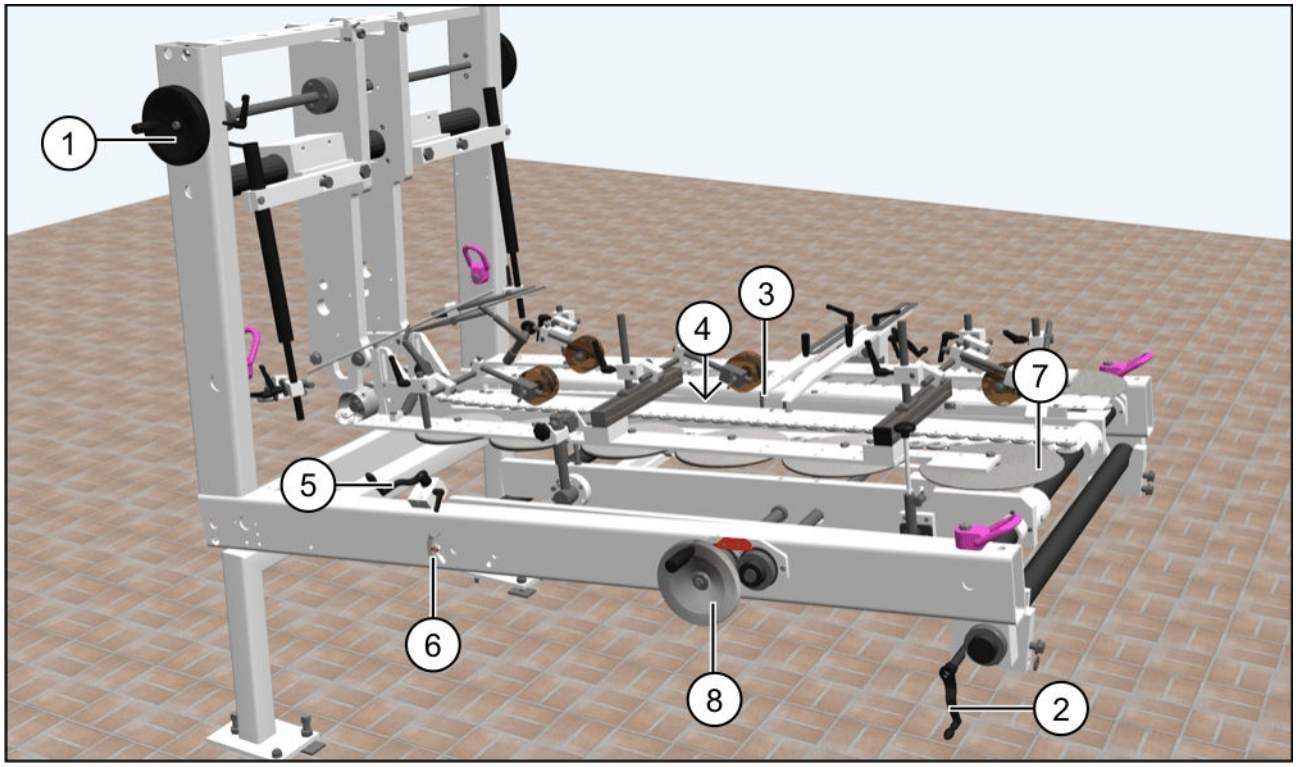




- ▶ Loosen the levers (1) and (2).
- ▶ Use the handwheel (3) to move the scale (4) to the zero position (red mark).
- ▶ Use handwheel (5) to adjust infeed roller to tube width according to the scale.
- ▶ Set the former plate holder (6) to the tube width according to the scale (7).
- ▶ Tighten the levers (1) and (8).
- ▶ Use the handwheel (3) to move the former plates outward into a slightly conical position (see arrows) one after the other.
- ▶ Tighten the levers (2).

5.3.5 Adjusting the side gusseted tube forming

Functional elements



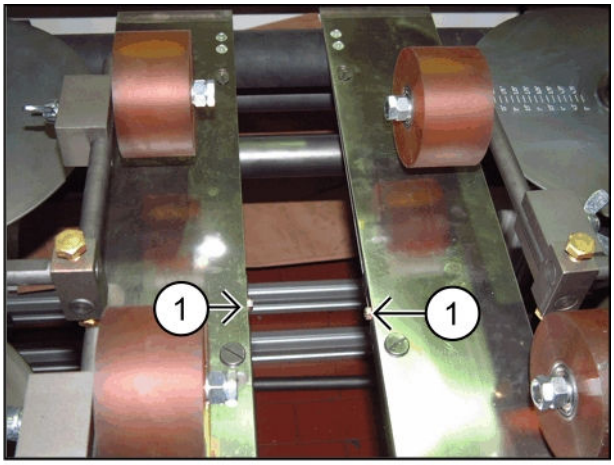
(1)	Handwheel with scale, infeed rollers	(2)	Hand crank, set the center of the tube
(3)	Top former plates	(4)	Bottom former plates
(5)	Hand crank, set the height	(6)	Scale, adjust the height
(7)	Folding discs	(8)	Hand crank, folding discs

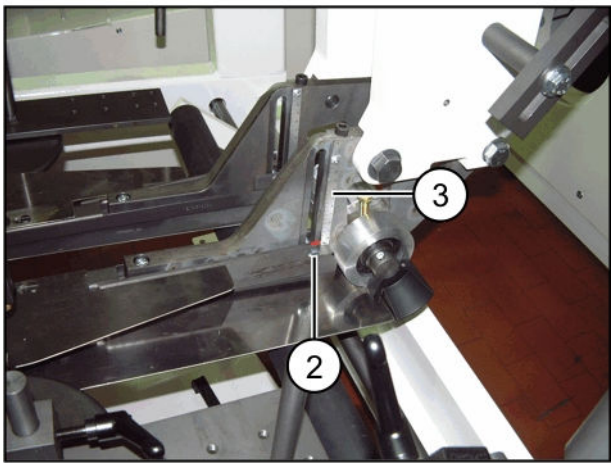
Adjusting the side gusset depth

Two successively arranged folding plates are pushed into the tubular material from outside to form the side gussets.

Adjusting the former plates

(1)	Screw
(2)	Clamping screw
(3)	Scale



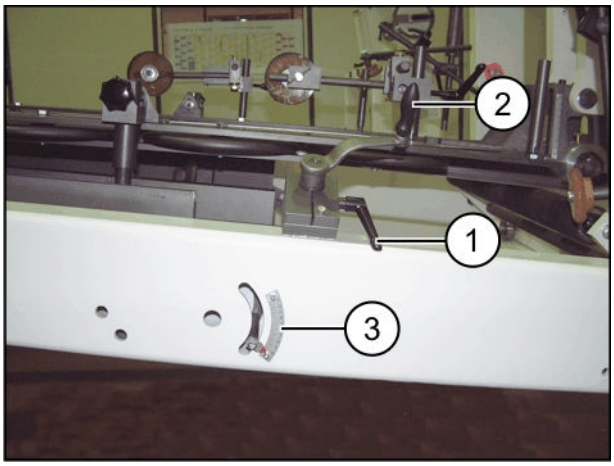


- ▶ Loosen the screws (1).
- ▶ Loosen the clamping screw (2) and adjust the former plates to the gusset depth according to the scale (3).

  Greater the slope = greater the side gusset depth
- ▶ Retighten all screws.

Folding discs - Adjusting the height

(1)	Locking lever
(2)	Crank handle
(3)	Scale

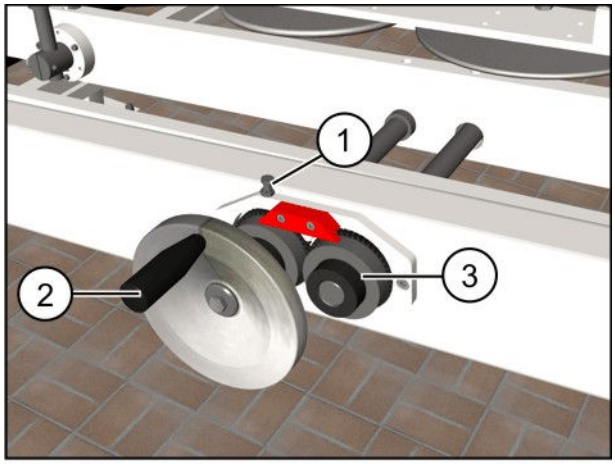


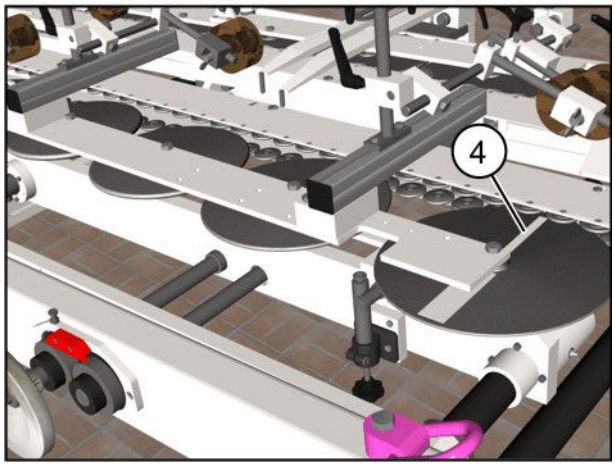
- ▶ Loosen the locking lever (1).
- ▶ Use the crank handle (2) to adjust the folding discs to the gusset depth according to the scale (3).

  The folding discs are located in the middle between the upper and lower former plates.
- ▶ Relock the locking lever.

Lateral adjustment of the folding plates

(1)	Latching bolt
(2)	Hand wheel
(3)	Gear wheel
(4)	Scale





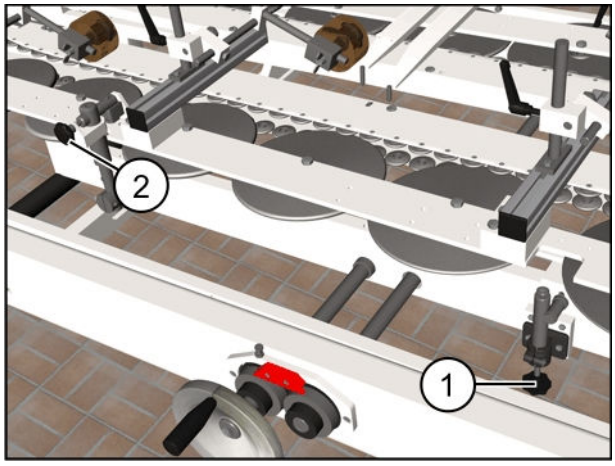
- Start the machine and let it run at base speed.
- Lift the latching bolt (1), turn by 90 degrees and set it down.

  The latching bolt is in its parked position.
- Turn the hand wheel (2) to jointly adjust the folding plates on the operator and drive side.
- Disengage the gear wheel (3).
- Turn the hand wheel (2) to adjust the folding plates on the operator side.
- Turn the gear wheel (3) to adjust the folding plates on the drive side.
- At the end of the tube forming unit, hold the folding plates with your hands for a short while in order to read the side gusset depth at the scale (4).
- Slide in the gear wheel (3) for engagement.
- Lift the latching bolt (1), turn by 90 degrees and set it down.

  The latching bolt is engaged. The gearbox is blocked. The side gusset depth cannot be adjusted.

Adapting the folding plates

(1)	Star handle
(2)	Star handle



- Use the star handle (1) to adapt the height of the folding plate to the production.
- Use the star handle (2) to adapt the lateral position of the folding plate to the production.

X000301783

Safe behavior

BEWARE

You can be injured when the web is running

Parts of your body, clothes, hair and objects may be pulled in and crushed. Always wear tight-fitting clothes and a hair net. Work without industrial safety gloves Keep a safe distance.

BEWARE

Injuries can be caused by moving machine parts

Hands may be crushed or sheared off. Never reach between moving system parts. Keep a safe distance.

BEWARE

Risk of injury from blades, circular knives or sharp-edged machine parts

Wear cut-proof protective gloves.

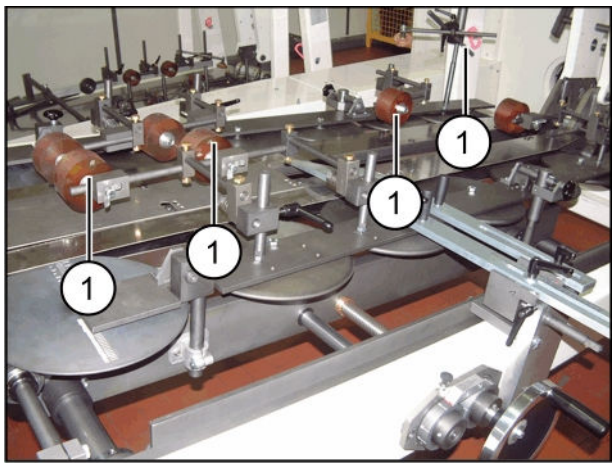
Adjusting the folding elements

The folding elements support the folding of the tube.

Adjust at base speed.

(1)	Folding element

▶ Adjust the folding elements (1) until the web passes the tube forming unit wrinkle-free.

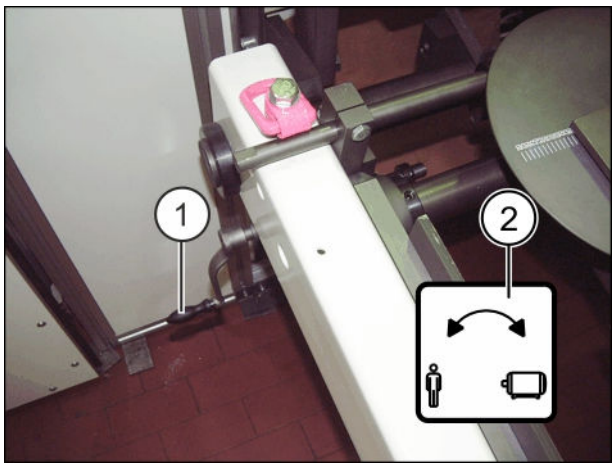


Setting up the tube forming unit laterally

Shift the tube forming unit in the transverse direction.

The setup can be carried out during production.

(1)	Crank handle
(2)	Label: Movement towards the operator or drive side



- ▶ Turn the crank handle (1).

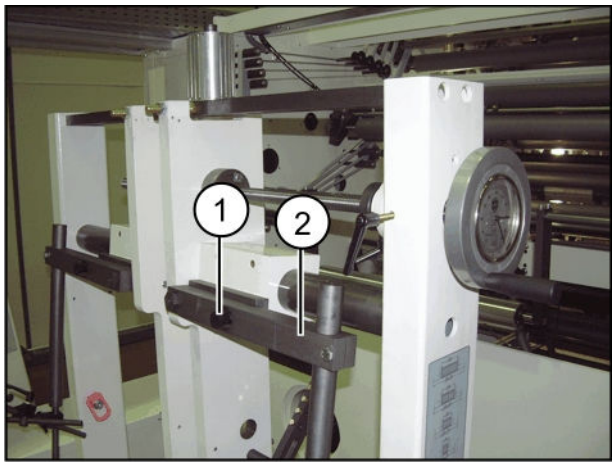
  Depending on the direction of rotation, the tube forming unit moves to the operator or drive side. See label (2)

- ▶ Shift the tube forming unit until the tube center is positioned in the machine center.

Raising the holder

When drawing the tube in or in the event of a jam, raise the holder with the press rolls and the folding elements.

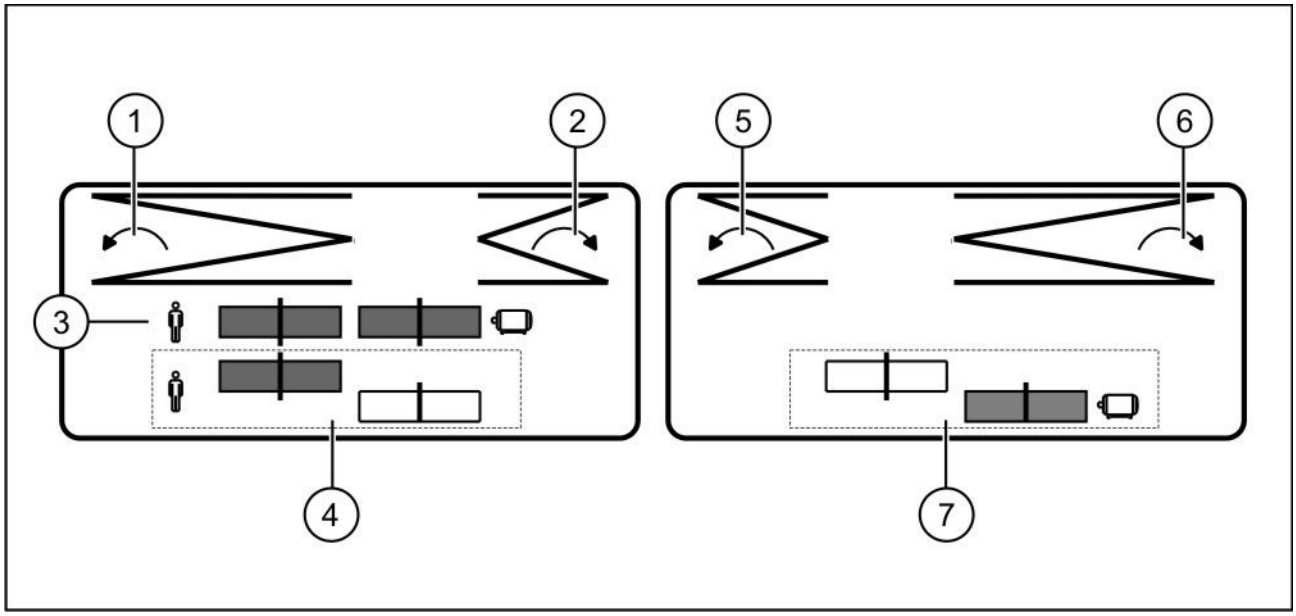
(1)	Latching bolt
(2)	Holder



- ▶ Pull out the latching bolt (1).
- ▶ Swing up the holder (2) by hand until the latching bolt engages above the holder.

5.3.6 Adjusting the side gusset depth

Graph caption



(1)	Direction of rotation, to the left, increases the side gusset depth	(2)	Direction of rotation, to the right, reduces the side gusset depth
(3)	Gear wheels engage, set up side gussets on either side	(4)	Right gear wheel disengaged, set up side gusset on operator side
(5)	Direction of rotation, to the left, reduces the side gusset depth	(6)	Direction of rotation, to the right, increases the side gusset depth
(7)	Disengage left gear wheel, set up side gusset on operator side		

Scale at the adjusting device

(1)	Pointer
(2)	Scale
(3)	Scale mark
(4)	+/- 0,5 millimeters [+/- 1⁄64 in.], gusset depth shifting per scale mark

Adjust the folding discs using the scales at the gear wheels.

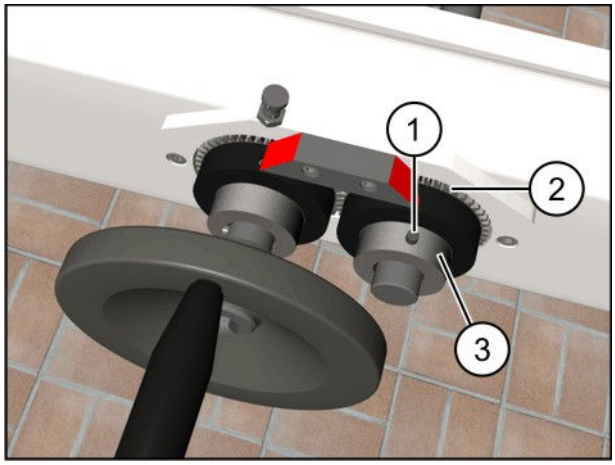
Turning the gear wheels and thus the scales by one scale mark shifts the folding disks by 0.5 millimeters [1⁄64 in.].

Depending on how the handwheel is turned and the folding disks are to be adjusted, the folding disks are shifted either to the drive side or the operator side.

Adjusting the side gusset depth

Disengaging the gear wheel

(1)	Screw
(2)	Gear wheel
(3)	Knurled surface

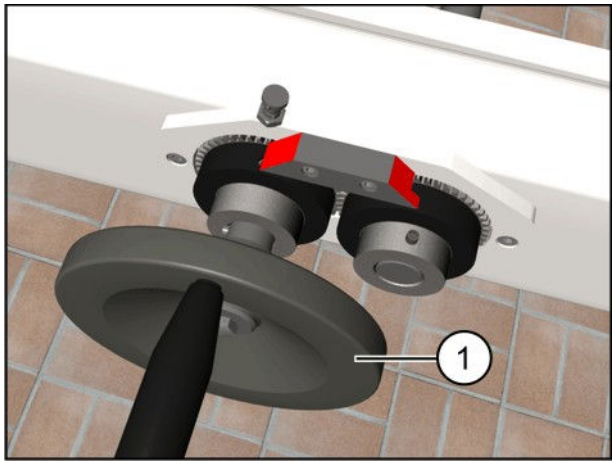


- ▶ Undo the screw (1) in the gear wheel (2).
- ▶ Disengage the gear wheel (2) at the knurled surface (3).
- ▶ Tighten the screw (1).
- ▶ Set up the side gusset depth on the operator and drive side.
- ▶ Slacken the screw (1).
- ▶ Engage the gear wheel (2).

Setting the side gusset depth on either side

Set the side gusset depth on either side simultaneously

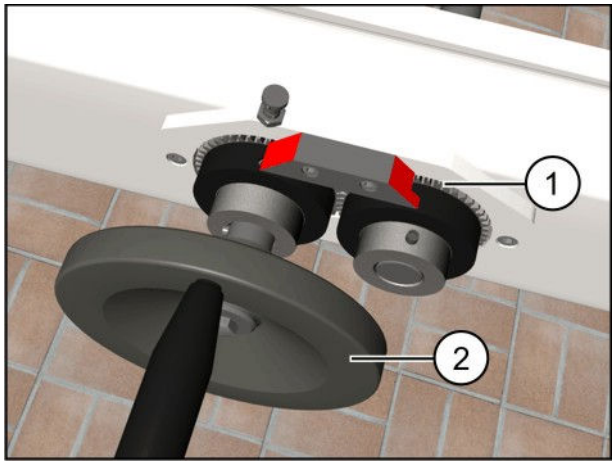
(1)	Hand wheel



- ▶ Turn the hand wheel (1).
  The folding plates on either side move out of their current positions simultaneously.
- ▶ Turn the hand wheel to the right.
  The side gusset depth is reduced.
- ▶ Turn the hand wheel to the left.
  The side gusset depth increases.

Setting the side gusset depth on the operator side

(1)	Gear wheel
(2)	Hand wheel



- ▶ Disengage the gear wheel (1).
- ▶ Turn the hand wheel (2).

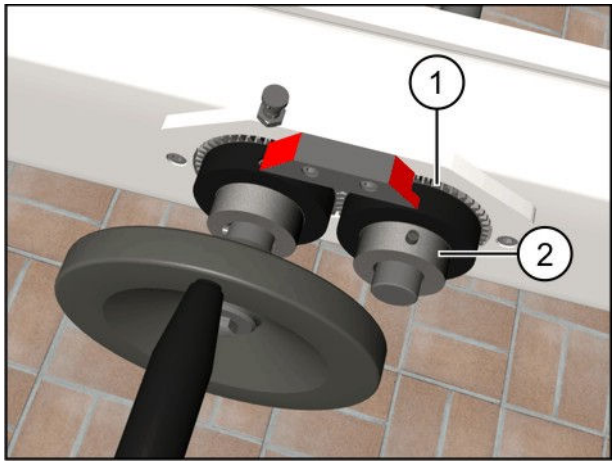
  The folding plates on the operator side move out of their current position.
- ▶ Turn the hand wheel to the right.

  The side gusset depth is reduced.
- ▶ Turn the hand wheel to the left.

  The side gusset depth increases.
- ▶ Engage the gear wheel.

Setting the side gusset depth on the drive side

(1)	Gear wheel
(2)	Knurled surface



- ▶ Disengage the gear wheel (1).
- ▶ Turn the gear wheel at its knurled surface (2).

  The folding plates on the drive side move out of their current position.
- ▶ Turn the gear wheel to the left.

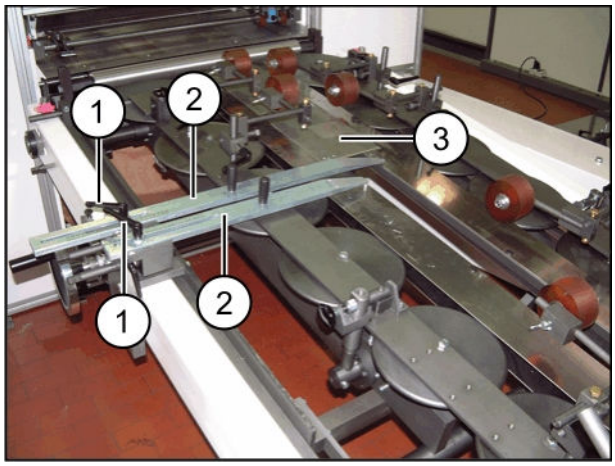
  The side gusset depth is reduced.
- ▶ Turn the gear wheel to the right.

  The side gusset depth increases.
- ▶ Engage the gear wheel.

5.3.7 Side gusset tube forming unit - Former plates

Install the former plate holder

(1)	Lever
(2)	Former plate holder
(3)	Plate

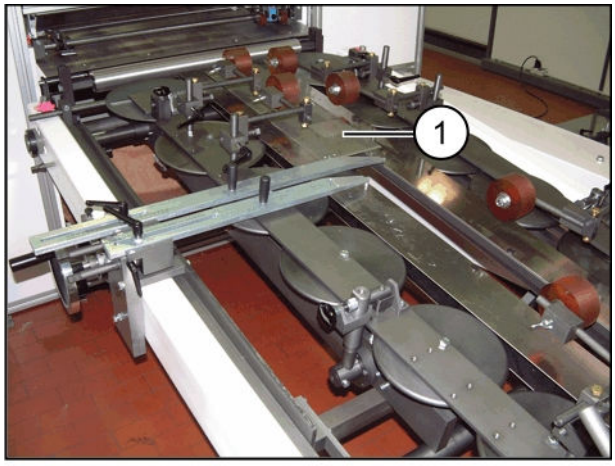


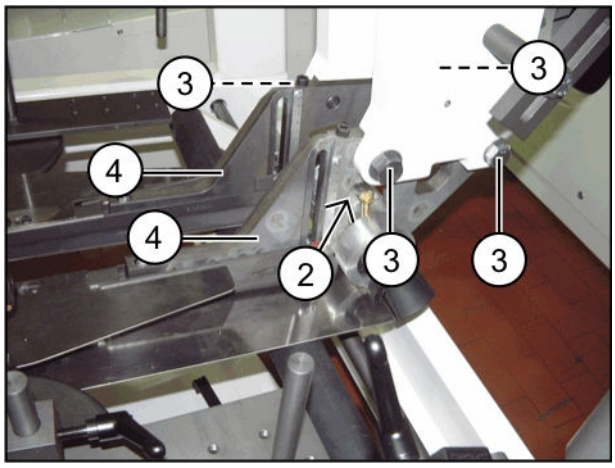
- Loosen the levers (1).
- Install the former plate holders (2) in accordance with the tube fold.
- Tighten the levers.
- If necessary, install the plate (3) for stabilizing the fold .

Changing the former plate

Change the former plate as required by the tube shape.

(1)	Plate
(2)	Infeed roller
(3)	Screw
(4)	Holder with former plate





- Remove the plate (1).
- Remove the infeed roller (2).
- Loosen the screws (3).
- Change the holder with former plate (4) to match the requirements of the tube width and gusset depth.
- Install the infeed roller.
- Retighten the screws.
- Install the plate if necessary.

5.4 Seam extruder

5.4.1 Longitudinal seam extruder 223

5.4.2 Transporting the frame 224

5.4.3 Switching on Automatic mode 225

5.4.4 Refilling resin – Safety instructions 225

5.4.5 Resin monitoring 226
Fitting the proximity sensor 226
Adjusting the level sensor 226
Checking for proper functioning 226

5.4.6 Thermocouples – Overview 229

5.4.7 Thermocouples 230

5.4.8 Heating elements 231
Heating elements - Assembly 231
Dismantling heating elements 231

5.4.9 Melt pressure sensor 232

5.4.10 Longitudinal seam monitoring – Drag rope 232

5.4.11 Longitudinal seam extruder - Collecting tank 233

5.4.1 Longitudinal seam extruder

Due to the characteristics of processing plastic resins, the barrel, screw, tools and the resin being processed can become hot during the process.

DANGER

Risk of injury due to hot surfaces, temperatures exceeding 60°C/140°F

Hands and other parts of your body can be burnt by excessive heat. Wear industrial safety gloves, safety shoes and long-sleeved clothing. Keep a safe distance.

DANGER

Risk of injury due to hot material.

The material emerging from the extruder's die has a temperature of more than 60°C/140°F.

Do not touch the extruded material with bare hands. Allow the extruded material to cool before touching it with bare hands.

Risk of injury due to extruded material spraying out of the die due to gas formation caused by the extruder overheating.

Monitor the temperature in the extruder. Never overheat the extruder.

Risk of burns

Fire hazard in the event of a material web standstill if the extruder die protrudes into the material web.

If the material web comes to a standstill immediately move the extruder out of the material web and into the basic position.

If there is an emergency stop, immediately move the extruder out of the material web and into the basic position.

WARNING

Risk of injuries due to poisonous fumes

Risk of injury due to toxic fumes caused by burning material.

If the material web comes to a standstill immediately move the extruder out of the material web and into the basic position.

If there is an emergency stop, immediately move the extruder out of the material web and into the basic position.

WARNING

Injuries can be caused by moving machine parts.

Your hands might be crushed or sheared. Do not reach between moving machine parts. Keep a safe distance.

Only move the extruder into the material web when nobody is between the extruder, the draw and the tube forming unit..

CAUTION

Risk of injury from electric shock if live components are touched

Disconnect the main power supply and secure it against unintentional reactivation.. Verify the safe isolation from supply. Ground and short-circuit the line part. Any adjacent parts that are still live must be covered. Only skilled electricians are permitted to work on the unit. Keep a safe distance.

CAUTION

Risk of injury from rotating gear wheels

Parts of your body, clothes, hair and objects can be pulled in and crushed. Do not open or remove protective devices. Wear tight-fitting clothes and a hair net. Work without industrial safety gloves. Keep a safe distance.

CAUTION

Risk ofjury from rotating chains and rotating gears and wheels

Parts of your body, clothes, hair and objects can be pulled in and crushed. Do not open or remove protective devices. Wear tight-fitting clothes and a hair net. Work without industrial safety gloves. Keep a safe distance.

CAUTION

When starting the extruder never stand directly in front of the barrel end or the tools..

When installing or removing tools and extruder components utilize lifting gear depending on the weight and ease of handling.

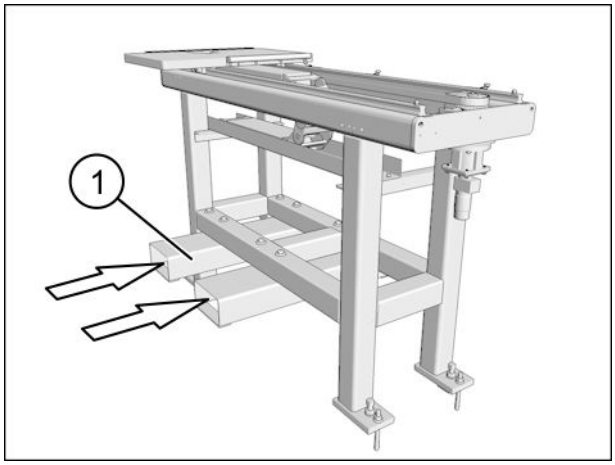
NOTE

Retighten the connecting components on tools and adapters after loosening and tightening them and after heating.

5.4.2 Transporting the frame

(1)	Transport device

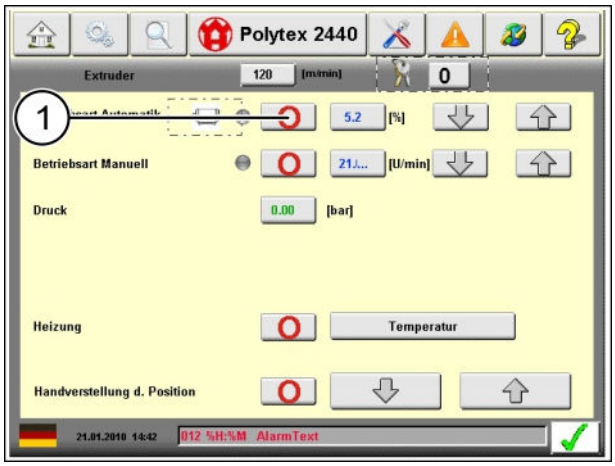
Transport the seam extruder using a fork-lift truck and the transport device (1).



5.4.3 Switching on Automatic mode

The description applies for machines provided with a seam extruder or a seam pasting unit.

(1)	'Extruder' screen page



- Depending on the machine equipment, switch on Automatic mode at the operator terminal, either on the 'Extruder' screen page (1) or the 'Seam' screen page.
- Hold down the MACHINE ON button until the extruder is in its working position and the machine starts automatically.

5.4.4 Refilling resin – Safety instructions

WARNING

Filling hopper at high height

A fall from heigh heights causes severe injuries to the entire body.

Fit railings or other fencings. Use a service platform. Wear a safety harness.

Use a ladder or a stepladder when refilling resin.

Do not bend over the ladder.

Never load the ladder beyond the specified limits.

Pay attention to the rating plates.

Always face the ladder when ascending and descending, holding onto the ladder with both hands.

Put the resin container down on the ladder platform without climbing on the ladder. Then, climb on the ladder to pour resin into the filling hopper.

Do not use machine parts as climbing aids.

Put on personal fall protection before starting any maintenance work at high heights.

Check and maintain ladders at regular intervals. The pertinent regulations regarding the intervals and the scope must be heeded.

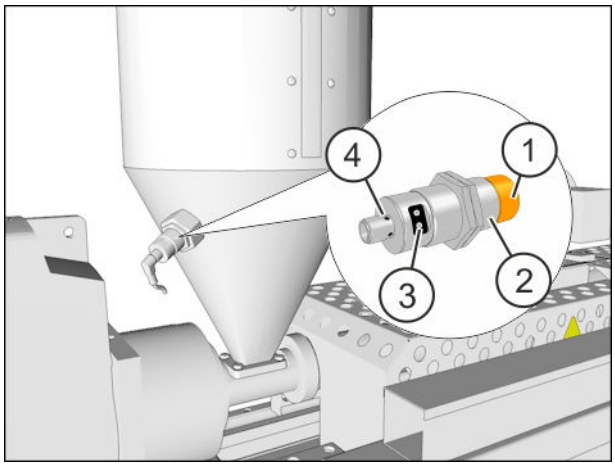
5.4.5 Resin monitoring

Fitting the proximity sensor

Screw the proximity sensor into the threaded bushing so that it is flush with the inner side of the filling hopper.

Screwing the proximity sensor in too far will lead to incorrect measurements.

(1)	Active area
(2)	Threaded bushing
(3)	Programming buttons
(4)	LED ring, yellow



▶ Screw the proximity sensor with the threaded bushing (2) into the filling hopper and secure with the check nut.

Adjusting the level sensor

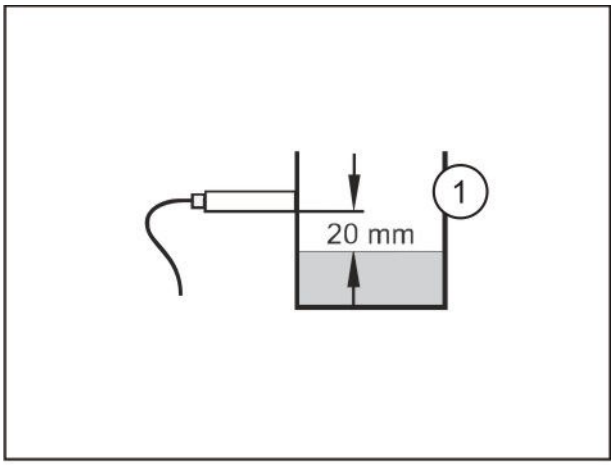
Buttons are used to adjust the responsivity. An empty and full adjustment must be carried out.

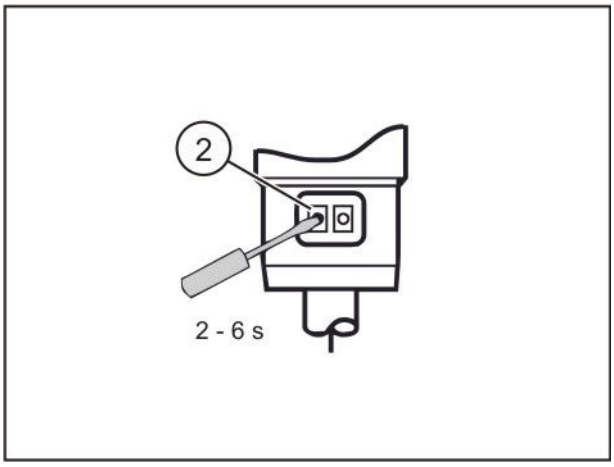
Empty adjustment

(1)	Container
(2)	OUT OFF

- Empty the container (1) until the level is at least 20 millimeter [25⁄32 Zoll] below the level sensor.
- Press the OUT OFF button for at least 2 seconds (max. 6 seconds).

  While holding down the button, the LED blinks slowly. After letting go of the button, the LED starts blinking again.

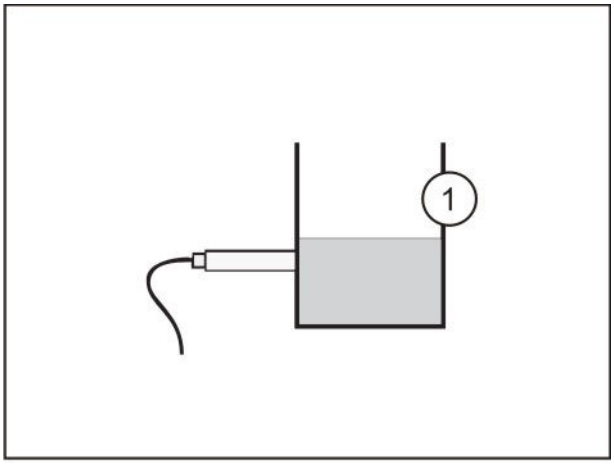




Full adjustment

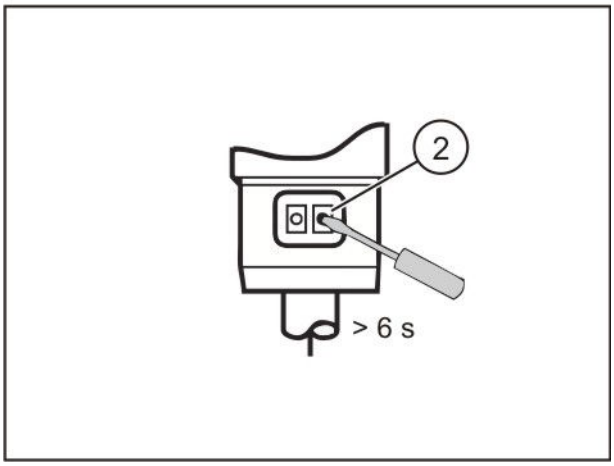
An empty adjustment must be carried out before doing the full adjustment! Use resin to carry out full adjustment!

(1)	Container
(2)	OUT ON



- Fill the container (1) until the active surface of the level sensor is covered.
- Press the OUT ON button for at least 6 seconds.

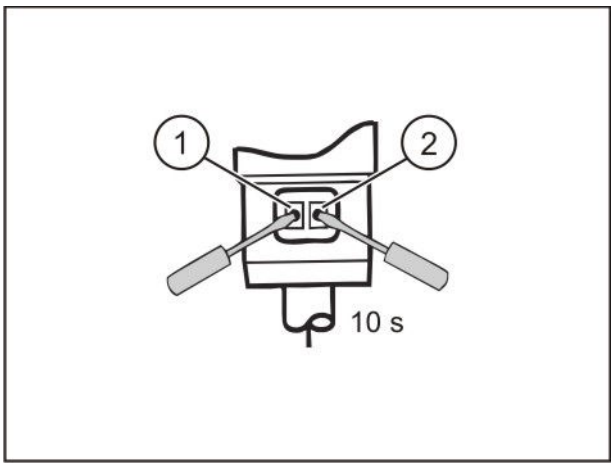
  While holding down the button, the LED starts to blink slowly and after 6 seconds it blinks faster. After letting go of the button, the LED lights up continously.



Locking / Unlocking

The level sensor can be secured against inadvertent readjusting.

(1)	OUT OFF
(2)	OUT ON

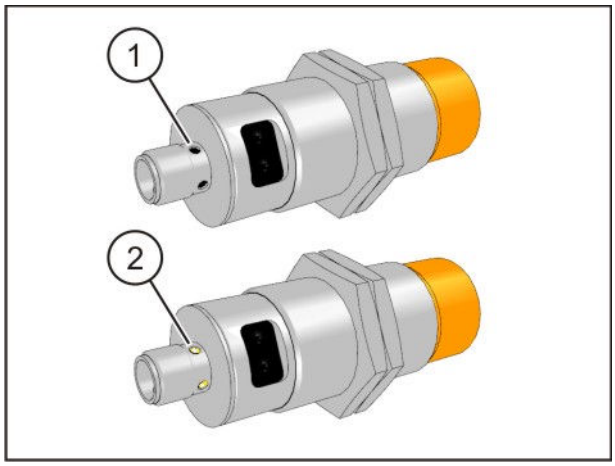


- For locking, press the OUT OFF (1) and OUT ON (2) buttons simultaneously for 10 seconds.

  Acknowledgement: The LED status (yellow) briefly changes (the lighting LED goes out briefly or the dark LED lights up briefly).
- Repeat this process for unlocking the level sensor.

Checking for proper functioning

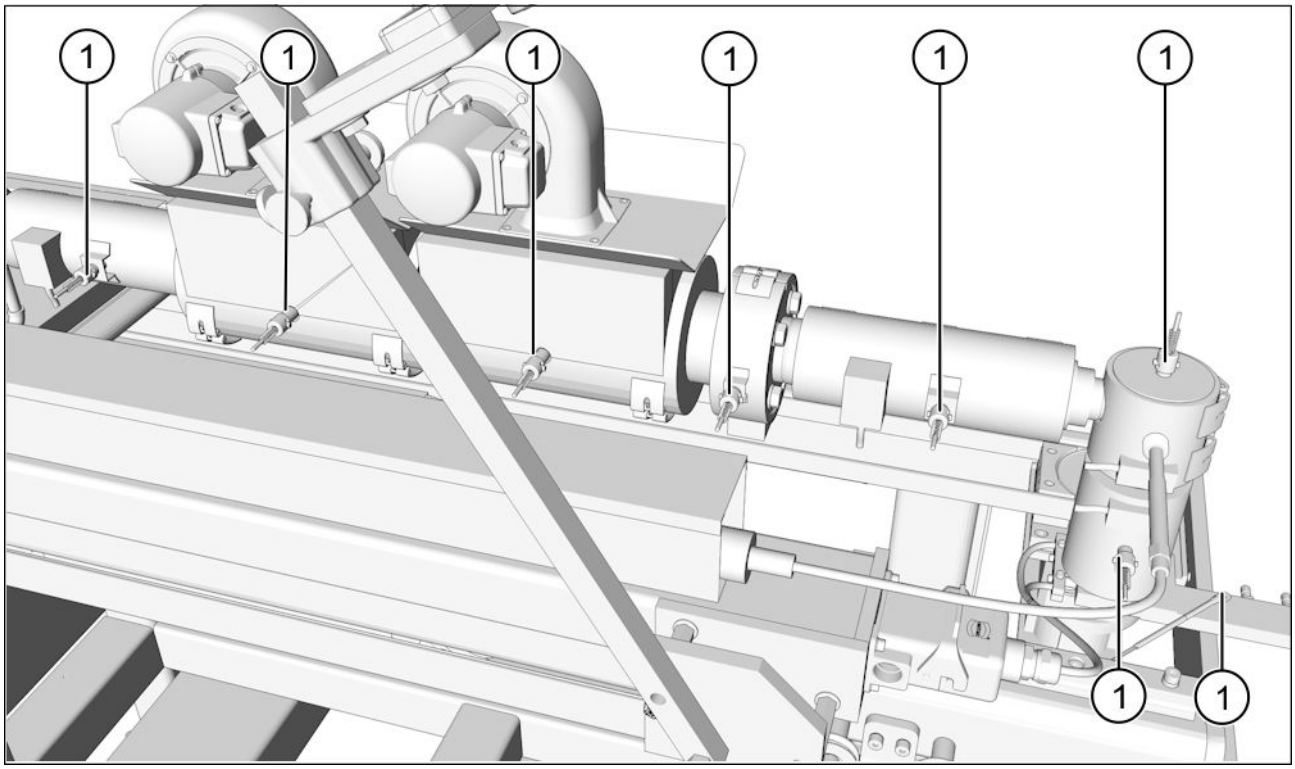
(1)	Yellow LED ring OFF
(2)	Yellow LED ring ON



▶ Take suitable measures to ensure the responsiveness of the level sensor.

LEDs OFF (1) = switching output blocked
LEDs ON (2) = switching output conductive

5.4.6 Thermocouples – Overview



(1)	Thermocouple		

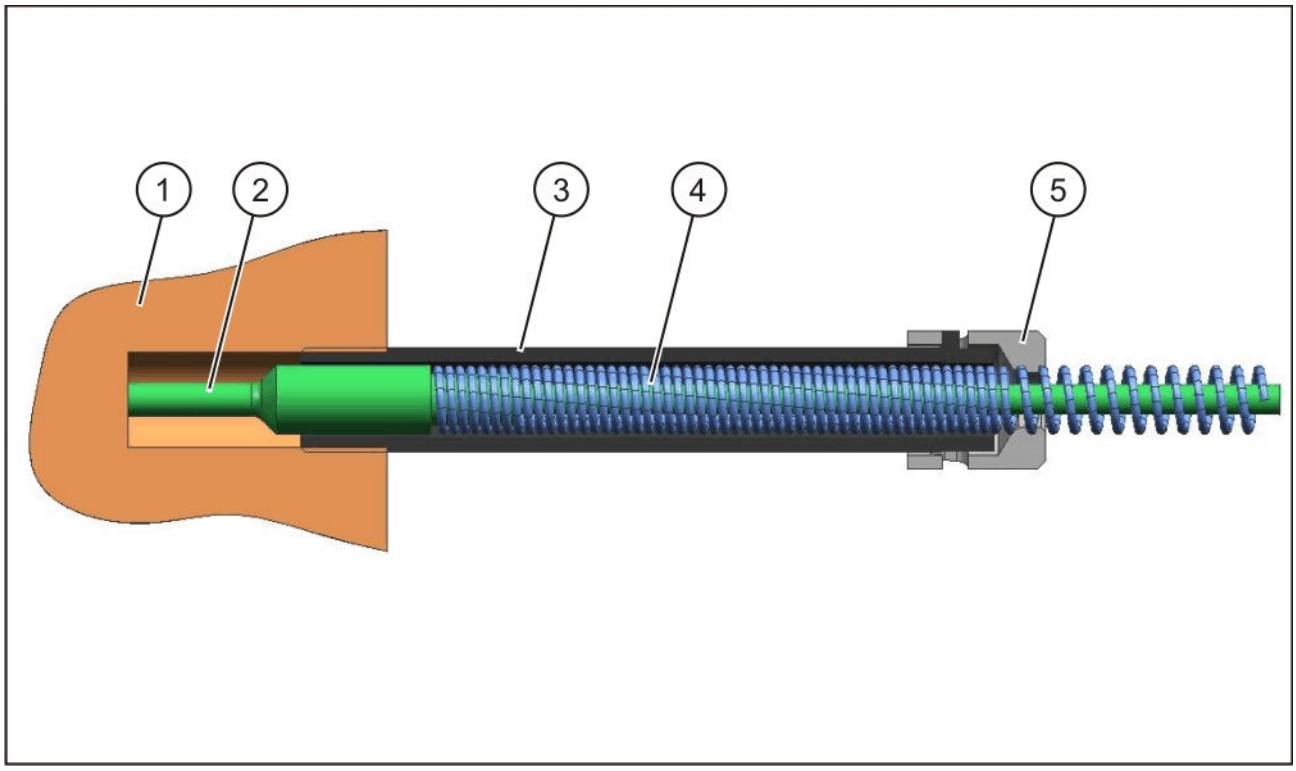
5.4.7 Thermocouples

NOTE

If a thermocouple is positioned incorrectly the set temperature cannot be recorded correctly. This can result in overheating.

The probe tip of the thermocouple must come into contact with the component. This is the only means of guaranteeing the correct temperature recording.

Carefully insert the thermocouple into the heat pipe until the probe tip comes into contact with the component. Make sure that the sleeve of the quarter-turn fastener is wound back by a few windings of the thread.. Tension the spring by closing the quarter-turn fastener.



(1)	Component	(2)	Thermocouple with precisely positioned probe tip in the component
(3)	Heat pipe	(4)	Spring
(5)	Quarter-turn fastener		

5.4.8 Heating elements

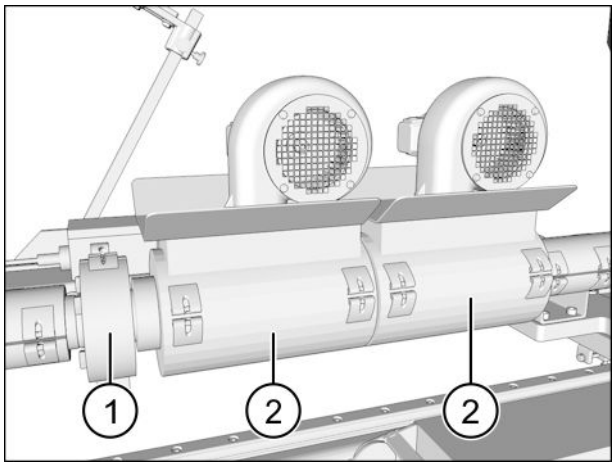
Heating elements - Assembly

Installation, operating behavior and dismantling of heating elements (ring heaters, heating-cooling sleeves, panel heaters)

NOTE

All heating elements will function properly only if they are firmly attached to the machine or to the tool for an optimal heat yield. With insufficient contact pressure, the emitted heat cannot be dissipated on the element to be heated. This results in a heat build-up inside the heating element which may destroy the heating band. The service life of a heating element is mainly determined by this factor.

(1)	Ring-type heater
(2)	Heating sleeve



Ring-type heaters and heating sleeves in standard design can be spread out up to the measure of the inner diameter so that they can be pushed over the machine cylinder from the side. Heaters with a diameter less than 80 millimeters [3 5⁄32 in.] should be provided with a hinge.

After mounting the heating elements, tighten the tensioning bolts in the clamping devices. Only then must the round-shaped heaters and the panel heaters be activated.

During the first startup of all heaaters until reaching the operating temperature, the tensioning bolts should each be checked for tight fit at several intervals. If necessary, retighten the tensioning bolts. If there is more than one tensioning bolt, retension them in an alternating manner until a tight fit is achieved.

Dismantling heating elements

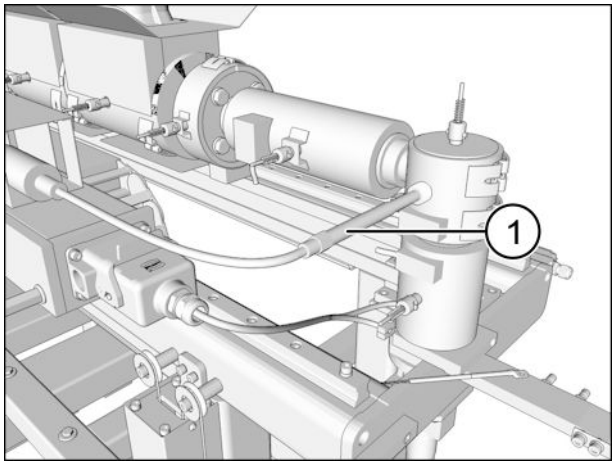
When dismantling heating elements intended to be reused make sure to prevent major deformation of the elements from occurring. Apply a thin coat of special temperature-resistant lubricant to the tensioning bolts to prevent corrosion from occurring.

The high temperature resistant anticorrosive prevents heat corrosion and baking on of tightly toleranced matching parts from occurring. Extremely high temperatures do not impair the lubricating properties.

5.4.9 Melt pressure sensor

(1)	Melt pressure sensor

The melt pressure sensor (1) measures the melt pressure in the extruder. The permitted melt pressure is set during commissioning and must not be changed. The maximum permitted melt pressure is 300 bar.



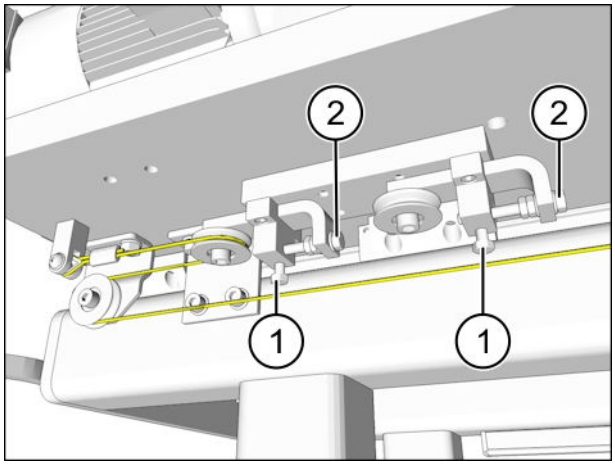
5.4.10 Longitudinal seam monitoring – Drag rope

Adjust the start position of the sensor for seam monitoring.

A device for adjusting the start position is provided for every course of the drag rope.

These devices render possible to adapt the start position of the sensors independently from each other, depending on the course of the drag rope.

(1)	Screw
(2)	Screw



- ▶ Slacken the screw (1).
- ▶ Turn the screw (2).

  The start position of the sensors changes.
- ▶ Turn the screw until the start position of the sensors coincides with the position of the nozzles in the applicator.
- ▶ Tighten the screw (1).

5.4.11 Longitudinal seam extruder - Collecting tank

When moving the extruder nozzle out of the longitudinal seam, drips of the left-over plastic melt leak from the nozzle.

Prevent the floor and machine parts from soiling.

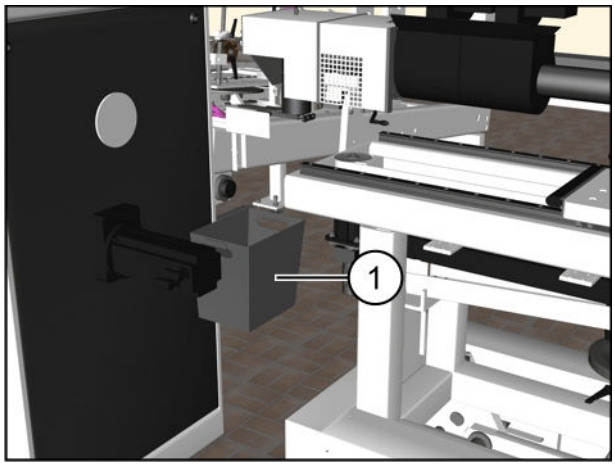
Only ever operate the extruder together with the collecting tray.

Clean the collecting tray at regular intervals.

Fill the collecting tray with water up to a half. In the water, the plastic melt quickly solidifies and does not stick on the walls of the collecting tray.

Line the collecting tray with pieces of the web. The plastic melt does not stick on the walls of the tray but on the pieces of the web so that it can easily be removed from the container after solidifying.

(1)	Collecting tray



- ▶ Fit the collecting tray (1) onto the mandrel at the side frame.
- ▶ Check and empty the collecting tray at regular intervals.

5.5 Main draw

5.5.1 Control voltage ... 235

5.5.2 Key switch ... 235

5.5.3 Machine ON ... 235
Requirements ... 235
Switching on the machine ... 235

5.5.4 Jog ... 236
Requirements ... 236
Jog ... 236

5.5.5 Switching on Automatic mode ... 225

5.5.6 Setting up the contact pressure ... 238
Squeeze roll ... 238
Seam cooling roller ... 238
Main draw ... 238

5.5.7 Main draw ... 239
Safe behavior ... 239
Functional elements ... 239

5.5.8 Main draw – Compressed air supply ... 240

5.5.9 Longitudinal seam monitoring ... 241

5.5.10 Seam cooling roller ... 241

5.5.11 Replacing the seam cooling roller ... 242
Removing the seam cooling roller ... 242
Mounting the seam cooling roller ... 242

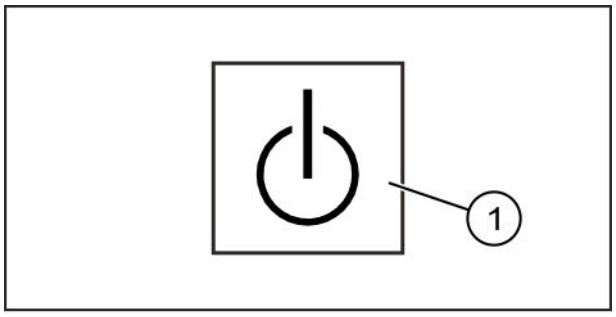
5.5.12 Supplying coolant ... 244

5.5.13 Nip roll ... 244

5.5.14 Cut edge blocking ... 245
Opening the cut edge blocking assembly ... 245
Changing the needle pattern ... 245
Adjusting the venthole diameter ... 245

5.5.1 Control voltage

(1)	CONTROL VOLTAGE ON/OFF



- ▶ Press the CONTROL VOLTAGE ON/OFF pushbutton to switch on the control voltage.

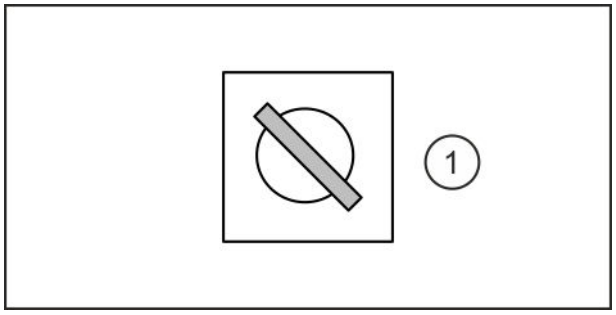
  The pushbutton flashes:
- ▶ The control voltage is switched off after an Emergency-Stop button has been hit. Press the pushbutton to switch the control voltage back on.
- ▶ Press the pushbutton to switch off the control voltage.

  The pushbutton is dark.
- ▶ Pressing the pushbutton during operation triggers the Emergency-Stop function.

5.5.2 Key switch

Lock and unlock the drive.

(1)	Key switch "Locked"



- ▶ Turn the key to the "CLEAR" or "0" position to unlock the drive.

  The key cannot be withdrawn.
  The drives can now be switched on.
- ▶ Turn the key to the "LOCKED" or "I" position to lock the drive.

  The key can be withdrawn.
  The drives cannot be switched on.
  Auxiliary devices can be switched on.

5.5.3 Machine ON

Requirements

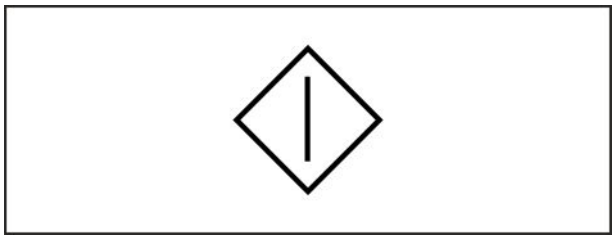
- Electrically monitored safety devices closed
- Control voltage is turned on
- Key switch is unlocked
- All errors and faults have been eliminated

Switching on the machine

The machine or the main drive is switched on.

For machines with more than one pushbutton applies: The pushbutton pressed first gets the switch-on authorization. The switch-on process can only be started via this pushbutton.

–	MACHINE ON, green



- ▶ Press and hold down the MACHINE ON pushbutton.

  A signal sounds for one second. The button lights up.

- ▶ Let go of the button.

  After two seconds, the machine is ready to be switched on. The button starts flashing.
  The ready to switch on status lasts 10 seconds and is indicated at the signaling device blinking green.

- ▶ Press the pushbutton again within 10 seconds.

  The ready to switch on status expires and the pushbutton goes dark unless the machine is restarted within 10 seconds.
  After dropping out, the ready to switch on status must be built up again.

  The button lights up.
  The machine starts at the selected speed and can be accelerated up to production speed.

5.5.4 Jog

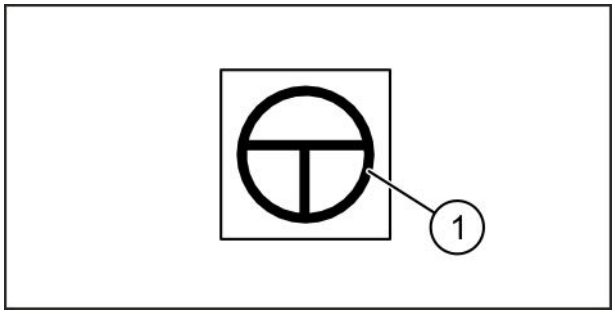
Requirements

- Electrically monitored safety devices are closed
- Control voltage is On
- All errors and faults have been eliminated

Jog

The 'Jog machine' or 'Jog module' sequence is described here.

(1)	JOG



- ▶ Press and hold down the JOG pushbutton.

  A signal sounds for one second.
- ▶ Let go of the pushbutton.

  The machine is ready to be switched on after two seconds. The pushbutton flashes:
- ▶ Press the button agaub within the next 10 seconds and hold it down.

  The relevant drive is turning and stops when letting go of the button.

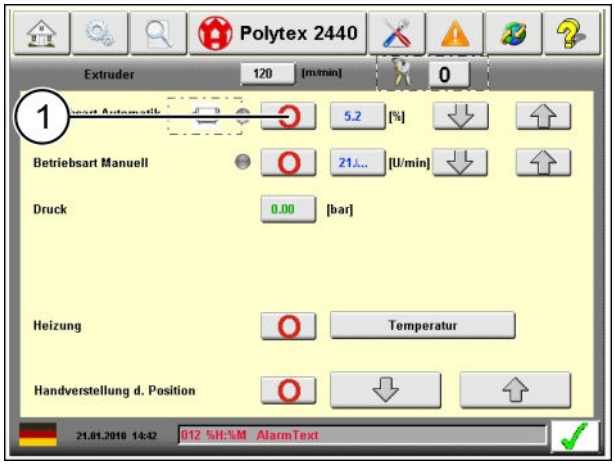
The jog state drops out and the button goes dark unless the machine is restarted within 10 seconds.

If the jog state expires, it must be restarted once again.

5.5.5 Switching on Automatic mode

The description applies for machines provided with a seam extruder or a seam pasting unit.

(1)	'Extruder' screen page

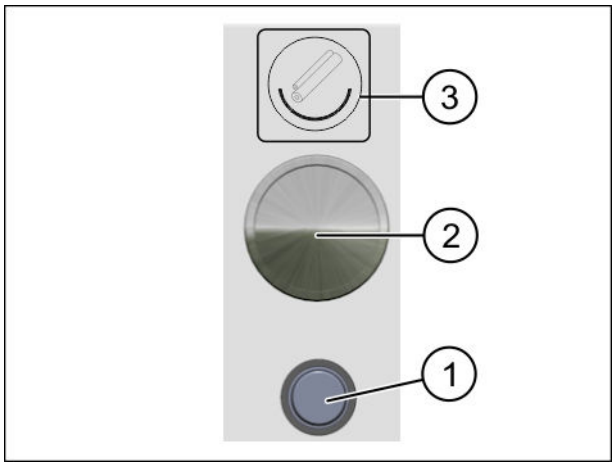


- ▶ Depending on the machine equipment, switch on Automatic mode at the operator terminal, either on the 'Extruder' screen page (1) or the 'Seam' screen page.
- ▶ Hold down the MACHINE ON button until the extruder is in its working position and the machine starts automatically.

5.5.6 Setting up the contact pressure

Squeeze roll

(1)	Knob for setting up the contact pressure
(2)	Pressure gauge
(3)	Symbol, squeeze roll



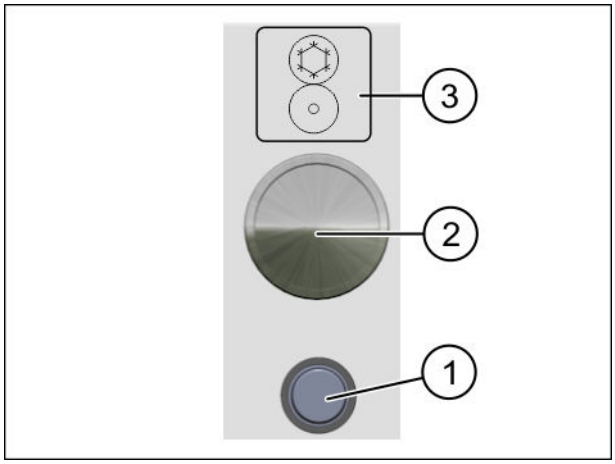
- Turn the knob (1) to set up the contact air pressure for the squeeze roll.

  Set up the air pressure so that as much air as possible is squeezed out of the tube.
  Set the air pressure to a value between 0 and 2 bar.

- Read the set air pressure at the pressure gauge (2).

Seam cooling roller

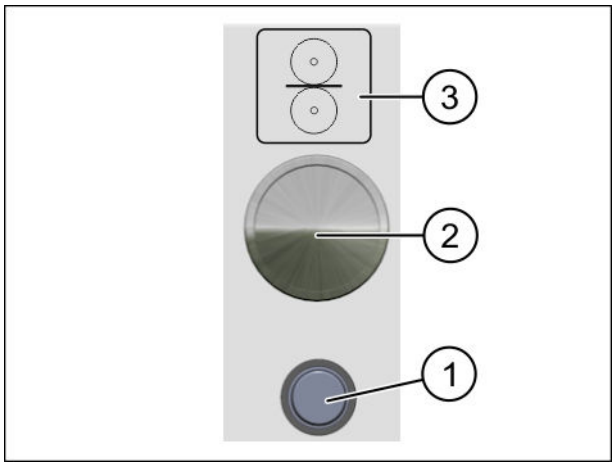
(1)	Knob
(2)	Pressure gauge
(3)	Symbol Seam cooling roller



- Turn the knob (1) to set up the contact air pressure for the seam cooling roller.
- Read the set air pressure at the pressure gauge (2).

Main draw

(1)	Knob
(2)	Pressure gauge
(3)	Draw symbol



- Turn the knob (1) to set up the contact air pressure for the cooled draw roller.
- Read the set air pressure at the pressure gauge (2).

5.5.7 Main draw

Safe behavior

CAUTION

Injuries can be caused by moving parts

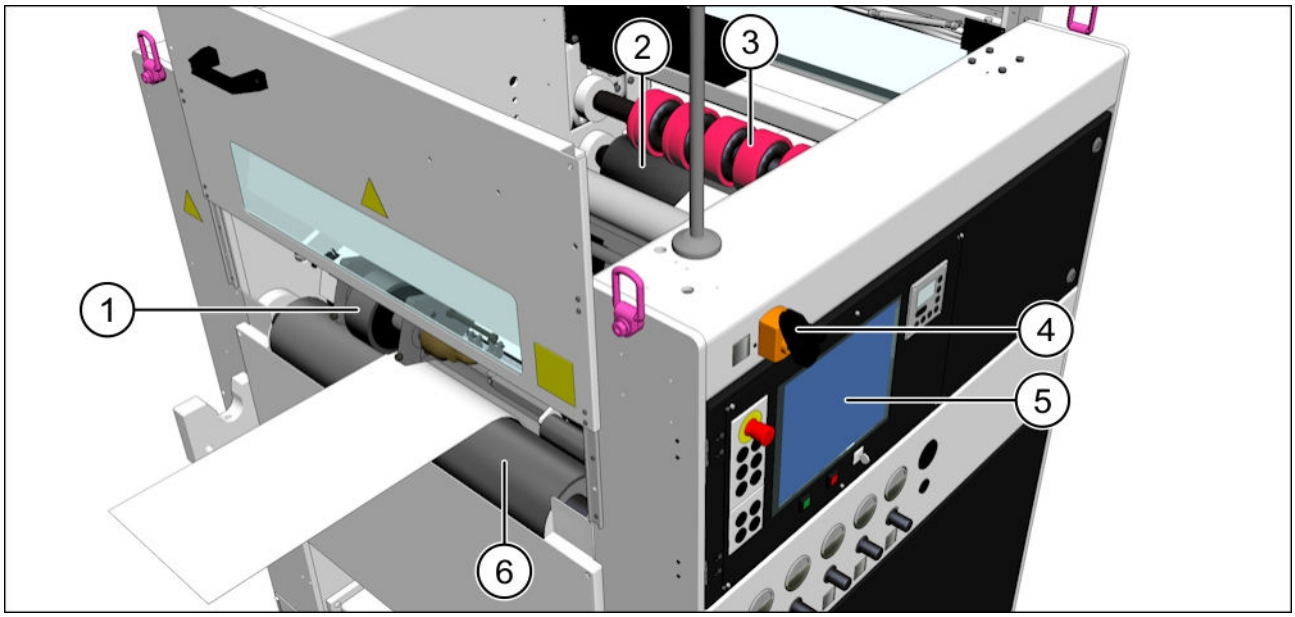
Hands might be crushed or sheared off. Do not reach in between moving components. Keep a safe distance.

CAUTION

Risk of injury from rotating rollers

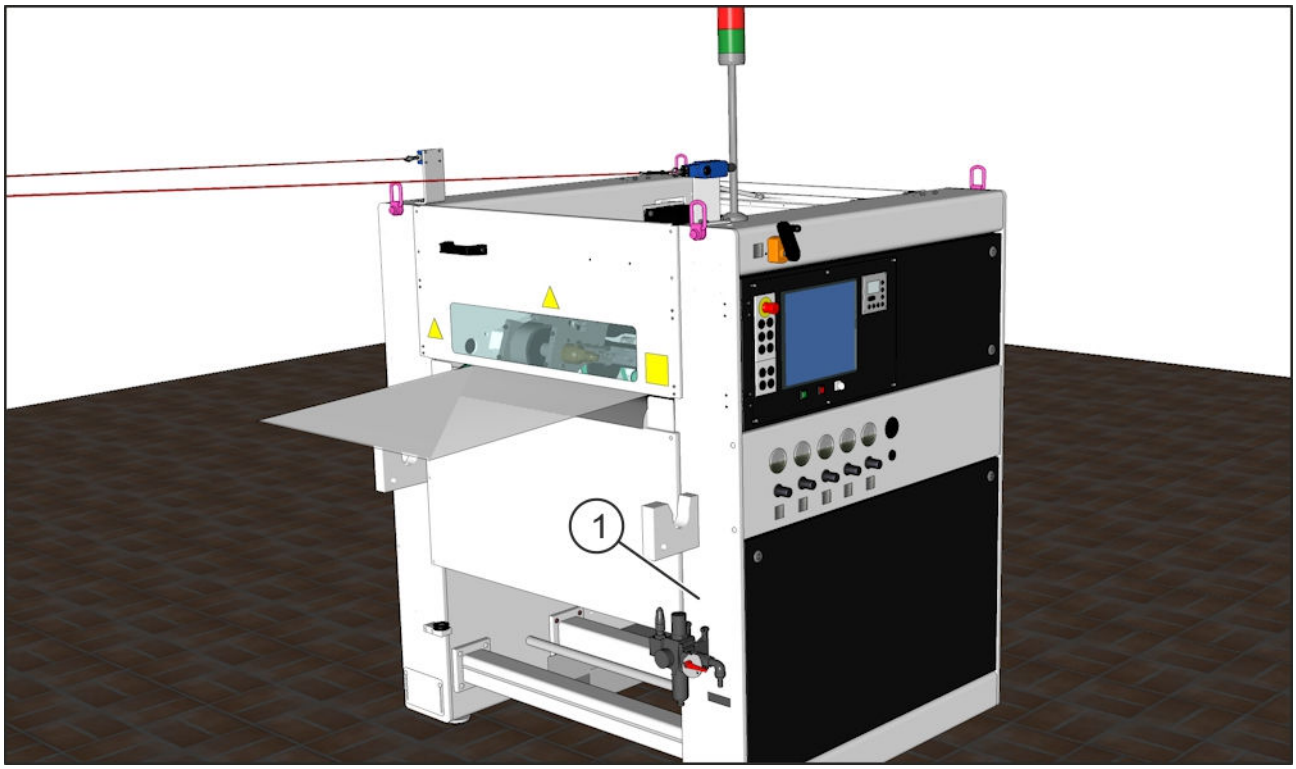
Parts of your body, clothes, hair and objects can be caught and crushed. Do not reach into the nip. Keep a safe distance.

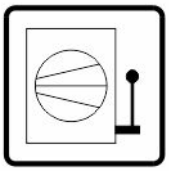
Functional elements



(1)	Seam cooling roller	(2)	Cooled lower draw roller
(3)	Split draw roller	(4)	Control element
(5)	Operator terminal	(6)	Cooled lower roller

5.5.8 Main draw – Compressed air supply



Item	Danger sign	Location	NOTE
(1)		Close to the compressed-air conditioner	Use the switch at the compressed-air conditioner to switch compressed air supply on and off.

5.5.9 Longitudinal seam monitoring

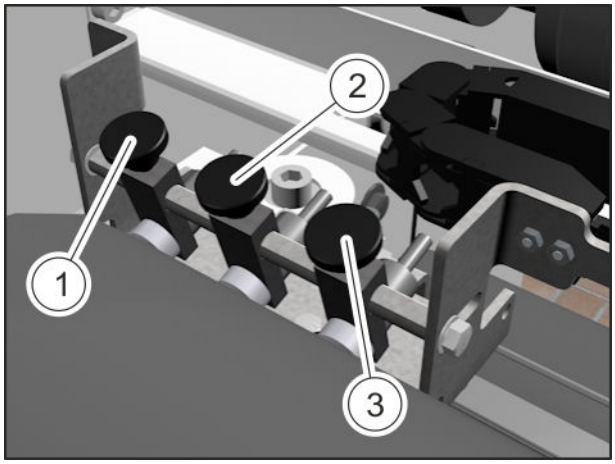
The mechanical controller for the longitudinal seam monitoring movement is W&H-set.

Once aligned with the glue line, there is no need to align the sensor position with the glue line once more.

The longitudinal seam monitoring device follows the movements of the nozzle head on the extruder.

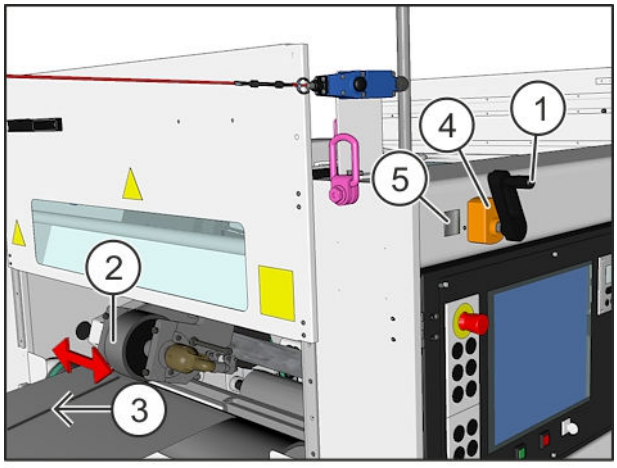
Sensor arrangement

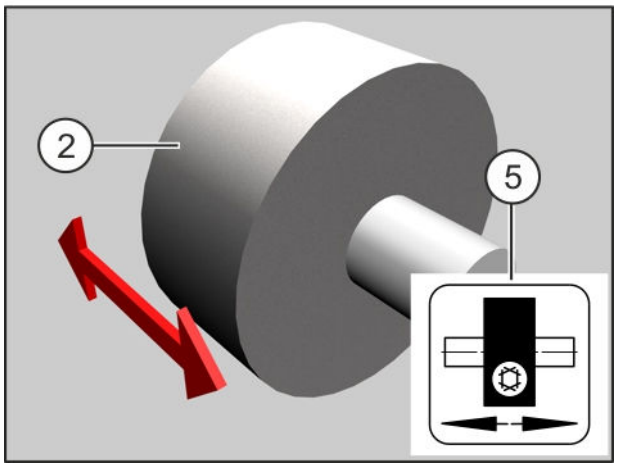
(1)	Longitudinal seam monitoring 1: Sensor on the extruder side
(2)	Longitudinal seam monitoring 2: Sensor centered between the other two sensors
(3)	Longitudinal seam monitoring 3: Sensor on the side of the operator panel



5.5.10 Seam cooling roller

(1)	Crank handle
(2)	Seam cooling roller
(3)	Seam
(4)	Display
(5)	Position seam cooling roller symbol





- ▶ Turn the crank handle (1) to set the seam cooling roller (2) to the seam (3) in the tube.

  The seam cooling roller cannot be moved towards the operator side beyond the middle of the machine.

- ▶ Read the position of the seam cooling roller from the display (4).

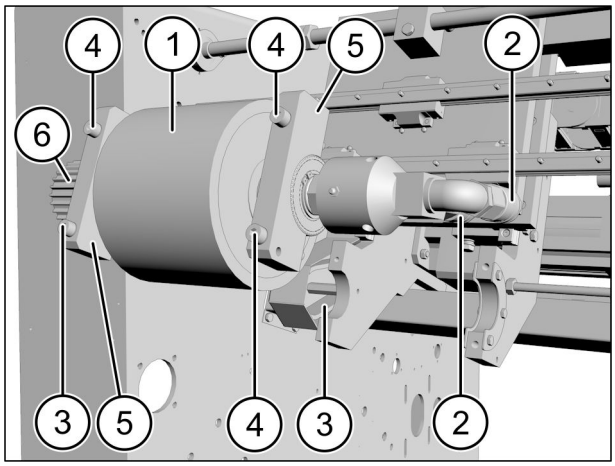
  The display shows values for your orientation, it does however not show a linear measure.

5.5.11 Replacing the seam cooling roller

Fit a seam cooling roller according to the format into the main draw.

Removing the seam cooling roller

(1)	Seam cooling roller
(2)	Coupling
(3)	Timing belt
(4)	Bolt
(5)	Fixing bracket
(6)	Timing belt gear



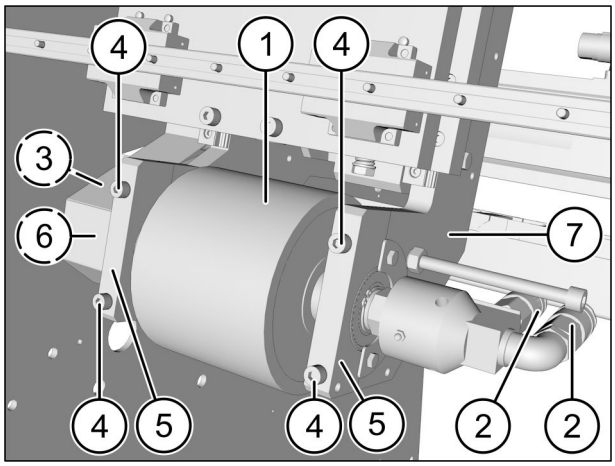
- ▶ Disconnect the hoses of the water supply and discharge of the seam cooling roller (1). Open the couplings (2) and pull the hoses out.
- ▶ Remove the guard above the timing belt (3).
- ▶ Loosen the clamping device of the seam cooling roller. Loosen the screws (4). Lift the fixing brackets (5) from the holders.
- ▶ Lift the seam cooling roller out of the main draw and tilt the timing belt gear (6) of the seam cooling roller out of the timing belt.

  If the timing belt gear cannot be pulled out of the timing belt, reduce the tension of the timing belt.

Select and mount a seam cooling roller of a different size range.

Mounting the seam cooling roller

(1)	Seam cooling roller
(2)	Coupling
(3)	Timing belt
(4)	Bolt
(5)	Fixing bracket
(6)	Timing belt gear
(7)	Holder



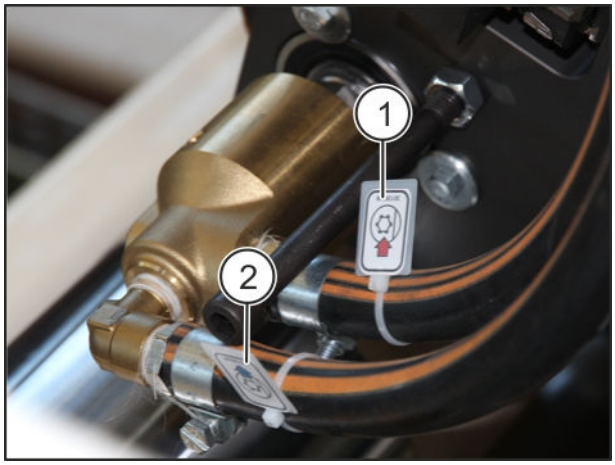
- ▶ Thread the timing belt gear (5) of the seam cooling roller into the timing belt (6).
- ▶ Place the seam cooling roller (1) into the holder (7).
- ▶ Secure the seam cooling roller in place. Place the fixing brackets (4) on the holder (7).
- ▶ Insert and tighten the screws (3).
- ▶ Connect the hoses of the cooling water supply and discharge to the seam cooling roller. Push the hoses into the couplings (2) until they engage.
- ▶ Tighten the loose timing belt.
- ▶ Mount the guard above the timing belt.

Replacing the seam cooling roller is completed.

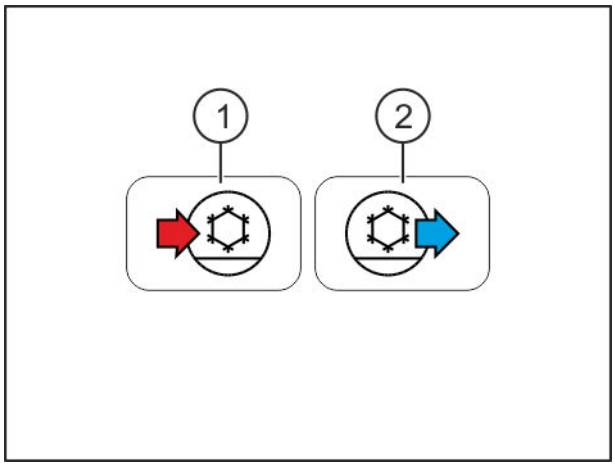
5.5.12 Supplying coolant

(1)	Supply hose
(2)	Return hose

Secure the coolant connections at the machine paying attention to the coding and re-establish whenever necessary.



(1)	Supply hose, warm coolant is running towards the temperature control unit
(2)	Return hose, cold coolant is running from the temperature control unit towards the cooling roller



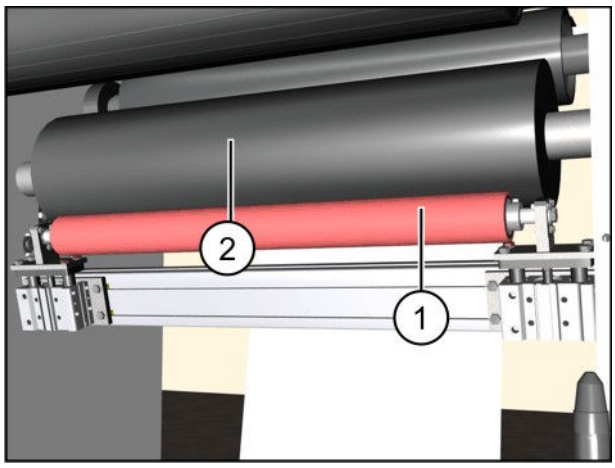
5.5.13 Nip roll

(1)	Nip roll
(2)	Idler roller

If the tube contains too much air, the tube edges tend to wander during production. The edges are no longer true to size.

The nip roll (1) creates a barrier to the air getting into the tube during the tube forming process. Downstream of the nip roll, the tube contains less air so that size accuracy of the edges is increased.

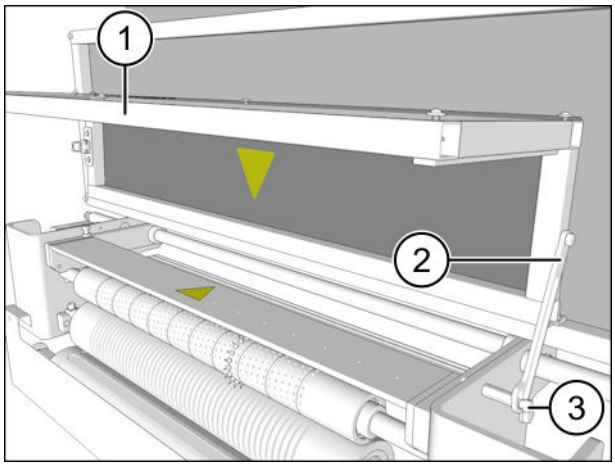
The nip roll is pressed against the idler roller (2).



5.5.14 Cut edge blocking

Opening the cut edge blocking assembly

(1)	Guard
(2)	Mounting link
(3)	Bolt



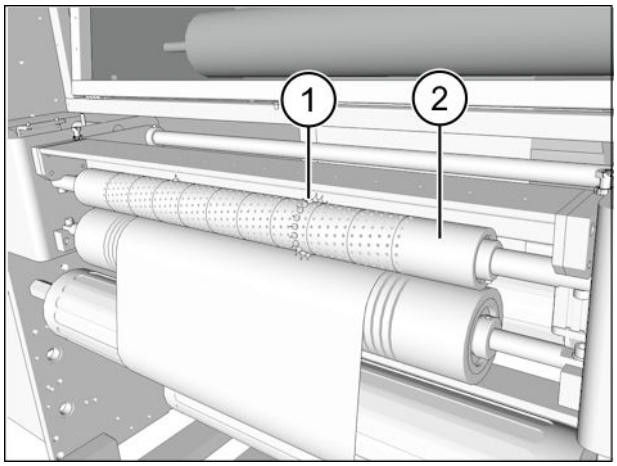
- ▶ Stop the machine.
- ▶ Fold out the guard (1) until it hits the stop and down until the mounting link (2) lashes into place.

  If necessary, press the lower end of the mounting link over the bolt (3) with the guard open.
- ▶ Carry out adjustments on the cut edge blocking assembly.
- ▶ For closing, lift the guard until the bolt (3) comes to a stop in the mounting link. Pull the mounting link forwards.
- ▶ Lower the guard slowly and close.

Changing the needle pattern

Defining the needle pattern in the film tube

(1)	Needle
(2)	Needle roller



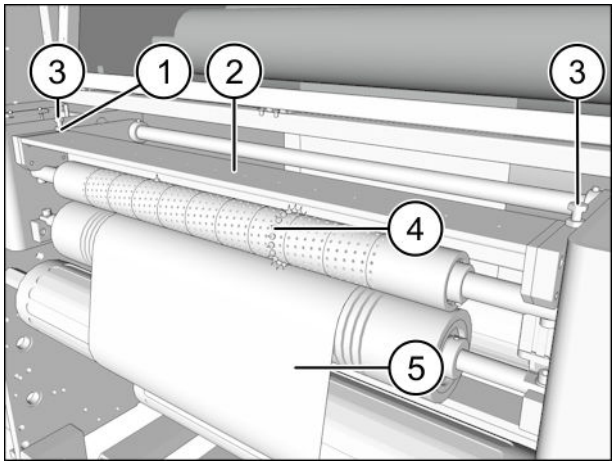
- ▶ Turn the needles (1) out of the needle roller (2) or additional needles into the needle roller.

  Arrange the needles on the needle roller as the product requires.

Adjusting the venthole diameter

Setting up the penetration depth

(1)	Nut
(2)	Cross-beam
(3)	Screw
(4)	Needle roller
(5)	Film tube



- ▶ Slacken the nuts (1) on either side of the cross-beam (2) and turn towards the head of the screw (3).
- ▶ Turn the screws on either side of the cross-beam.

  Turning the screws clockwise: lifts up the needle roller (4). The holes in the film tube (5) get smaller and blocking is reduced.
  Turning the screws counterclockwise: lowers the needle roller. The holes in the film tube get greater and blocking is higher.
  Turn the screws on either side of the needle roller in or out to the same extent to prevent the needle roller from tilting. Tilting results in an uneven needle pattern in the film tube and may cause premature wear to components.
- ▶ Run the machine and, if necessary, reposition the needle roller.
- ▶ Turn and tighten the nuts in the cross-beam.

5.6 Web guide control

5.6.1 Web guide control, ekr....248
Safe behavior....248
Control console....248
Display - Operation display....248
Selecting the mode....248
Setting the set position....248
Correcting the controller gain....248

5.6.2 Web guide control - Material setup....254
Display - Material set-up selection....254
Display - Material set-up....254
Selecting the material setup....254
Material set-up - Web edge controller....254
Material setup - Web center control....254

X000011237

5.6.1 Web guide control, ekr

Safe behavior

CAUTION

Risk of injury from infeed nips

Parts of your body, clothes, hair and objects can be pulled in and crushed. Always wear tight-fitting clothes and a hairnet. Work without industrial safety gloves.Keep a safe distance.

CAUTION

Injuries can be caused by moving parts

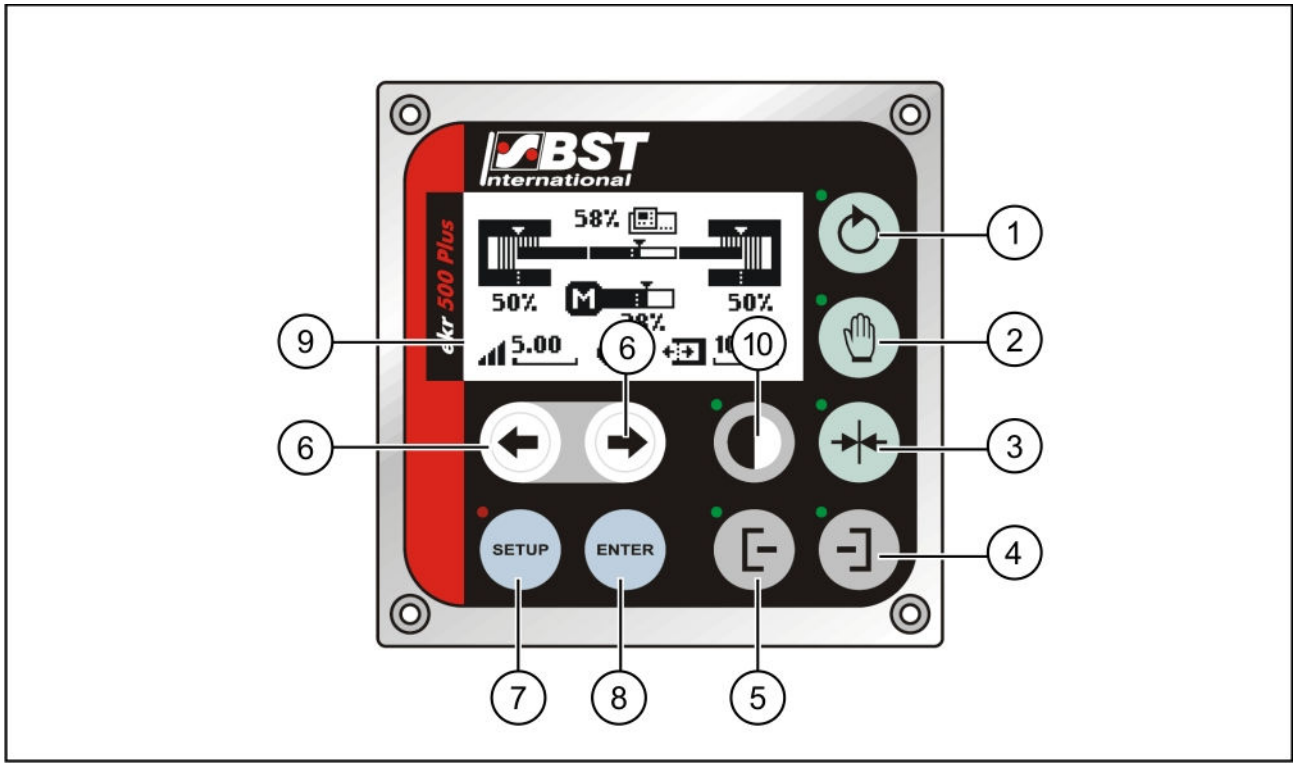
Hands might be crushed or sheared off. Do not reach in between moving components. Keep a safe distance.

CAUTION

Risk of injury from rotating rollers

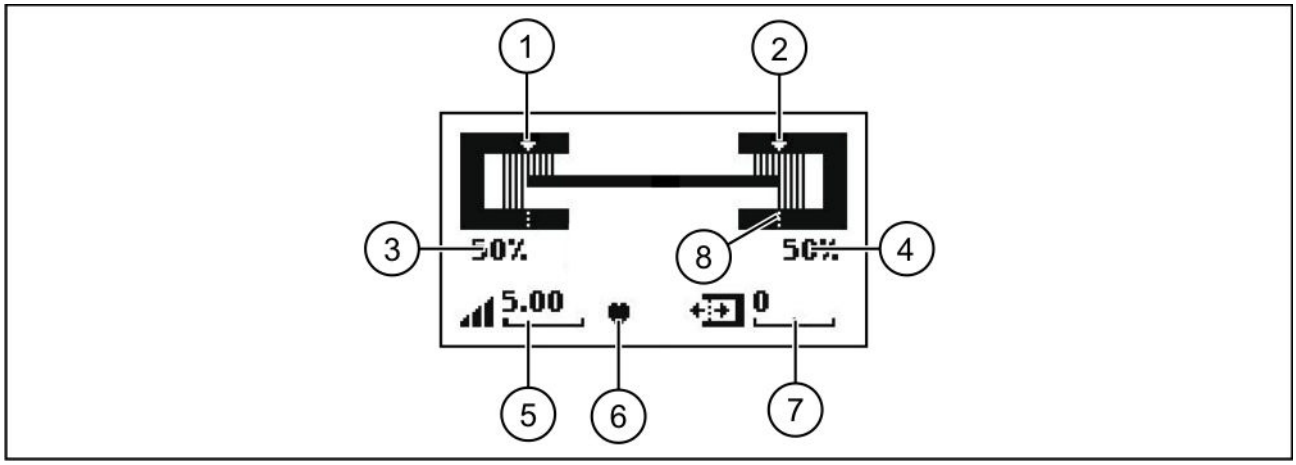
Parts of your body, clothes, hair and objects can be pulled in and crushed. Do not reach into the nip. Keep a safe distance.

Control console



(1)	AUTOMATIC on/off, automatic operation of the web guide control	(2)	MANUAL MODE on/off, manual operation of the web guide control
(3)	MID-POSITION on/off, mid-position of the web guide control	(4)	CONTROL RIGHT on/off, control to the right web edge
(5)	CONTROL LEFT on/off, control to the left web edge; CONTROL RIGHT and CONTROL LEFT pressing both simultaneously activates the web guide control	(6)	RIGHT arrow and LEFT arrow in manual mode: Web guide control to the right, web guide control to the left; in automatic mode: Set position to the left, set position to the right; in setup: Increase value, decrease value
(7)	SETUP, in manual mode: Access set-up; in automatic mode: Adjust the controller gain	(8)	ENTER, save value and exit setup
(9)	Display	(10)	Line / contrast sampling (not in operation)

Display - Operation display

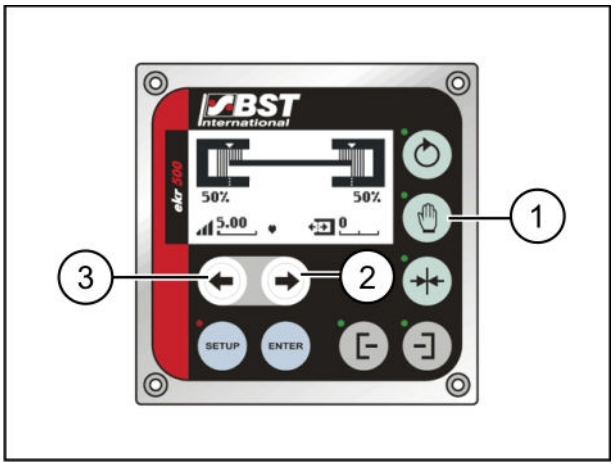


(1)	Set position sensor 1	(2)	Set position sensor 2
(3)	Cover sensor 1	(4)	Cover sensor 2
(5)	Controller gain	(6)	Status report, heart beats, the system is OK
(7)	Numerical display of the set position	(8)	Middle of the measurement area

Selecting the mode

Switching on manual mode

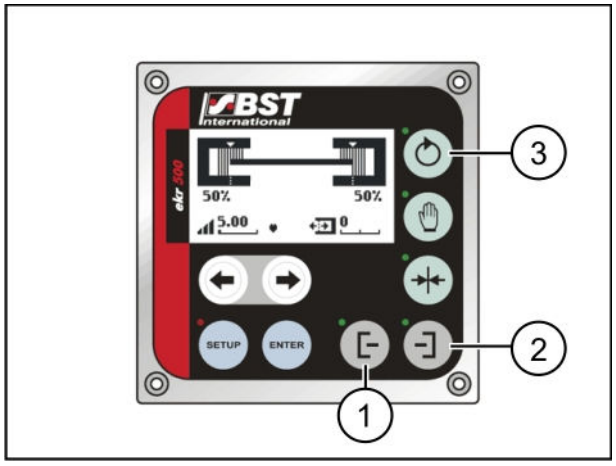
(1)	MANUAL MODE
(2)	RIGHT
(3)	LEFT



- ▶ Press the pushbutton (1) to switch on manual mode.
- ▶ Press pushbutton (2) or (3) to move the web guide control unit to the desired position.

Switching on automatic mode

(1)	CONTROLLER TO THE LEFT
(2)	CONTROLLER TO THE RIGHT
(3)	AUTOMATIC MODE



▶ Press pushbutton (1) or (2) to select the type of scanning.

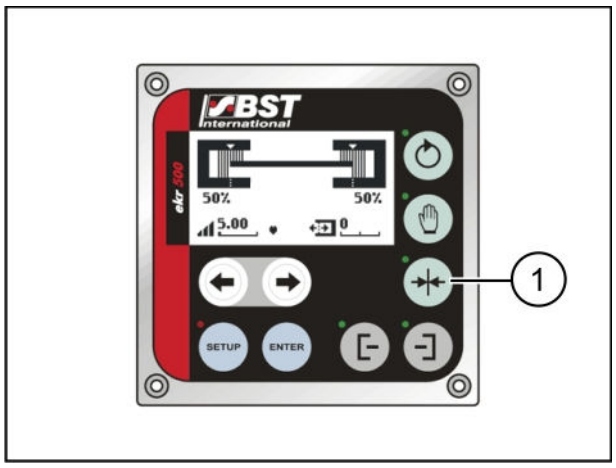
Press the pushbutton (1) to switch on the controller moving towards the left web edge. Press the pushbutton (2) to switch on the controller moving towards the right web edge. Press both pushbuttons to switch on the controller moving towards the web center.

▶ Press the pushbutton (3) to switch on automatic mode.

Switching on mid-position mode

When in mid-position mode the web guide control unit will be moved to the mid-position automatically.

(1)	MID-POSITION

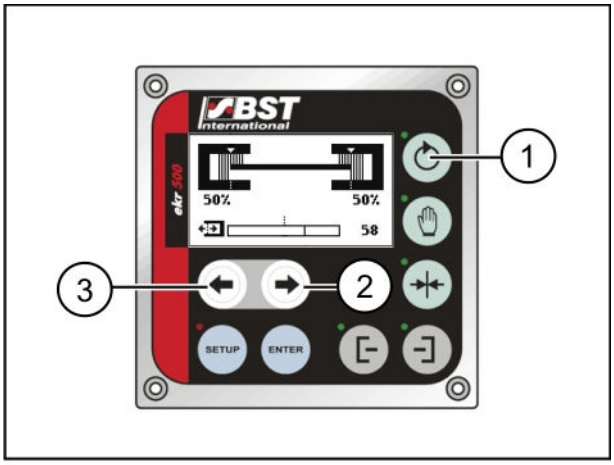


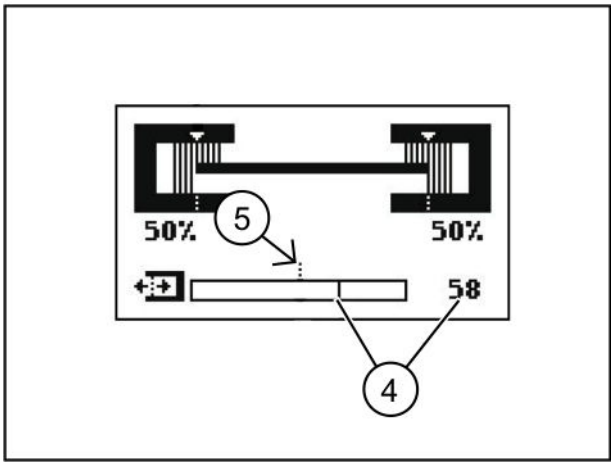
▶ Press the pushbutton (1) to move the web guide control unit to the mid-position.

Setting the set position

Set the set position of the material web.

(1)	AUTOMATIC
(2)	RIGHT
(3)	LEFT
(4)	Set position
(5)	Middle of the measurement area





- Press the pushbutton (1) to activate the automatic mode.
- Press the pushbutton (2) or (3) to activate the set position.

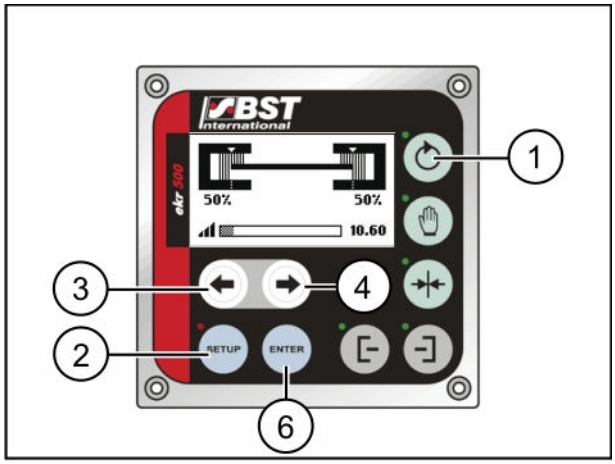
  If a pushbutton is not pressed then the adjustment exits automatically.

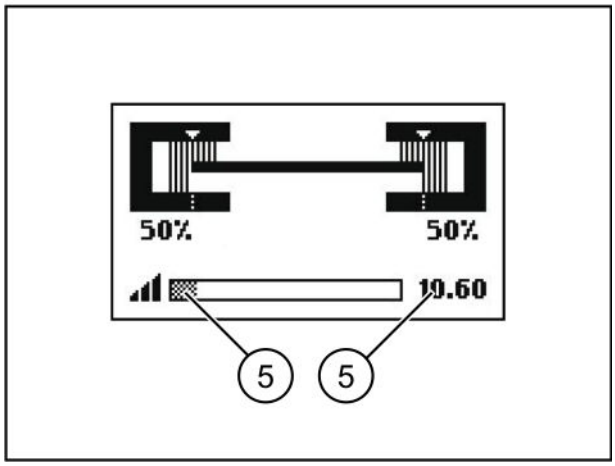
  The display shows:
  the current set position (4) as a bar graph and a numerical value,
  and the middle of the measurement area (5).

Correcting the controller gain

Correct the controller gain if the web guide control shakes or control fluctuations occur.

(1)	AUTOMATIC
(2)	SETUP
(3)	LEFT
(4)	RIGHT
(5)	Controller gain
(6)	ENTER





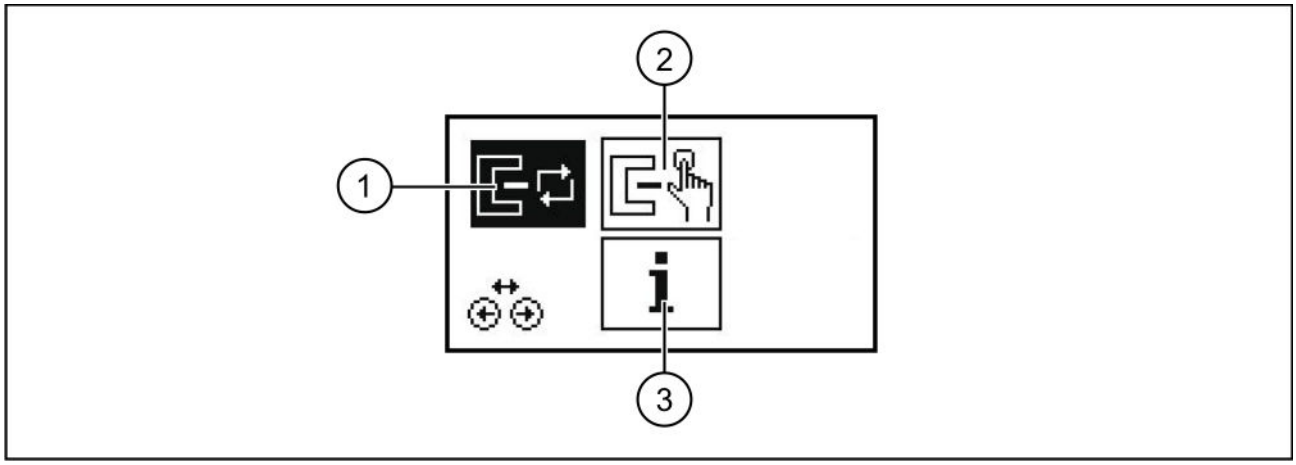
- ▶ Press the pushbutton (1) to activate the automatic mode.
- ▶ Press the pushbutton (2) to start the adjustment.
- ▶ Press the pushbuttons (3) and (4) until the web guide control works without shaking or fluctuations.

  The display shows: the controller gain (5) as a bar graph and a numerical value.
- ▶ Press the pushbutton (6) to exit the adjustment.

  If a pushbutton is not pressed then the adjustment exits automatically.

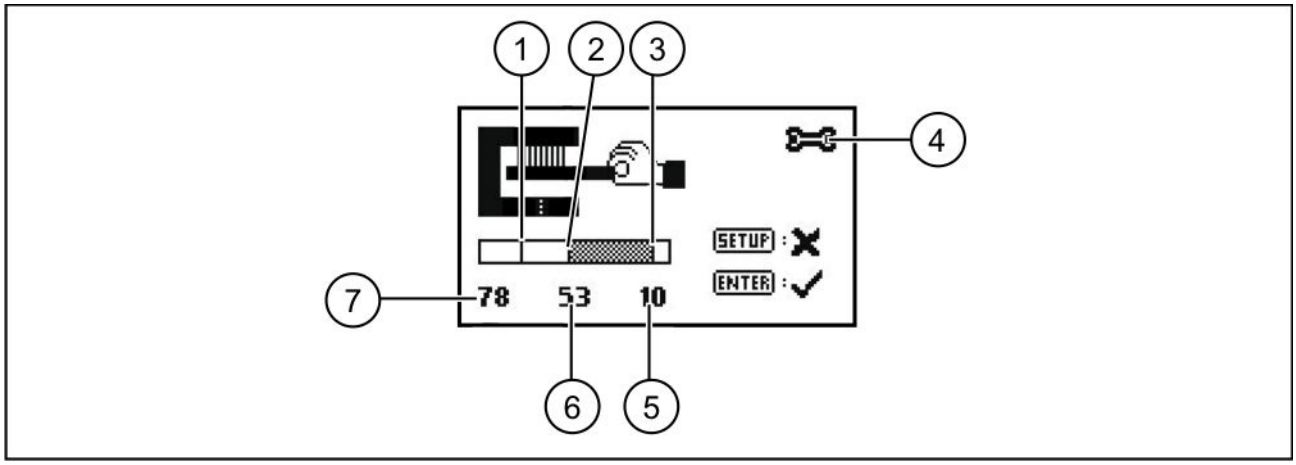
5.6.2 Web guide control - Material setup

Display - Material set-up selection



(1)	Automatic material set-up	(2)	Manual material set-up
(3)	Information, display system information		

Display - Material set-up



(1)	Dark value	(2)	Current value
(3)	Light value	(4)	Manual material set-up activated
(5)	Lowest light value	(6)	Current value
(7)	Largest dark value		

Selecting the material setup

BEWARE

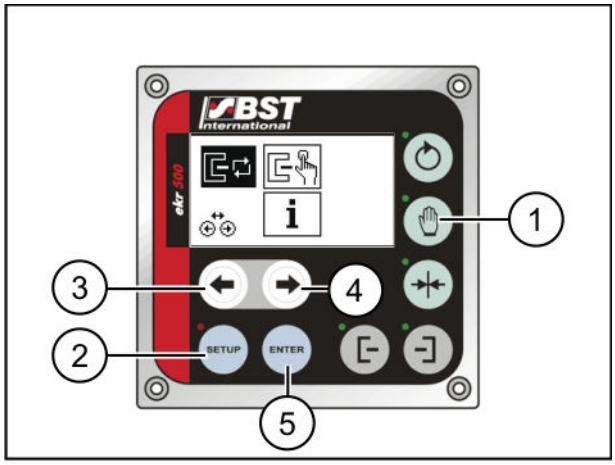
Risk of injury from the edges of the running material web.

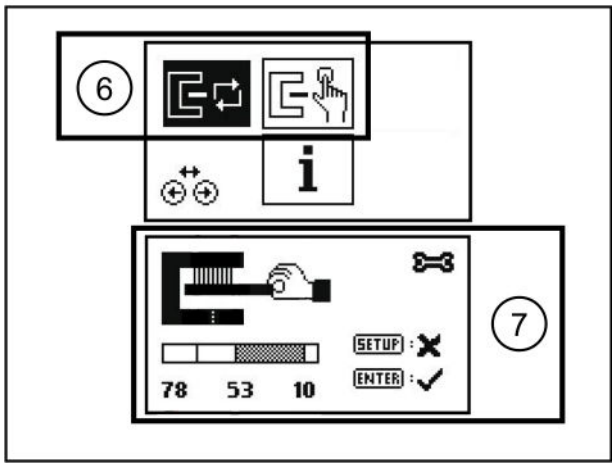
Do not perform the material setup on a running machine.

Stop the machine to perform the material setup.

During the material setup the web guide control is automatically adjusted to the properties of the material to be sampled.

(1)	MANUAL MODE
(2)	SETUP
(3)	LEFT
(4)	RIGHT
(5)	ENTER
(6)	Select the material setup
(7)	Light and dark values of a sensor





- ▶ Press the pushbutton (1) to activate the manual mode.
- ▶ Press the pushbutton (2) to begin the selection of the material setup.
- ▶ Press the pushbutton (3) or (4) to select the desired material setup.

  Automatic material setup or manual material setup.
- ▶ Press the pushbutton (5) to call up the selected material setup (6).

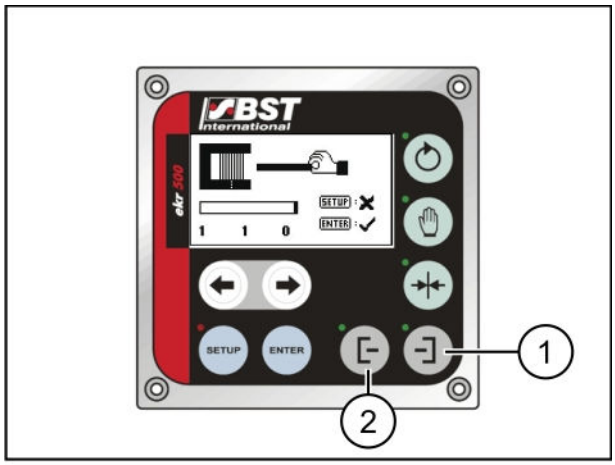
  The material setup is activated.
  The display (7) displays the light and dark values of a sensor.

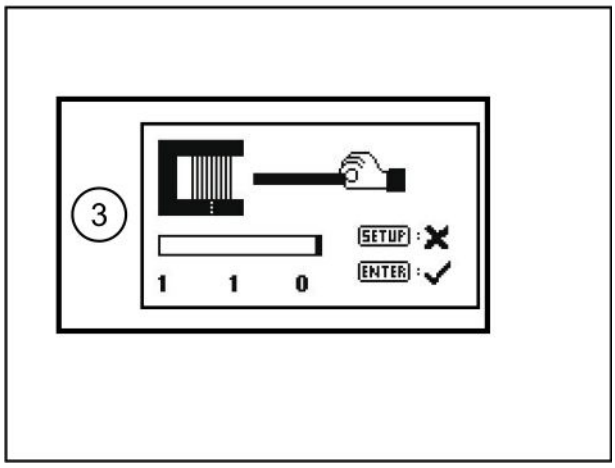
Material set-up - Web edge controller

Automatic material set-up for a web edge.

Preparing the material set-up

(1)	RIGHT CONTROL
(2)	LEFT CONTROL
(3)	Display the process for determining the light and dark values





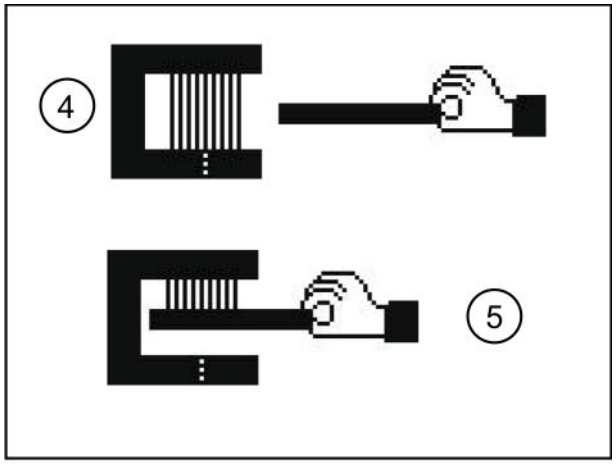
- Clean the sensors.

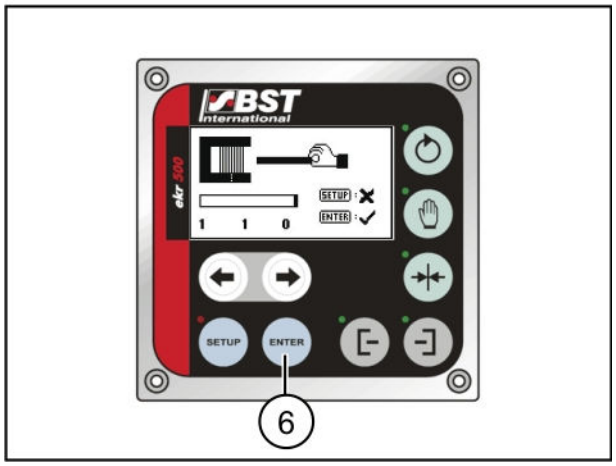
  Observe the measurement window.
- Press the pushbutton (1) or the pushbutton (2) to select the sensor for which the material setup is to be performed.
- Activate the automatic material setup.

  The display (3) displays the process for determining the light and dark values.
  Follow the indications.

Performing the material setup

(4)	Material web outside the measurement area
(5)	Material web within the measurement area
(6)	ENTER





- ▶ Remove the material web from the measurement area (4) of the sensor.

  The measured value is displayed as a light value.

- ▶ Hold the material web to be processed in the measurement area (5) of the sensor for at least 2 seconds.

  The measured value is displayed as a dark value.

- ▶ Press the pushbutton (6) to exit the material setup.

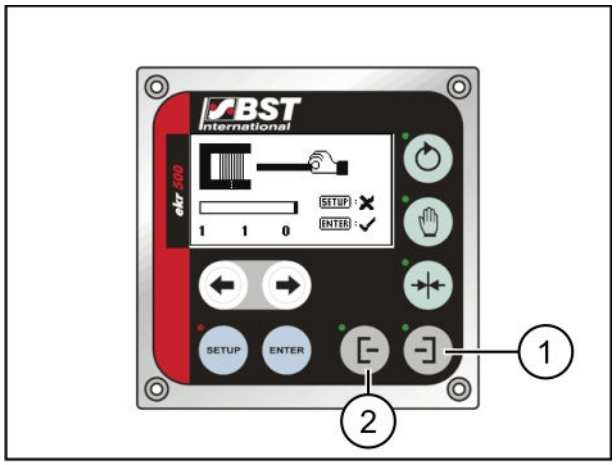
  The light and dark values measured by the sensor are saved.
  The material setup exits.

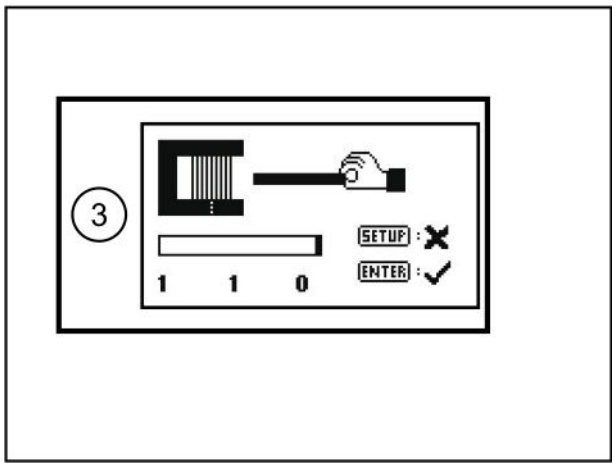
Material setup - Web center control

Automatic material set-up for the web center control.

Preparing the material setup

(1)	RIGHT CONTROL
(2)	LEFT CONTROL
(3)	Display the process for determining the light and dark values





- ▶ Clean the sensors.

  Observe the measurement window.
- ▶ Press the pushbutton (1) and the pushbutton (2) to select both sensors.
- ▶ Activate the automatic material setup.

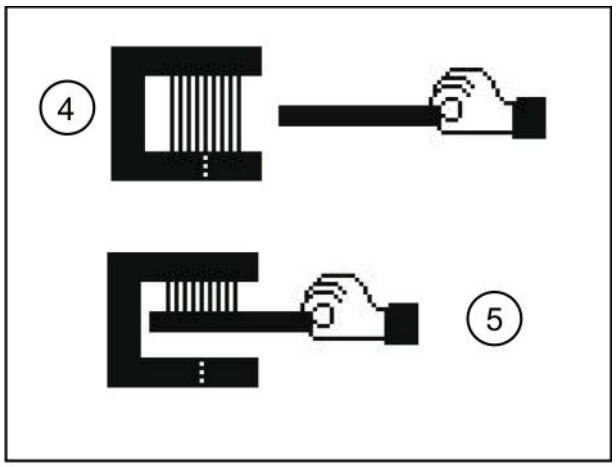
  The display (3) displays the process for determining the light and dark values.

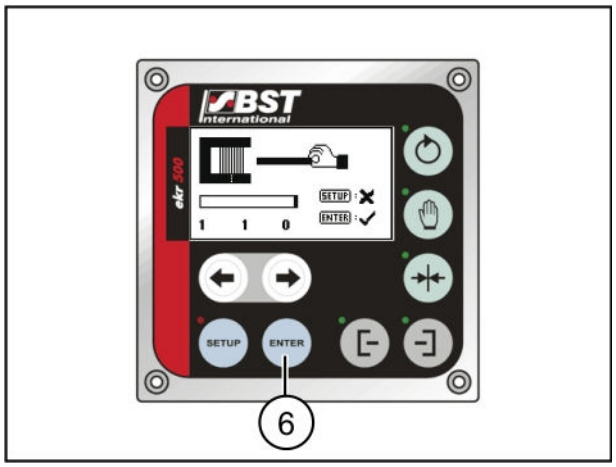
Follow the information shown for "Sensor 1".

Performing the material setup

Material setup for sensor 1.

(4)	Material web outside the measurement area
(5)	Material web within the measurement area
(6)	ENTER





- ▶ Remove the material web from the measurement area (4) of the sensor.

  The measured value is displayed as a light value.

- ▶ Hold the material web to be processed in the measurement area (5) of the sensor for at least 2 seconds.

  The measured value is displayed as a dark value.

- ▶ Press the pushbutton (6).

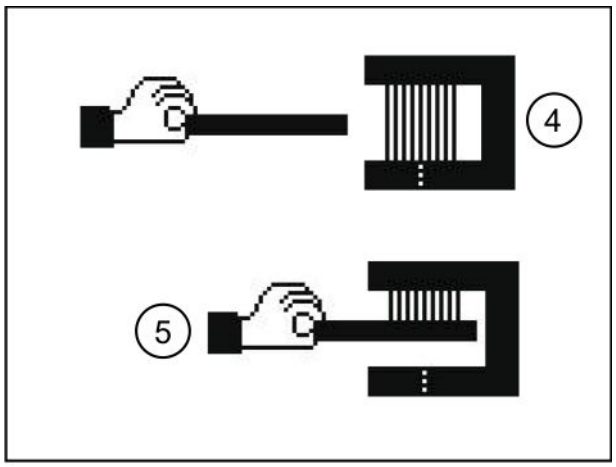
  The light and dark values measured by the sensor are saved.
  The display displays the process for determining the light and dark values.

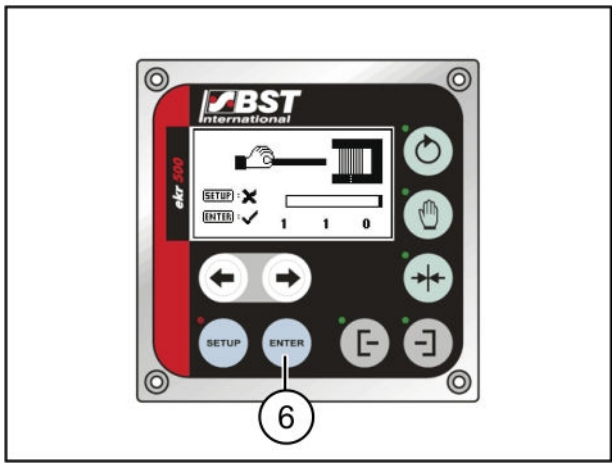
Follow the information shown for "Sensor 2".

Performing the material setup

Material setup for sensor 2.

(4)	Material web outside the measurement area
(5)	Material web within the measurement area
(6)	ENTER





- ▶ Remove the material web from the measurement area (4) of the sensor.

  The measured value is displayed as a light value.

- ▶ Hold the material web to be processed in the measurement area (5) of the sensor for at least 2 seconds.

  The measured value is displayed as a dark value.

- ▶ Press the pushbutton (6) to exit the material setup.

  The light and dark values measured by the sensor are saved.
  The material setup exits.

5.7 Winder

5.7.1 Control voltage ... 262
Switching the control voltage on/off ... 262
Stopping the winding station ... 262

5.7.2 Change winding shaft ... 263

5.7.3 Fixing the winding core ... 265

5.7.4 Starting the winder ... 266
Operating mode selector switch ... 266
Threading up the film ... 266
1. Performing a roll change ... 266
Starting quality production ... 266

5.7.5 Web sag on startup ... 270

5.7.6 Automatic step sequence during a roll change ... 271
Safety instructions for a roll change ... 271
Sequence ... 271

5.7.7 Moving the film reel ... 272

5.7.8 Film roll delivery ... 273

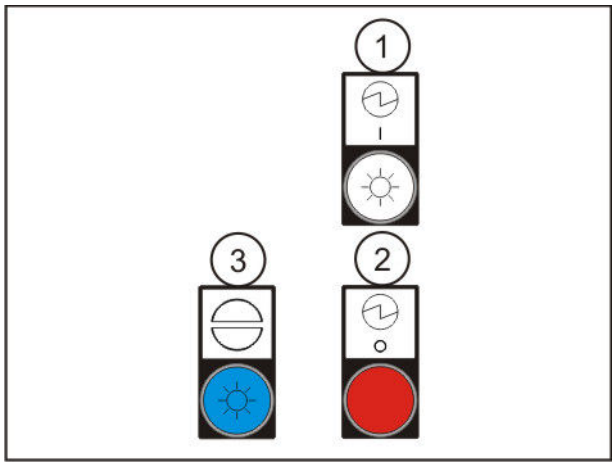
5.7.9 Shut-down sequence when producing tubular film ... 274

5.7.1 Control voltage

Switching the control voltage on/off

The control voltage to the respective winding station is enabled via the RESET pushbutton.

(1)	CONTROL VOLTAGE ON
(2)	CONTROL VOLTAGE OFF
(3)	RESET

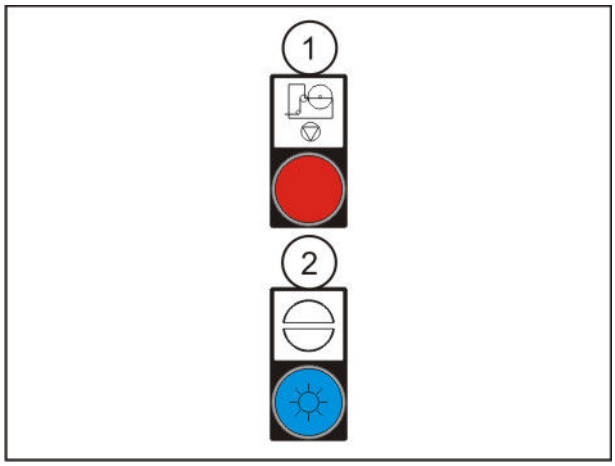


- ▶ Press the (1) pushbutton in control console +R82A to switch the control voltage to the winder on.
- ▶ After waiting 10 seconds, use the button (3) to enable the control voltage on the respective winding stations.

  The signal lamp in the pushbutton (3) goes out. The drives in the winding stations are ready to be started.

Stopping the winding station

(1)	WINDING STATION STOP
(2)	RESET



- ▶ Stop the winding station using the pushbutton (1).
  - *The winding station stops immediately.*
  - *The control voltage on the winding station is off.*
  - *The other winding station and the draw will continue to run.*
- ▶ Press the (2) pushbutton to switch the control voltage to the winding station back on.

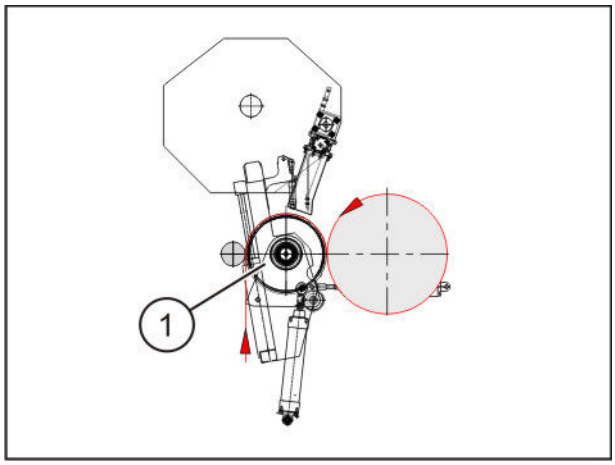
  Then, the winding station drives can be re-started.

5.7.2 Change winding shaft

For changing over for example from a 3" winding shaft to a 6" winding shaft and vice versa, the winding shaft placed into the roll change system before must be removed by manual step functions.

Swiveling the roll change system

(1)	Roll change system

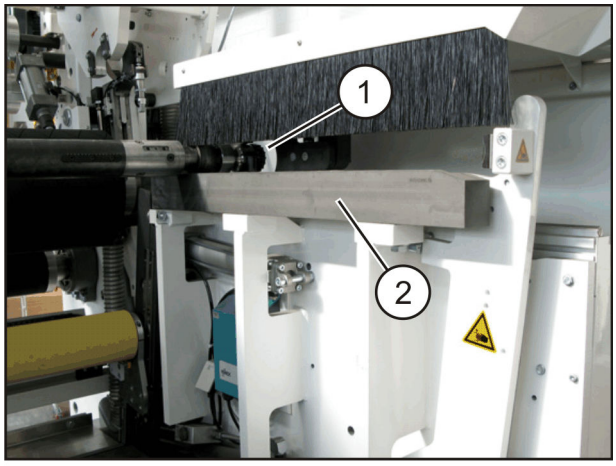


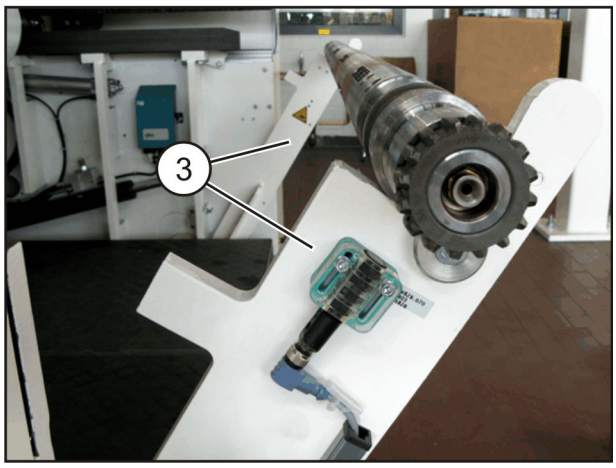
▶ Turn the SWIVEL ROLL CHANGE SYSTEM swivel switch in position ['↓'] to swivel the roll change system together with the winding shaft from the pick-up position to the transfer position.

In the pick-up position the winding shaft is taken to force it against the contact roller and then swiveled in the transfer position. In the transfer position the winding shaft is placed down on the mounting rails of the parallel guide.

Traversing the roll carriage

(1)	Roll carriage including winding shaft
(2)	Mounting rails of the parallel guide
(3)	Lowering arms





- Turn the PARALLEL GUIDE swivel switch in position ['←'] to allow the roll carriage to take the winding shaft.
- Turn the ROLL CHANGE SYSTEM swivel switch in position ['↑'] to swivel the roll change system without the winding shaft back from the transfer to the pick-up position.
- Move the roll carriage including the winding shaft back until the lowering arms (3).

  The winding shaft rolls into the pick-up recesses of the lowering arms.
- Use the DELIVERY ARM swivel switch on control console +R85 to swivel the winding shaft with the lowering arms out.
- Take the winding shaft out of the pick-up recesses of the lowering arms.

After removal of the winding shafts the winding shaft magazine or the winding shaft support can be refitted with new winding shafts.

5.7.3 Fixing the winding core

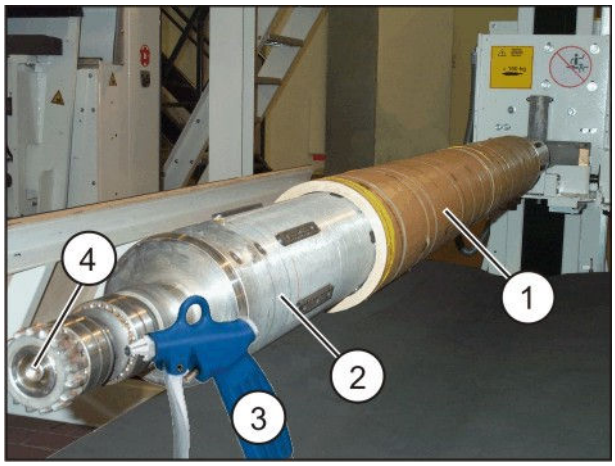
CAUTION

Risk of injury from compressed air

Never aim the air gun at yourself or other persons.

Do not use the air gun but to fix the winding core.

(1)	Winding core
(2)	Winding shaft
(3)	Air gun
(4)	Winding shaft filling valve



- Push the winding core (1) on the winding shaft (2) which is clamped in place on one side and align.
- Fit the air gun (3) to the winding shaft filling valve (4) and briefly pressurize the winding shaft.

  The winding shaft's clamping elements are pressed against the winding core to fix it.

5.7.4 Starting the winder

Operating mode selector switch

Selecting the startup mode

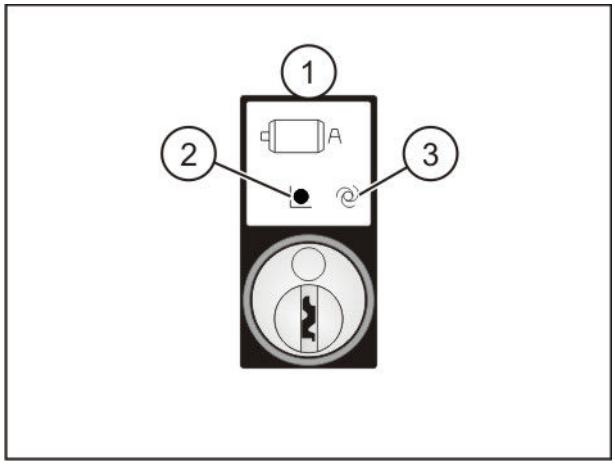
For safety reasons, the following restrictions apply when using the 'Startup' operating mode:

Depending on the product, the winder should be started at the lowest possible thread-up speed.

The web tension generated by the surface drive is limited to 150 [N].

The center drive cannot be switched on.

(1)	WINDING STATION A key switch
(2)	'Startup' mode
(3)	'Production' mode



- ▶ Turn the key switch (1) into position (2) to be able to enter the safety cut-out mat within the winding area for threading up the web.

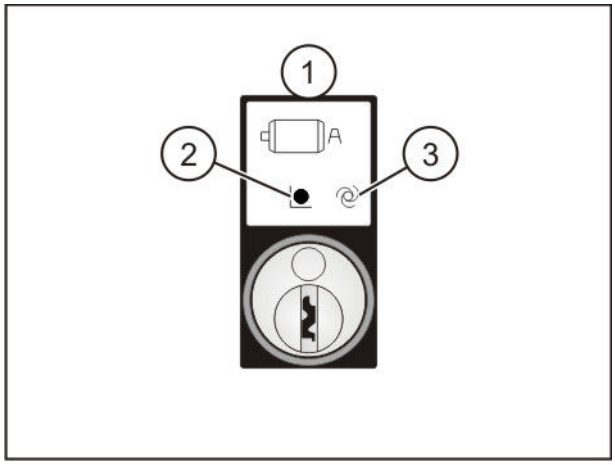
  All dangerous movements are blocked, the surface drive continues to run.

- ▶ After initial winding of the tubular or flat film the safety cut-out mat in the safety area should be left immediately.

After threading up the web and starting the winding station, change over to the 'Production' mode.

Selecting the 'Production' mode

(1)	WINDING STATION A key switch
(2)	'Startup' mode
(3)	'Production' mode

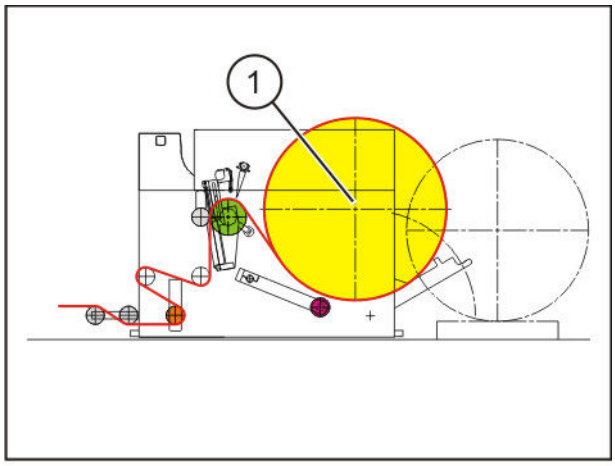


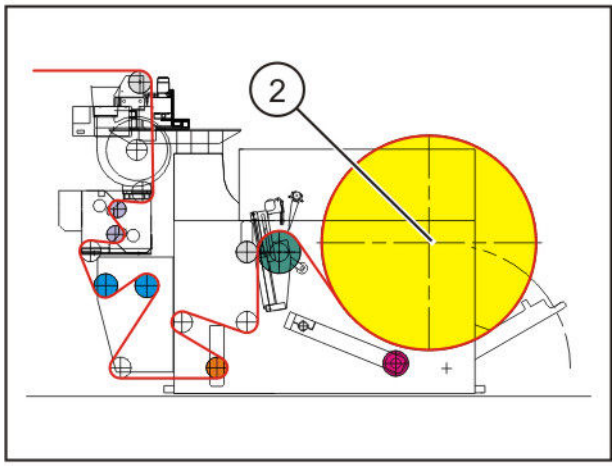
- ▶ Turn key switch (1) into position (3) to be able to run up the line to production speed.

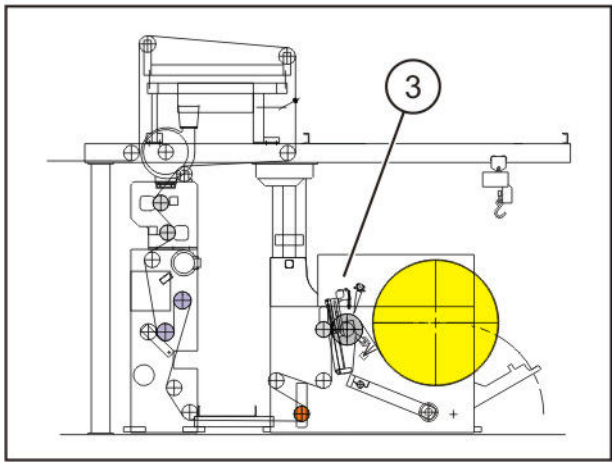
Run the line up to production speed.

Threading up the film

(1)	Mono winder without chill rolls
(2)	Mono winder with non-driven chill roll
(3)	Mono winder with draw







- Tie up the free end of the threading tape with the tubular film coming from the extrusion line's film haul-off unit.
- Release the winding station via the WINDING STATION SECURED key switch.
- Switch on the winder using the WINDER ON pushbutton.
- Use the threading tape to manually thread the tubular film into winding station.
- Keep the web taut and use a film cutting knife to cut it right at the winding core.

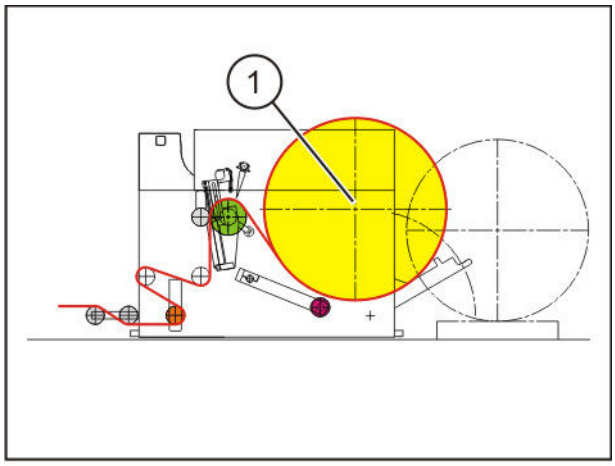
  The tubular film will be caught and wound up by the rotating winding core.
- Use the LAY-ON ROLLER ON pushbutton to load the lay-on roller to the contact roller.
- Even out the web sag between the film haul-off unit and the contact roller by pressing the SPEED UP pushbutton.

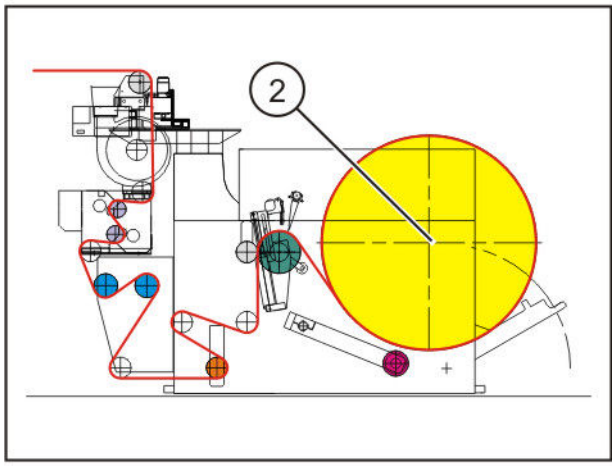
  Release the pushbutton the moment the web sag is evened out.

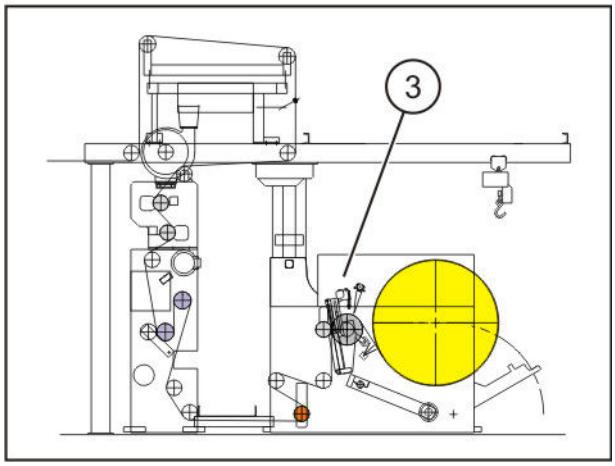
  The layflat tubular film will be wound up at startup speed.
  Reaching the starting speed completes the production make-ready process.

1. Performing a roll change

(1)	Mono winder without chill rolls
(2)	Mono winder with non-driven chill roll
(3)	Mono winder with draw







- ▶ Perform a reel splice when everything is running stable. To do this, press the START REEL SPLICE pushbutton.
- ▶ Take the film roll and the waste produced during startup out of the winder and insert a new winding shaft with winding cores into the winding shaft inserting device.
- ▶ Set the operating mode selector switch to 'Production' mode.
- ▶ Run the line up to production speed.

Starting quality production

Upon reaching the setpoints in the 'Startup/Changeover' mode, change over to the 'Production' mode.

(1)	'Status' submenu
(2)	PRODUCTION button

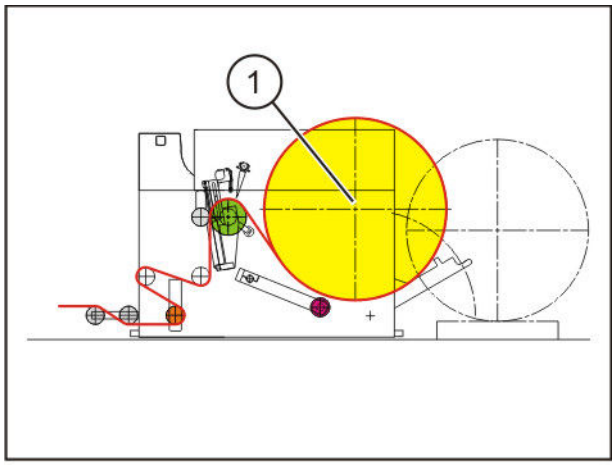
▶ Press the PRODUCTION button in the 'Status' submenu of the 'Job' menu.

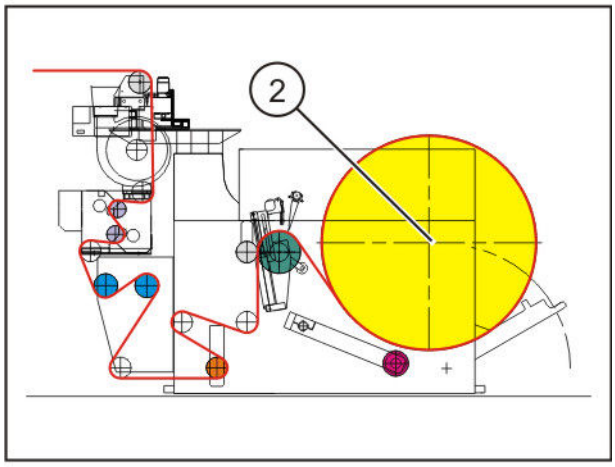
Quality production starts after the first reel splice, separately from winding station A and B.

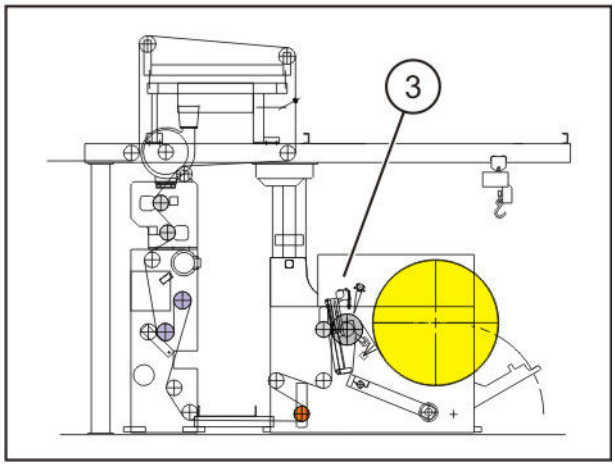
5.7.5 Web sag on startup

When starting the line a web sag may occur between the draw and the winding station. The drive does not run web-tension controlled

(1)	Mono winder without chill rolls
(2)	Mono winder with non-driven chill roll
(3)	Mono winder with draw







- ▶ Hold the SPEED UP WEB pushbutton to eliminate the web sag.

  While holding the pushbutton the drive of the winding station is running about 10 % faster.

- ▶ Release the SPEED UP WEB pushbutton immediately the moment the web sag is evened out.

  There is a risk of web breakage, If the pushbutton is held for too long .

5.7.6 Automatic step sequence during a roll change

Safety instructions for a roll change

WARNING

Risk of Injury through cutting blades, circular knives or sharp-edged machine parts

Wear cut-resistant safety gloves.

DANGER

Risk of injury through cutting blade

If a roll change has been aborted, it is forbidden to reach into the area between the vacuum cutover drum and the film roll unless the winding station is secured. For troubleshooting, the WINDING STATION A/B key switch must be in the 'Winding station secured' position.

Sequence

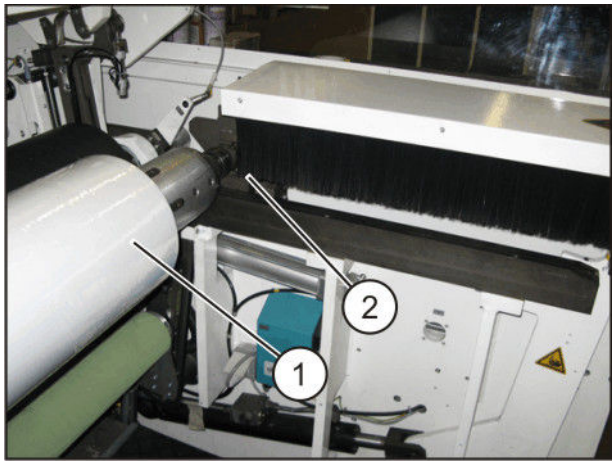
Upon tripping the roll change process the following sequence proceeds automatically:

- Upon starting the roll change a horn sounds in intervals and the blinker lamp flashes. The film roll is in its winding position and is pressed up against the vacuum cutover drum.
- The winding shaft magazine is being turned from its insert position to the intermediate position and into its transfer position. It transfers the winding shaft with the pasted winding core on to the roll change lever. In the roll change lever the winding shaft is locked in place and accelerated up to the web speed with the aid of a winding shaft accelerator.
- The winding shaft is pressed up against the vacuum cutover drum and thus against the film web.
- To fix the film web on the drum surface a vacuum is generated within the vacuum cutover drum having an effect on the film through small holes in the drum shell.
- The web cut is actuated. The cutting assembly integrated in the vacuum cutover drum applies a right-angled cut to separate the web.
- The cutting pulse resets the roll meter counter to zero.
- The new leading edge of the film is caught by the sticky tape on the winding core and rewound.
- The brake on the parallel guide is released. The catches of the parallel guide retract the film roll again and convey it up to the end of the carrier rail. Doing so, the film roll decelerates.
- After the catches have been opened the film roll is transferred to the lowering arms.
- The roll change lever moves the winding shaft with the new film roll from the roll start position into the transfer position and places it down there on the carrier rails of the parallel guide.
- The catches of the parallel guide move back to the new winding shaft. The catches grasp the winding shaft and press the film roll up against the vacuum cutover drum using the preselected pressure force.
- The roll change lever moves back into the take-over position.

5.7.7 Moving the film reel

The roll carriage in the parallel guide is used to displace the film reel between the surface roller and the delivery arms. The winding shaft rolls over the winding shaft bearing onto the mounting rails.

(1)	Film reel with winding shaft
(2)	Winding carriage



- Turn the PARALLEL GUIDE swivel switch into the '→' position to withdraw the film reel fitted with the winding shaft (1) from the contact roller.

  The roll carriage (2) displaces the film reel fitted with the winding shaft up to the delivery arms.

- Turn the PARALLEL GUIDE swivel switch into the ['←'] position to allow the roll carriage to be advanced towards the contact roller.

  The parallal guide forces the film reel against the contact roller at a preset contact pressure.

5.7.8 Film roll delivery

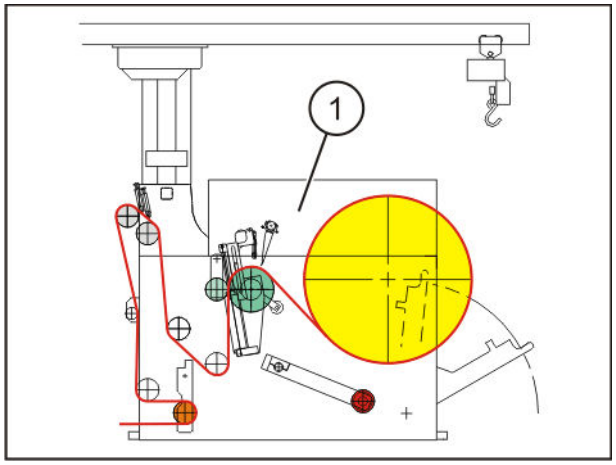
WARNING

There is a risk of being crushed when film roll delivery takes place.

The delivery area for finished rolls must be clear of personnel and objects.

The machine operator must make sure that the delivery area is secure before lowering the film roll. The machine operator must be able to see the delivery area during the complete lowering process.

(1)	Winding station



- Turn the LOWERING ARM swivel switch into the 'Down' position for film roll delivery.

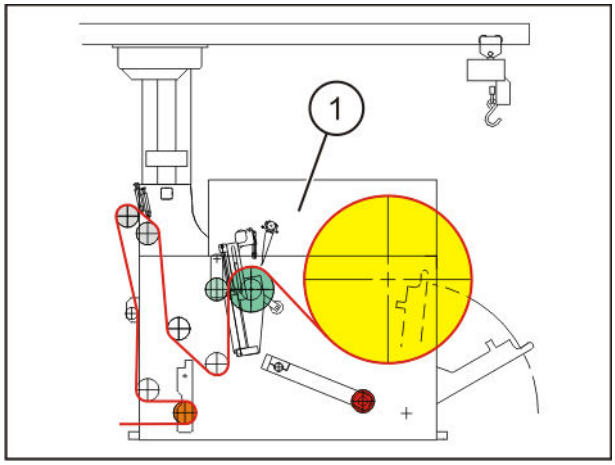
  The lowering arms swivel the film roll hydraulically from the take-up position into the delivery position and place it down there, e.g. on a pallet. The film roll fitted with the winding shaft rolls out of the lowering arms.

- Swivel the lowering arms back up to the take-up position by turning the LOWERING ARM swivel switch in the 'Up' position.
- Place a new winding shaft with spliced winding cores into the receiving recesses of the lowering arms.
- Use the hoisting gear to take up the winding shaft, transport it to the winding shaft magazine and insert it there.
- Swivel the lowering arms completely back into their stop position by turning the LOWERING ARM swivel switch into the 'Up' position once again

5.7.9 Shut-down sequence when producing tubular film

The tubular film is wound up on winding station (A).

(1)	Winding station A



- Carry out a reel splice on winding station A (1). Move the finished reel back to the lowering arms.
- Lower the film reel with the lowering arms.
- Remove the film reel.
- Swivel the lowering arms back again.
- Reduce the web speed of the entire line to the lowest possible threading speed depending on the product to be produced.
- Run the production capacity of the extrusion line down to '0'. The tubular film is torn off at the die head.
- Before the end of the tubular film reaches the infeed idler roller for web guide control, simultaneously switch off the drive on winding station A, the drive for the spreader roller and for the web draw.

The remaining tubular film can be left in the winding station for the next initial winding process in winding station A.

6 Production

6.1 Messages
6.1.1 Alarms

6.2 Web travel
6.2.1 Safe behavior
6.2.2 Unwinding unit
6.2.3 Micro-perforator, outer web
6.2.4 Continuous embossing unit, outer web
6.2.5 Main draw
6.2.6 Cooling zone

6.3 Seam extruder
6.3.1 Extrusion lines

6.1 Messages

6.1.1 Alarms 277

6.1.1 Alarms

ID	Fault message	Troubleshooting
(0)	Feedback loop of Emergency-Stop contactors faulty	Unlatch the Emergency-Sstop button and acknowledge the fault. Check the Emergency-Stop button for proper functioning and replace, if necessary.
(1)	Emergency-Stop button hit	Unlatch the Emergency-Sstop button and acknowledge the fault. Check the Emergency-Stop button for proper functioning and replace, if necessary.
(3)	Low temperature Heater Extruder	Check the temperature setting, replace the temperature sensor, if necessary
(4)	Excess temperature Heater Extruder	Check the temperature setting, replace the temperature sensor, if necessary
(5)	Inverters not initialized	Check or switch on control voltage
(6)	Fault: Virtual Master: {vMaster}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(7)	Fault: Feed-in Drive systems: {Feed-in}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(8)	Fault: Converter Draw Main drive: {Main}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(9)	Fault: Converter Draw downstream of the unwind Outer: {DrawA}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(10)	Fault: Converter Draw downstream of the unwind Inner: {DrawB}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(11)	Group alarm Motor protection switch	Improper setting - set to rated motor current. Motor defective - replace. Sluggischness - check motor, shaft, gear wheel, spindle for smooth running.
(12)	Group alarm MCB(s)	Fuses in switch cabinet +E25 have blown, identify the cause
(13)	Group alarm MCB(s) Extruder	Fuses in switch cabinet +E25 have blown, identify the cause
(14)	Fault: Inverter Embossing unit: {Embossing}	Fault number ==> see B&R's user manual. Read the fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(15)	Guard door open Draw	Close the guard door. Check the safety shutdown switches on the guard door for proper functioning. Set up or replace, if necessary.

ID	Fault message	Troubleshooting
(16)	Guard door open Web processing	Close the guard door. Check the safety shutdown switches on the guard door for proper functioning. Set up or replace, if necessary.
(17)	Compressed air monitor	Minimum pressure fallen below. Check compressed air supply
(18)	Fault Climate control unit E25	Read the display, identify and clear the fault using the subvendor documentation.
(19)	Winder not ready	Winder Establish the ready-for-operation status, switch on the winder
(20)	Fault: Web break Inner web	Paste on the web. Check the web break detector for proper functioning, replace, if necessary.
(21)	Fault: Applicator Dabber cross pasting	Read the display, identify and clear the fault using the subvendor documentation.
(22)	Fault: Inverter Draw roller: {Draw roller}	Fault number ==> see B&R's user manual. Read the fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(23)	Fault: Inverter Seam cooling: {Seam cooling}	Fault number ==> see B&R's user manual. Read the fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(24)	Fault: Inverter Seam extruder: {Extruder}	Fault number ==> see B&R's user manual. Read the fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(25)	Fault: Safety shutdown board Extruder	Acknowledge fault. Check the safety shutdown board for proper functioning, replace, if necessary.
(26)	Brake Unwind Inner web switched Off	Switch on Brake Unwind Inner web
(27)	Brake Unwind Outer web switched Off	Switch on Brake Unwind Outer web
(28)	Feedback loop of guard doors faulty	Control voltage switched back on too early after switch-off (waiting time at least 5 seconds), check feedback loop of guard doors for proper functioning
(29)	No control voltage	Switch on the control voltage.
(30)	Locking of drives	Locking of the drives by a key switch (+R20)
(31)	Extruder not in start position	Move the extruder to the start position, check the limit switch position, replace, if necessary.
(32)	Fault: Web break Outer web	Paste on the web. Check the web break detector for proper functioning, replace, if necessary.
(33)	Quick stop	Machine has stopped with a quick stop
(34)	Extruder moved too far, limit position reached	Move the extruder to the start position, check the limit switch position, replace, if necessary.
(35)	Group alarm	General indicator if a fault has occurred

ID	Fault message	Troubleshooting
(36)	{Extruder} Automatic mode faulty	Extruder screw does not turn, check extruder drive
(37)	Fault: Climate control unit Winder	Read the display, identify and clear the fault using the sub-vendor documentation.
(38)	Fault: Pressure Extruder	Clean application head nozzles, replace, if necessary. Melt pressure sensor, replace, if necessary.
(39)	Fault: Gusset forming tower	Clear fault at the gusset forming tower.
(40)	Fault: Print mark detector	Align print mark detector, readjust contrast at the sensor, replace, if necessary.
(41)	Fault: Inverter Oscillation: {Oscillation}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(42)	Gusset forming tower: Max. dancer roller position reached	Advance web until the dancer roller is in the mid-position.
(43)	Gusset forming tower: Manual mode selected	Switch on automatic mode
(44)	Draw swung out	Swing in draw
(45)	Fault: Rewind unit	Clear fault at the rewind.
(46)	No resin in the extruder	Refill resin, check filling level sensor, replace, if necessary.
(47)	Web break, dancer roller in its end position	Advance web until the dancer roller is in the mid-position.
(48)	Laser not ready	Laser Establish the ready-for-operation status, switch on the laser.
(49)	Setup mode active	Switch off setup mode
(50)	Unwind not ready	Unwind Establish the ready-for-operation status, switch on the unwind.
(51)	Fault: safety PLC	Watch-Dog for safety PLC G99.L05 faulty, check safety PLC G99.L05 im + E25, replace, if necessary.
(52)	Fault: Plausibility check Guard doors	Check the guard doors, check the guard door switches for proper functioning, replace, if necessary
(53)	Fault: Emergency stop chain	Unlatch the Emergency-Sstop button and acknowledge the fault. Check the Emergency-Stop button for proper functioning and replace, if necessary.
(54)	Fault: Plausibility check Rope pull switch	Unlock the rope pull switch, set up the rope pull switch, replace, if necessary
(55)	Fault: Unwind	Clear the fault at the unwind.
(56)	Fault: Inverter Sensing roller: {Sensing roller}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.

ID	Fault message	Troubleshooting
(57)	Fault: Check Seam	Set up the monitored temperature range (min./max.). Align the sensor, replace, if necessary.
(58)	Fault: Inverter Counter roller Interm. Perforator: {CR_Per}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(59)	Fault: Inverter Counter roller Interm. embossing unit: {CR_Emb}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(60)	Punch not ready	Punch Establish ready-for-operation status, switch on the punch.
(61)	Fault: Inverter Cycle Interm. Perforator OS: {Cycle_Per_OS}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(62)	Fault: Inverter Tool Interm. Perforator OS: {Tool_Per_OS}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(63)	Fault: Inverter Cycle Interm. Perforator DS: {Cycle_Per_DS}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(64)	Fault: Inverter Tool Interm. Perforator DS: {Tool_Per_DS}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(65)	Fault: Inverter Cycle Interm. Embossing unit DS: {Cycle_Emb_DS}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(66)	Fault: Inverter Tool Interm. Embossing unit DS: {Tool_Emb_DS}	Fault number ==> see the B&R/ ACOPOS user manual Diagnosis in panel Read fault text and inform W&H's Service Department. Disconnect the controller from the mains, if necessary (main switch Off/On), restart the machine.
(67)	Longitudinal seam Hot Melt not ready	Longitudinal seam Hot Melt Establish ready-for-operation status, switch on Longitudinal seam Hot Melt.
(68)	Fault: Hot Melt Longitudinal seam	Read the display, identify and clear the fault using the subvendor documentation.
(69)	Fault: Web break Main draw	Paste on the web. Check the web break detector for proper functioning, replace, if necessary.
(70)	Fault: Web monitoring Tube former	Check tube position in the tube former, check web sensor in the tube former for proper functioning
(71)	Fault: Oscillation	Check winder oscillation, sluggishness of the spindle, limit switch proximity sensors

ID	Fault message	Troubleshooting
(72)	Fault: Inline printing unit	Check the Flexa 600, the error message is displayed on the panel
(73)	Emergency-Stop button hit Inline printing unit	Check the Emergency-button on the Flexa 600
(74)	Set-up mode: Inline printing unit	Set-up mode: Inline printing unit
(75)	Web travel error: Inline printing unit	Check the web for proper traveling through the Flexa printer, activate or deactivate the printing unit at the POLYTEX
(90)	Guard door open Perforator Outer web	Close the guard door. Check the safety shutdown switches on the guard door for proper functioning. Set up or replace, if necessary.
(91)	Guard door open Perforator Outer web	Close the guard door. Check the safety shutdown switches on the guard door for proper functioning. Set up or replace, if necessary.
(92)	Guard door open Embossing unit	Close the guard door. Check the safety shutdown switches on the guard door for proper functioning. Set up or replace, if necessary.
(93)	Guard door open Intermittent tool	Close the guard door. Check the safety shutdown switches on the guard door for proper functioning. Set up or replace, if necessary.
(94)	Guard door open Convergence	Close the guard door. Check the safety shutdown switches on the guard door for proper functioning. Set up or replace, if necessary.

ID	Warning	Troubleshooting
(1)	Unwind Outer web: BST End position reached	Ply ran out of the edge sensor range, realign web
(2)	Unwind Inner web: BST End position reached	Ply ran out of the edge sensor range, realign web
(3)	Min. filling level Extruder	Confirm warning==> Decision: refill or run until empty.
(4)	Pressure Extruder > {MaxPress} bar	Acknowledge warning==> Extruder pressure exceeds set warning pressure
(5)	Extruder heater to be switched off soon	Acknowledge warning==> Heater will not be switched off
(6)	Min. reel diameter Inner web	Acknowledge warning==> Minimum reel diameter reached, change reel
(7)	Min. reel diameter Outer web	Acknowledge warning==> Minimum reel diameter reached, change reel
(10)	BST Manual adjustment selected at extruder	Deselect manual adjustment for changeover to automatic mode
(11)	Temperature reduction was selected at the extruder	Deselect temperature reduction
(13)	Max. number of cycles Punch	Reduce machine speed

ID	Warning	Troubleshooting
(14)	Inline printing unit Continuous rotation ready	Activate continuous rotation on the Flexa 600

6.2 Web travel

6.2.1 Safe behavior 284
Web thread-up - Using step-ladder 284
Threading in the web 284

6.2.2 Unwinding unit 285

6.2.3 Micro-perforator, outer web 286

6.2.4 Continuous embossing unit, outer web 287

6.2.5 Main draw 288

6.2.6 Cooling zone 288

6.2.1 Safe behavior

Web thread-up - Using step-ladder

CAUTION

Risk of injury if you fall

Always fit railings or other safety barriers. Use a working platform. Put on safety harness.

When threading the web, use a step-ladder or step.

Only load a step-ladder based on the given values.

Pay attention to the information signs.

Always face the step-ladder when ascending and descending, holding onto the ladder with both hands.

Do not use machine parts as climbing aids.

Wear safety equipment to prevent falling when carrying out maintenance work at great heights.

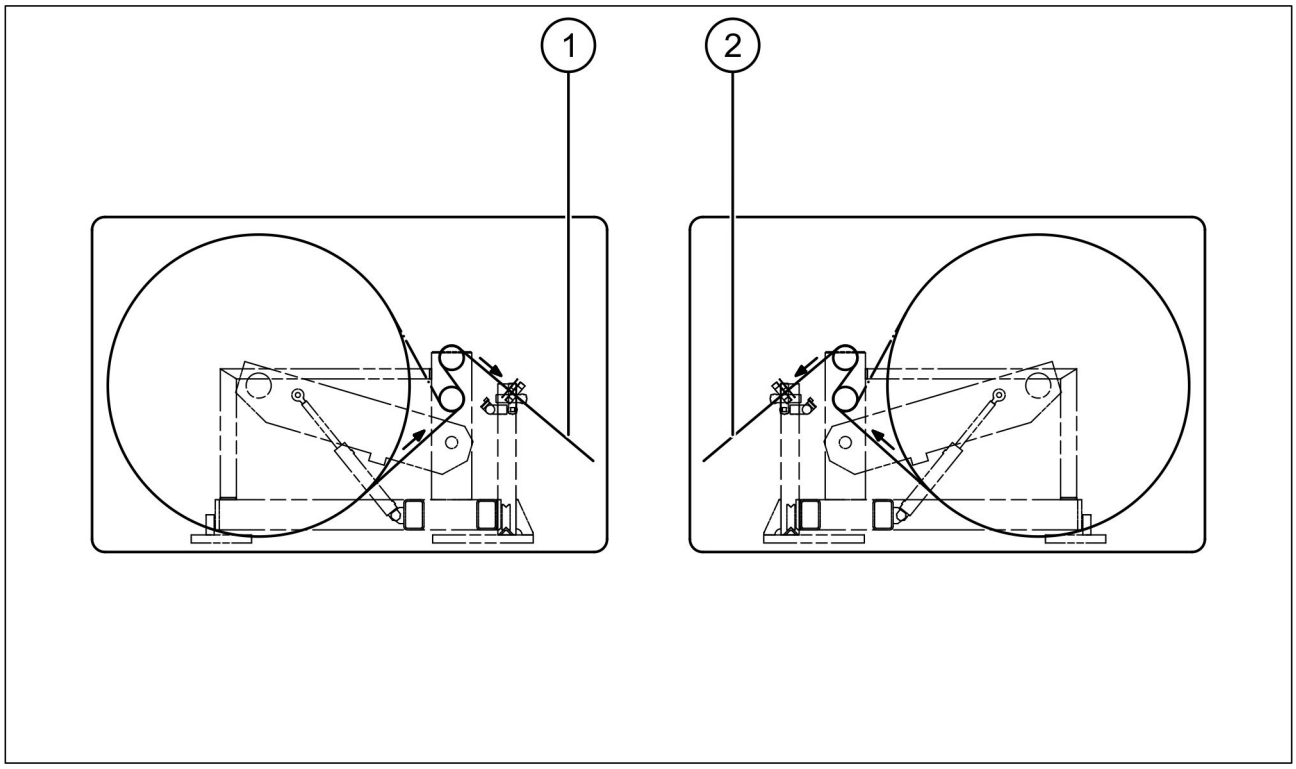
Check and maintain step-ladder at regular intervals. Adopt the intervals and extent from the relevant regulations.

Threading in the web

Always have a second person present when threading in the material web.

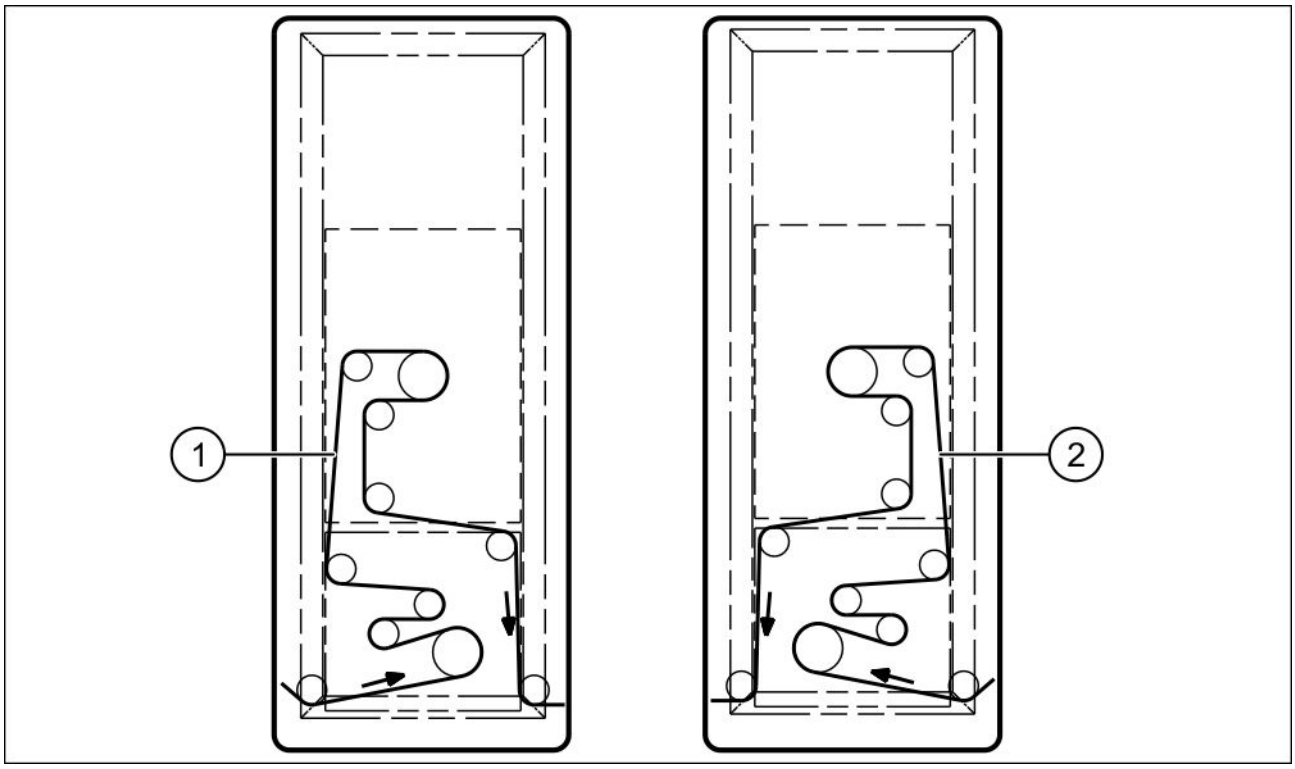
In case of a web break and when changing material, glue the material web to be threaded in to the material web already in the machine and using this pull it through the machine. This avoids the need for a complete re-threading of the web.

6.2.2 Unwinding unit



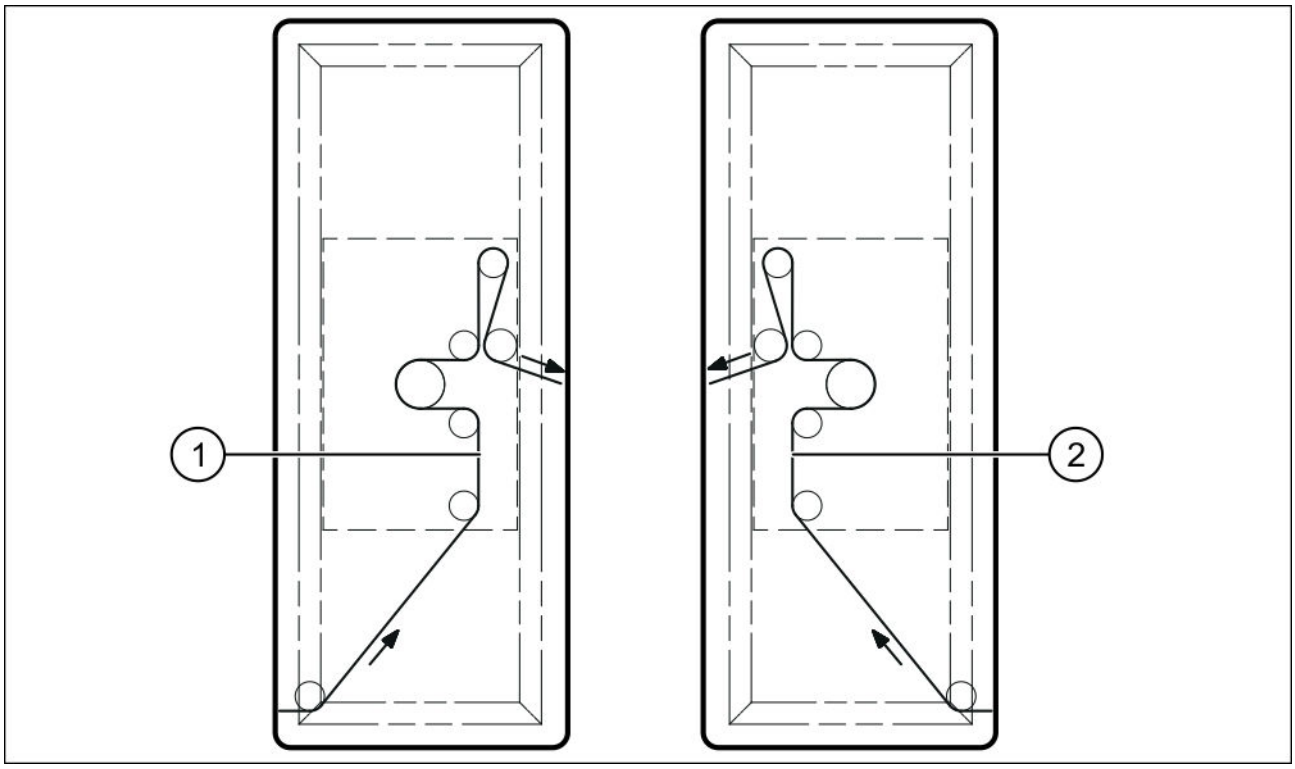
(1)	Web, view from the operator side	(2)	Web, view from the drive side

6.2.3 Micro-perforator, outer web



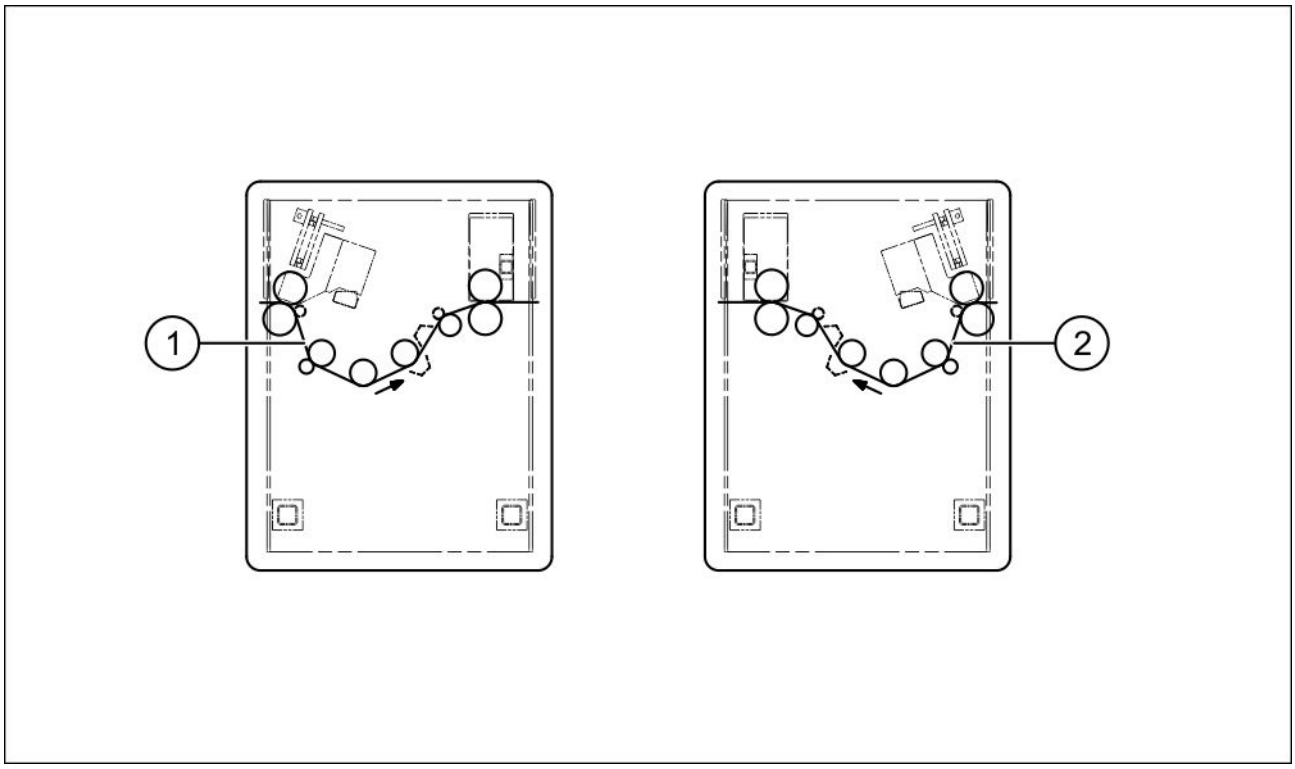
(1)	Web, view from the operator side	(2)	Web, view from the drive side

6.2.4 Continuous embossing unit, outer web



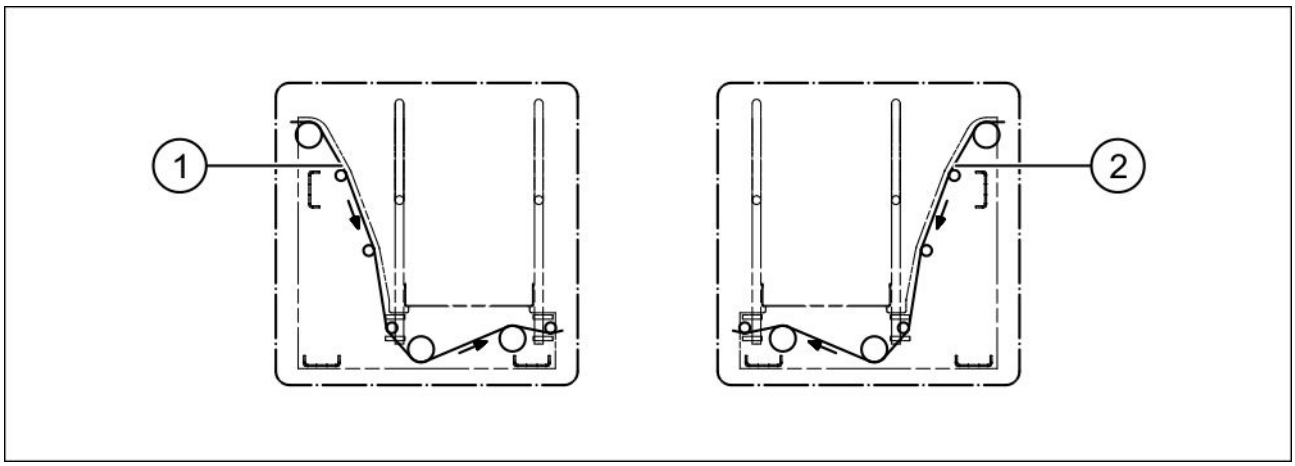
(1)	Web, view from the operator side	(2)	Web, view from the drive side

6.2.5 Main draw



(1)	Web, view from the operator side	(2)	Web, view from the drive side

6.2.6 Cooling zone



(1)	Web, view from the operator side	(2)	Web, view from the drive side

6.3 Seam extruder

6.3.1 Extrusion lines 290
Heated line parts 290
Plastic melt 290
Production 290
Cleaning 290
Risk of being pulled in through rotating chains and belts 290
Risk of being pulled in by the rollers 290
Risk of being pulled in at infeed nips 290
Danger of being cut 290
Danger of falling 290
Electricity 290
Melt pressure sensor 290
Melt temperature 290
Melt pressure 290
Safety information in the sub-vendor documentation 290

6.3.1 Extrusion lines

The machine represents the state of the art and it conforms to the latest safety standards. Nevertheless, using the machine can cause danger to life and limb of the operating personnel. Only trained and qualified personnel may operate the machine. The machine operating company must provide the operating personnel with regular training in safety procedures.

Heated line parts

CAUTION

Risk of injuries due to hot surfaces, temperatures exceeding 60°C / 140°F

Protective clothing must be worn such as protective gloves against being burnt and tight-fitting clothes with long sleeves.

To prevent dangerous situations from occurring, put up warning signs reading 'HOT MACHINE PARTS'.

WARNING

Fire hazard through contact with inflammable material

Make sure that no inflammable materials such as textiles, grease, paraffin, oils or hydraulic oils come into contact with hot surfaces.

WARNING

Risk of injuries due to tripping over heated up machine parts

Do not climb over the extruder connecting point and the die connecting point but slowly.

Plastic melt

CAUTION

Burns resulting from escaping plastic melt

Protective clothing must be worn against escaping hot melt and fumes that might endanger your health, i.e. industrial safety gloves, safety goggles and your clothes should be close-fitting with long sleeves.

WARNING

Injuries due to tripping over melt residue on surfaces and paths

Properly remove melt residues immediately.

Provide melt residue collecting containers and position them in accordance with the regulations governing safe movement.

Production

CAUTION

Sickness and fatigue through ascending gasses

Make sure that the operator's workplace is sufficiently ventilated.

Make sure that the time spent near to the melt is minimized.

Always abide by the details given in the safety data sheets.

Cleaning

WARNING

Skin, eye and breathing tract irritation due to incorrect cleaning agents

Only use recommended cleaning agents and mark cleaning agents.

Ensure for sufficient room ventilation.

Always abide by the details given in the safety data sheets.

Risk of being pulled in through rotating chains and belts

DANGER

Risk of injuries due to rotating chains, belts, gears and wheels

Parts of your body, clothes, hair and objects can be pulled in and crushed. Do not open or remove protective devices. Always wear tight-fitting clothes and a hair net. Work without industrial safety gloves.

Keep a safe distance.

Risk of being pulled in by the rollers

DANGER

Risk of injury from rotating rollers

Parts of your body, clothes, hair and objects can be pulled in and crushed. Always wear tight-fitting clothes and a hairnet. Work without industrial safety gloves. Do not reach into the nip.

Keep a safe distance.

X000037755

Risk of being pulled in at infeed nips

DANGER

Injuries can be caused by infeed nips

Parts of your body, clothes, hair and objects can be pulled in and crushed. Always wear tight-fitting clothes and a hairnet. Work without industrial safety gloves.

Keep a safe distance.

Danger of being cut

WARNING

Injuries through cutting blades or circular knives

Wear cut-proof safety gloves when threading the film web and when replacing the blades or knives.

Danger of falling

DANGER

You can be injured if you fall

Never use machine and railing elements, especially on platforms, as a climbing aid.

Always secure platform openings using suitable measures such as covers or safety fences to prevent anyone from stepping into or falling through them.

Utilize suitable climbing aids and platforms when threading up the film or when performing assembly work.

Electricity

DANGER

Injuries due to electric shock when assembling and removing the heaters

Disconnect the power supply for the heaters before removal and secure against being switched back on. Check the safe isolation from supply. Ground and short-circuit the machine part.

Alternatively, have an electrician switch off the load disconnect switch for the heating zone or the automatic circuit-breaker for the heater.

Keep a safe distance.

Melt pressure sensor

DANGER

Burns due to hot parts and melt

Wear personal protective clothing such as protective gloves against burns and safety goggles or face protection against hot melt splashes when dismantling the melt pressure sensor.

Only carry out work on the melt pressure sensor when the extruder is depressurized. Switch off extruder. Wait at least 30 minutes until beginning the dismantling.

Observe the operating instructions from the component manufacturer.

Melt temperature

WARNING

Burns due to hot surfaces or plastic melt

The maximum permitted melt temperature (monitoring and temperature limiting through temperature control system) may not be exceeded.

Hot machine parts must be protected by safety guards. Hot surfaces which are required to operate the machine and which cannot be covered must be labeled accordingly.

Melt pressure

WARNING

Risk of injuries due to protruding machine parts

The maximum permitted melt pressure (monitoring and pressure limiting through pressure control system) may not be exceeded.

Safety information in the sub-vendor documentation

The supplier documentation in the 'Sub-vendor documentation' contains the supplier's own safety information. These must be observed by the operating personnel of the machine operating company.

7 Maintenance instructions

7 Maintenance instructions

7.1 Scope of maintenance works ... 296
Scope of maintenance works ... 296
Cleaning ... 296
Inspection ... 296
Lubrication ... 296
Time intervals ... 296
Sub-vendor documentation ... 296

7.2 Maintenance staff ... 297

7.3 Safety instructions ... 297
Switching off the machine ... 297
Power-driven technical equipment ... 297
Removing safety devices ... 297
Overhead installation work ... 297
Cleaning agents: ... 297

7.4 Corrosion protection measures ... 298
Hints for using cleaning agents ... 298
Removing the corrosion protection coating ... 298
Restoring the corrosion protection coating ... 298

7.5 Corrosion protection ... 299

7.1 Scope of maintenance works

Scope of maintenance works

The chart gives you an overview of the maintenance works described in this instruction manual. Maintenance is divided into works to be carried out at regular intervals or product-related works.

Cleaning the machine	Describes cleaning works to be carried out at regular intervals to prevent machine downtimes and maintain the machine in a good functional condition.
Maintaining the machine	Describes maintenance works to be carried out at regular intervals to prevent machine downtimes and maintain the machine in a good functional condition.
Lubricating the machine	Describes lubrications to be carried out at regular intervals to prevent machine downtimes and maintain the machine in a good functional condition.
Performing repair works	Describes various production-related works on the machine which are necessary to restore the functional condition of the machine.

Cleaning

- Keep the machine clean.
- Remove production-related soiling according to the specified intervals.
- Perform basic cleanings at regular intervals.

Inspection

Malfunctions due to insufficient or improper maintenance are liable to cause expensive repairs and extended machine down time.

Maintenance should be carried out according to the time intervals specified.

Lubrication

Malfunctions due to insufficient or improper lubrication will result in high cost of repairs and long machine downtimes.

Carry out lubrications according to the lubrication chart and the lubricant specifications.

Lubricants from various manufacturers may be used provided their properties are the same.

Time intervals

The time intervals are recommendations. They depend on the daily period of use and the operating conditions and should be customized, if necessary.

Sub-vendor documentation

The chapter under the heading "Sub-vendor documentation" contains more detailed information on cleaning, routine maintenance, lubrication, repair work and maintenance of the machine.

7.2 Maintenance staff

Only qualified maintenance staff as described under the heading Qualification of the personnel is allowed to do maintenance work on the machine.

Operators must be informed before starting any maintenance work.

Maintenance work should be carried out at the intervals specified so that the machine can be operated in a safe manner.

7.3 Safety instructions

For all maintenance works, the following safety instructions apply. The maintenance staff must heed the safety instructions and follow the measures to prevent dangerous situations for persons from occurring.

Switching off the machine

DANGER

Risk of injury from movements at and in the machine

Risk of personal injury from machine movements when working on the machine when switched on.

Switch off the machine and secure against being switched back on inadvertently. Works should only be carried out when the machine is switched off.

DANGER

Electricity

Risk of life-threatening injuries when getting in contact with live components.

Disconnect the power supply and secure against being switched back on. Check the safe isolation from supply. Ground and short-circuit the machine part.Any adjacent parts that are still live must be covered. Only electrically skilled personnel are allowed to carry out works on the unit. Keep a safe distance.

Potential risks for persons may occur during maintenance work. During maintenance work the machine must be switched off and cut off from power supply. Prevent the machine from being switched back on inadvertently. Only the person who switched off the machine is allowed to switch the machine back on. Cordon off the working area.Make sure that the hazardous area is clear of persons if the machine must be moved or power supply be switched on during maintenance. Maintenance may not be continued unless the machine is switched off again, cut off from power supply and secured against being switched back on.

Power-driven technical equipment

Adjustments to power-driven technical equipment on the machine and maintenance work on moving parts is prohibited.

Disengage the power-driven technical equipment before starting any maintenance work.

Removing safety devices

If any of the provided safety devices must be removed to carry out the required maintenance work, they must be refitted immediately after maintenance work has been finished. After refitting, check the safety devices for proper functioning.

Refit and close all removed or opened safety devices immediately after completion of the maintenance work.

Overhead installation work

Always use the climbing aids or work platforms intended for this purpose that meet the safety requirements.

Do not use machine parts as climbing aids.

Cleaning agents:

CAUTION

Cleaning agents

Cleaning agents may cause skin, eye and respiratory tract irritations.

Always observe the safety data sheet applicable to the product when handling cleaning agents. Observe the health and safety instructions on the product label. Wear safety clothing!Ensure for sufficient room ventilation.

7.4 Corrosion protection measures

Hints for using cleaning agents

NOTE

All of the used cleaning agents and cleaning waste are classified as hazardous waste.

Hazardous waste must be disposed of in accordance with local regulations and laws.

Make sure that the cleaning agents being used do not corrode the plastic and rubber surfaces and dissolve the splices.

Removing the corrosion protection coating

Factory-applied corrosion protection coating

The factory-applied corrosion protection coating is a concentrate made from organic corrosion inhibitors, bound in a carrier medium of naphthenic mineral oil. The corrosion protection coating is free from solvents, nitrites, phosphates and chromium. It contains no PCBs or PCTs. Product name: CORTEC VpCI 329 and CORTEC VpCI 329 F.

Use an alkaline based cleaner (tenside water) to clean off the corrosion protection coating applied to all non-treated surfaces in the factory or remove by wiping off before initial machine startup.

Restoring the corrosion protection coating

If the machine is to be shut down for a long period, all treated surfaces must be protected against corrosion.

When cleaning a non-treated surface, corrosion protection on the cleaned surface should be restored.

7.5 Corrosion protection

Special corrosion protection measures must be taken if the relative humidity on site permanently exceeds 80 percent or daily temperature fluctuations in the factory hall exceed 10 degrees or aggressive dusts deposit on the machine.

Weeky corrosion protection is required for all untreated surfaces (see 'Cleaning' and 'Lubrication') not scheduled for regular cleaning and lubrication.

After cleaning, apply a thin coat of oil to the untreated surfaces. Use a suitable oil for spraying or brushing.

Use silicone oils for surfaces getting in contact with the material to be converted (food hygiene). For all other surfaces, CLP 46 oil must be used.

Proceed as follows:

- Wipe untreated surfaces to remove oil residues and dirt.
- Apply an evenly thin coat of oil to the cleaned surfaces. Wipe off excessive oil.

8 Maintenance

8.1 Cleaning the machine

8.1.1 Unwind – Web break detector
8.1.2 Web guide control
8.1.3 Sliding doors
8.1.4 Micro-perforator
8.1.5 Embossing unit
8.1.6 Gusseted tube forming unit
8.1.7 Longitudinal seam extruder
8.1.8 Main draw
8.1.9 Longitudinal seam monitoring
8.1.10 Main draw - Slide protection
8.1.11 Oscillation
8.1.12 Contact roller
8.1.13 Center drive
8.1.14 Cross cutting system
8.1.15 Hydraulic unit
8.1.16 Delivery arms
8.1.17 Electrostatic charging and discharging

8.2 Maintaining the machine

8.2.1 Winding shaft
8.2.2 Unwind
8.2.3 Unwind – Web break detector
8.2.4 Web guide control
8.2.5 Draw and dancer roller
8.2.6 Sliding doors
8.2.7 Micro-perforator
8.2.8 Embossing
8.2.9 Gusseted tube forming unit
8.2.10 Longitudinal seam extruder
8.2.11 Main draw
8.2.12 Main draw - Slide protection
8.2.13 Compressed-air conditioner
8.2.14 Equipment
8.2.15 Parallel guide
8.2.16 Contact roller
8.2.17 Hydraulic unit
8.2.18 Pneumatic system
8.2.19 Air conditioner for the switch cabinet

8.3 Lubricating the machine

8.3.1 Lubricants
8.3.2 Sliding doors
8.3.3 Draw and dancer roller
8.3.4 Embossing
8.3.5 Gusseted tube forming unit
8.3.6 Longitudinal seam extruder
8.3.7 Longitudinal seam monitoring
8.3.8 Main draw
8.3.9 Oscillation
8.3.10 Dancer roller
8.3.11 Center drive
8.3.12 Hydraulic unit

8.1 Cleaning the machine

8.1.1 Unwind – Web break detector 303

8.1.2 Web guide control 303

8.1.3 Sliding doors 304
Sliding door 304
Limit switch 304

8.1.4 Micro-perforator 305

8.1.5 Embossing unit 305

8.1.6 Gusseted tube forming unit 306
Guide elements 306
Roller 306

8.1.7 Longitudinal seam extruder 307
Rail 307
Applicator 307

8.1.8 Main draw 308
Guide rail - Positioning the seam cooling roller 308
Guide rail - Pivoting the seam cooling roller 308
Seam cooling roller 308

8.1.9 Longitudinal seam monitoring 309

8.1.10 Main draw - Slide protection 310
Slide protection, infeed section 310
Slide protection, outfeed section 310

8.1.11 Oscillation 312

8.1.12 Contact roller 312
Contact roller drive 312
Contact roller coating 312
Light barrier on the winder pull-in gap 312

8.1.13 Center drive 314

8.1.14 Cross cutting system 314

8.1.15 Hydraulic unit 315

8.1.16 Delivery arms 315

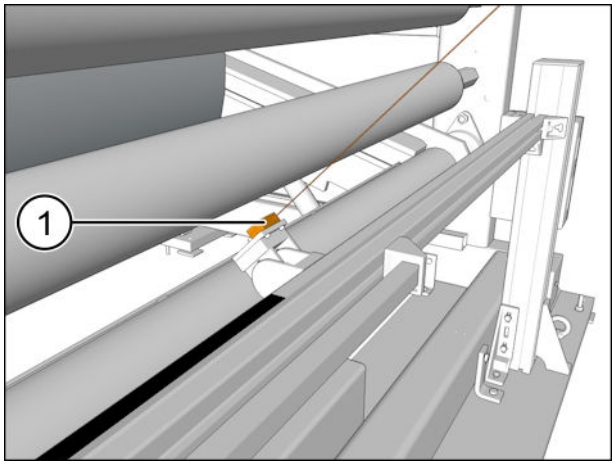
8.1.17 Electrostatic charging and discharging 316

8.1.1 Unwind – Web break detector

Cleaning the sensor

(1)	Sensor

▶ Clean the sensor (1).
Interval: 200 hours
Use: clean, soft cloth

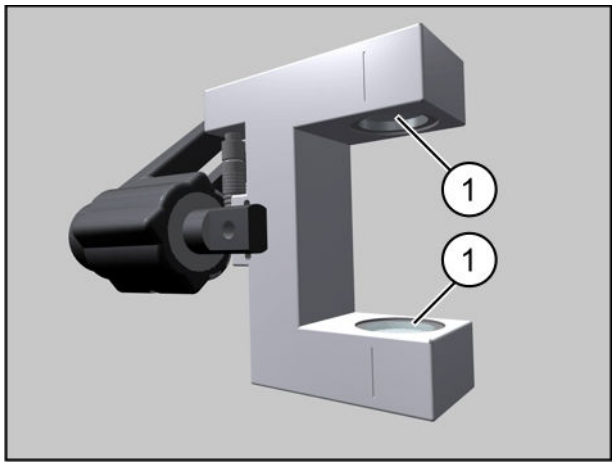


8.1.2 Web guide control

Cleaning the sensor

(1)	Sensor

▶ Clean the sensor (1).
Interval: 200 hours
Use: clean, soft cloth



After cleaning the sensor, carry out a material setup.

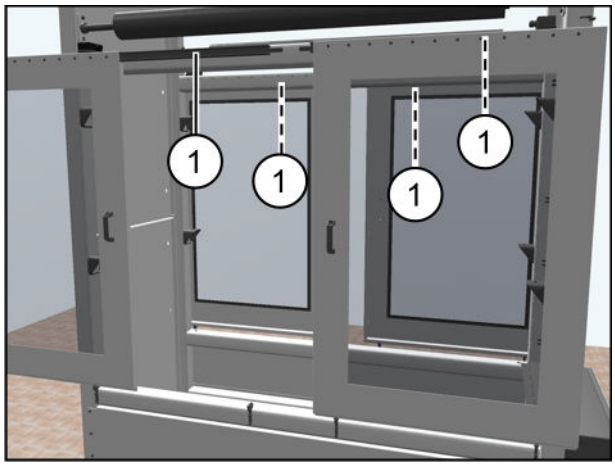
See section Web guide control - Material setup.

8.1.3 Sliding doors

Sliding door

Cleaning the telescope rail

(1)	Telescope rail



▶ Clean the telescope rail (1).

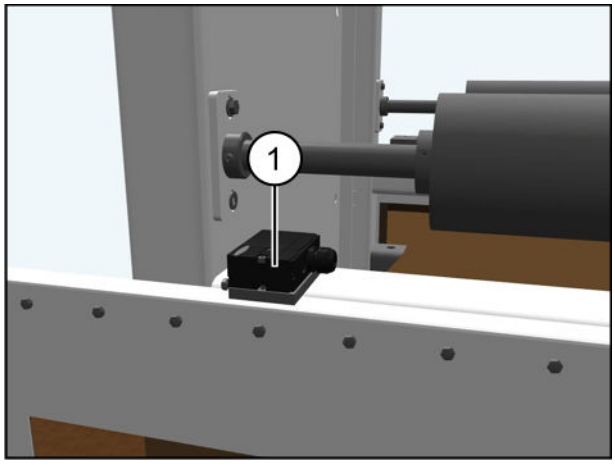
Interval: 1,000 hours
Use: clean, soft cloth

Limit switch

Cleaning the limit switch

Clean all limit switches regularly to ensure that they function correctly.

(1)	Limit switch



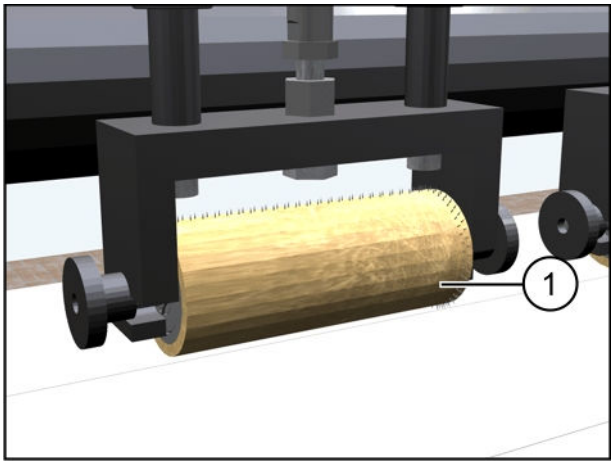
▶ Clean the limit switch (1).

Interval: 200 hours
Use: clean, soft cloth

8.1.4 Micro-perforator

Cleaning the needle roller

(1)	Needle roller



▶ Clean the needle roller (1).

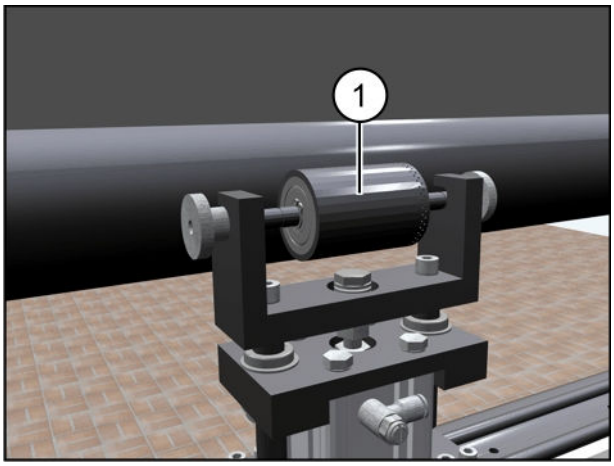
Interval: 500 hours
Use: vacuum air, coarsely bristled soft brush

Take care when cleaning not to bend the needles.

8.1.5 Embossing unit

Cleaning the embossing roller

(1)	Embossing roller



▶ Clean the embossing roller (1).

Interval: 500 hours
Use: vacuum air, coarse, soft brush

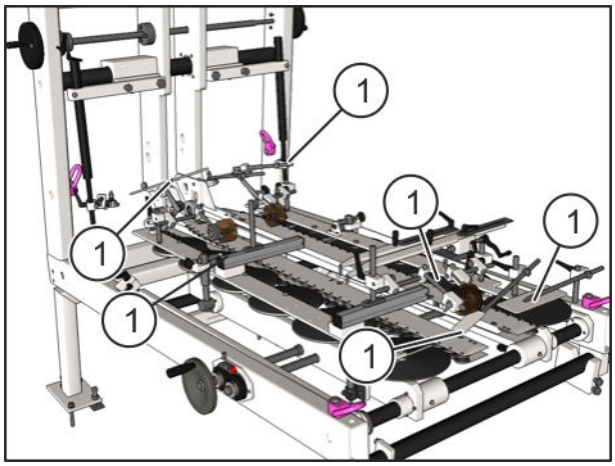
Take care when cleaning not to bend the needles.

8.1.6 Gusseted tube forming unit

Guide elements

Cleaning the guide elements

(1)	Guide elements



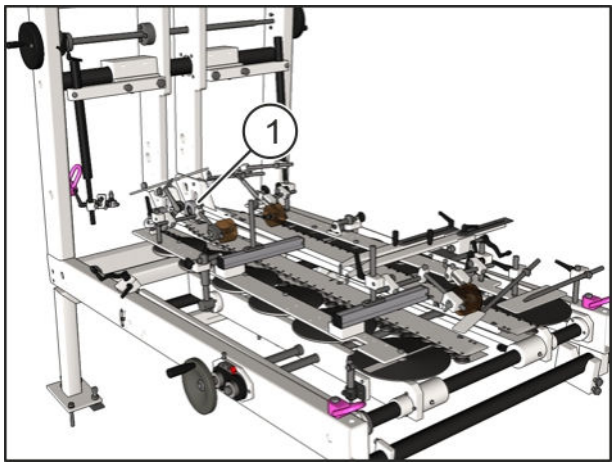
▶ Clean the guide elements (1).

Interval: 40 hours
Use: damp, clean cloth, plastic scraper

Roller

Cleaning the roller

(1)	Roller



▶ Clean the roller (1).

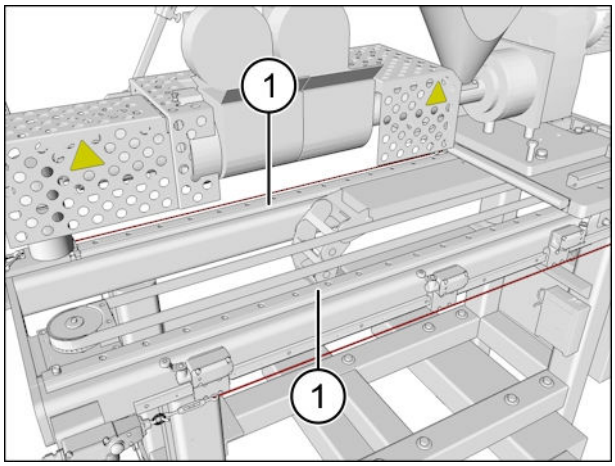
Interval: 500 hours
Use: clean, soft cloth

8.1.7 Longitudinal seam extruder

Rail

Cleaning the rail

(1)	Rail

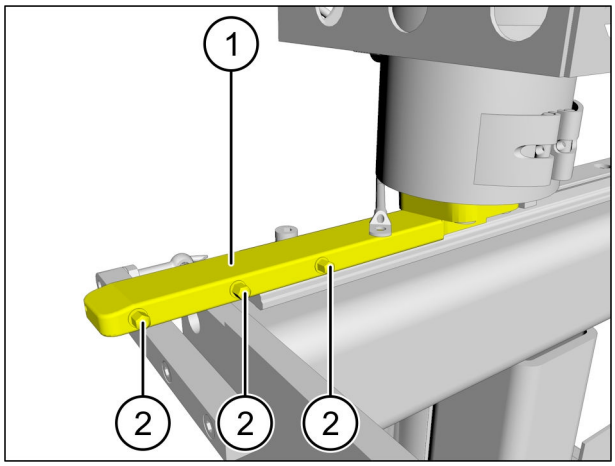


▶ Clean the rail (1) every 8 hours.
Use: neat cloth

Applicator

Cleaning the applicator and nozzles

(1)	Applicator
(2)	Nozzle



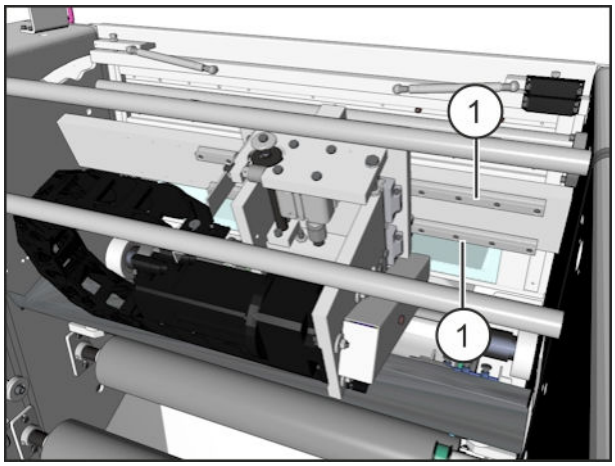
▶ Clean the applicator (1) and nozzles (2) before starting the machine.
Use: brass wire brush, copper spatula, wooden scraper and cleaning wool
Remove melt residue completely.
If required, stop the production for cleaning the applicator and nozzles.

8.1.8 Main draw

Guide rail - Positioning the seam cooling roller

Cleaning the guide rail

(1)	Guide rail



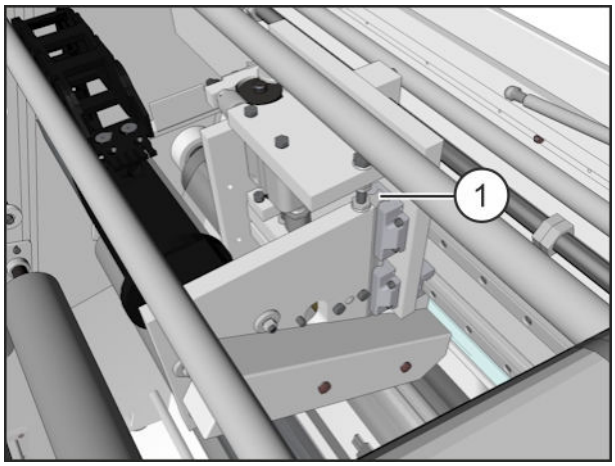
▶ Clean the guide rails (1).

Interval: 1,000 hours
Use: clean cloth

Guide rail - Pivoting the seam cooling roller

Cleaning the guide rail

(1)	Guide rail



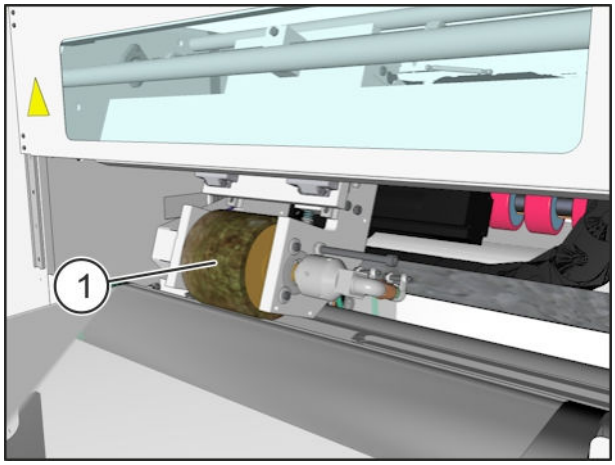
▶ Clean the guide rails (1).

Interval: 1,000 hours
Use: clean cloth

Seam cooling roller

Cleaning the roller

(1)	Roller



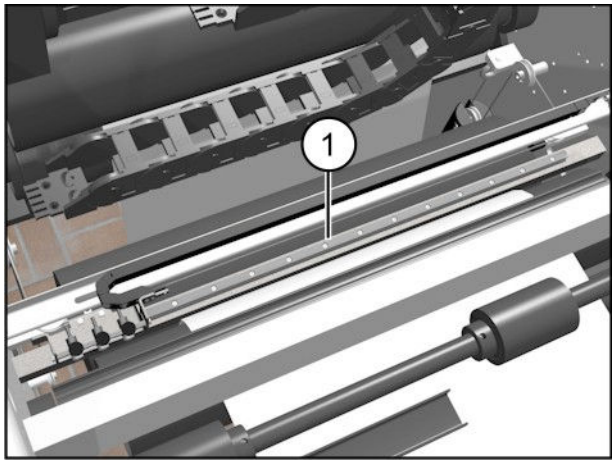
▶ Clean the roller (1).

Interval: 500 hours
Use: clean cloth

8.1.9 Longitudinal seam monitoring

Cleaning the rail

(1)	Rail



▶ Clean the rail (1) every 8 hours.

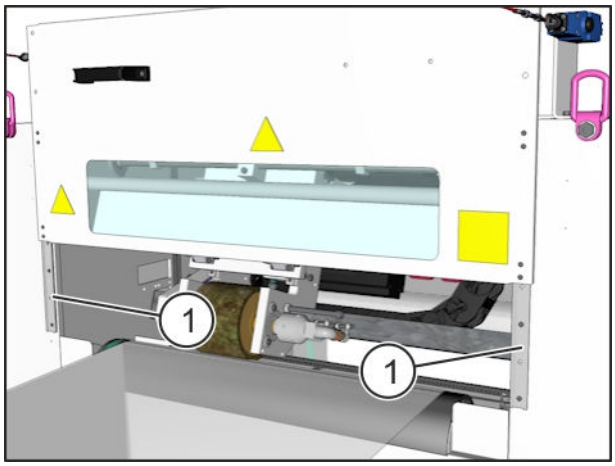
Use: clean cloth

8.1.10 Main draw - Slide protection

Slide protection, infeed section

Cleaning the telescope rail

(1)	Telescope rail



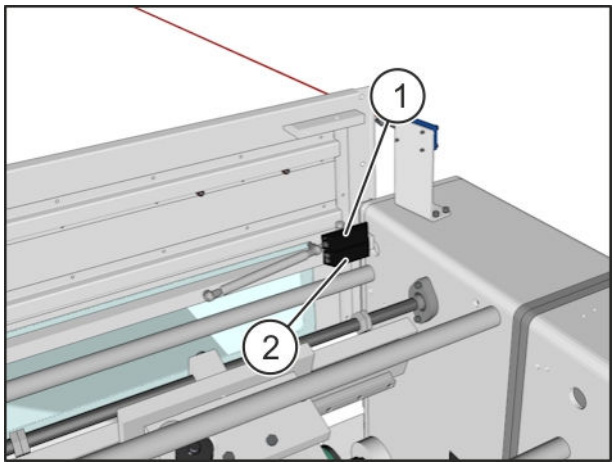
▶ Clean the telescope rail (1).

Interval: 1,000 hours
Use: clean, soft cloth

Cleaning the safety limit switches

All safety limit switches need cleaning at regular intervals to ensure proper functioning.

(1)	Sensor
(2)	Magnet



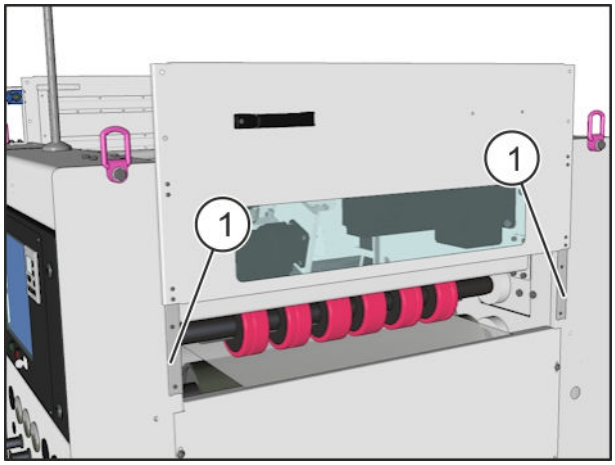
▶ Clean the sensor (1) and the magnets (2).

Interval: 200 hours
Use: clean, soft cloth

Slide protection, outfeed section

Cleaning the telescope rail

(1)	Telescope rail



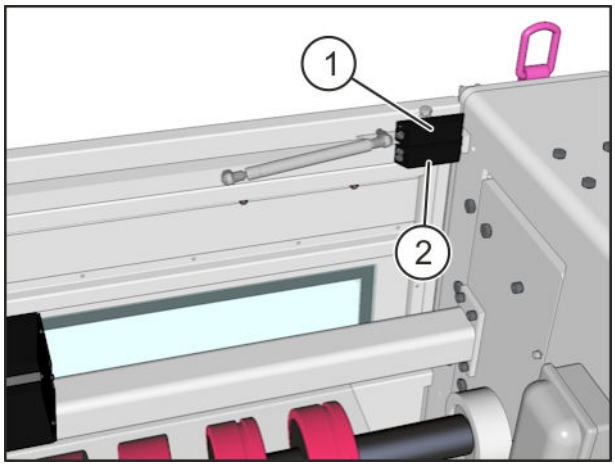
▶ Clean the telescope rail (1).

Interval: 1,000 hours
Use: clean, soft cloth

Cleaning the safety limit switches

All safety limit switches need cleaning at regular intervals to ensure proper functioning.

(1)	Sensor
(2)	Magnet



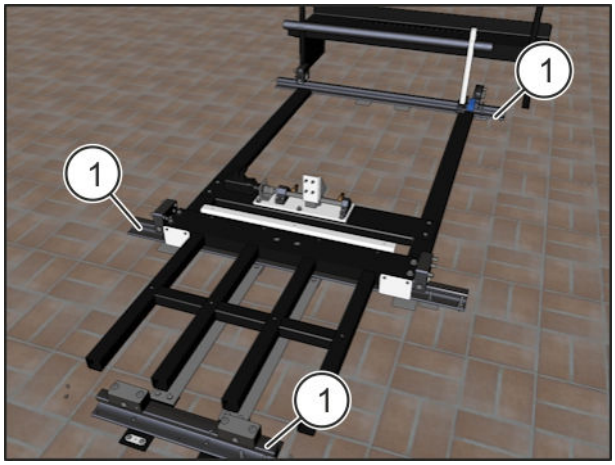
▶ Clean the sensor (1) and the magnets (2).

Interval: 200 hours
Use: clean, soft cloth

8.1.11 Oscillation

Cleaning the rail

(1)	Rail



▶ Clean the rails (1).

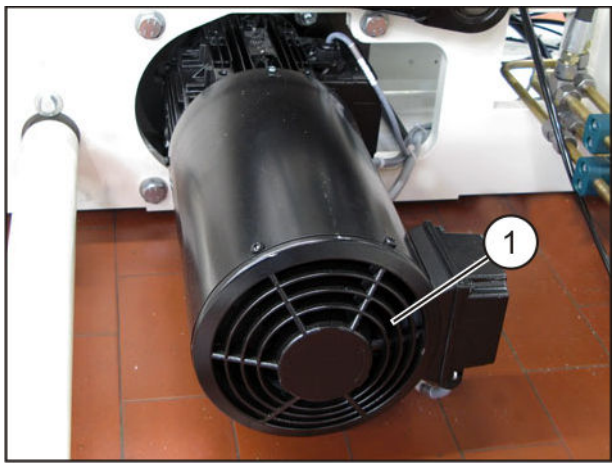
Interval: 2,000 hours
Use: clean, soft cloth

8.1.12 Contact roller

Contact roller drive

Clean the vent grating

(1)	Vent grating



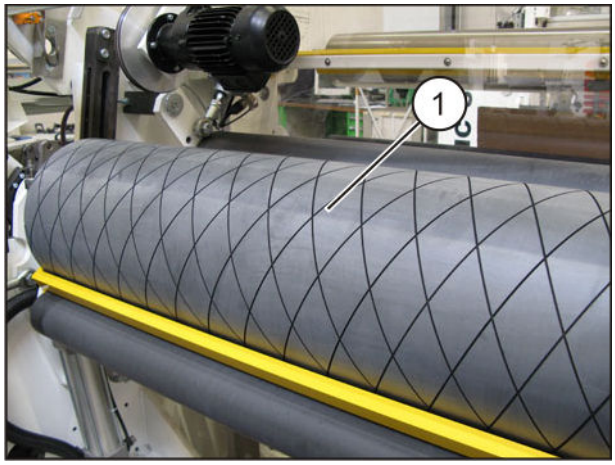
▶ Clean the vent grating (1) of the contact roller drive.

Interval: 3,000 hours
Use: Large brush, vacuum-cleaner and cleaning cloth
The cleaning interval can be reduced or extended, depending on the working environment.

Contact roller coating

Clean the surface

(1)	Contact roller



- Clean roller coating of the contact roller (1).

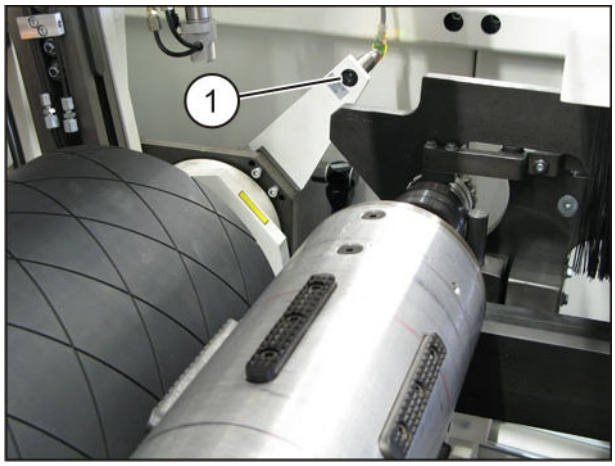
  Interval: 1,000 hours
  Use: Cleaning cloth, soft brush, soap solution, alcohol (spirits)

  The cleaning interval can be reduced or extended, depending on the working environment.

Light barrier on the winder pull-in gap

Clean the transmitter / receiver

(1)	Light barrier



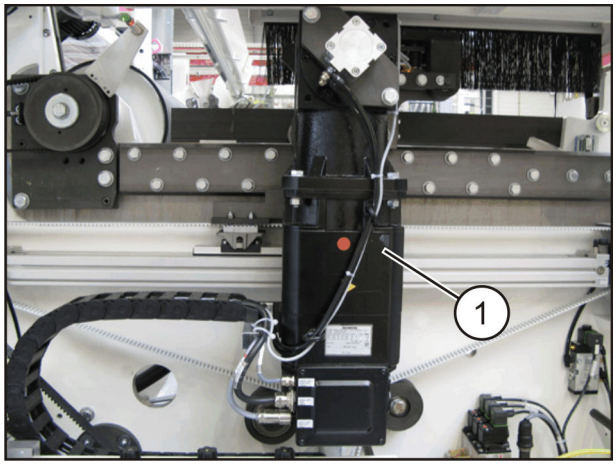
- Clean the transmitter and receiver of the light barrier (1) between the contact roller and the film winder.

  Interval: 1,000 hours
  Use: Cleaning cloth, glass cleaning agent
  The cleaning interval can be reduced or extended, depending on the working environment.

8.1.13 Center drive

Cleaning the air vent grid

(1)	Center drive



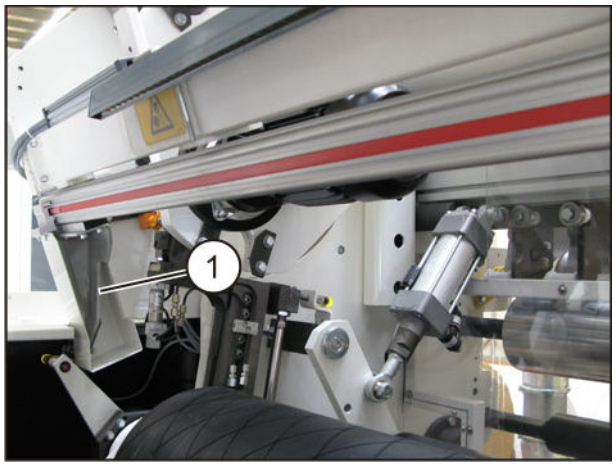
▶ Clean the air vent grid at the center drive (1).

Interval: 3,000 hours
Use: vacuum cleaner, brush, cleaning cloth
Depending on the ambient and production conditions the cleaning interval may be shorter or longer.

8.1.14 Cross cutting system

Cleaning the cutting blade

(1)	Cutting blade



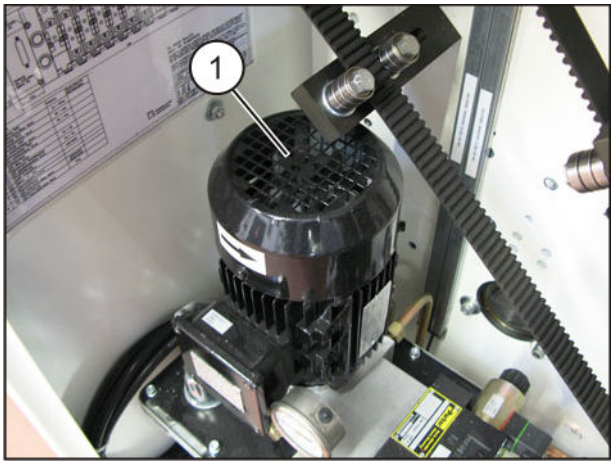
▶ Clean the cutting blade (1).

Interval: 1000 hours
Use: brush, cleaning cloth, alcohol (spirit)
Depending on the ambient and production conditions the cleaning interval may be shorter or longer.

8.1.15 Hydraulic unit

Clean the vent grating

(1)	Vent grating



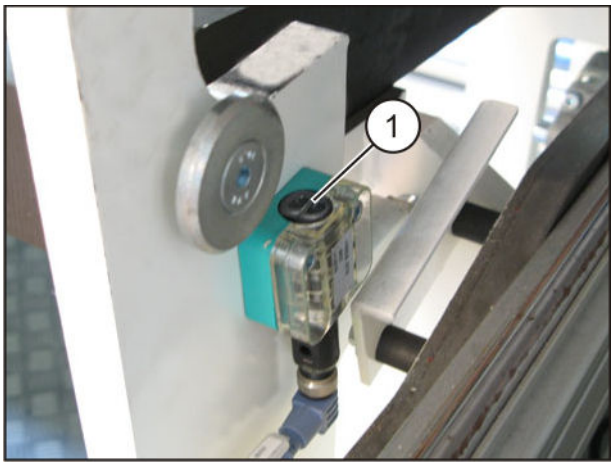
▶ Clean the vent grating (1) of the hydraulic unit drive.

Interval: 3,000 hours
Use: Vacuum cleaner, brush, cleaning cloth
The cleaning interval can be reduced or extended, depending on the working environment.

8.1.16 Delivery arms

Clean the sensors

(1)	Sensor



▶ Clean the sensors (1) on both delivery arms.

Interval: 1,000 hours
Use: Cleaning cloth, glass cleaning agent
The cleaning interval can be reduced or extended, depending on the working environment.

X000058124

8.1.17 Electrostatic charging and discharging

Cleaning the electrodes

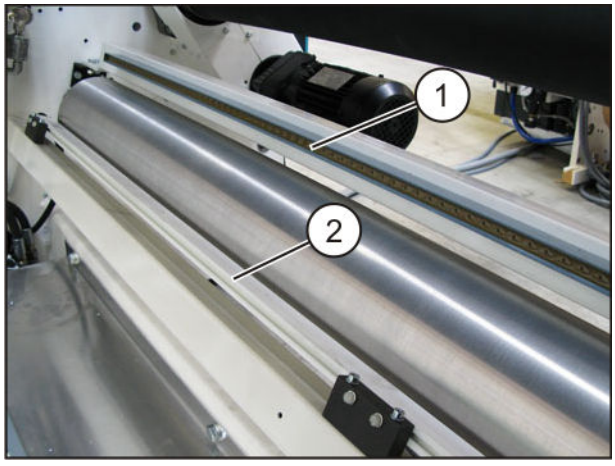
WARNING

Risk of personal injury resulting from electric shocks.

Turn the WINDING STATION SECURED key switch to I = SECURED to switch off the high-voltage generator and the control voltage. Necessary maintenance work should only be carried out by a professional electrician when the machine is at a standstill.

For further information please refer to the relevant sub-vendor documentation.

(1)	Charging electrode
(2)	Discharge electrode



▶ Clean the electrodes (1).

Interval: 500 hours
Use: vacuum air, soft synthetic brush, cleaning solvent when contaminations contain grease
Depending on the ambient and production conditions the cleaning interval may be shorter or longer.

8.2 Maintaining the machine

8.2.1 Winding shaft....319

8.2.2 Unwind....319
Hydraulic pump....319
Hydraulic cylinder....319
Hydraulic system....319

8.2.3 Unwind – Web break detector....322

8.2.4 Web guide control....322

8.2.5 Draw and dancer roller....323
Timing belt....323
Sensor....323

8.2.6 Sliding doors....324

8.2.7 Micro-perforator....324

8.2.8 Embossing....325
Tools....325
Timing belt....325

8.2.9 Gusseted tube forming unit....326

8.2.10 Longitudinal seam extruder....326
Motor....326
Gearbox....326
Barrel....326
Screw....326
Return pressure bearing....326
Heating equipment....326
Timing belt....326

8.2.11 Main draw....330
Timing belt - Seam cooling roller....330
Timing belt - Split upper draw roller....330
Timing belt - Lower draw roller....330
Seam cooling roller....330
Draw roller....330

8.2.12 Main draw - Slide protection....332
Slide protection, infeed section....332
Slide protection, outfeed section....332

8.2.13 Compressed-air conditioner....333

8.2.14 Equipment....334

8.2.15 Parallel guide....334

8.2.16 Contact roller....334

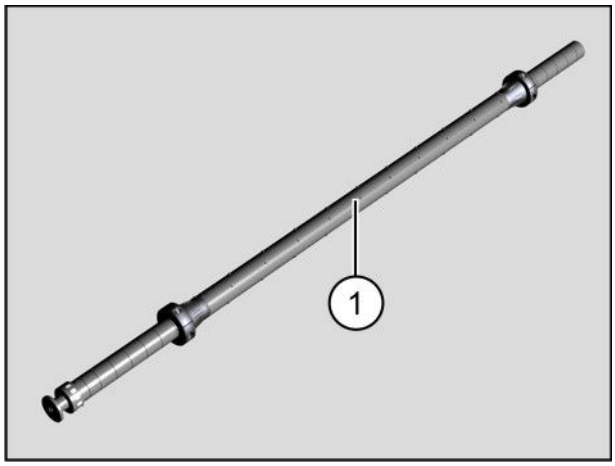
8.2.17 Hydraulic unit....335
Hoses and threaded joints....335
Hydraulic oil filter....335

8.2.18 Pneumatic system....339

8.2.19 Air conditioner for the switch cabinet....0

8.2.1 Winding shaft

(1)	Winding shaft



▶ Carry out a thorough visual inspection of the winding shaft (1).

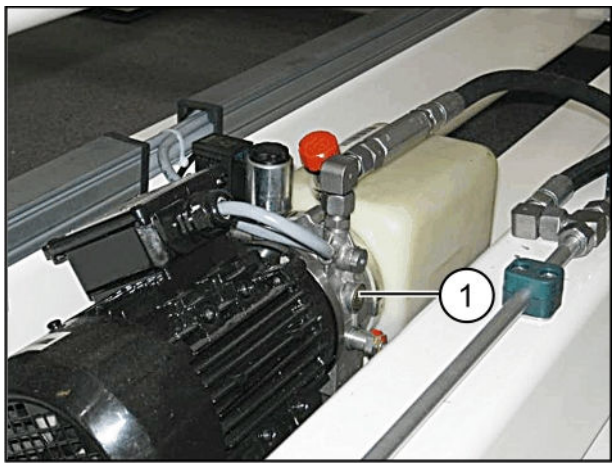
Interval: 400 hours
Replace immediately whenever necessary.

8.2.2 Unwind

Hydraulic pump

Checking the hydraulic pump

(1)	Hydraulic pump



▶ Check the Hydraulic pump (1) every 500 hours.

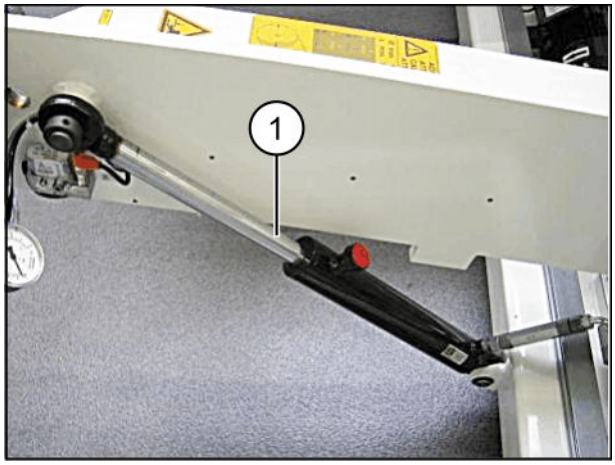
▶ Replace worn seals if necessary.

X000047276

Hydraulic cylinder

Checking the hydraulic cylinder

(1)	Hydraulic cylinder



- Check the hydraulic cylinder (1) every 500 hours.
- Replace worn seals if necessary.

Hydraulic system

Checking the hydraulic system

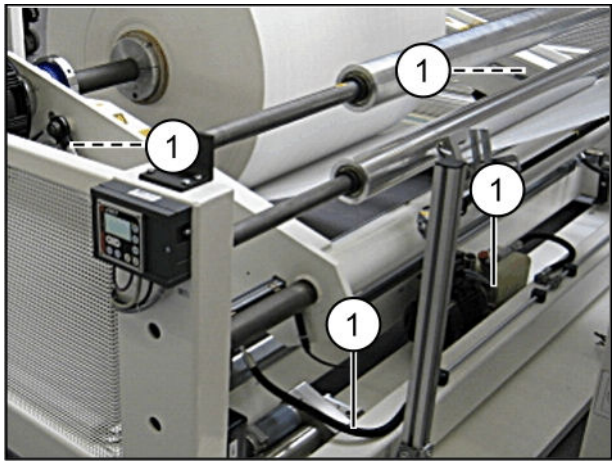
CAUTION

Liquid under pressure

Spraying out of hydraulic fluid

Never loosen the screw fittings, hoses or components as long as the equipment is under pressure!

(1)	Hydraulic system



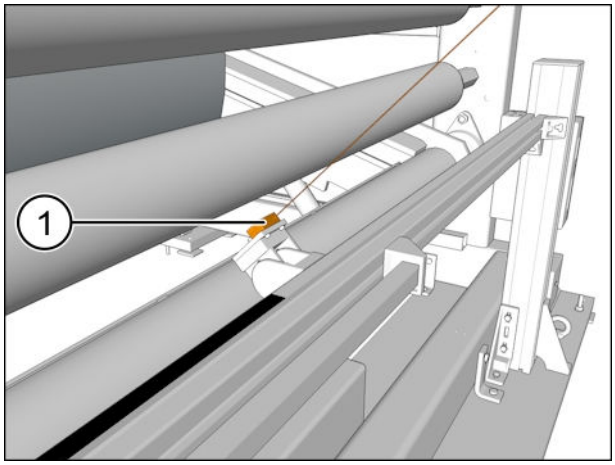
- Check the hydraulic system (1) every 500 hours.
- Replace destroyed or hardened sealing elements where necessary.

  Look for leakages in the pipe system
  Leakages at connection points, which are sealed with soft seals (O-rings), cannot be cleared by tightening, as the sealing elements in such cases are either destroyed or hardened.
  Check hydraulic fluid
  The timing depends on the operating conditions, particularly on the average operating temperature.
  80 degrees Celsius (176 degrees Fahrenheit) is taken as a maximum value for hydraulic fluids with a low water content.
  Do not mix different hydraulic fluids. Mixing can lead to chemical reactions, such as the formation of foam, resinification, or similar.
  Monitor operating temperature
  An operating temperature which increases over time indicates increasing friction and thus wear or a leakage.
  Check system and control pressure settings
  Repeated readjustment in order to reach the intended pressure indicates wear.
  Check pressure, runback and ventilator filters
  Check working of control and monitoring components (e.g. manometer)

8.2.3 Unwind – Web break detector

Checking the sensor

(1)	Sensor



- Check the sensor (1), its bolting and plug connections.

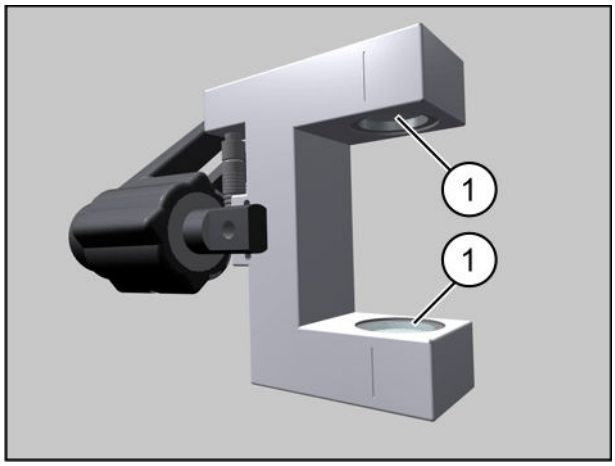
  Interval: 800 hours

- If necessary, readjust the sensor when out of tune or replace the sensor when defective.

8.2.4 Web guide control

Checking the sensor

(1)	Sensor



- Check the sensor (1).

  Interval: 1,000 hours

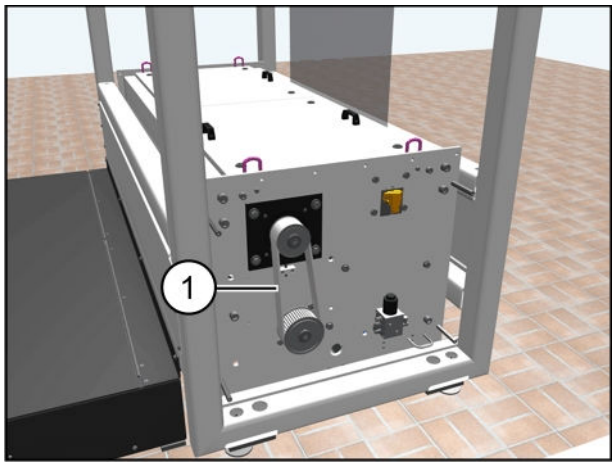
- If necessary, replace the sensor if it is defective or readjust the sensor.

8.2.5 Draw and dancer roller

Timing belt

Checking the timing belt

(1)	Timing belt

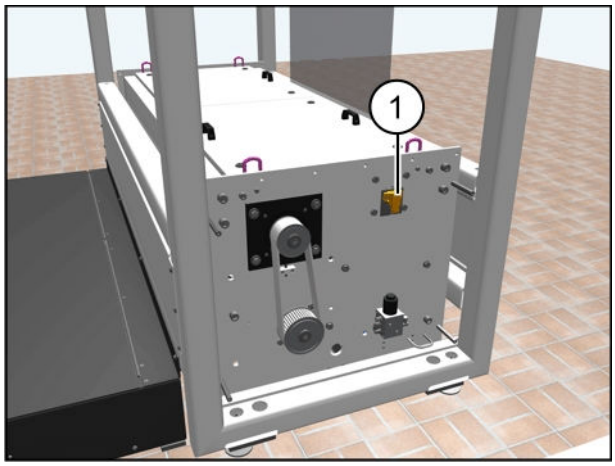


- ▶ Check the timing belt (1).
  Interval: 1,000 hours
- ▶ If necessary, re-tension the timing belt when slack or replace the timing belt when worn down.
  Timing belt tension: 65 Hz

Sensor

Checking the sensor

(1)	Sensor

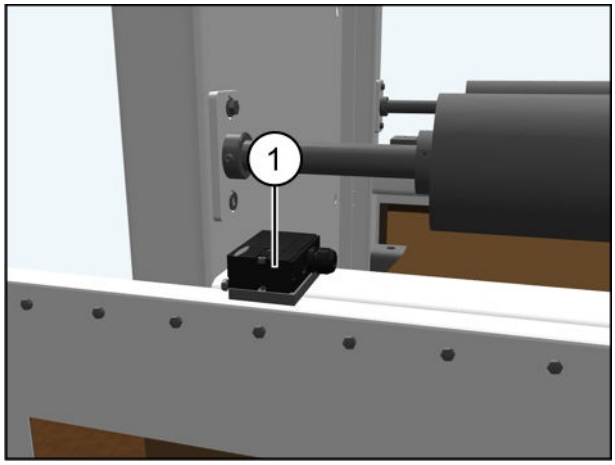


- ▶ Check the sensor (1).
  Interval: 800 hours
- ▶ If necessary, replace the sensor if defective or readjust the sensor.

8.2.6 Sliding doors

Checking the limit switch

(1)	Limit switch

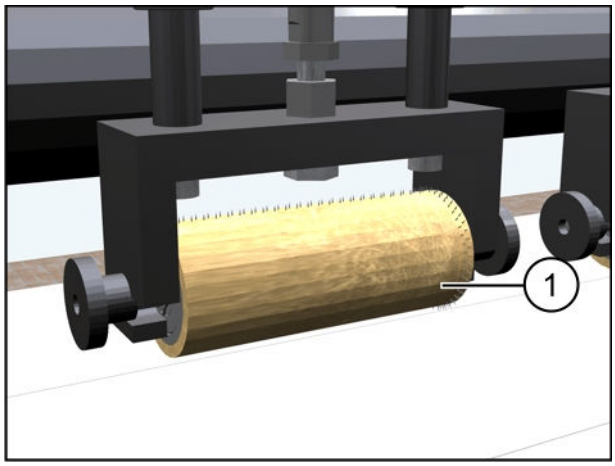


- Check the limit switch (1).
  Interval: 1,000 hours
- Check that the guard lock and the activator are firmly fitted..
- Check the input cabling and cable connections.
- If necessary, readjust the limit switch or immediately replace a defective limit switch.

8.2.7 Micro-perforator

Checking the needle roller

(1)	Needle roller



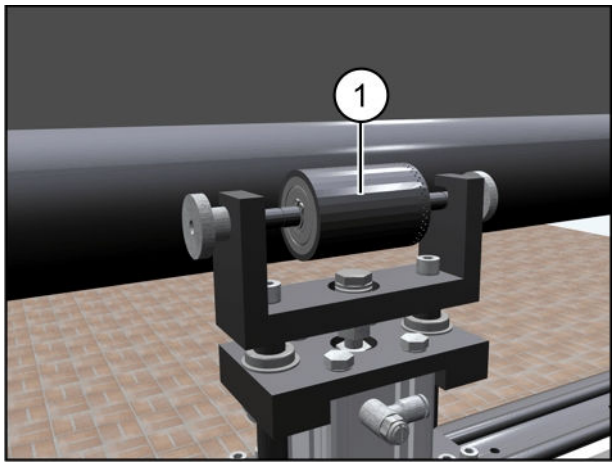
- Check the needle bearing (1).
  Interval: 200 hours
- If necessary, replace the needle roller, e.g. where the needles are bent.

8.2.8 Embossing

Tools

Checking the embossing roller

(1)	Embossing roller

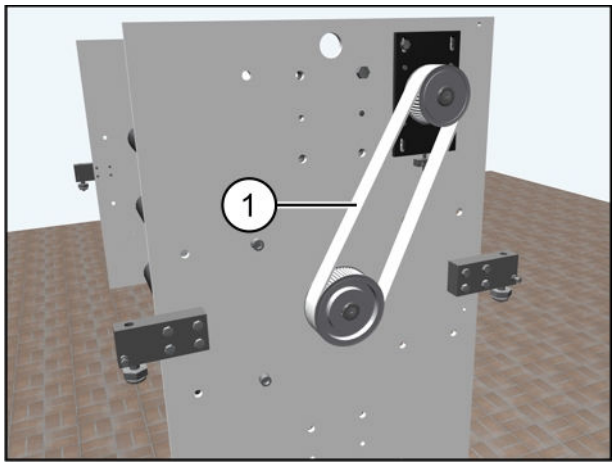


- ▶ Check the embossing roller (1).
  Interval: 200 hours
- ▶ Replace the embossing roller whenever necessary.

Timing belt

Checking the timing belt

(1)	Timing belt

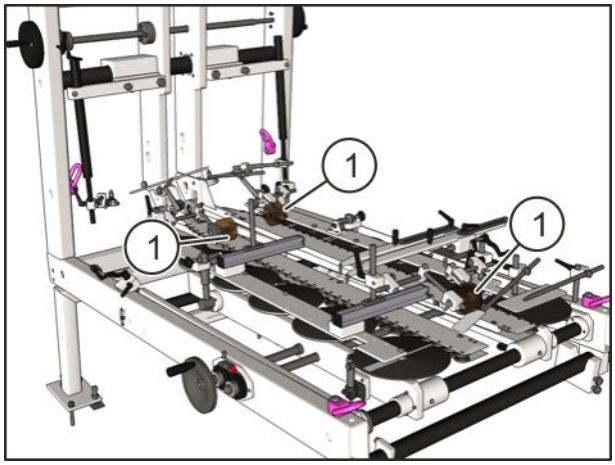


- ▶ Check the timing belt (1).
  Interval: 1,000 hours
- ▶ Re-tension slack timing belt or replace worn timing belt whenever necessary.
  Timing belt tension: xx Hz

8.2.9 Gusseted tube forming unit

Checking the press roll

(1)	Press roll



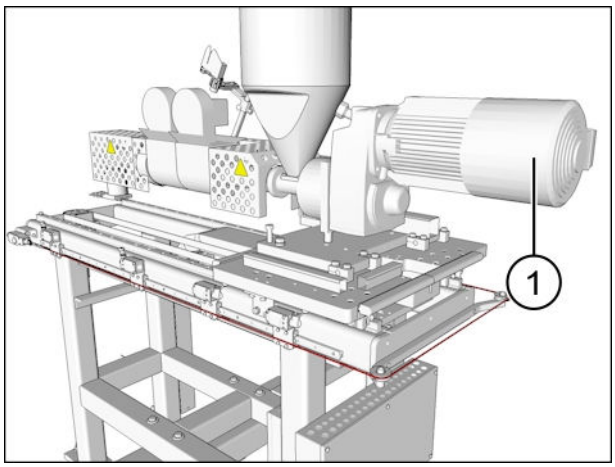
- Check the press rolls (1).
  Interval: 1,000 hours
- Replace whenever necessary.

8.2.10 Longitudinal seam extruder

Motor

Checking the motor

(1)	Motor

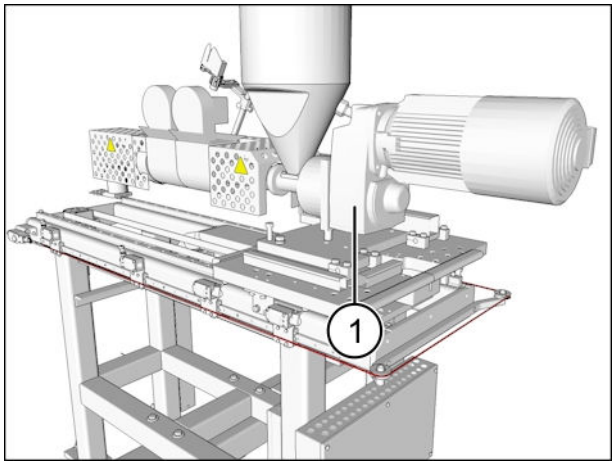


- Check the motor (1) for smooth running and appropriate temperature every 500 hours.
  See the manufacturer's description.

Gearbox

Checking the gearbox

(1)	Gearbox



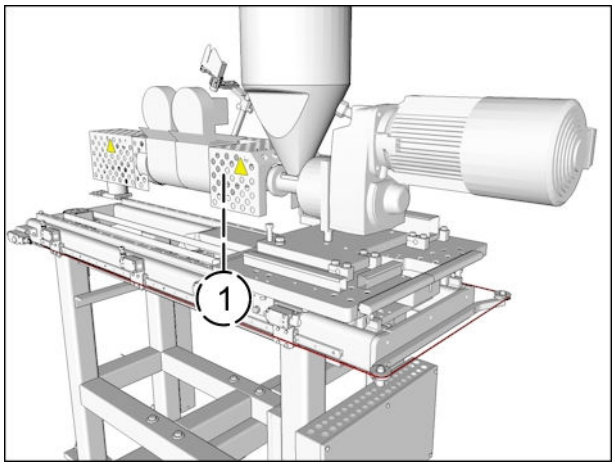
- Check the gearbox (1) for smooth running, leaks and appropriate temperature every 2,000 hours.

  See the manufacturer's description.

Barrel

Checking the barrel

(1)	Barrel

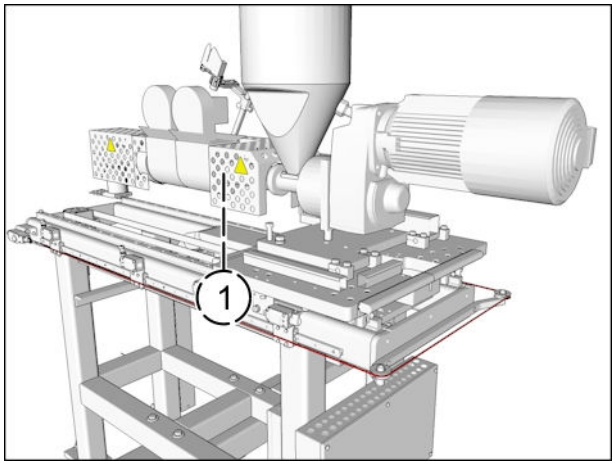


- Check the barrel for wear (1) every 2,000 hours.

Screw

Checking the screw

(1)	Screw

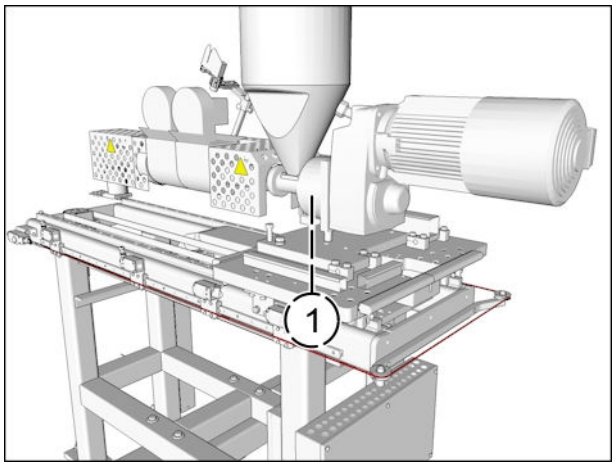


▶ Check the screw for wear (1) every 2,000 hours.

Return pressure bearing

Checking the back pressure bearing

(1)	Bearing

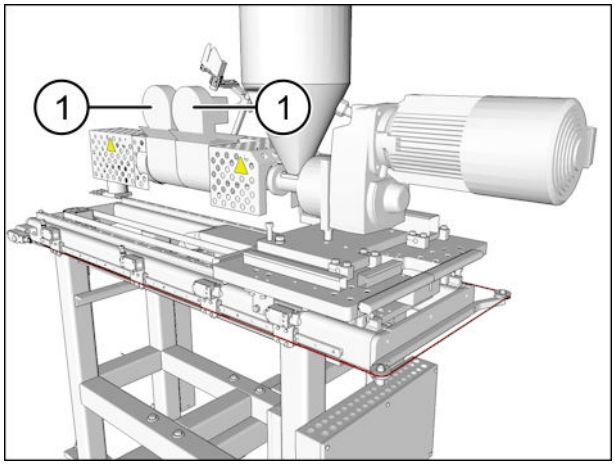


▶ Check the back pressure bearing for wear (1) every 2,000 hours.

Heating equipment

Checking the heaters

(1)	Heaters



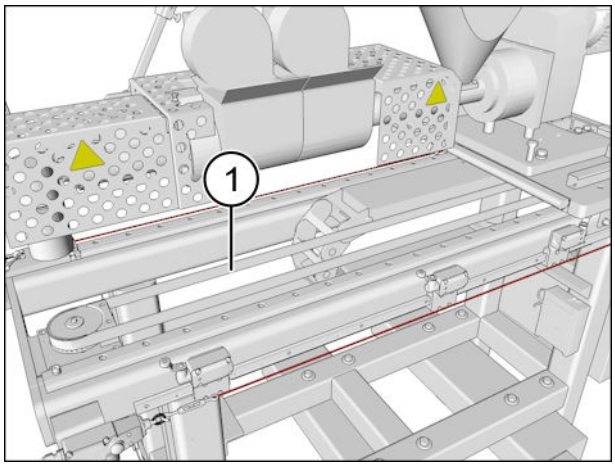
- Check the heaters for proper functioning (1) every 8 hours.

  Constantly monitor the heaters during operation.

Timing belt

Checking the timing belt

(1)	Timing belt



- Check the timing belt (1) every 1,000 hours.
- If necessary, tighten the timing belt when slack or replace the timing belt when worn.

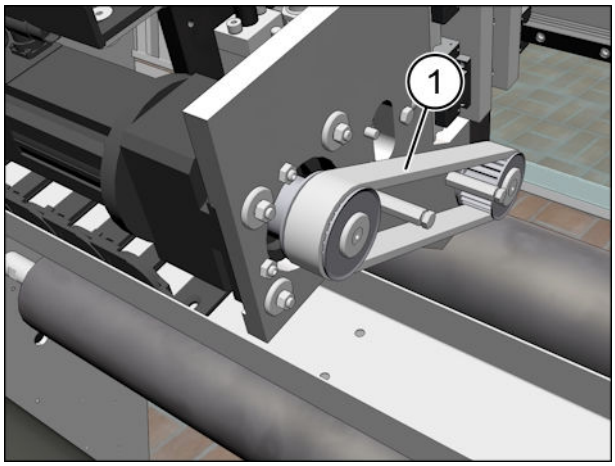
  Timing belt tension: 12 Hz

8.2.11 Main draw

Timing belt - Seam cooling roller

Checking the timing belt

(1)	Timing belts

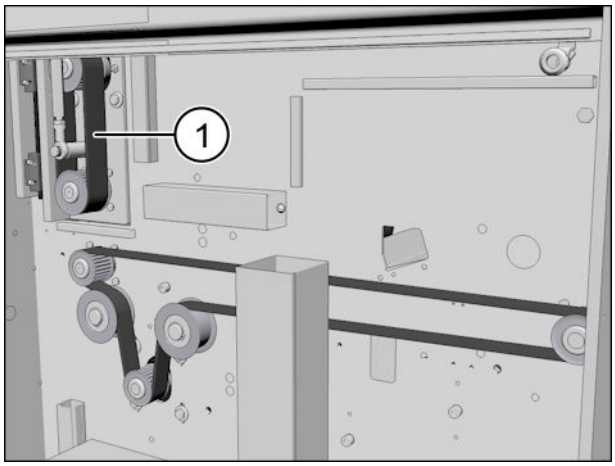


- ▶ Check the timing belt (1).
  Interval: 1,000 hours
- ▶ If necessary, tighten the timing belt when slack or replace the timing belt when worn.

Timing belt - Split upper draw roller

Checking the timing belt

(1)	Timing belts



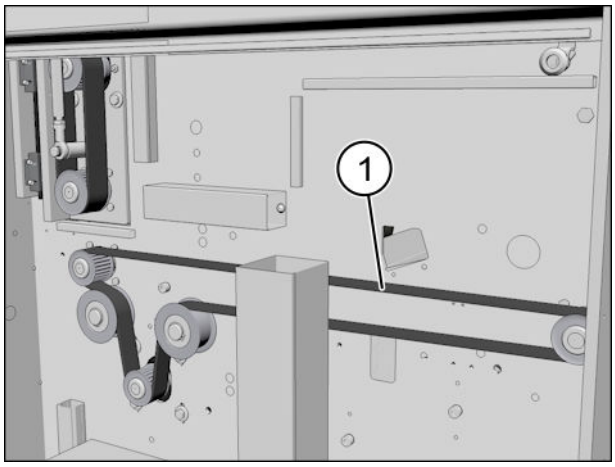
- ▶ Check the timing belt (1).
  Interval: 1,000 hours
- ▶ If necessary, tighten the timing belt when slack or replace the timing belt when worn.

Timing belt - Lower draw roller

Timing belt for the drive of the lower draw roller and the chill roll

Checking the timing belt

(1)	Timing belt

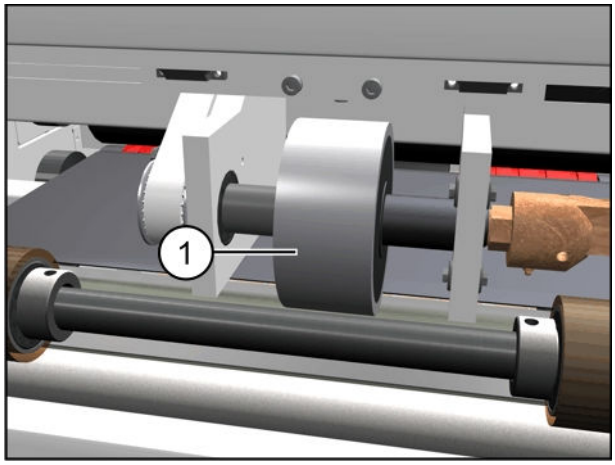


- ▶ Check the timing belt (1).
  Interval: 1,000 hours
- ▶ If necessary, tighten the timing belt when slack or replace the timing belt when worn.

Seam cooling roller

Checking the roller

(1)	Roller



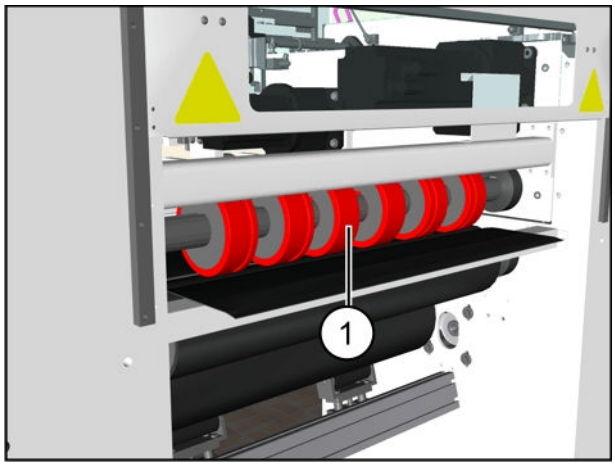
- ▶ Check the roller (1).
  Pay particular attention to the condition of the surface.
  Interval: 1,000 hours
- ▶ Where the roller has difficulty in turning, replace the bearings where necessary. Replace any roller with a damaged surface.

Draw roller

Checking the roller

Check all segments of the roller.

(1)	Roller



- Check the roller (1).

  Pay particular attention to the condition of the surface.

  Interval: 1,000 hours

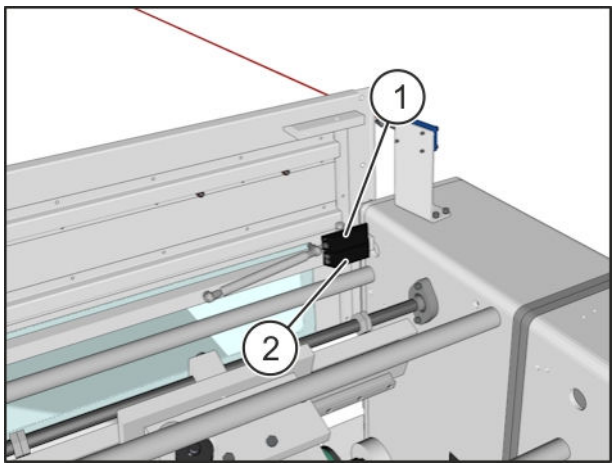
- Where the roller has difficulty in turning, replace the bearings where necessary. Replace individual roller segments with damaged surface where necessary.

8.2.12 Main draw - Slide protection

Slide protection, infeed section

Checking the sensor

(1)	Sensor
(2)	Magnet



- Check the sensor (1).Check the magnets (2) for presence.

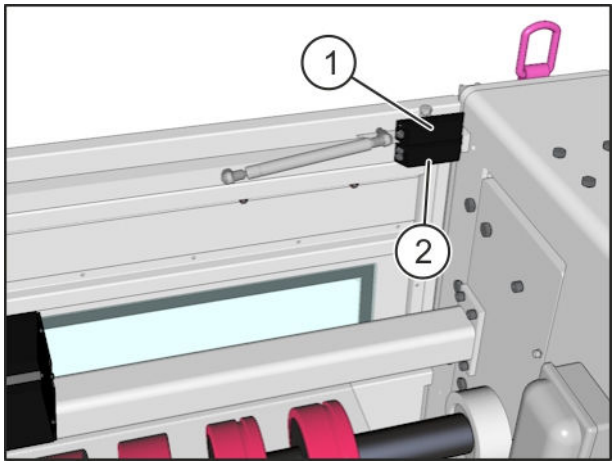
  Interval: 1,000 hours

- Replace the sensor when defective or readjust the sensor whenever necessary.Fit a new magnet whenever necessary.

Slide protection, outfeed section

Checking the sensor

(1)	Sensor
(2)	Magnet



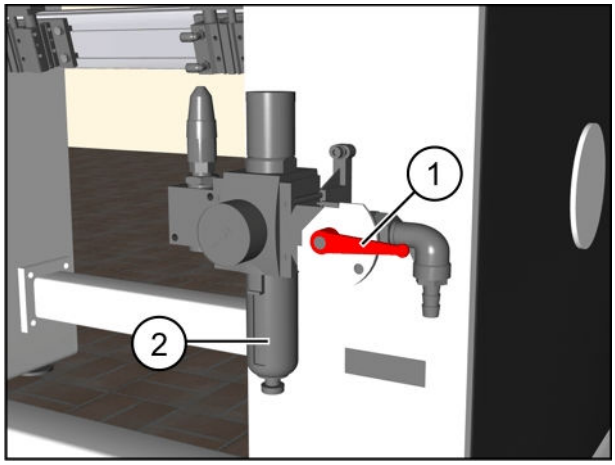
- Check the sensor (1).Check the magnets (2) for presence.

  Interval: 1,000 hours

- Replace the sensor when defective or readjust the sensor whenever necessary.Fit a new magnet whenever necessary.

8.2.13 Compressed-air conditioner

(1)	Main shut-off valve
(2)	Filter pressure regulator



The compressed-air conditioner consists of a lockable main shut-off valve (1) and a filter pressure regulator (2). To operate the unit the valve must be opened.

The filter pressure regulator, consisting of a filter with water trap and pressure regulator valve with manometer, has the task of cleaning and removing liquid and solid constituents from the compressed air (the resource) as well as controlling the air pressure. Condensation water collected in the semi-automatic water trap is drained every time the pressure is turned off.

The pressure regulator reduces the network pressure to the required operating pressure and maintains this value as a constant setting regardless of the amount of air consumed or pressure fluctuations.

8.2.14 Equipment

Pneumatic, hydraulic and electrotechnical equipment needs maintaining

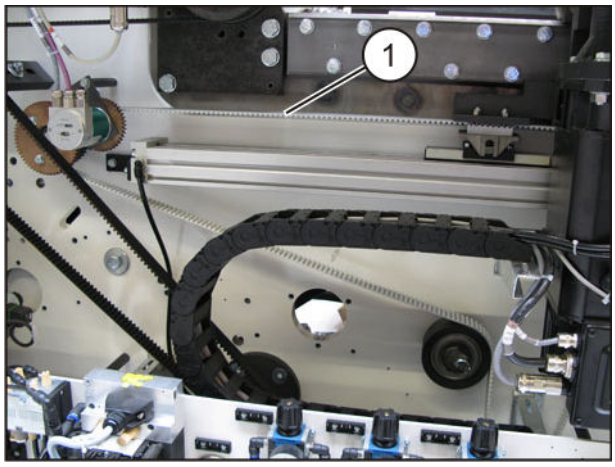
Every technical equipment must be maintained in compliance with the legal regulations and the manufacturer's instructions.

For the maintenance instructions please refer to the manufacturers' manuals. For these manuals please refer to the chapter under the heading 'Sub-vendor documentation.

8.2.15 Parallel guide

Re-tension the timing belt

(1)	Timing belt



- ▶ Check the tensioning of the timing belts (1) mounted on both sides of the winder.

  Interval: 3,000 hours
  Use: Belt pre-tension measuring device
  The frequency setting of the oscillating leg of the timing belt must be F = 18 Hz.

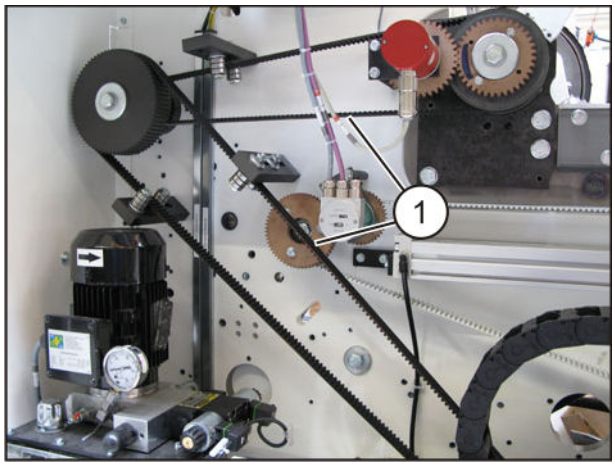
- ▶ Position the measuring unit up against the leg of the timing belt and then use a tool to set the section of the belt to be measured into oscillatory movement.

- ▶ Change the timing belt if it shows signs of wear at the edges or if cracks have appeared.

8.2.16 Contact roller

Re-tensioning the timing belt

(1)	Timing belt



- ▶ Check the timing belt (1) for proper pre-tensioning.

  Interval: 3,000 hours
  Use: Belt frequency meter
  The frequency setting of the oscillating leg of the timing belt must be F = 18 Hz.

- ▶ If necessary, re-tension the timing belt via the eccentric take-up pulley.

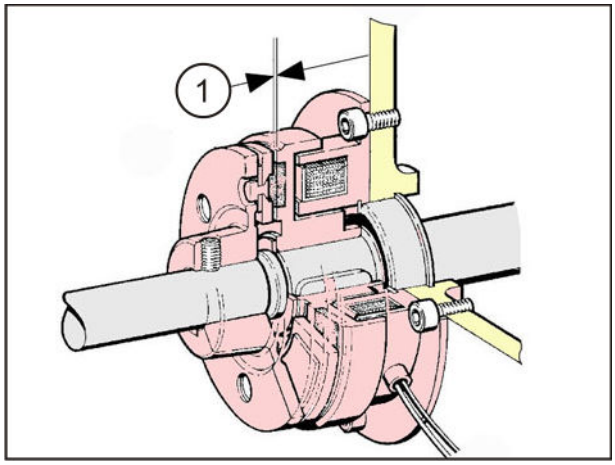
- ▶ Change the timing belt if it shows signs of wear at the edges or if cracks have appeared.

Checking the electromagnetic brake

DANGER

The electromagnetic brake is a safety device and used for quickly braking the contact rollers. It may only be operated when in perfect working order.

(1)	Air gap



- Check the air gap (1) in the electromagnetic brake.

  Interval: 5,000 hours
  Set the air gap as specified in the manufacturer's instruction manual.

- Set the air gap, if necessary.

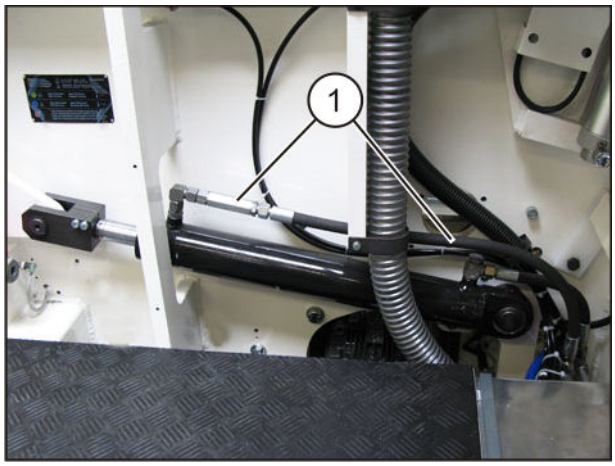
8.2.17 Hydraulic unit

Hoses and threaded joints

Checking the hydraulic hoses

Tighten the threaded joint to the hydraulic hoses to a tightening torque of 120 -150 N m.

(1)	Hydraulic hoses



- Check all of the hydraulic hoses (1) for mechanical damage or leaks.

  Interval: 3,000 hours

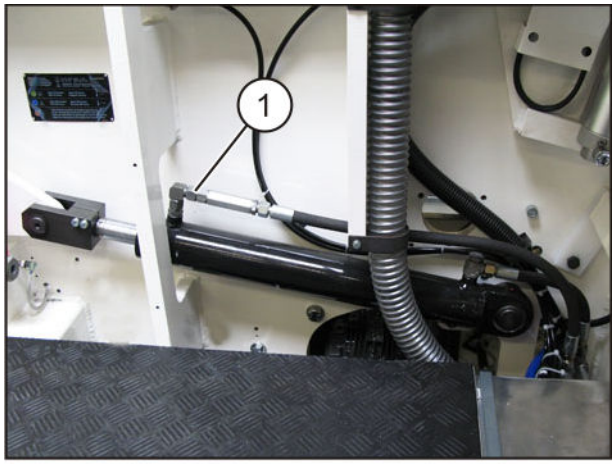
- In case of visible damages (mechanical damages, leaks) the respective hydraulic hose should be replaced immediately.

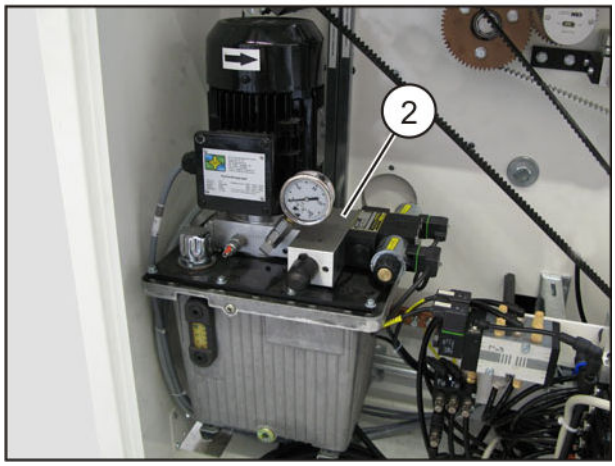
X000058332

Checking the threaded joints

Tighten the threaded joint to the hydraulic hoses to a tightening torque of 120 -150 N m.

(1)	Threaded joints
(2)	Hydraulic unit





- Check the threaded joints (1) at all of the hydraulic components for leaks.

  Interval: 3,000 hours

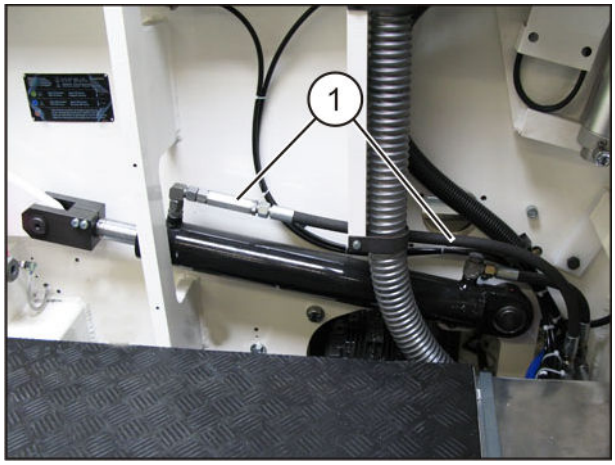
- Tighten the threaded joints at the hydraulic components whenever necessary.

X000058332

Replacing hydraulic hoses

Tighten the threaded joint to the hydraulic hoses to a tightening torque of 120 -150 N m.

(1)	Hydraulic hoses



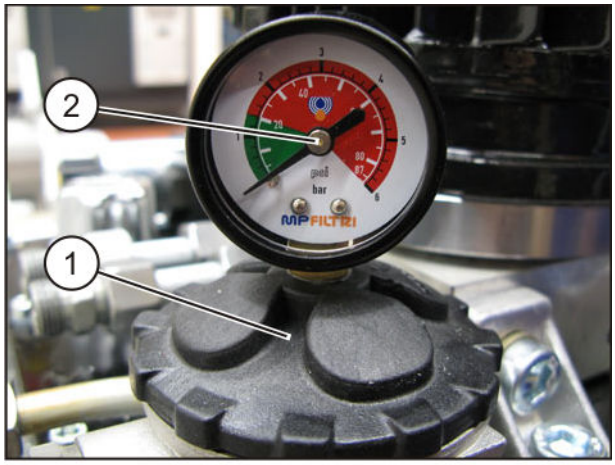
- As a rule, all of the hydraulic hoses (1) should be replaced every 4 years.

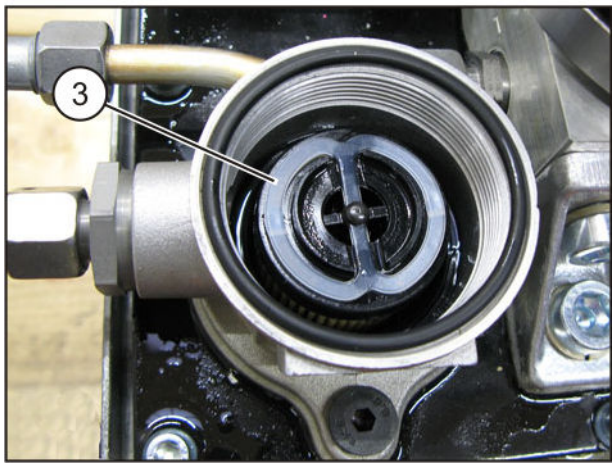
  Interval: 20,000 hours

Hydraulic oil filter

Replacing a filter cartridge

(1)	Hydraulic oil filter
(2)	Contamination indicator
(3)	Filter cartridge





- Check the hydraulic oil filter (1) for cleanliness using the contamination indicator (2).

  Interval: 3,000 hours
  If the indicator needle of the soiling indicator is in the red zone, the filter cartridge (3) must be replaced.

- Unscrew the cover with the soiling indicator and pull out the filter cartridge (3) using the straps.

- Replace soiled filter cartridge with a new one and screw the cover back on.

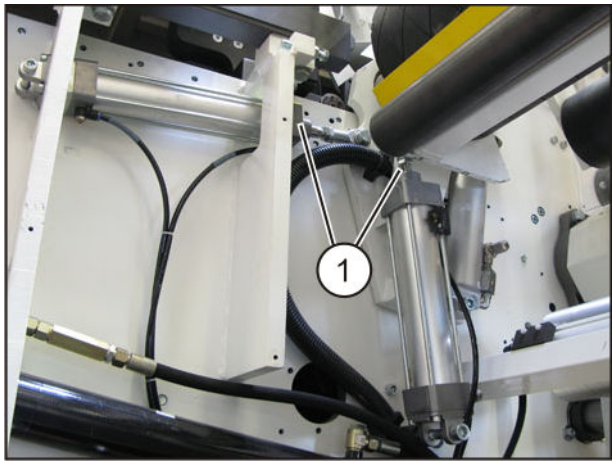
For further information please refer to the relevant Sub-vendor documentation.

The appropriate filter can be ordered with the supplier as the original spare part.

8.2.18 Pneumatic system

Checking the piston rod guides

(1)	Piston rod guide



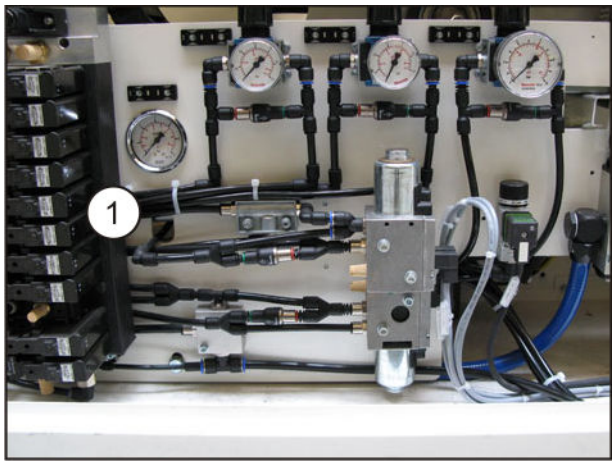
- Check the piston rod guides (1) in the pneumatic cylinders for wear.

  Interval: 3,000 hours

- If necessary, replace damaged piston rod guides as specified in the manufacturer's instruction manual.

Replacing the sealing elements

(1)	Pneumatic valves



- Replace the sealing elements irrespective of the switching capacity of the pneumatic valves (1).

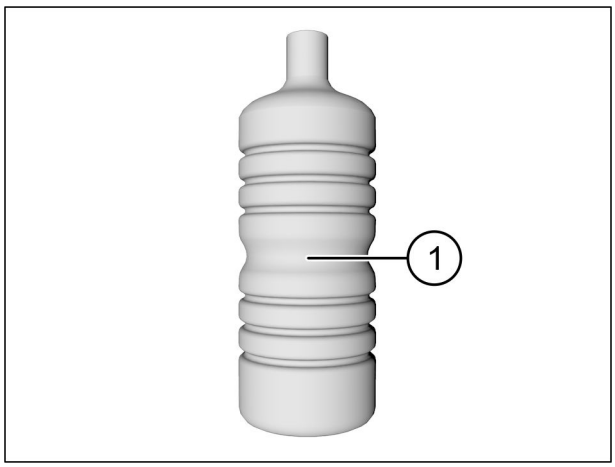
  Interval: 20,000 hours

  The manufacturer's instructions must be adhered to when replacing the sealing elements.

8.2.19 Air conditioner for the switch cabinet

Draining the condensation water container

(1)	Condensation water container



- Check the filling level of the condensation water container (1) in the switch cabinet and empty, if necessary.

  Interval: 400 hours
  Use: -
  Depending on the ambient and production conditions the cleaning interval can be shorter or longer.

8.3 Lubricating the machine

8.3.1 Lubricants....341
General....341
Lubricating instructions....341
Color code caption on the machine....341
Details on lubricants....341

8.3.2 Sliding doors....347

8.3.3 Draw and dancer roller....347

8.3.4 Embossing....347

8.3.5 Gusseted tube forming unit....348
Lubricating the guide....348
Greasing the spindle....348
Greasing the bearing....348
Greasing the joints....348
Adjustment unit Former plates – Greasing the spindle....348

8.3.6 Longitudinal seam extruder....349
Bearing....349
Return pressure bearing....349
Gearbox....349

8.3.7 Longitudinal seam monitoring....351

8.3.8 Main draw....351
Guide cart - Positioning the seam cooling roller....351
Guide cart - Raising and lowering the seam cooling roller....351
Spindle - Positioning the seam cooling roller....351
Spindle - Turning the seam cooling roller vertically....351
Gear wheel - Turning the seam cooling roller vertically....351
Gearbox....351
Slide protection, infeed section....351
Slide protection, outfeed section....351

8.3.9 Oscillation....355

8.3.10 Dancer roller....355

8.3.11 Center drive....356

8.3.12 Hydraulic unit....356

8.3.1 Lubricants

General

Proper lubrication even under difficult conditions provides the following advantages:

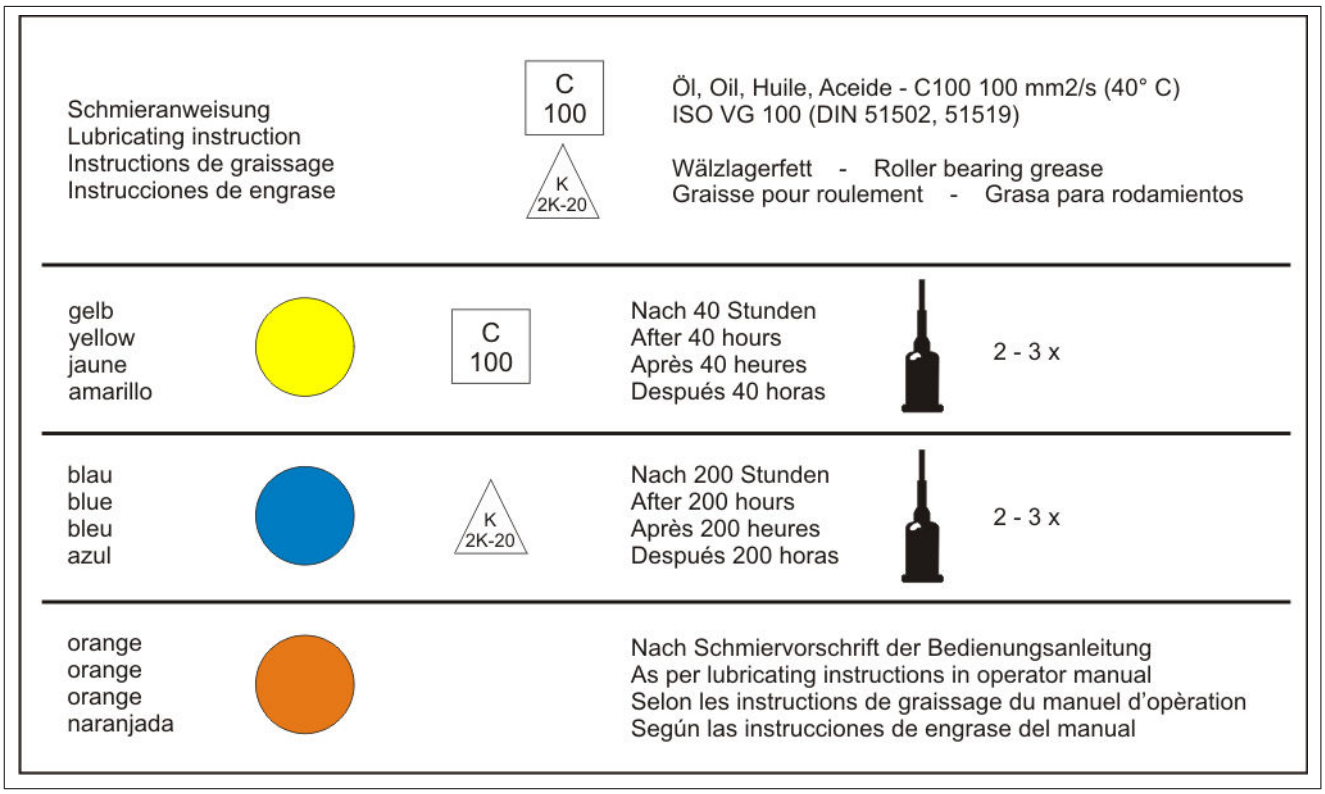
- increase of machine's service life
- malfunctions are avoided
- reduction of production downtimes

Lubricating instructions

- Proper lubrication of machines calls for cleanliness.
- As far as central lubrication sytems and gearboxes are concerned the oil level must be checked and replenished as and when necessary. Oil grades and lubricating intervals are to be noted from the respective manufacturer's instructions.
- Free flow chains must be kept clean. Clean chain from abrasions, resinifications and contamination using e.g. petroleum. Afterwards, lubricate the chains using oil or spray with special chain sprays. When lubricants liable to react with solvent vapors are used, we recommend to use molybdenum disulfide-based lubricants for chains.
- Apart from a few exceptions, the bearings are provided as roller bearings and equipped with a cover plate as necessary. They are durably lubricated so that relubricating is not required. If the machine features gearbox housings the roller bearings are sufficiently supplied with oil by means of the grease gun and the splash lubrication system resp.
- All Lubricating points are color-coded. A lubrication chart is attached to the machine indicating the lubricants to be used and the frequencies of all changes.
- The oil grades and lubricating intervals for components supplied by sub-contractors can be taken from the manufacturer's instruction manual.

X000302307

Color code caption on the machine



Details on lubricants

Quality of lubricants

The lubricants supplied by the lubricant manufacturers must comply with the properties stipulated under the code description as per DIN standard 51502/51506 or the API classification.

We cannot be held responsible for the quality of the lubricants.

Procuring lubricants

The following list contains an exemplary assortment of lubricants.
If a listed lubricant cannot be procured locally, use the code description to find and procure a comparable lubricant with the support of a local specialist retailer.

Oils

Never mix polyglycol-based oils (PG), polyalphaolefin-based oils (PAO) and synthetic hydrocarbon (HC) based oils with other oil grades.

(1)	Abbreviation according to DIN standard 51502 and DIN 51506
(2)	Flash point at °C or higher
(3)	Pourpoint at °C or lower
(4)	API classification
(5)	SAE viscosity classification

(1)	(2)	(3)	Castrol	Klüber	Mobil	Shell
HL 10	125	-20	Castrol Hyspin AWS 10	–	Mobil Velocite Oil No. 6	Shell Morlina S2 M32
HL 32	175	-18	Castrol Hyspin AWS 32	–	Mobil DTE Oil Light	Shell Tellus S2 M32
HL 46	185	-15	Castrol Hyspin AWS 46	–	Mobil DTE Oil Medium	Shell Morlina S2 B46
HL 68	240	-21	Castrol Hyspin AWS 68	–	Mobil Vactra No. 2	Shell Morlina S2 B68
HLP 10	125	-20	Castrol Hyspin AWS 10	–	Mobil DTE 21	Shell Morlina S2 M32

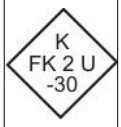
(1)	(2)	(3)	Castrol	Klüberoil	Mobil	Shell
HLP 32	161	-36	Castrol Hyspin AWS 32	HLP 32	Mobil DTE 21	Shell Tellus S2 M32
HLP 46	225	-33	Castrol Hyspin AWS 46	HLP 46	Mobil DTE 25	Shell Tellus S2 M46
HLP 68	237	-30	Castrol Hyspin AWS 68	HLP 68	Mobil DTE 26	Shell Tellus S2 M68
VDL 100	227	-15	Castrol Aircol PD 100	–	Mobil Rarus 427	Shell Corena S2P, Shell Corena S4P,
CLP 32	175	-18	Castrol Hyspin AWS 32	HLP 32	Mobil DTE 24	Shell Tellus S2 M32, Shell Tegula V 32
CLP 46	175	-15	Castrol Hyspin AWS 46	Klüberoil GEM 1-46 N	Mobil DTE 25	Shell Tellus S2 M46
CLP 68	225	-24	Castrol Alpha SP 68	Klüberoil GEM 1-68 N	Mobilgear 600XP68	Shell Omala S2 G68
CLP 100	230	-24	Castrol Alpha SP 100	Klüberoil GEM 1-100 N	Mobilgear 600XP100	Shell Omala S2 G100
CLP 150	235	-21	Castrol Alpha SP 150	Klüberoil GEM 1-150 N	Mobilgear 600XP150	Shell Omala S2 G150
CLP 220	235	-21	Castrol Alpha SP 220	Klüberoil GEM 1-220 N	Mobilgear 600XP220	Shell Omala S2 G220
CLP 320	240	-15	Castrol Alpha SP 320	Klüberoil GEM 1-320 N	Mobilgear 600XP320	Shell Omala S2 G320
CLP 460	240	-15	Castrol Alpha SP 460	Klüberoil GEM 1-460 N	Mobilgear 600XP460	Shell Omala S2 G460
CLP 680	240	-15	Castrol Alpha SP 680	Klüberoil GEM 1-680 N	Mobilgear 600XP680	Shell Omala S2 G680
CGLP 220	240	-9	Castrol Magna SW-D 220	Lamora D 220	Mobil Vectra Oil No. 4	Shell Tonna S2 M220, Shell Tonna S3 M220
CLP HC 150	220	-25	Castrol Alphasyn EP150	Klübersynth GEM 4-150 N	Mobil SHC 629	Shell Omala S4 GX150
CLP HC 220	220	-40	Castrol Alphasyn EP220	Klübersynth GEM 4-220 N	Mobil SHC 630	Shell Omala S4 GX220

(1)	(2)	(3)	Castrol	Klüber	Mobil	Shell
CLP PG 150	229	-41	Castrol Alphasyn PG150	Klübersynth GH 6-150, Syntheso HT 150	Glygoyle 22	Shell Omala S4 WE150
CLP PG 220	262	-39	Castrol Alphasyn PG220	Klübersynth GH 6-220	Glygoyle 30	Shell Omala S4 WE220
CLP PG 460	250	-23	Castrol Alphasyn PG460	Klübersynth GH 6-680	Glygoyle 460	Shell Omala S4 WE460
CLP PAO 68	220	-39	Castrol Alphasyn HTX 68	–	–	–
CLP PAO 150	220	-39	Castrol Alphasyn HTX 150	–	–	–
CLP PAO 220	220	39	Castrol Alphasyn HTX 220	Klübersynth GH 4-220	Mobil Gear XMP220	–
(4)	(5)					
GL 5	75W / -90	–	Manual TAF -x	–	Mobilube 1SHC 75W-90	Shell S6 AXME Spirax 75W-90

Greases

(1)	Code designation as per DIN standard 51502
(2)	Worked penetration
(3)	Operational temperature range at approx. °C

(1)	(2)	(3)	Castrol	Klüber	Mobil	Shell
K 2K-20	265-295	-20 +125	Castrol Spheerol AP 2	Centoplex 2	–	Shell Gadus S2 V100 2
KP 2K-20	265-295	-20 +120	Castrol Spheerol EPL 2	Centoplex 2 EP, Klüberplex BE 31-222	Mobilux EP 2	Shell Gadus S2 V100 2
KP 2K-40	–	-40 +120	–	Polylub WH 2	–	–

(1)	(2)	(3)	Castrol	Klüber	Mobil	Shell
	265-295	-30 +140	Castrol Tribol GR-2 PD	Klüber Centoplex 2	Mobilux EP 2	Shell Gadus S2 V100 2
	265-295	-30 +250	–	Barrierta L55/2	–	–
	220-250	-20 +100	Castrol Spheer-ol AP 3	Centoplex 3	–	Shell Gadus S2 V100 3
	400-430	-20 +100	Castrol CLS-Grease	Klüberplex GE 11-680	Mobilux EP 004	–

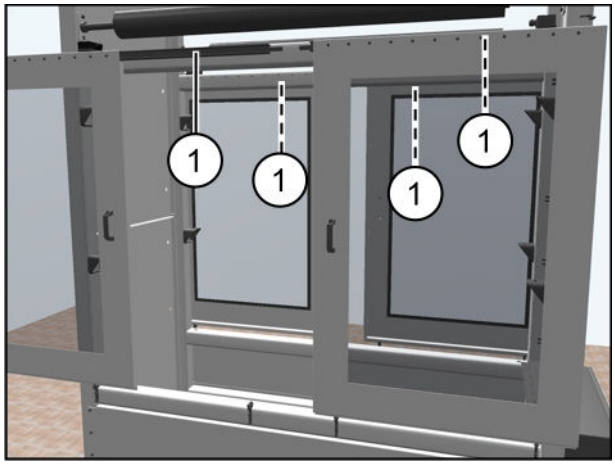
List of manufacturers/suppliers

Castrol	Deutsche Castrol Vertriebsgesellschaft mbH Max-Born-Str. 2 22761 Hamburg
Klüber	Klüber Lubrication Deutschland KG Geisenhausenerstr 7 81379 München
Mobil	Exxon Mobil Central Europe Holding GmbH Kapstadtring 2 22297 Hamburg
Shell	Deutsche Shell Aktiengesellschaft Suhrenkamp 71 - 77 22284 Hamburg

8.3.2 Sliding doors

Greasing the telescope rail

(1)	Telescope rail

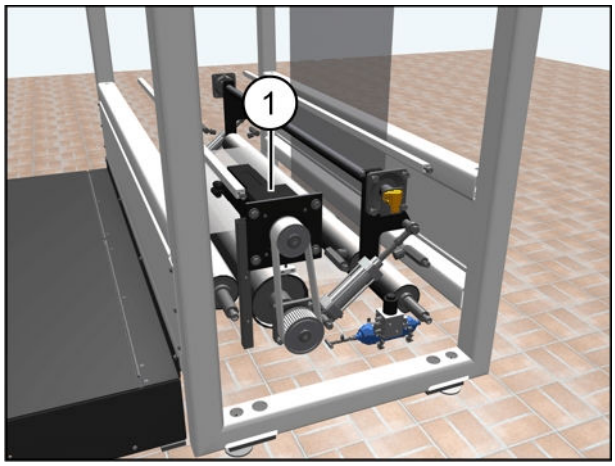


▶ Grease the telescope rail (1).

Interval: 1,000 hours
Use: KP 2 K-20, e.g. lithium grease for roller bearings

8.3.3 Draw and dancer roller

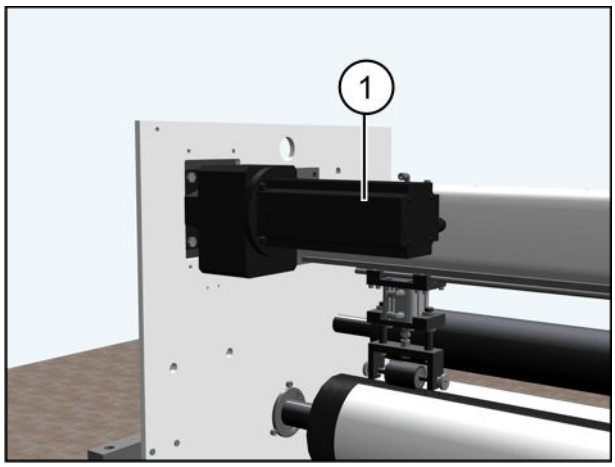
(1)	Gearbox



The gearbox (1) has life-long lubrication. An oil change is not required here.

8.3.4 Embossing

(1)	Gearbox

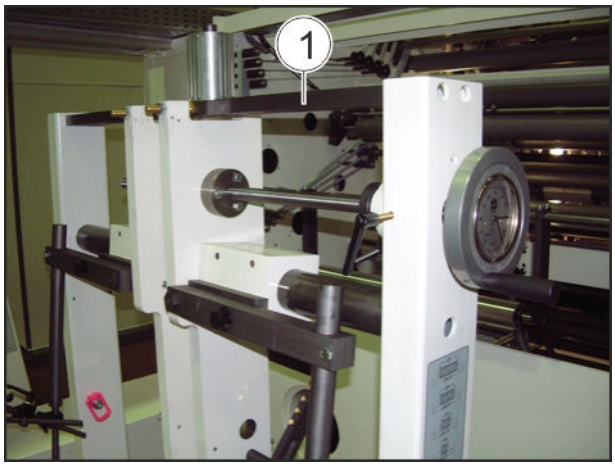


The gearbox (1) has life-long lubrication. An oil change is not required here.

8.3.5 Gusseted tube forming unit

Lubricating the guide

(1)	Guide

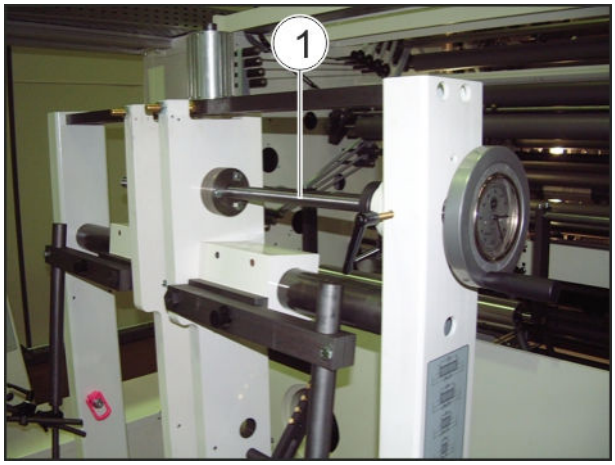


▶ Lubricate the guide (1).

Interval: 400 hours
Use: CLP 100

Greasing the spindle

(1)	Spindle

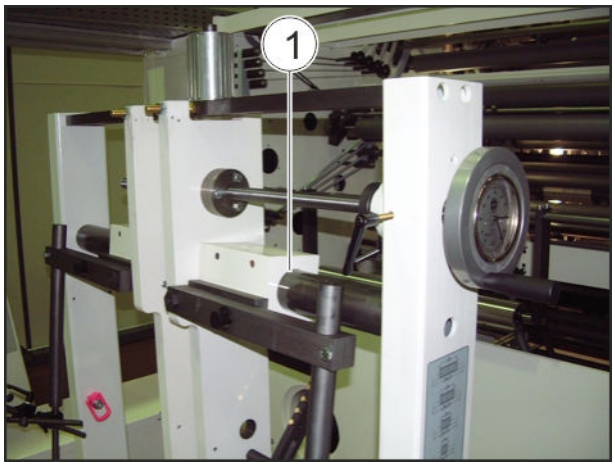


▶ Grease the spindle (1).

Interval: 400 hours
Use: K2K-20

Greasing the bearing

(1)	Bearing

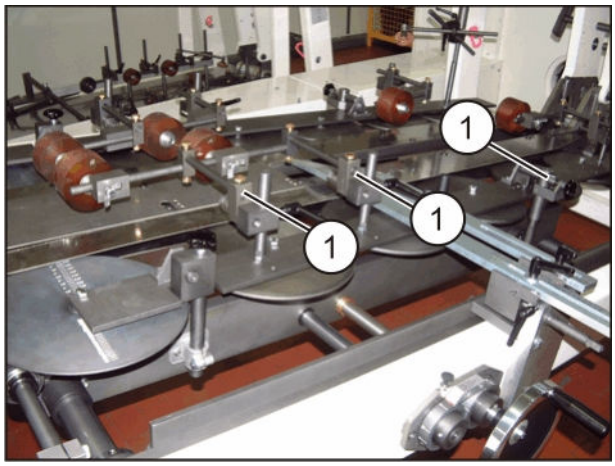


▶ Grease the bearing (1).

Interval: 200 hours
Use: KP 2 K-20

Greasing the joints

(1)	Joint

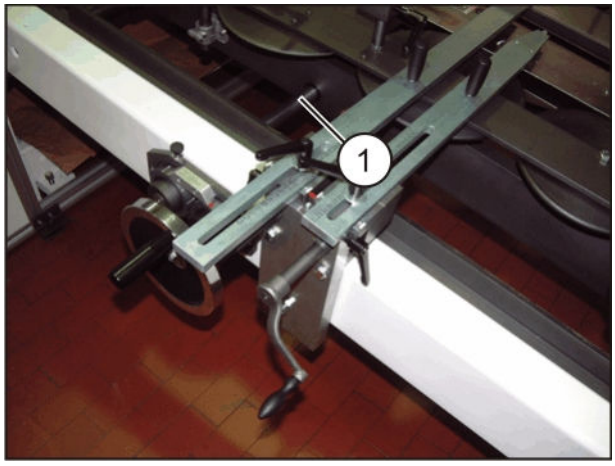


▶ Grease the joints (1).

Interval: 200 hours
Use: K2K-20

Adjustment unit Former plates – Greasing the spindle

(1)	Spindle



▶ Grease the spindle (1).

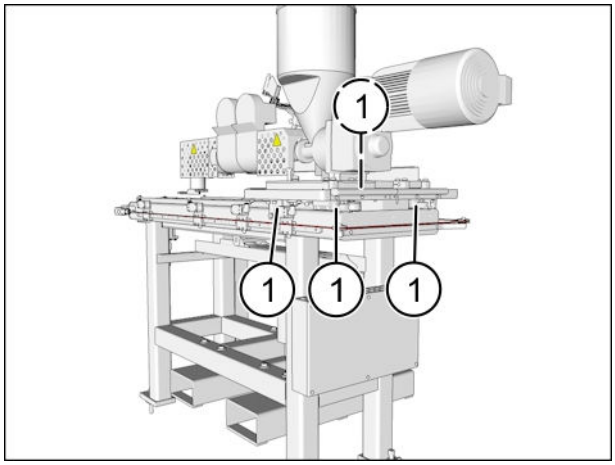
Interval: 400 hours
Use: K2K-20

8.3.6 Longitudinal seam extruder

Bearing

Greasing the ball bearing carriage

(1)	Ball bearing carriage



▶ Grease the ball bearing carriage (1) every 2,000 hours.

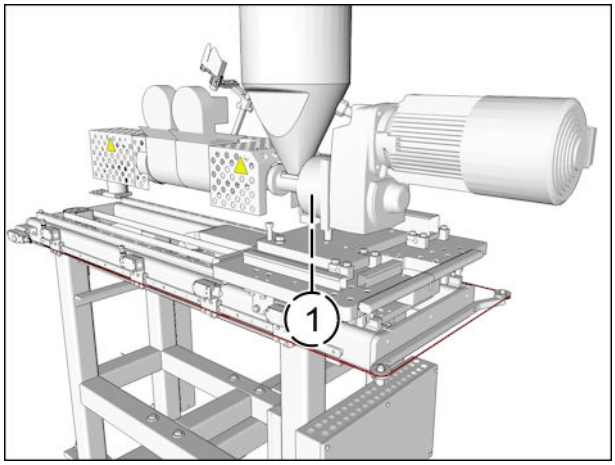
If necessary, more often.
Immediately grease the ball bearing carriage, if no grease film is generated when moving the extruder.

Use: KP2K-20

Return pressure bearing

Greasing the back pressure bearing

(1)	Back pressure bearing



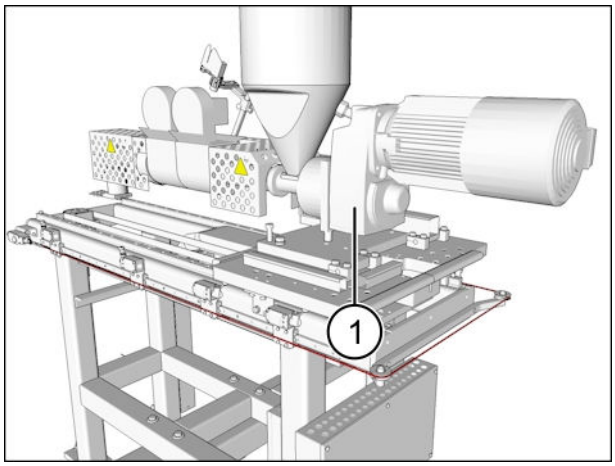
▶ Grease the back pressure bearing (1) every 8,000 hours.

Grease the back pressure bearing every 2 years at the latest.
Use: SKF roller bearing grease LGMT-3
Fill 30 up 50% of the space in the bearing with grease.
Replace the sealing ring.

Gearbox

Gearbox - Correcting the oil level

(1)	Gearbox

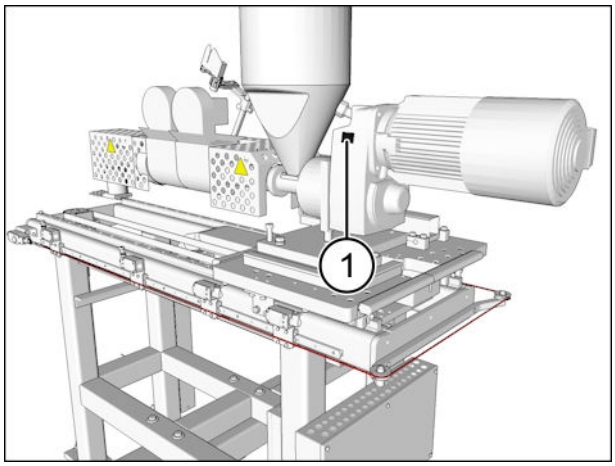


▶ Correct the oil level in the gearbox (1) every 2,000 hours.

Observe the oil level mark.
Use: CLP 220

Gearbox Carrying out an oil change

(1)	Gearbox [rating plate]



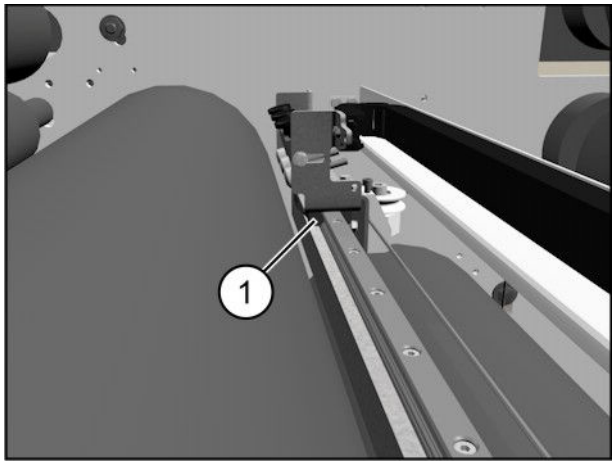
▶ Change the oil in the gearbox (1) every 8,000 hours.

Change the oil every 2 years at the latest.
Observe the oil level mark.
Use: CLP 220
Filling rate: see rating plate

8.3.7 Longitudinal seam monitoring

Greasing the ball bearing carriage

(1)	Ball bearing carriage



▶ Grease the ball bearing carriage (1) every 2,000 hours.

If necessary, more often.
Immediately grease the ball bearing carriage, if there is no grease film when moving the extruder.

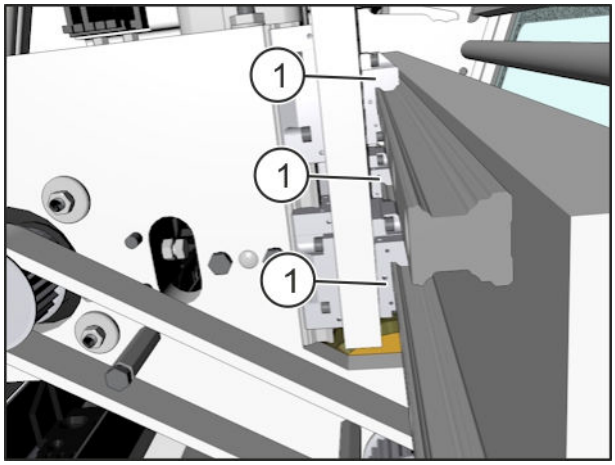
Use: KP2K-20

8.3.8 Main draw

Guide cart - Positioning the seam cooling roller

Greasing the guide cart

(1)	Guide cart



▶ Greasing the guide cart (1).

Interval: 2,000 hours

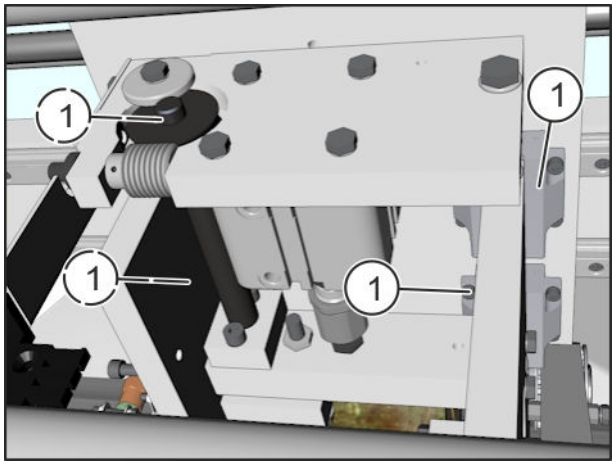
If the movement does not generate a grease film on the guide rails, the ball bearing carriage should be greased immediately.

Use: KP2K-20

Guide cart - Raising and lowering the seam cooling roller

Greasing the guide cart

(1)	Guide cart



▶ Greasing the guide cart (1).

Interval: 2,000 hours

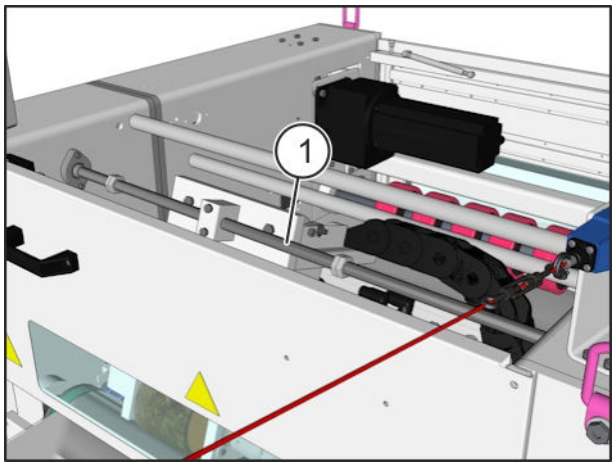
If the movement does not generate a grease film on the guide rails, the ball bearing carriage should be greased immediately.

Use: KP2K-20

Spindle - Positioning the seam cooling roller

Greasing the spindle

(1)	Spindle



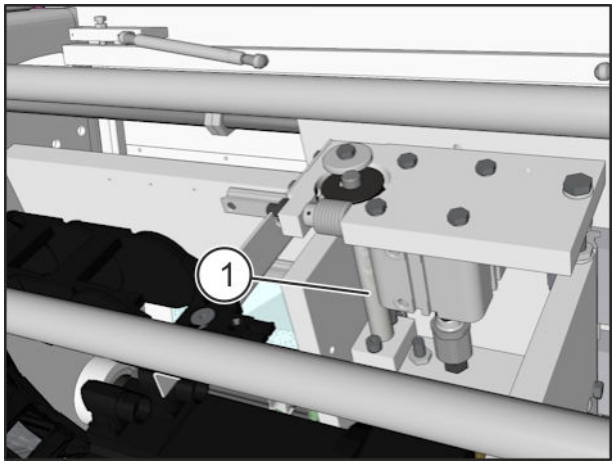
▶ Grease the spindle (1).

Interval: 500 hours
Use: K2K-20

Spindle - Turning the seam cooling roller vertically

Greasing the spindle

(1)	Spindle



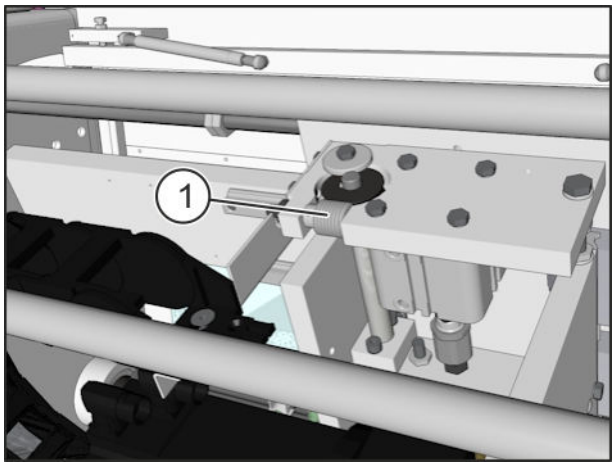
▶ Grease the spindle (1).

Interval: 500 hours
Use: K2K-20

Gear wheel - Turning the seam cooling roller vertically

Greasing the gear wheel

(1)	Gear wheel

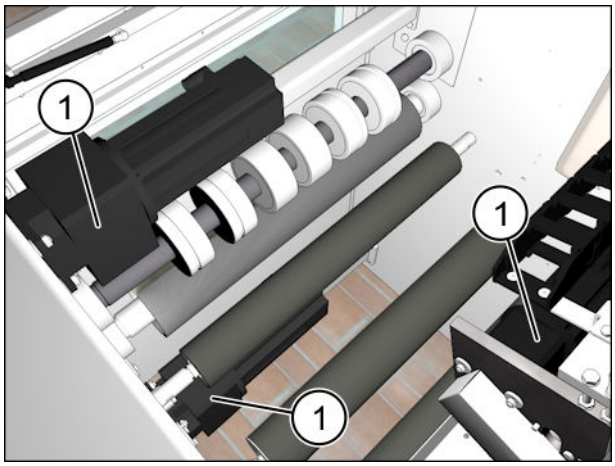


▶ Grease the gear wheel (1).

Interval: 2,000 hours
Use: KP 2 K-20

Gearbox

(1)	Gearbox

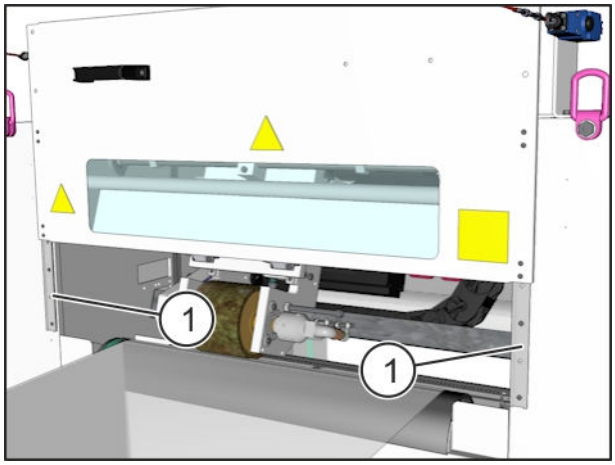


The gearbox (1) is lifetime lubricated. An oil change is not necessary.

Slide protection, infeed section

Greasing the telescope rail

(1)	Telescope rail



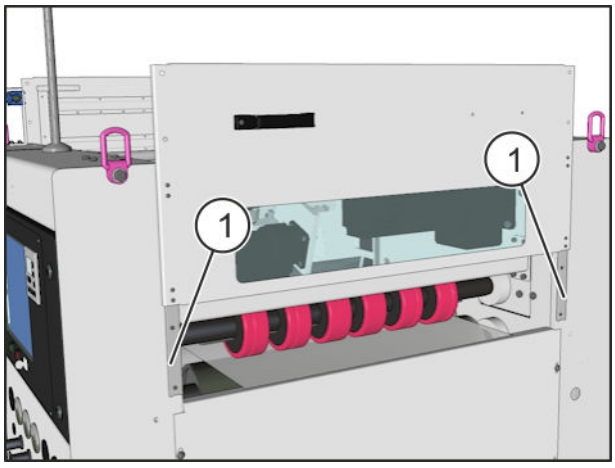
▶ Grease the telescope rail (1).

Interval: 1,000 hours
Use: KP 2 K-20, e.g. lithium base grease for roller bearings

Slide protection, outfeed section

Greasing the telescope rail

(1)	Telescope rail



▶ Grease the telescope rail (1).

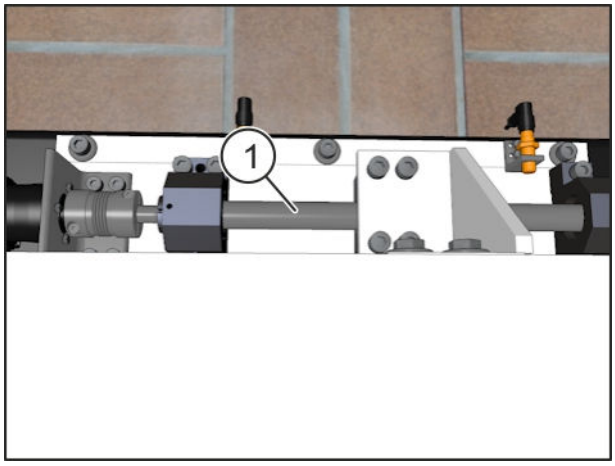
Interval: 1,000 hours
Use: KP 2 K-20, e.g. lithium base grease for roller bearings

8.3.9 Oscillation

Greasing the spindle

At the outfeed section of the winder.

(1)	Spindle



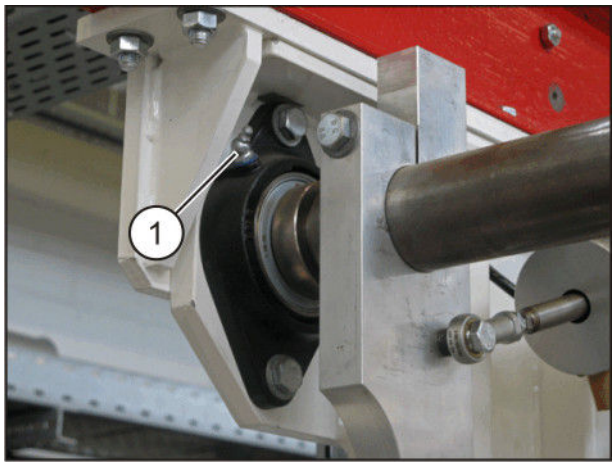
▶ Grease the spindle (1).

Interval: 2,000 hours
Use: K2K-20

8.3.10 Dancer roller

Greasing the flange bearings

(1)	Flange bearing



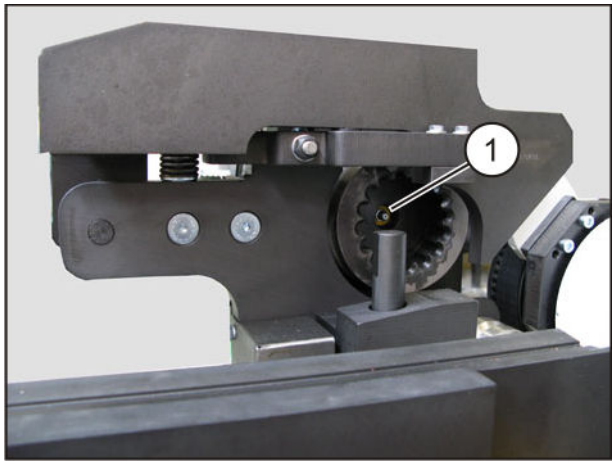
▶ Grease the flange bearings (1) on either side of the dancer roller.

Interval: 3,000 hours
Use: Lube grease K2K-20, grease gun

8.3.11 Center drive

Lubricating the coupling

(1)	Coupling

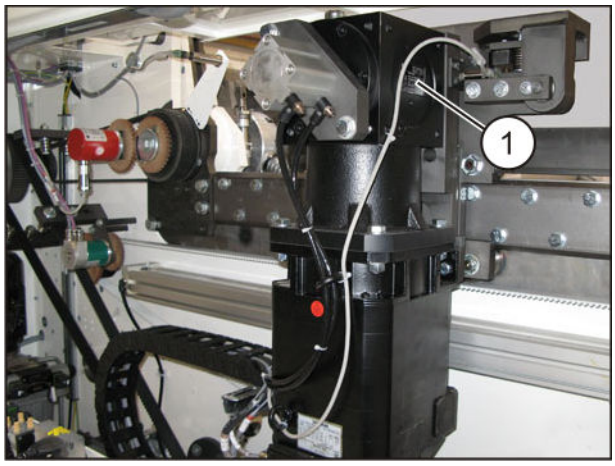


▶ Lubricate the coupling (1) in the bevel gear drive by use of an oil press.

Interval: 1,000 hours
Use: CLP 100 gearbox oil

Change the oil

(1)	Bevel gear drives



▶ Carry out an oil change.

Interval: 15,000 hours
Use: Fully synthetic gearbox oil GH6-032 (provided by KLÜBER)
Oil level: to the lower edge of the test opening
Filling quantity: approx. 0.7 liters

8.3.12 Hydraulic unit

Check the oil level

(1)	Hydraulic unit

▶ Check the oil level in the hydraulic unit (1).

Interval: 2,000 hours

▶ If necessary, refill with oil.

Use: HLP 46
Oil level: to the middle of the oil level indicator

Changing hydraulic oil

(1)	Hydraulic unit

▶ Change the hydraulic oil in the hydraulic unit (1).

Interval: 6,000 hours
Use: HLP 46
Oil level: up to middle of oil level gauge

Filling volume: approx. 13 liters

9 Doing maintenance work

9.1 General maintenance
- 9.1.1 Safe behavior
- 9.1.2 Safety-relevant components
- 9.1.3 Emergency-Stop switch
- 9.1.4 Emergency pull wire switch
- 9.1.5 Safety light curtain
- 9.1.6 Safety switch with guard locking
- 9.1.7 Safety switch without guard locking
- 9.1.8 Warning devices
- 9.1.9 Screws
- 9.1.10 Roller covers
- 9.1.11 Timing belts
- 9.1.12 Universal joint shafts
- 9.1.13 Proximity sensors
- 9.1.14 Filter
- 9.1.15 Pneumatic cylinder
- 9.1.16 Establishing a remote diagnostics connection

9.2 Winder
- 9.2.1 Replacing the cut-off knife

9.1 General maintenance

9.1.1 Safe behavior....360
General....360
Switching to neutral....360
Power-driven technical equipment....360
Machine movements....360
Removing safety devices....360
Overhead installation work....360

9.1.2 Safety-relevant components....361

9.1.3 Emergency-Stop switch....362

9.1.4 Emergency pull wire switch....362

9.1.5 Safety light curtain....363

9.1.6 Safety switch with guard locking....363

9.1.7 Safety switch without guard locking....364

9.1.8 Warning devices....364

9.1.9 Screws....365

9.1.10 Roller covers....365
General....365
Storing....365
Cleaning....365
Operation....365

9.1.11 Timing belts....366
General....366
Tightening the timing belt....366
Tightening the timing belt using a spring balance....366

9.1.12 Universal joint shafts....368
General information....368
Prepare the universal joint shafts....368
Fitting the universal joint shafts....368

9.1.13 Proximity sensors....370

9.1.14 Filter....370
Basics....370
Cleaning....370

9.1.15 Pneumatic cylinder....371

9.1.16 Establishing a remote diagnostics connection....372

9.1.1 Safe behavior

General

NOTE

Make sure to avoid accidents

All of the safety devices and EMERGENCY-STOP features on the machine should be checked regularly when running at base speed.

You must abide by the pertinent regulations and safety requirements.

Repair work must be carried out carefully by skilled and authorized personnel so that operation can take place safely.

Repair work must only be carried out by skilled personnel.

Before starting any repair work the operating personnel must be informed accordingly. Appoint a supervisor.

After finishing repair works, always retighten slackened screw connections and properly reinstall previously removed guard doors and covers.

Switching to neutral

CAUTION

Take appropriate measures to prevent any dangers to personnel from arising.

Repair work should be carried out only with the machine at a standstill or when power supply is cut off.

During repair work, all switching on/off procedures must be carried out as stipulated in the instruction manual.

If the machine is completely switched off for repair purposes, it must be secured against inadvertent starting.

Lock the main command devices and withdraw the keys.

Attach a danger sign to the main switch.

Switch machine parts to neutral.

Check disconnected parts for presence of power, then ground and short out them.

Insulate adjacent parts that are energized.

Power-driven technical equipment

Adjustments to power-driven technical equipment on the machine and repair work on moving parts is prohibited.

Disengage the power-driven technical equipment before starting any repair work.

Machine movements

Appropriate precautions must be taken to protect the personnel from accidents if it becomes necessary to switch the machine on and to move certain parts so that the repair work can be carried out.

If it is imperative for repair purposes to move certain parts or to cut off power supply, all safety measures must be taken to protect the personnel from accidents.

Continue repair work only after the machine has been moved and the main switch has been switched off.

Removing safety devices

If any of the provided safety devices must be removed to enable necessary repair work, they must be refitted immediately after maintenance work has been finished. After refitting, check the safety devices for proper functioning.

Refit and close all removed or opened safety devices immediately after completion of the repair work.

Overhead installation work

Always use the climbing aids or work platforms intended for this purpose that meet the safety requirements.

Do not use machine parts as climbing aids.

9.1.2 Safety-relevant components

Only qualified persons are allowed to carry out such work.
Do not continue to use the machine if the safety-relevant components are faulty.

Safety-relevant components have a maximum service time of 20 years.

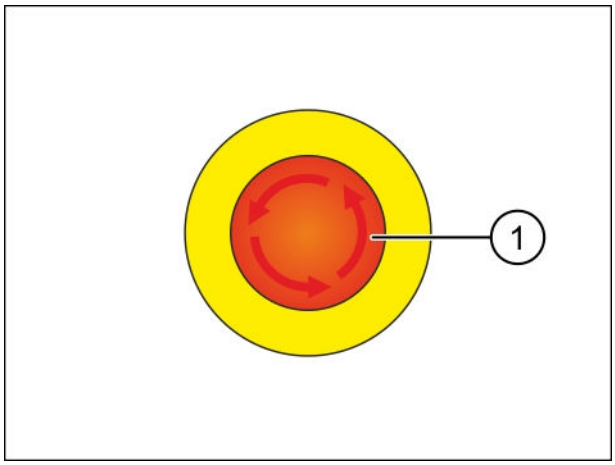
Replace safety-relevant components with equal components or components of equal quality.

If you have any questions please do not hesitate to consult W&H's Service Department.

9.1.3 Emergency-Stop switch

Only qualified persons are allowed to carry out such works.
Do not continue to use the machine if the safety device is faulty.

(1)	Emergency-Stop switch



Check the Emergency-Stop switch (1) mechanically:

- it is attached safely
- the equipment has not been tampered with or does not show visible damage
- the cables are properly connected
- Emergency-Stop clicks into place when actuated

Check the Emergency-Stop switch electrically:

- the running machine stops
- the control voltage is switched off
- the message is displayed in the operator terminal

Replace the Emergency-Stop switches whenever necessary.

9.1.4 Emergency pull wire switch

Only qualified persons are allowed to carry out such work.
Do not continue to use the machine if the safety device is faulty.

Check the emergency trip wire and the emergency pull-wire switch mechanically to make sure that:

- they are attached safely
- the equipment has not been tampered with or does not show visible damage
- the cables are properly connected

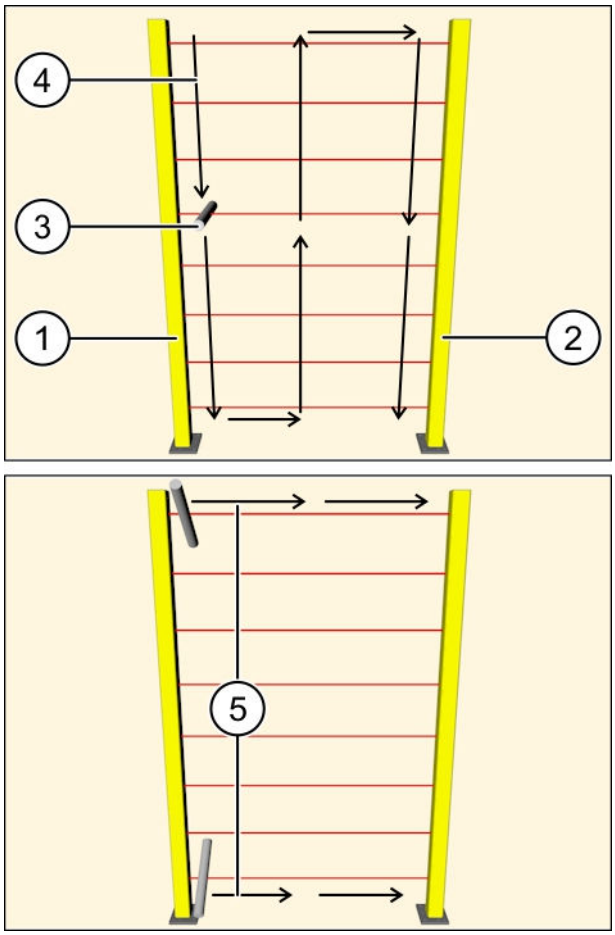
Check the emergency pull-wire switch electrically to make sure that:

- the running machine stops
- the control voltage is switched off
- the message is displayed in the operator terminal

Replace the emergency trip wire and the emergency pull-wire switch whenever necessary.

9.1.5 Safety light curtain

(1)	Transmitter
(2)	Receiver
(3)	Test rod
(4)	Directional movement Test 1
(5)	Directional movement Test 2



- ▶ Check the safety light curtain - transmitter (1) and receiver (2) using the test rod (3) supplied together with the equipment.
- ▶ The signal lamp on the receiver shows a green light.
- ▶ First test: Move the test rod slowly through the safety light curtain (4).
- ▶ Second test: Move the test rod slowly along the edges of the safety light curtain (5).
- ▶ During the test the signal lamp on the received should show steady red light.
- ▶ Readjust the safety light curtain if out of tune or replace, if defective.

9.1.6 Safety switch with guard locking

Only qualified persons are allowed to carry out such work.
Do not continue to use the machine if the safety device is faulty.

Check the safety shutdown switch mechanically to make sure that:

- it is attached safely
- the equipment has not been tampered with or does not show visible damage
- the cables are properly connected

Check the safety shutdown switch electrically to make sure that:

- the guard cannot be opened while the machine is running
- the machine cannot be started when the guard door is open.

Replace the safety shutdown switch whenever necessary.

9.1.7 Safety switch without guard locking

These works should only be carried out by persons who are specially trained for this.
Do not continue to use the machine if the safety device is faulty.

Check the safety switch mechanically to make sure that:

- it is attached safely
- the equipment has not been tampered with or does not show visible damage
- the cables are properly connected

Check the safety switch electrically to make sure that:

- the running machine stops

Replace the safety switch, if necessary.

9.1.8 Warning devices

(1)	Flashing light
(2)	Horn

Check the acoustic and optical warning devices for proper functioning at regulator intervals and replace whenever necessary.

9.1.9 Screws

- Check all screws under dynamic load for tight fit at regular intervals and tighten to the specified torque, if necessary. Make sure to tighten the individual screws to their specified torques.
- Apply a thin coat of threadlocker to all the screws as they have been secured before. The type of threadlocker to be used can be taken from the relevant sub-assembly drawing.
- Use new lock nuts for reassembly to-secure the screw connections as they have been secured before.
- After reassembly, the screw connections should generally be secured like they have been secured before.

Diameter	Diameter Wrench size	Torque [Nm] Property class 8.8	Torque [Nm] Property class 10.9
M6	10	10	15
M8	13	25	36
M10	17	49	72
M12	19	85	125
M16	24	210	310
M20	30	425	610
M24	36	730	1050

9.1.10 Roller covers

General

Improper storage conditions have an adverse effect on the properties of the roller covers and leads to a reduction in the service life of a roller cover.

In order to ensure that the covers have a long service life, please observe the following instructions on storage, cleaning and operation. Please refer to DIN 7716 (ISO 8331) for other requirements regarding the storage and cleaning of the roller covers.

Storing

Rollers	Should be stored in their transport case and supported on studs. Do not apply or fit coverings. Turn rollers through 180 degrees, turning 'soft' qualities more frequently than 'hard' ones.
Room heating	Screen rollers from heating elements.
Temperature	5°C - 20°C
Humidity	Max. 65% relative humidity.
Lighting	Protect from direct sunlight and strong artificial light with high UV content.
Ozone	Storage areas must be absolutely free of ozonic equipment (e.g. pre-treater, electric motors).
Other	Solvents, fuels and other chemicals must not be stored in the same area as rollers.

Cleaning

Coated rollers	Use solvents which are also contained in the inks, adhesives etc. in use. Clean the roller covers with a little bit of solvent keeping the cleaning time to a minimum, then wipe dry with a clean cloth. Never use hard objects for cleaning.
Rubberized rollers	Only ever use soap, warm water and a lint-free cloth when cleaning rubberized rollers. The manufacturer recommends the use of effective and gentle cleaning agents.
Counter rollers	Remove deposits or contaminations, if any.

Operation

Relieve pressure immediately when roller covers running under pressure come to a standstill. Dry running of the rollers must be avoided. Regular maintenance of the roller coverings prevents major damage and downtimes.

9.1.11 Timing belts

General

NOTE

Be careful to avoid excessive wear and belt breakage.

Tightening the timing belts results in a changed position of the tools.

Check the tool position after tightening the belts.

Rectify the position of the tools as required.

Tightening the timing belt

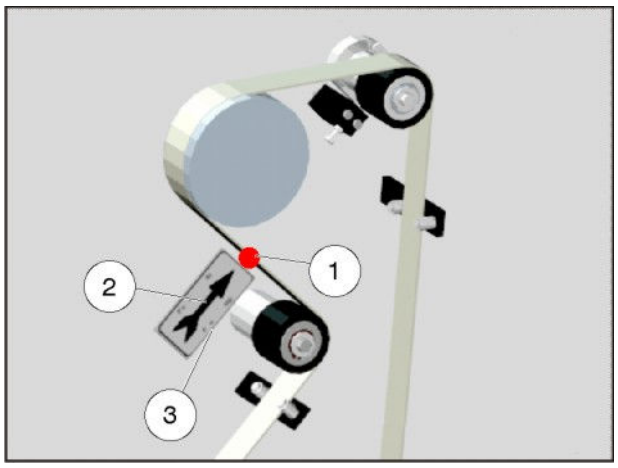
NOTE

Only tighten new belts to the specified frequency.

Do not re-use belts after use in order to ensure operational reliability.

If timing belts are to be re-used measure belt frequency prior to removing it from the unit and restore the same frequency after re-installing the belt back in place.

(1)	Measuring point
(2)	Arrow pointing to the measuring point
(3)	Frequency for a new timing belt



- ▶ Pre-tighten the timing belt using an appropriate tightener.
- ▶ Check timing belt for proper alignment and correct as necessary.
- ▶ Measure the belt frequency at measuring point (1) using the tensiometer.
- ▶ Tighten new timing belt to the specified frequency.
- ▶ When installing a used belt, tighten the belt to the frequency measured prior to removing it.

Tightening the timing belt using a spring balance

This adjustment can only be made if Fv = N (vertical force in Newton) is mentioned on the timing belt's rating plate.

NOTE

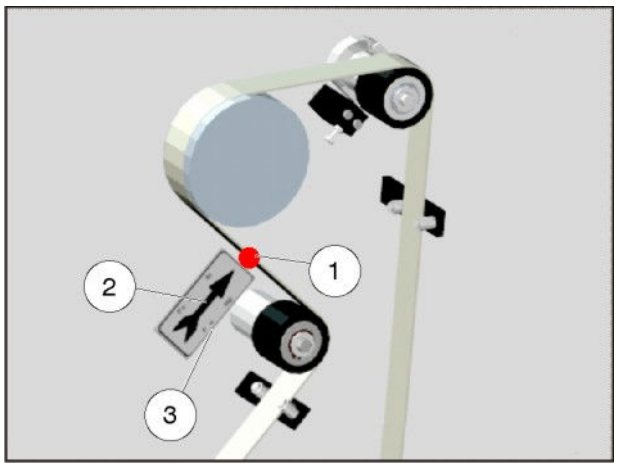
After that tighten the new timing belt to the specified force.

Do not re-use belts after use in order to ensure operational reliability.

If timing belts are to be re-used measure the belt force prior to removing it from the unit and restore the same force after re-installing the belt back in place.

Use spring balances suitable of measuring forces up to 3000 N.

(1)	Measuring point
(2)	Arrow points to the measuring point
(3)	Value of the vertically applied force to a new timing belt



- ▶ Pre-tighten the timing belt using an appropriate tightener.
- ▶ Check timing belt for proper alignment and correct as necessary.
- ▶ Fit the spring balance at the measuring point (1).
- ▶ Retract the spring balance vertically from the timing belt until a deformation of the timing belt becomes visible. Then, read the reached force at the spring balance.
- ▶ Tighten a new timing belt until the spring balance shows the specified value.
- ▶ When installing a used belt, tighten it to the value measured prior to removing it.

9.1.12 Universal joint shafts

General information

NOTE

Avoid wear

Check the assembly and installation position of the universal joint shaft.

X000017673

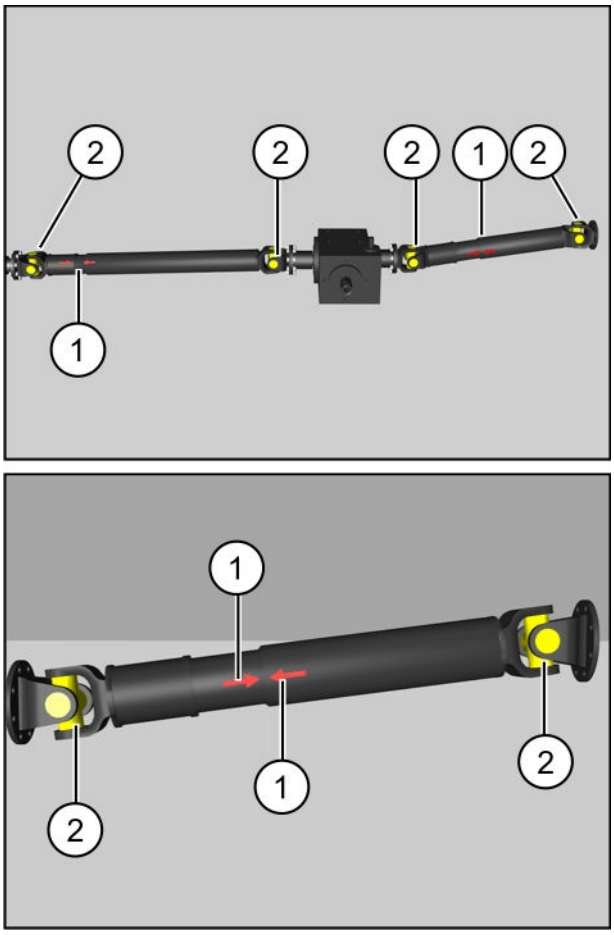
Prepare the universal joint shafts

Do not separate and replace the universal joint shafts on the wedge profile between each other.

Do not remove the balance weights.

Swapped universal joint shafts and balance weights that have been removed reduce the balancing quality. The path of the universal joint shafts disturbed.

(1)	Marking
(2)	Fork of the universal joint



- Clean the flange surfaces

  Only clean flange surfaces can transfer the desired torque.

- Assemble the universal joint shafts according to the following markings (1).

- If two or more universal joint shafts are arranged one after another, screw the forks (2) of the universal joint turned at 90 degrees to one another on the counter flange of the drive or the intermediate bearing.

  The forks turned against each other ensure a quieter path of the articulated shafts.

Fitting the universal joint shafts

With horizontally running universal joint shafts, mount the shorter end of the universal joint shaft to the drive side.

Fix the universal joint shaft with screws of property class 10.9.

With counter flanges that have through holes, used self locking nuts of property class 10.

With counter flanges with threads, insert the screws with locking washers and screw locking device.

Equally tighten the four screws crosswise using a pneumatic screwdriver.

Tightening torque:

- M5 screws: 8 Nm
- M6 screws: 14 Nm
- M8 screws: 35 Nm
- M10 screws: 69 Nm
- M12 screws: 120 Nm

9.1.13 Proximity sensors

Proximity sensors are positioning sensors which detect initial, end or defined machine part positions.

- After replacing proximity sensors, resetting is required.
- For setting up the proximity sensors, see the manufacturer's instructions.
- Before removing the proximity sensor, measure its distance from the cam switch/casing.
- Replace the proximity sensor and set up the distance noted down before.

9.1.14 Filter

Basics

- If the ambient air contains coarse fluff, filters should be installed in all of the machine's separately driven fans, cooling and climate control units.
- Filters when installed reduce the cooling capacity.
- Exhausters must never be operated without filters.

Cleaning

CAUTION

Dust can harm your respiratory tract!

Working on the device without wearing a breathing mask will result in your health being harmed.
Wear a breathing mask (fine dust mask).
Wear eye protection goggles (tight-fitting safety goggles) if severe dust formation occurs.
Remove all sources of ignition (e.g. cigarette lighter).

- Check filter for dirt at regular intervals. Clean filter immediately if it appears to be dirty.
- Tap the filter to remove the dirt. Do not brush or wash out the filter.
- Use a vacuum cleaner to carefully remove any remaining dirt from the filter.
- Check the filter for damages after cleaning. Replace the filter when least damages are detected.
- Replace the filter if it cannot be cleaned.
- Replace the filter every two years at the latest.

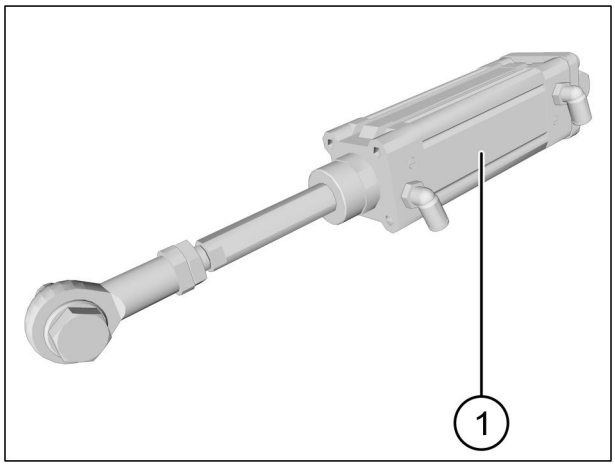
9.1.15 Pneumatic cylinder

The inspection interval depends on the stroke frequency.

(1)	Pneumatic cylinder

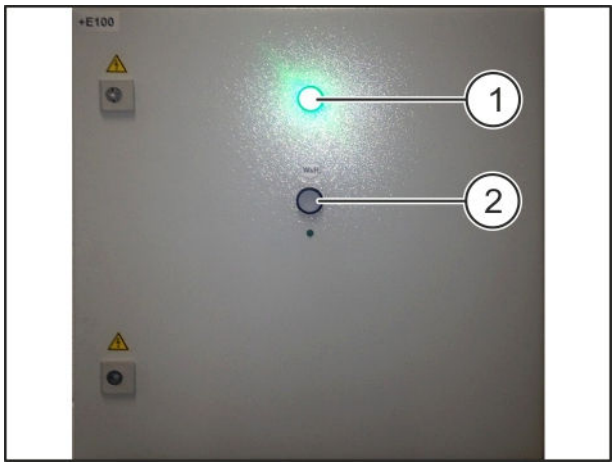
▶ Check the condition of the pneumatic cylinder (1) every 200 hours.

▶ Replace the pneumatic cylinder whenever necessary.



9.1.16 Establishing a remote diagnostics connection

(1)	Signal lamp: Supply voltage
(2)	W&H, establish a VPN connection



- The 'Supply voltage' signal lamp (1) flashes.

  The supply voltage for the +E100 or +E101 switch cabinet is available.

- Press the W&H button (2).

  The pushbutton flashes: The router (mGuard) establishes a Internet connection.
  The pushbutton flashes: A VPN connection (Virtual Private Network connection) is established.

9.2 Winder

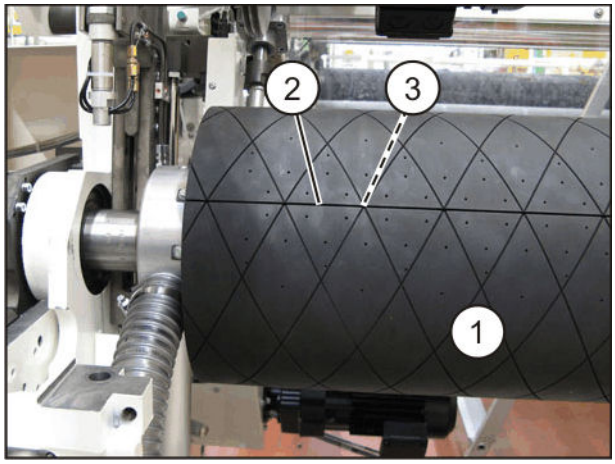
9.2.1 Replacing the cut-off knife....374
Cross cutting knife....374
Preparatory work....374
Safety brake....374
Moving holding plates....374
Loosening the cut-off knives....374
Replacing the cut-off knife....374
Locking the cut-off knives in place....374
Relocating the holding plates....374
Finishing work....374

9.2.1 Replacing the cut-off knife

Cross cutting knife

The vacuum cutover drum equipped is with a pneumatically driven cutting system within the drum body. During a drum change the cross cutting knife temporarily protrudes from the drum shell's nip to cut the film web. In case of declining cutting quality the individual knife elements should be replaced.

(1)	Contact roller
(2)	Nip
(3)	Cross cutting knife



Preparatory work

To be able to replace the knives the divided vacuum rotary connection should be dismantled on both sides. To do this, the winder should be in a safe state.

Dismantling the vacuum rotary connection

⚠ CAUTION

Risk of injury from the cut-off knife

Due to the pressure in the pneumatic system the cut-off knife may move in an uncontrolled manner.

Bleed the pneumatic system completely before starting dismantling.

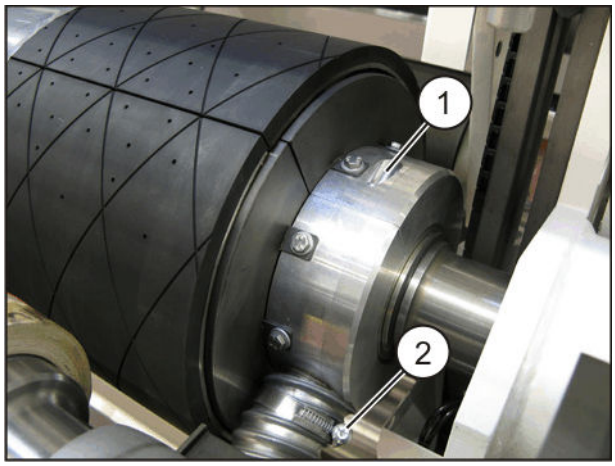
⚠ WARNING

Risk of injury from machine movements

The control voltage may cause the machine to move in an uncontrolled manner.

Turn the key switch WINDING STATION SECURED in position 'I = Winding station secured' and withdraw the key until the repair work is completed.

(1)	Rotary connection
(2)	Hose clip



- ▶ Loosen the hose clip (2) at the vacuum hose.
- ▶ Pull the vacuum hose off the socket on the rotary connection (1) and place down.
- ▶ Screw two socket head screws out of the rotary connection.

  Hold the lower half of the rotary connection with one hand.
- ▶ Remove the upper half first, then remove the lower half and put aside.

Safety brake

The safety brake in the cut-off assembly prevents the cut-off knives in the nip from moving in an uncontrolled manner. The safety brake must be pneumatically vented when changing the cut-off knives.

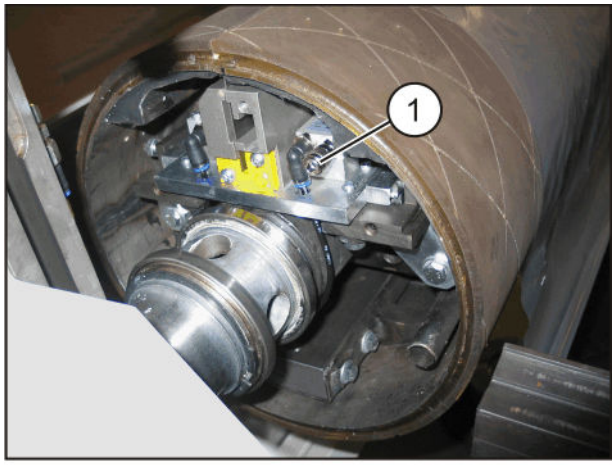
DANGER

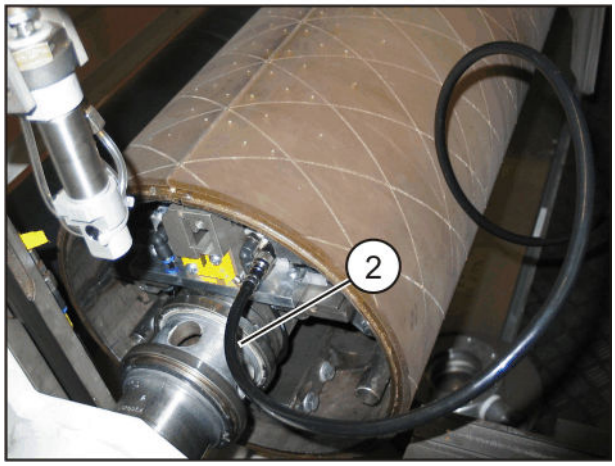
Risk of severe injuries from protruding cut-off knives!

Wear cut-resistant industrial safety shoes.

Take care not to touch the extremely sharp tips of the cut-off knives!

(1)	Compressed-air connection
(2)	Compressed-air hose





- Connect the supplied compressed air hose (2) to the compressed-air connection (1) on the face side and connect it to the compressed-air supply on site.

  The pressurized brake cylinder releases the connecting rod with the cut-off knives to move.

Moving holding plates

The holding plates secure the knife bar and the retainer beam in place to each other. Before replacing the cut-off knives the holding plates on both sides should be moved from the working position into the maintenance position.

DANGER

Risk of severe injuries from protruding cut-off knives!

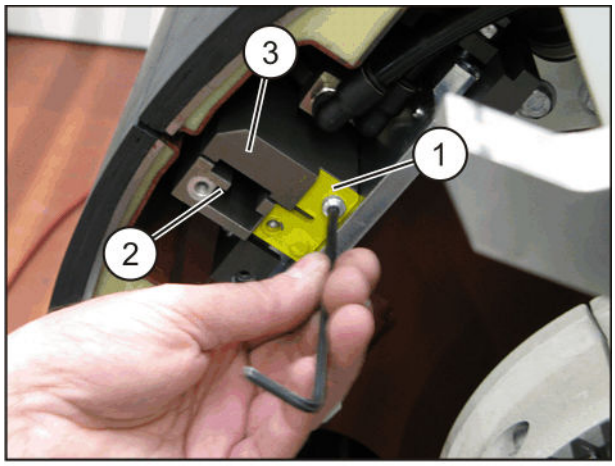
Wear cut-resistant industrial safety shoes.

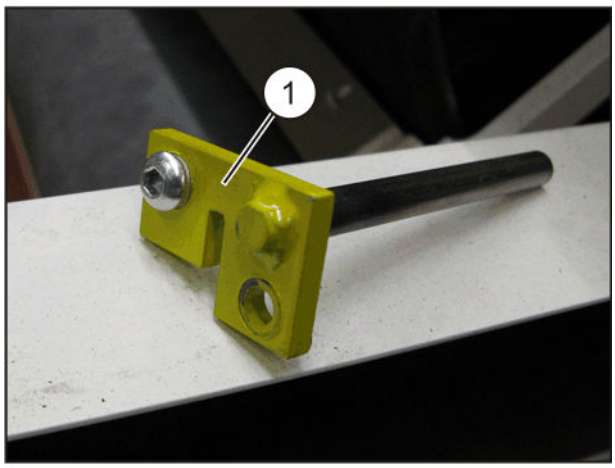
Take care not to touch the extremely sharp tips of the cut-off knives!

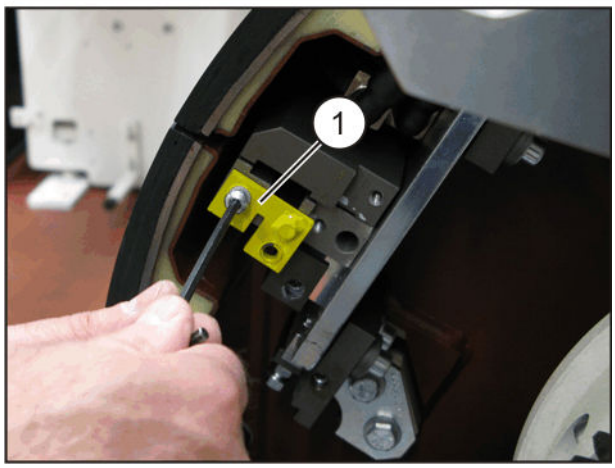
(1)	Holding plate with bolt
(2)	Knife bar
(3)	Retainer beam

- Use an Allen wrench to turn out both socket head screws provided to attach the holding plate (1) to the knife bar and the retainer beam.
- Remove the holding plate and attach it to the retainer beam so that the welded bolt presses the connecting rod with the cut-off knives back. Use the upper threaded hole in the knife bar for this.

  The bolt forces the connecting rod with the cut-off knives into the cutting position. The knife tips protrude from the nip.
- Move the holding plate with the bolt on the opposite side of the cut-off assembly in the same way.







Loosening the cut-off knives

Via the connecting rod in the knife bar, the cut-off knives are moved in the axial direction. The knives slide with their curved recess over roller bearings which are attached to the retainer beam. This raises the cut-off knives. For dismantling, the retainer beam must be moved to the side until the roller bearings are outside the recesses.

DANGER

Risk of severe injuries from protruding cut-off knives!

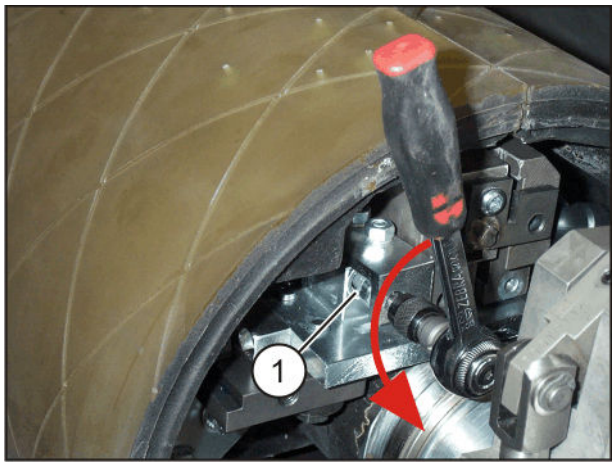
Wear cut-resistant industrial safety shoes.

Take care not to touch the extremely sharp tips of the cut-off knives!

(1)	Setscrew M6

▶ Turn the setscrew (1) anticlockwise until it hits the stop to move the retainer beam to the side.

The retainer beam is moved to the side via a locking bar and the cut-off knives are released for extraction.



X000059481

Replacing the cut-off knife

DANGER

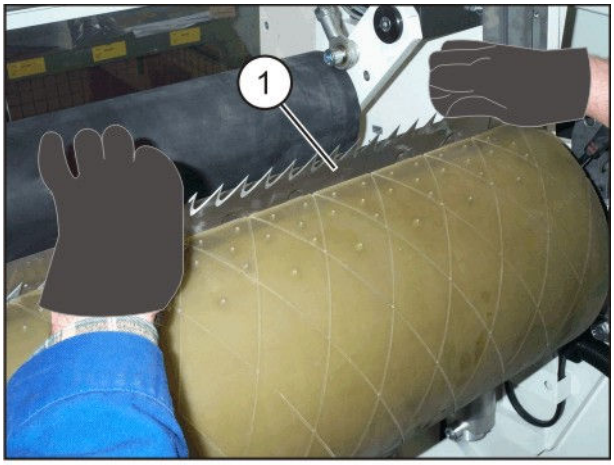
Risk of cuts from extremely sharp cut-off knives.

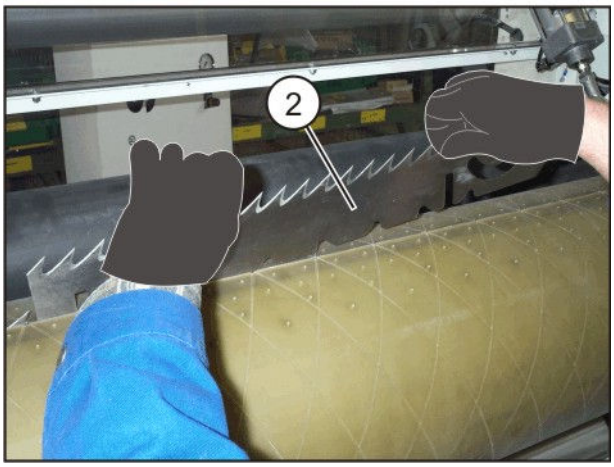
Lift the cut-off knife with both hands or with the aid of two clamping tongs out of the nip and place it down on a safe underlay.

Wear cut-resistant safety gloves.

Wear cut-resistant industrial safety shoes.

(1)	Used cut-off knife
(2)	New cut-off knife





- Have new cut-off knives ready for replacement on a safe underlay.
- Take the used cut-off knife (1) with both hands (alternatively with two clamping tongs) to pull it carefully upwards and out of the nip, then place it down safely.
- Insert the new cut-off knife (2) immediately. This procedure should be followed for the next one.

  It should be possible to insert the cut-off knives smoothly through the nip between the knife bar and the holding bar.

  After replacement of all the cut-off knives, they must be locked in place again.

Locking the cut-off knives in place

DANGER

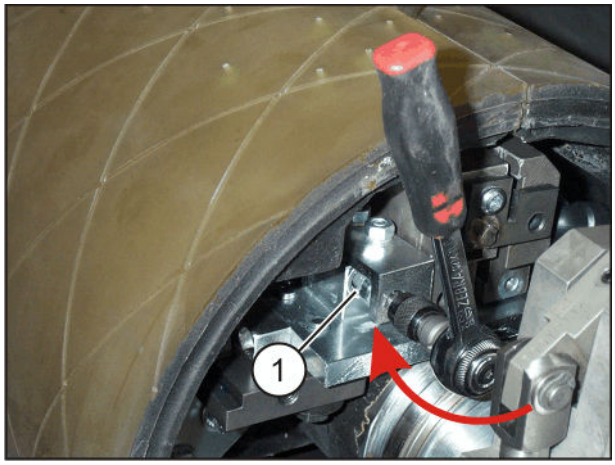
Risk of severe injuries from protruding cut-off knives!

Wear cut-resistant industrial safety shoes.

Take care not to touch the extremely sharp tips of the cut-off knives!

The cut-off knives should be easily movable and in proper alignment with each other.

(1)	Setscrew



- Turn the setscrew (1) carefully clockwise until it hits the stop to move the retainer beam to the side.

  Never use force!

  The knife carrier and the retainer beam get together again. The roller bearings at the retainer beam immerse in the curved recesses of the cut-off knife.

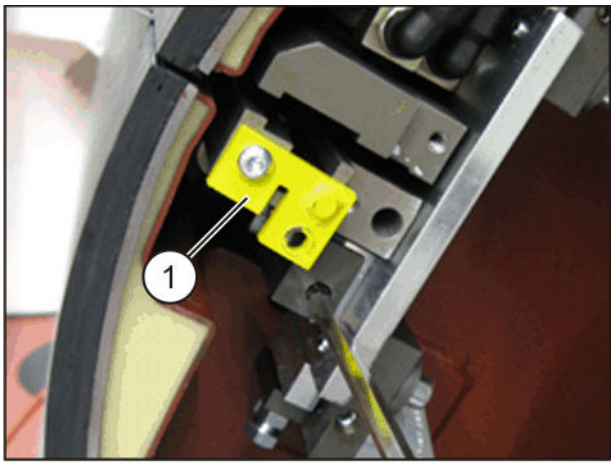
- Use a long Allen wrench or screwdriver to force the connecting rod in the knife carrier back into its initial position.

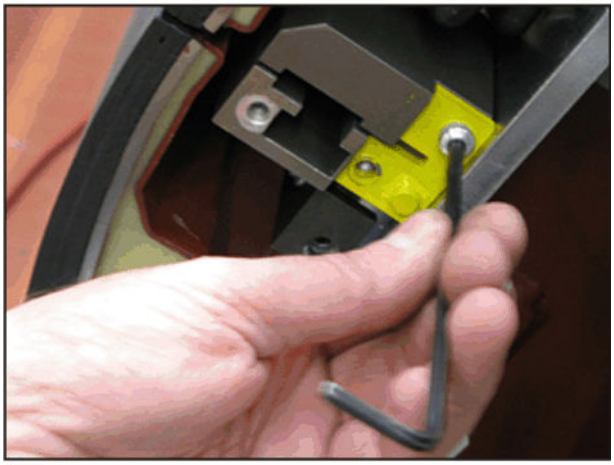
  The cut-off knives are lowered again by this.

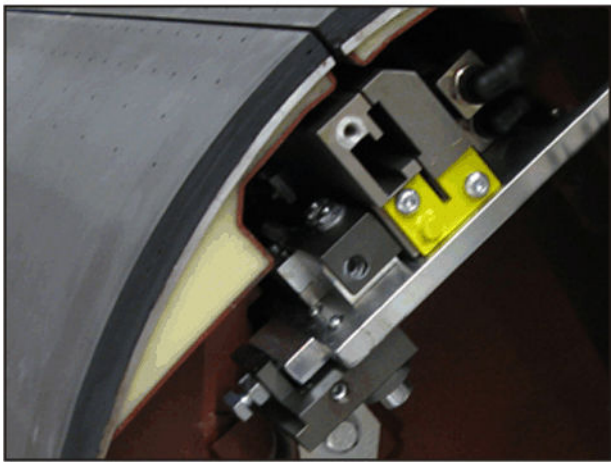
Relocating the holding plates

(1)	Holding plate

▶ Relocate the holding plates (2) on both sides of the cut-off assembly from their maintenance position into their working position. Use one screw each to fasten the holding plate to the knife carrier and to the retainer beam.

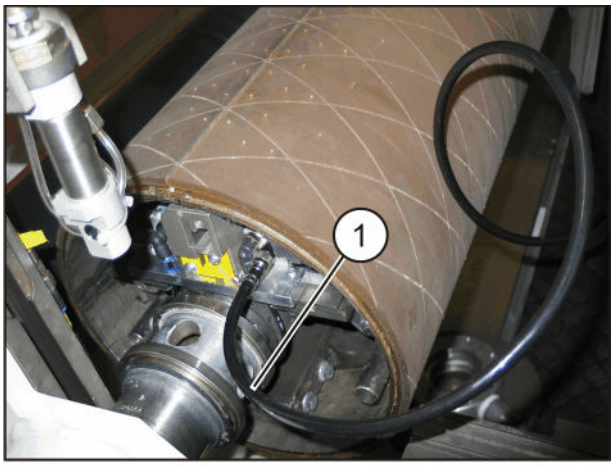


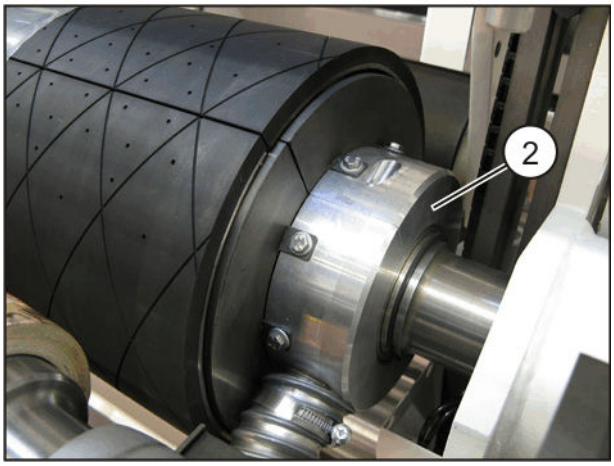




Finishing work

(1)	Compressed-air hose
(2)	Rotary connection





- ▶ Remove the compressed-air hose (1) from the compressed-air connection.
- ▶ Refit the rotary connections (2) incl. the cover on both face sides of the roll body.

Replacement of the cut-off knives is now completed.

10 Basic setting

10 Basic setting

10.1 Longitudinal seam monitoring 386
Positioning the sensor 386
Aligning the sensors 386
Setting up longitudinal seam monitoring 386
Setting up the sensitivity of longitudinal seam monitoring 386

10.2 Setting up the seam cooling roller – Setting up the idle stroke 392

10.3 Setting up the seam cooling roller 393
Safety instructions and general information 393
Aligning the seam cooling roller horizontally 393
Aligning the seam cooling roller vertically 393
Setting the contact pressure 393

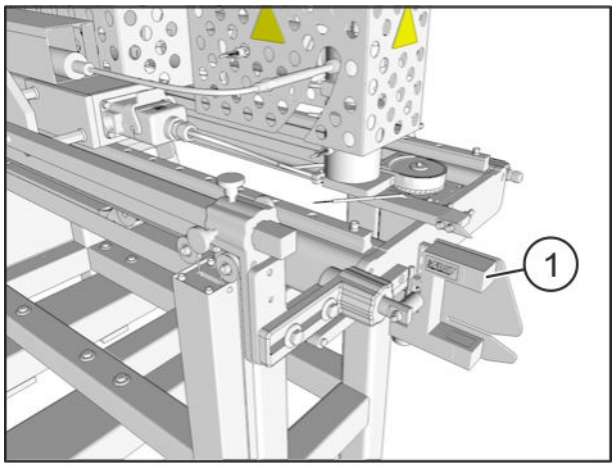
10.4 Oscillation 398

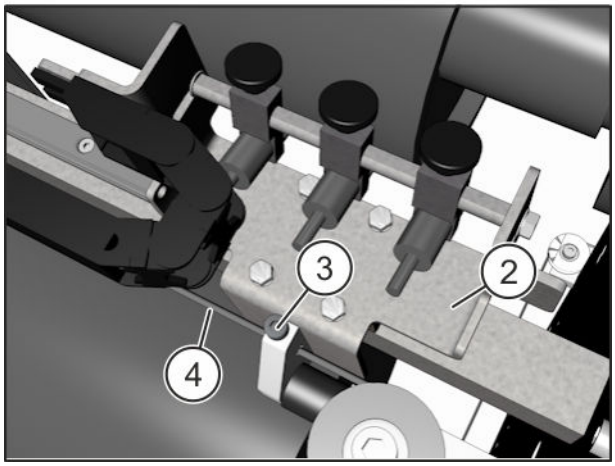
10.1 Longitudinal seam monitoring

Positioning the sensor

Preparing the positioning process

(1)	Sensor
(2)	Slide
(3)	Screw
(4)	Rope



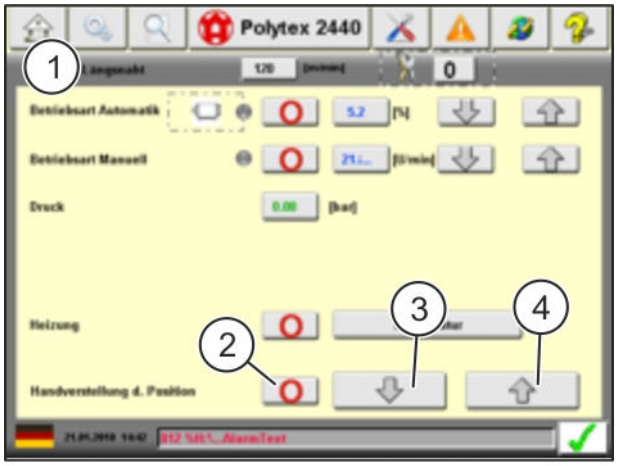


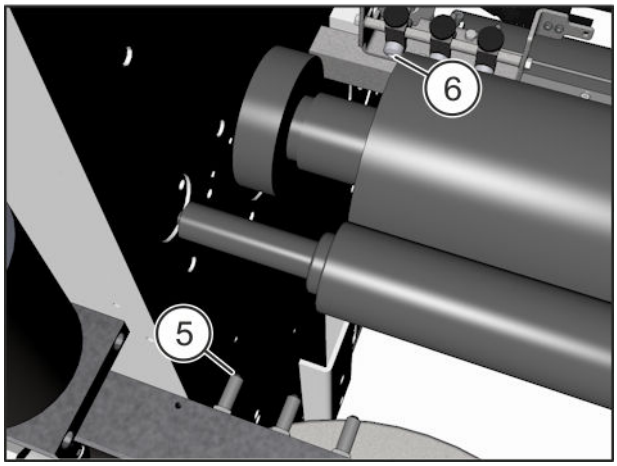
- ▶ Fold up the sensor (1) of the web guide control unit at the applicator.
- ▶ Slacken the screw (3) in the main draw at the slide (2) of the sensors.

  When running the seam extruder, the slide is no longer carried along by the rope (4).

Running the longitudinal seam extuder

(1)	'Longitudinal seam' screen page
(2)	Manual position adjustment YES/NO
(3)	LONGITUDINAL SEAM EXTRUDER TOWARDS THE OPERATOR SIDE
(4)	LONGITUDINAL SEAM EXTRUDER TOWARDS THE DRIVE SIDE
(5)	Nozzle
(6)	Sensor



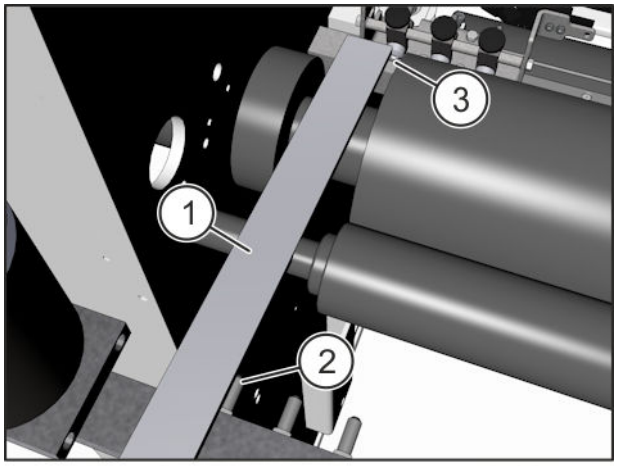


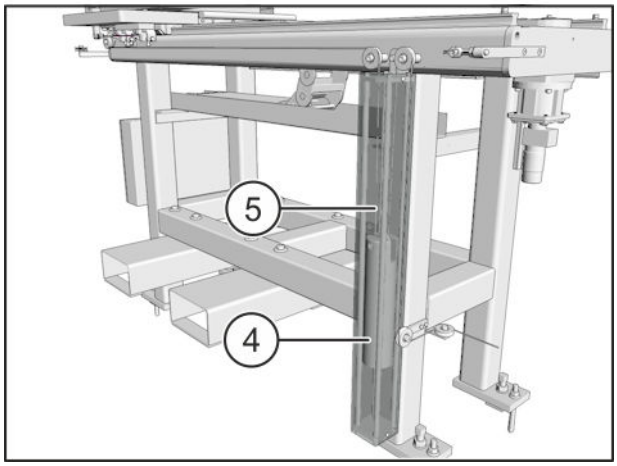
- ▶ Open the 'Overview' screen page at the operator terminal.
- ▶ Press the LONGITUDINAL SEAM button.

  The 'Longitudinal seam' screen page (1) pops up.
- ▶ Switch on 'Manual position adjustment' (2) mode.
- ▶ Use button (3) and button (4) to run the longitudinal seam extruder until the middle of the exit opening in the nozzle (5) points at the middle of the sensor (6).

Aligning the sensor

(1)	Ruler
(2)	Nozzle
(3)	Sensor
(4)	Weight
(5)	Rope



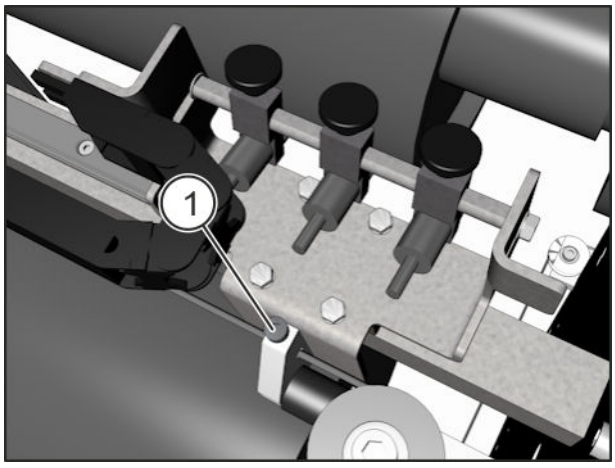


▶ Use a long ruler (1) to check the nozzle (2) for proper positioning to the sensor (3).

For defining the position of the nozzle to the sensor, hold up the ruler at a machine component which aligns with the machine frame. The middle of the exit opening in the nozzle should point at the middle of the sensor. In this position, the weight (4) should touch the floor and the rope (5) should be tensioned.

Finishing the positioning process

(1)	Screw

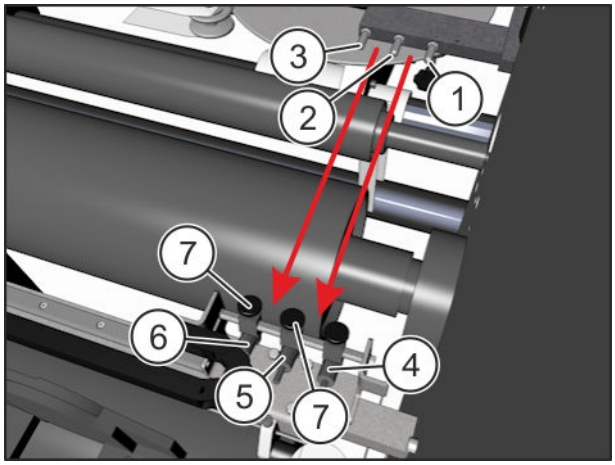


▶ Tighten the screw (1).

Aligning the sensors

Align optional sensors

(1)	Nozzle
(2)	Nozzle
(3)	Nozzle
(4)	Sensor
(5)	Sensor
(6)	Sensor
(7)	Knurled screw



- ▶ Measure the spacings between the exit openings in the nozzles (1) and (2) and between the nozzles (2) and (3).

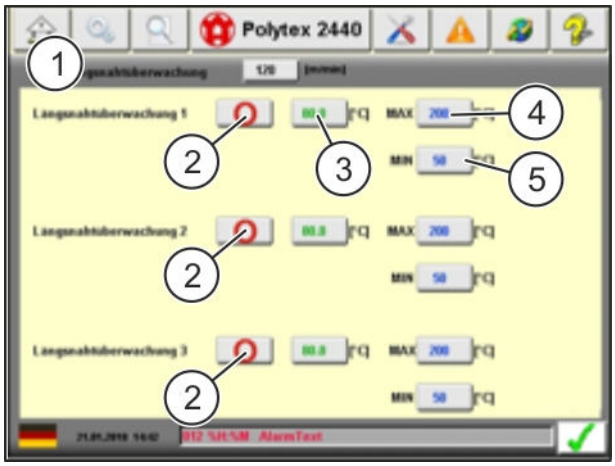
  Use the ascertained values to set up the spacings between the midpoints of the sensors (4) and (5) and between the sensors (5) and (6).
- ▶ Slacken the knurled screws (7).
- ▶ Move sensor (5) and sensor (6) to the positions defined before.

  Do not slacken and displace the sensor (4).
- ▶ Tighten the knurled screws.

Setting up longitudinal seam monitoring

Set up the temperature range to be covered by the longitudinal seam monitor for evaluation. When leaving this temperature range, a message appears and the machine decelerates.

(1)	'Longitudinal seam monitoring' screen page
(2)	Longitudinal seam monitoring ON/OFF
(3)	'Current temperature' display field
(4)	'Maximum temperature' text field
(5)	'Minimum temperature' text field



- ▶ Open the 'Overview' screen page at the operator terminal.
- ▶ Press the LONGITUDINAL SEAM MONITORING button.

  The 'Longitudinal seam monitoring' screen page (1) pops up.
- ▶ Switch on longitudinal seam monitoring (2) for the existing longitudinal seams.

  The relevant sensors are switched on.
- ▶ Read the currently ascertained temperature of the longitudinal seam from display field (3).

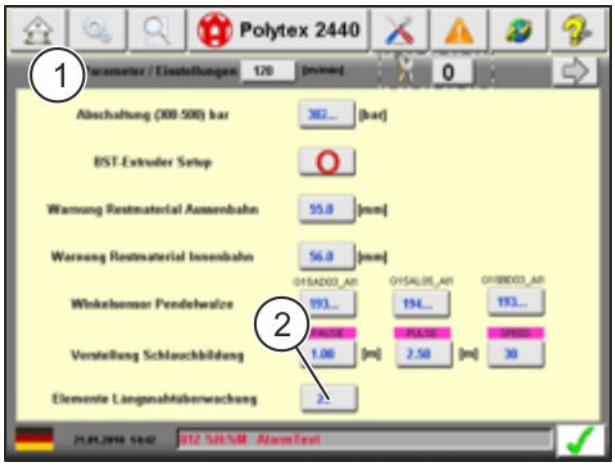
  If the temperatures display great differences, the sensors are differently aligned with the longitudinal seams. In this case, the sensors should be repositioned.
- ▶ Enter the maximum temperature into the text field 'MAX' (4).

  Enter a high temperature here. The temperature caps the monitored temperature range.
- ▶ Enter the minimum temperature into the text field 'MIN' (5).

  Enter the temperature of the longitudinal seam that should not be fallen below. This is the temperature of the longitudinal seam which ensures proper bonding of the tube. The temperature marks the lower limit of the monitored temperature range.

Setting up the sensitivity of longitudinal seam monitoring

(1)	'Parameters/settings' screen page
(2)	'Elements for longitudinal seam monitoring' text field



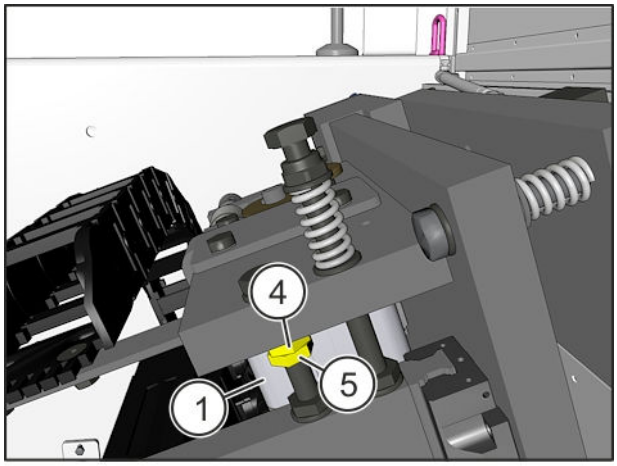
- ▶ Open the 'Parameters/settings' screen page at the operator terminal (1).
- ▶ Enter the number of measurements to be used as the values for ascertaining the temperature of the longitudinal seam into the text field 'Elements for longitudinal seam monitoring' (2).

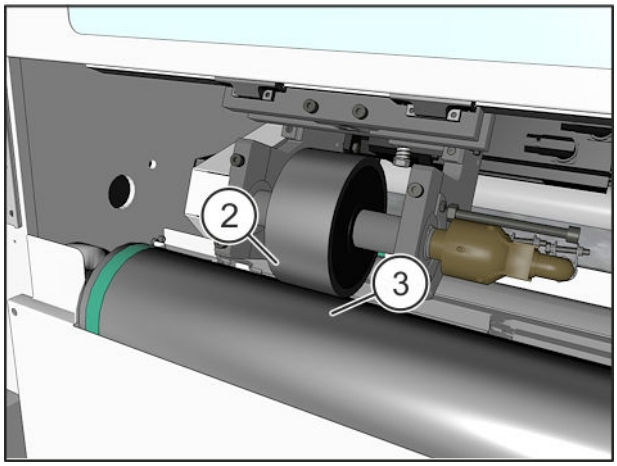
  High values reduce the measuring accuracy. The ascertained temperatures differ from the actual temperature of the longitudinal seam. Due to general measuring inaccuracies, small values will result in varying temperatures triggering false messages if the temperature leaves the set temperature range.
- ▶ Ascertain a value considering both criteria.

10.2 Setting up the seam cooling roller – Setting up the idle stroke

Setting up the idle stroke for the pneumatic cylinder.

(1)	Cylinder, seam cooling roller up and down
(2)	Seam cooling roller
(3)	Counter roller
(4)	Nut
(5)	Nut





- ▶ Extend the cylinder (1).

  The seam cooling roller (2) rests on the counter roller (3).

- ▶ Slacken nut (4) and nut (5).
- ▶ Turn the nuts downwards for limiting the idle stroke of the cylinder 1.

  Limit the idle stroke until the seam cooling roller does no longer rest on the counter roller when extending the cylinder.

- ▶ Tighten the nuts against each other.

10.3 Setting up the seam cooling roller

Basic setting for the seam cooling roller.

Safety instructions and general information

CAUTION

Lowering of the seam cooling roller.

Your hands may be caught between the seam cooling roller and the counter roller when lowering the seam cooling roller.

Do not reach in between the rollers. When setting up the seam cooling roller, use a large piece of paper to be able to maintain a sufficient distance from the seam cooling roller when lowered.

If it is found during production that the longitudinal seam is of poor quality, the seam cooling roller should be checked for proper positioning and a basic setting should be made, if necessary.

Make a basic setting after replacing the seam cooling roller.

The basic setting should be made according to the following sequence.

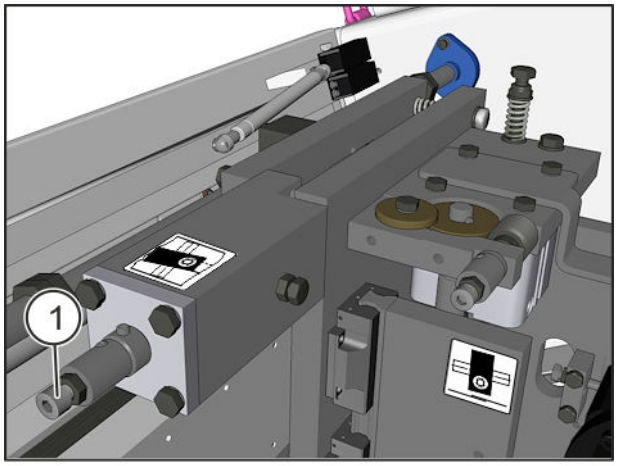
Aligning the seam cooling roller horizontally

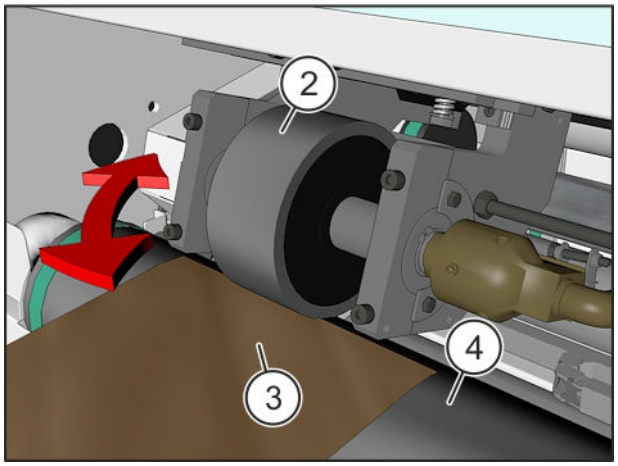
Check the seam coling roller for proper positioning to the counter roller and adjust.

Requirements:

The contact pressure of the seam cooling roller is set.

(1)	Screw
(2)	Seam cooling roller
(3)	Piece of paper
(4)	Counter roller





- Turn the screw (1).

  The holder with the seam cooling roller (2) is pivoted horizontally. See figure.

- Put a piece of paper (3) between the seam cooling roller and the counter roller (4).

- **CAUTION**
  Lowering of the seam cooling roller.
  Your hands may be caught. Do not reach in between the rollers.

  Lower the seam cooling roller.

  An indentation of the seam cooling roller appears on the paper.
  The indentation of the seam cooling roller should extend on the paper as a line parallel to the indentation of the counter roller.
  If the indentation does not leave a line on the paper but only a dot, the seam cooling roller needs vertical aligning.

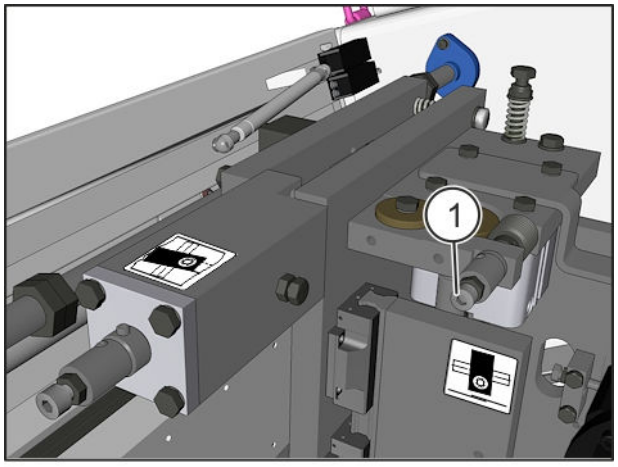
- Turn the screw (1) until both indetations on the paper are in parallel and on top of each other.

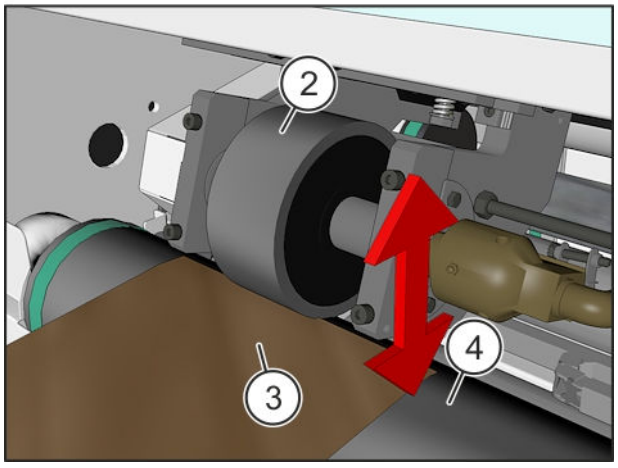
- Pivot the seam cooling roller away from the counter roller.

Aligning the seam cooling roller vertically

Check and adjust the spacing between the seam cooling roller and the counter roller.

(1)	Screw
(2)	Seam cooling roller
(3)	Piece of paper
(4)	Counter roller





- Turn the screw (1).

  The holder with the seam cooling roller (2) is pivoted vertically. See figure.

- Put a piece of paper (3) between the seam cooling roller and the counter roller (4).

- **CAUTION**
  Lowering of the seam cooling roller.
  Your hands may be caught. Do not reach in between the rollers.

  Lower the seam cooling roller.

  An indentation of the seam cooling roller appears on the paper.
  The indentation of the seam cooling roller should extend on the paper as a line and parallel to the indentation of the counter roller.
  If the indentation leaves no line parallel to the counter roller, the seam cooling roller needs horizontal aligning.

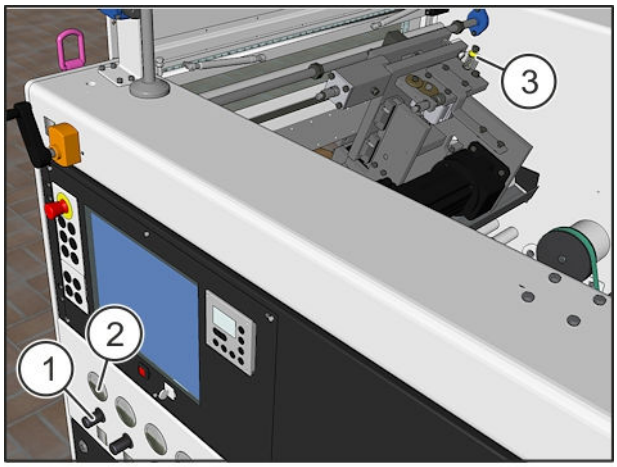
- Turn the screw (1) until both indentations on the paper are parallel and on top of each other and the indentation in the paper has the same depth over the entire length.

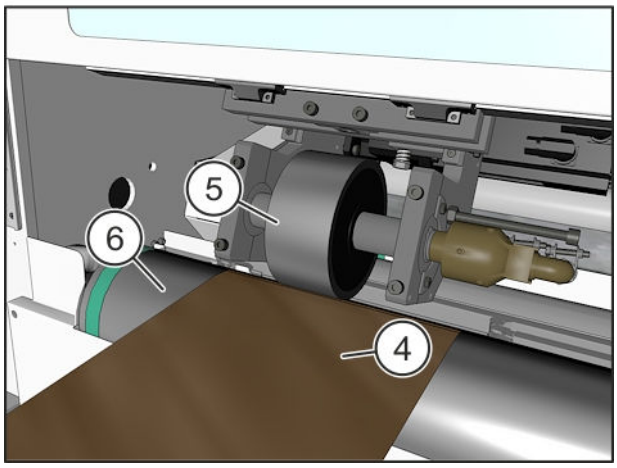
- Pivot the seam cooling roller away from the counter roller.

Setting the contact pressure

Set up the counter pressure to the contact pressure.

(1)	Knob
(2)	Pressure gauge
(3)	Nut and pressure spring
(4)	Paper
(5)	Seam cooling roller
(6)	Counter roller





- Turn the knob (1).

  The contact pressure of the seam cooling roller changes.

- Set the contact pressure to 1 bar (14.5 psi).

  Read the contact pressure from the pressure gauge (2).

- Turn the nut (3).

  The initial stress of the pressure spring changes and thus the counter pressure on the contact pressure of the seam cooling roller.

- Put a piece of paper (4) between the seam cooling roller (5) and the counter roller (6).

- **CAUTION**
  Lowering of the seam cooling roller.
  Your hands may be caught. Do not reach in between the rollers.

  Lower the seam cooling roller.

- Move the piece of paper two and fro between the seam cooling roller and the counter roller.

  If the paper:
  can be moved easily and without resistance between the rollers, the counter pressure should be increased.
  can be moved against resistance, the counter pressure is properly set.
  cannot or only be moved against great resistance, the counter pressure should be reduced.

- Turn the nut (3) until the paper can be moved against resistance between the seam cooling roller and the counter roller.

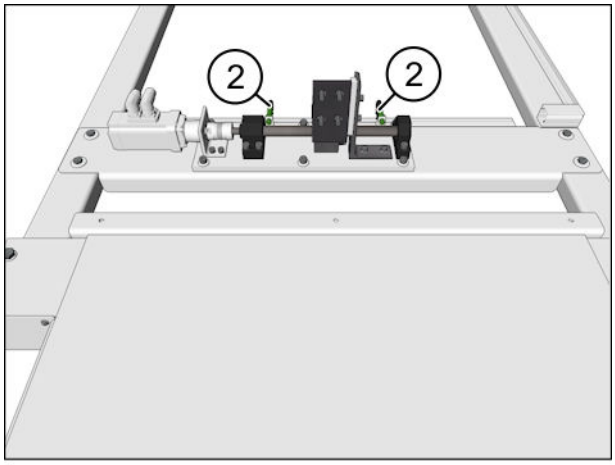
- Remove the paper and adjust the contact pressure to the production conditions.

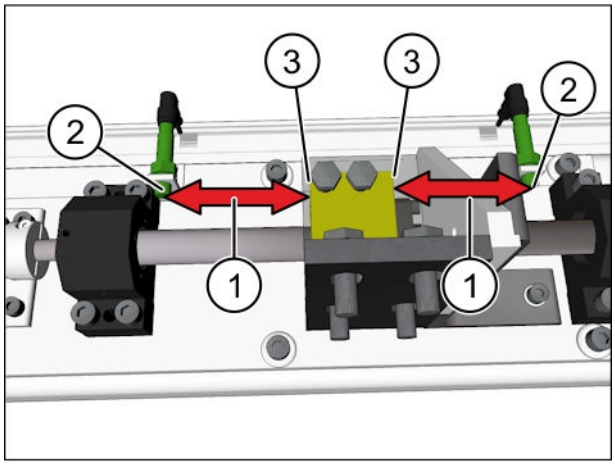
10.4 Oscillation

Setting the zero positon is required only after dismantling and reassembling the winder and the oscillating assembly.

(1)	Clearance
(2)	Sensor
(3)	Screw nut housing

▶ During the assembly, set up the winder and the oscillating assembly so that the clearance (1) between the mid-position of the sensors (2) and the respective edge of the screw nut housing (3) are the same.





After setting the zero position for the winder and the oscillating assembly, save this position via the 'Oscillation' screen page.

11 Putting the machine out of operation

11 Putting the machine out of operation

11.1 Assembly - Safety 401

11.2 Assembly instructions 0
Measuring the installation site 402
Transferring the reference lines 402
Installation 402
Securing the machine 402

11.3 Dismantling 403
Putting the machine out of operation 403
Dismantling 403

11.4 Recommissioning 404

11.5 Transportation 405
Instructions for safe transportation 405
Hoisting gear 405
Storing the machine 405

11.6 Corrosion protection 407
Corrosion protection 407
Cleaning 407

11.7 Disposal 408

11.1 Assembly - Safety

DANGER

Falling load

Falling loads may cause life-threatening injuries.

Do not walk below the suspended load. Wear a hardhat and industrial safety shoes. Keep a safe distance.

DANGER

Risk of falling at stairs and platforms

Insufficiently secured stairs and platforms are falling hazards. Risk of fatal injuries.

Fit railings or other safety barriers. Use a working platform. Put on safety harness.

DANGER

Heavy objects

Heavy objects may cause severe injuries when falling to the ground

Your body might be crushed. Wear a safety helmet and industrial safety shoes. Keep a safe distance.

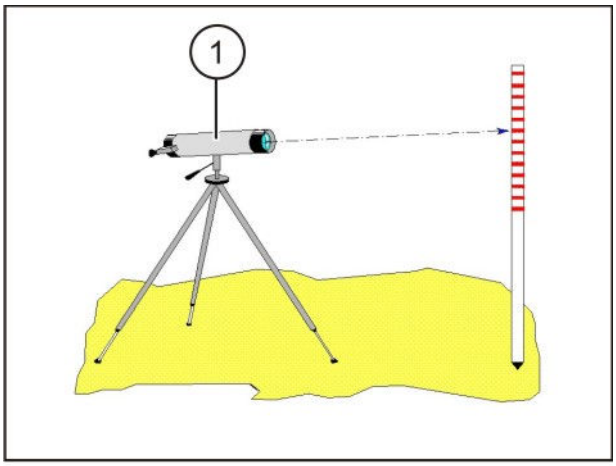
Follow the manufacturer's instructions for crane systems, lift trucks, fork lift trucks, cherry pickers and cargo harnesses.

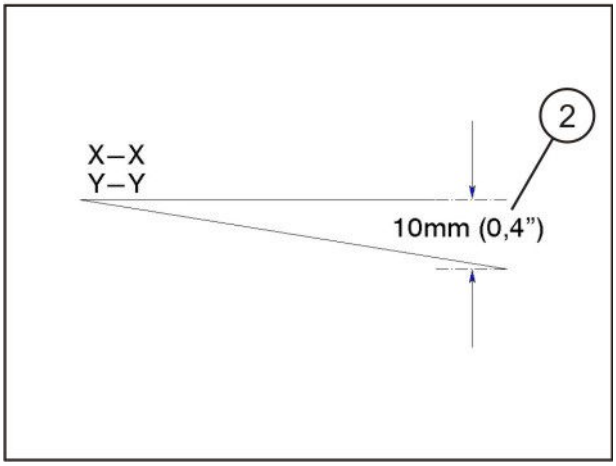
Do not exceed the permitted load bearing capacity.

11.2 Assembly instructions

Measuring the installation site

(1)	Measuring device
(2)	Level difference



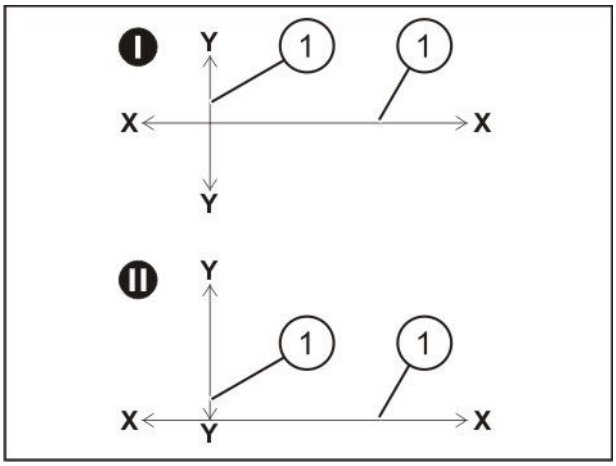


- Inspect the installation site for unevenness and level differences.
- Use an appropriate measuring device (1).

  The level difference (2) along the measured paths X-X and Y-Y must not exceed 10millimeters (0.4”), over the length and width of the machine.

Transferring the reference lines

(1)	Cord



- Transfer reference lines X-X and Y-Y (1) from the installation and foundation loading plan to the installation site using cords in accordance with option I or II.
- Secure the cords in place so that they cannot be moved.

Installation

NOTE

Poorly aligned machinery will result in reduced production quality.

Machines operated in unison must always be aligned with each other.

Great care must be taken when undertaking all of the work.

The machine must be installed in accordance with details given in the installation and foundation loading plan.

Securing the machine

NOTE

Avoid damage during operation.

At the end of installation, tighten the securing nuts and lock.

Details regarding the securing components to be used such as dowels, mounting angles, etc., can be found in the installation and foundation loading plan.

11.3 Dismantling

Putting the machine out of operation

- Run the machine until empty before putting it out of operation.
- Remove all operating materials.
- Switch off the mains switch to de-energize the machine and secure it against being switched back on inadvertently.
- Disconnect the machine from the mains.
- Disconnect the compressed air supply and secure it against being switched back on inadvertently.
- Depressurize the hydraulic units.
- Disconnect the machine from the water connection.
- Leave the machine to cool down.
- Clean the machine.
- Preserve the machine if it is to be shut down for a longer period of time.
- Turn the locking bolts out of the floor.
- Lock the machine in place to prevent dangers to employees and third parties from occurring.
- Mark the shut off machine accordingly.

Dismantling

Dismantling is carried out as follows:

- Components that can be re-used
- Secondary feedstock
- Harmful substances
- Residual waste

11.4 Recommissioning

Before recommissioning of the machine after a longer downtime, after reconstruction or reinstallation, the following safety devices must be checked by an industrial safety officer for proper functioning:

- Emergency-Stop devices
- safety locks
- guard door chains
- two-hand switching mechanisms
- safety shutdown mats
- safety light grid

11.5 Transportation

Instructions for safe transportation

Transportation - General

- You must abide by the pertinent regulations and safety requirements.
- Transport the machine with care in order to prevent danger to persons and damage to the machine.
- Transport the machine and the machine parts with load transporting tools of sufficient load bearing capacity.
- Use a cargo handling gear when a crane is used for machine transport.
- The side walls or attached machine parts may not be pressed together by angled ropes or chains.
- When transporting the machine on a cargo area, secure the machine against tipping and slipping.
- Do not place the machine on its protective hoods or drives.

Personnel

- The operator must be duly trained, instructed and authorized in the transportation and assembly including lifting operations.
- If necessary, a banksman should be duly trained, instructed and authorized.

Dimensions and weights

- See the installation and foundation load drawings for the dimensions and weights of the machine and individual modules.

Sling points

- See the installation drawing, the foundation load drawing, the transport and handling plan or the rating plates on the respective units for the sling points and the load bearing capacity.
- The machine itself must not be transported but on the intended sling points.As an alternative, round slings may also be used.
- Load hooks of hoisting gears should not be hooked into the load directly.
- Do not attach any additional sling points to the machine by welding, firing or drilling.

Packaged machine

- Observe the attached symbols for transportation and storage when the machine is packaged.
- If the machine is packaged then only attach the loading harness to the locations indicated.

After the assembly

- After the installation, transport protections should be removed, marked and put aside for later dismantling.
- Before initial startup, any parts that were removed prior to transportation must be refitted.
- Any transport damage must be reported to the manufacturer and the transport company immediately.

Hoisting gear

DANGER

Falling load

Falling loads may cause life-threatening injuries.

Do not walk below the suspended load. Wear a hardhat and industrial safety shoes. Keep a safe distance.

DANGER

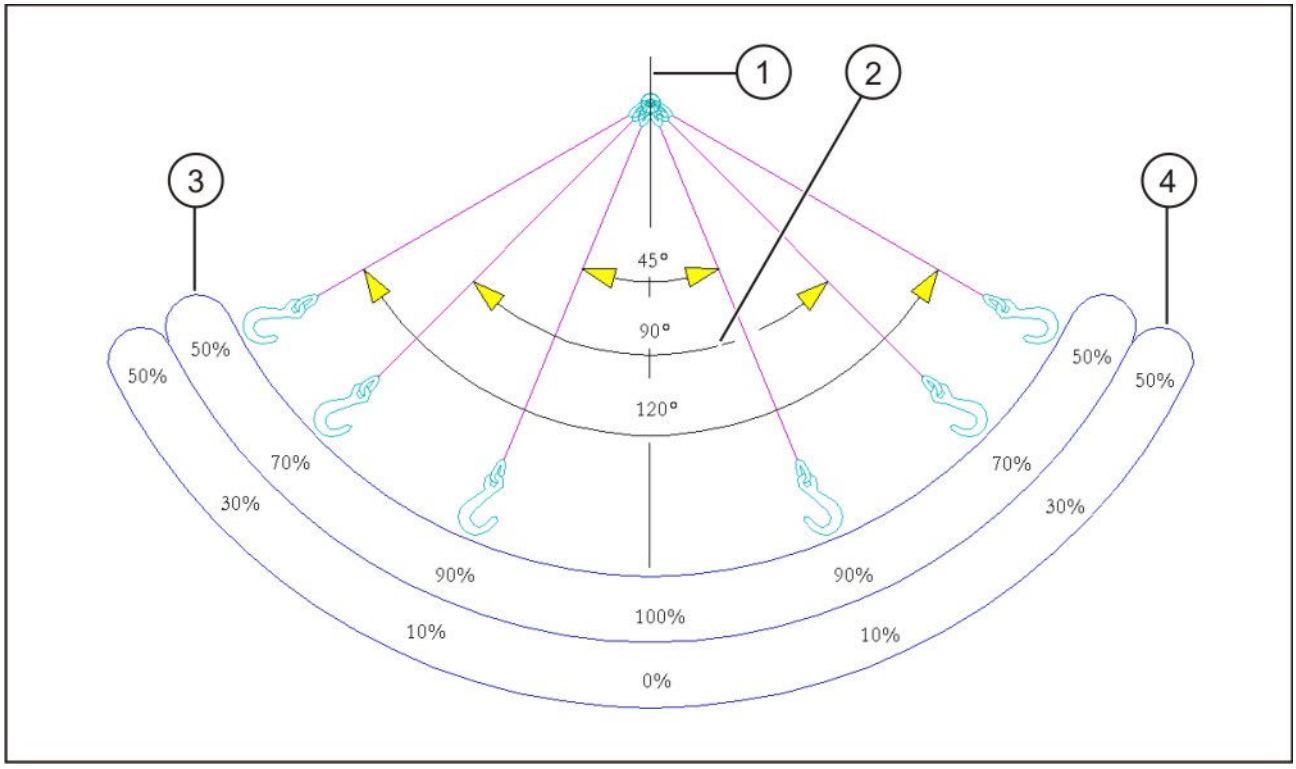
Moving machine parts

Risk of crushing or shearing parts of your body.

Never reach in between moving machine parts. Do not bypass the safety arrangement. Keep a safe distance.

During assembly, dismantling and loading works, make sure to use hoisting gears and load suspension devices with sufficient load bearing capacity.

Note that when using a cargo harness, the load bearing capacity will be reduced when the lifting cables are spread out.



(1)	Reference line for spreading angle, loading and reduction	(2)	Spreading angle of the lifting cables (in angular degrees)
(3)	Load bearing capacity (in percent) dependent on the spreading angle	(4)	Load reduction (in percent)

Storing the machine

- Store the machine in a closed location until final assembly.
- When storing the machine outside, cover it with tarpaulins so that condensed water can run off.
- Do not damage or open sea-worthy packaging of the machine for transport and storage.
- Place the machine on water-proof underlays in order to protect it against the effects of soil humidity.
- Always store electrical devices in closed, dry and frost-free rooms.
- Store loose parts included in the delivery in a dry location.
- Leave the machine to be stored in assembly condition. The manufacturer is not liable for damages resulting from an incorrectly disassembly machine.
- Observe the symbols attached to the packaging in order to ensure correct storage of the machine.

11.6 Corrosion protection

Corrosion protection

The corrosion protection applied in the factory is a concentrate made from organic corrosion inhibitors, bound in a carrier medium of naphthenic mineral oil. The corrosion protection is free from solvents, nitrites, phosphates and chromium. It contains no PCBs or PCTs. Product name: CORTEC VpCI 329 and CORTEC VpCI 329F.

If the machine is to be shutdown for a long period, all corrosion-prone surfaces must be protected against corrosion.

If a surface protected against corrosion has to be cleaned off during production or when maintenance or service work is being carried out, it anti-corrosion protection must be re-applied to this surface.

Open gearwheels must be treated in accordance with the lubricating instructions.

Cleaning

BEWARE!

All of the used cleaning agents and cleaning waste are classified as hazardous waste.

Hazardous waste must be disposed of in accordance with local regulations and laws.

Only work with substances that will not attack plastic or rubber surfaces and will not dissolve the adhesive positions on the conveyer belts.

Use an alkaline based cleaner (tenside water) to clean off the corrosion protection applied in the factory to all non-treated surfaces or remove by wiping off before initial machine startup.

11.7 Disposal

If necessary, free the machine from hazardous substances in compliance with the pertaining regulations.

Drain and dispose of all lubricating fluids according to the pertaining regulations.

The materials should be disposed of according to the local waste disposal regulations.

12 Spare parts and documents

12 Spare parts and documents

12.1 PDE - Parts and Documentation Explorer 411

12.1 PDE - Parts and Documentation Explorer

Spare parts lists and further documentation can be found in the PDE - Parts and Documentation Explorer on the supplied USB flash drive.

The USB flash drive is located in the folder of the instruction manual.

The PDE contains the following documents:

- instruction manuals
- circuit diagrams
- mechanical drawings
- spare parts lists
- sub-vendor documentation
- other documents (optional)